FDDT

Fraud Document Detection Tool

Complete project documentation

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FDDT — Fraud Document Detection Tool

(Formerly the "Document Authenticator Tool" / DocAuth. The product name is a display-level rename only — the repo folder, `docauth` database/containers and code identifiers are unchanged. Its single source is `APP_NAME` / `APP_FULL_NAME` in `backend/app/core/config.py` and `frontend/src/lib/appInfo.ts`.)

AI-powered document authentication, fraud detection, and case-management platform. Full documentation — architecture, every pipeline stage, database schema, API reference and deployment notes — is in `docs/`.

What it does: users submit a case of supporting PDF documents (vendor invoices, school documents, quotations, travel claims, payment evidence) — each upload is validated first (PDF only, ≤ 10 MB, not corrupted, not password-protected). A Celery pipeline OCRs each document (Azure Document Intelligence), then classifies it and extracts its fields (Azure OpenAI, behind the `LLMService` abstraction — see `app/services/llm_service.py`). It then runs field validation, issuer verification, cross-document consistency checks, and PDF forensics (metadata, Error Level Analysis, copy-move, duplicate detection, visual-inconsistency review, signature/stamp detection). A transparent, weighted rules engine turns the results into an explainable risk score, and the case goes to a Reviewer L1 who approves, rejects or escalates it; an escalated case moves to the Reviewer L2 tier, which resolves it. Every step is written to an append-only audit log, and reviewers can export a per-case forensic report as PDF.

Stack: FastAPI + PostgreSQL (Row-Level Security, PgBouncer) + Celery/Redis (three queues, fair-share, global Azure rate limits) backend; React + TypeScript (Vite) frontend with login, case queue, case detail (live processing status, overlays, risk reasons, audit timeline), new-case/upload, company settings and platform-admin screens. Multi-tenant: several client companies share one deployment, each seeing only its own data.

Prerequisites

- Docker Desktop (for Postgres, Redis and PgBouncer)
- Python 3.11+ (developed/tested against 3.14)
- Node.js 20+ (developed/tested against 24)

Setup

```
# 1. Start Postgres + Redis + PgBouncer
docker compose up -d

# 2. Backend: create venv, install deps, configure env
cd backend
python -m venv .venv
./.venv/Scripts/activate        # Windows
# source .venv/bin/activate     # macOS/Linux
pip install -r requirements.txt
cp .env.example .env            # edit if you changed any docker-compose ports/creds

# 3. Apply migrations
alembic upgrade head

# 4. Seed the platform admin (admin@example.com / ChangeMe123!)
python seed.py

# 5. Run the API
uvicorn app.main:app --reload
```

API docs: http://localhost:8000/docs

```
# 6. Celery workers — one per processing queue, each in its own terminal
#    (scale each independently; see docs/processing-queues.md)
cd backend
./.venv/Scripts/activate   # or the equivalent for your shell
celery -A app.tasks.celery_app worker -Q extraction_queue -n extraction@%h --pool=threads -c 4 --
loglevel=info
celery -A app.tasks.celery_app worker -Q vision_queue     -n vision@%h     --pool=threads -c 4 --
loglevel=info
celery -A app.tasks.celery_app worker -Q forensics_queue,celery -n forensics@%h --pool=threads -c 4 --
loglevel=info
# optional: the scheduler (nightly usage reconciliation, queue metrics log line)
celery -A app.tasks.celery_app beat --loglevel=info
```

On Linux/macOS use `./start-worker.sh extraction|vision|forensics` and `./start-beat.sh` (pool sizes from `*_WORKER_CONCURRENCY`; forensics uses prefork, one process per core). On Windows use `--pool=threads` (prefork doesn't work there). Or run them in Docker: `docker compose --profile workers up -d --scale forensics-worker=3`. Without the workers, uploaded documents sit at `processing_status: "pending"` forever — the API only enqueues the jobs.

PgBouncer (recommended): `docker compose up -d` also starts PgBouncer on :6432 (transaction pooling). Set `DATABASE_POOLER_HOST=localhost` in `backend/.env` to route the app through it (each process then keeps no pool of its own). See `docs/processing-queues.md` for the connection arithmetic, fair-share scheduling and the calibrated Azure rate limits (`python -m scripts.calibrate_llm_tokens` re-measures real token usage).

Multi-tenancy: `admin@example.com` is a *platform admin* (no company). Create a company under **Platform › Companies** and its users under **Platform › Users**, then sign in as one of them to submit cases. The running app connects to Postgres as the RLS-enforced `fddt_app` role created by the migration (`DATABASE_APP_*` in `.env`). See `docs/multi-tenancy.md`.

```
# 7. In a third terminal: frontend
cd frontend
npm install
npm run dev
```

Open http://localhost:5173 and sign in with `admin@example.com` / `ChangeMe123!`. See `frontend/README.md` for details, and `docs/deployment/production-deployment.md` for the production plan on Azure (architecture, service tiers, costs, runbook). The demo is live on Azure: https://docauth-web.icysand-160befae.eastus.azurecontainerapps.io/

Login: admin@example.com / tTHt0pRc1ISDF6!9

Verifying the document processing pipeline

1. Confirm `AZURE_DOCUMENT_INTELLIGENCE_ENDPOINT` / `_KEY` and `AZURE_OPENAI_KEY` / `_ENDPOINT` / `_DEPLOYMENT_NAME` are set in `backend/.env` (see `.env.example`) and the Celery worker (step 6 above) is running.
2. In the UI (as a company user): submit a new case, upload a PDF, then open the case detail page. Each document shows a status badge (Pending → Processing… → Complete/Failed) that updates on its own every few seconds while still in flight (the page polls `GET /cases/{id}` — no manual refresh needed). Once complete, the document type, its core fields (issuer/date/amount/reference #), and every other field the model found are shown inline, each flagged `uncertain` if the model wasn't confident.
3. To inspect the raw result directly (e.g. to review extraction quality against a source document, including Arabic text) instead of via the UI:

```
# Case detail via API — includes document_type/processing_status/
# extracted_fields per document:
curl http://localhost:8000/cases/<case_id> -H "Authorization: Bearer <token>"

# Or straight from Postgres, including the full OCR text
# (documents.ocr_text) that the LLM actually saw — useful for
# checking OCR accuracy on its own, independent of the LLM step:
docker exec docauth-postgres psql -U docauth -d docauth -c \
  "SELECT document_type, processing_status, extracted_fields, ocr_text FROM documents WHERE id =
'<document_id>';"
```

4. A failed document shows its `processing_error` inline in the UI (and in `documents.processing_error` / the `audit_log` table) instead of silently staying incomplete.
5. `backend/tests/test_document_processing.py` covers the task's success/ failure/unknown-id logic against mocked OCR/LLM services (no live Azure calls, no Celery worker needed) — run it with the rest of the suite via `pytest`. It does not (and cannot) verify real-world OCR/LLM accuracy — that has to be checked against actual sample documents, per step 3 above.

Case report export

Reviewers can click **Export report** on Case Detail to generate a standalone PDF (`POST /cases/{id}/reports` ; history at `GET /cases/{id}/reports`). Sections, in order: 1 Case Details, 2 Executive Summary, 3 Explainable Findings, 4 All Checks, 5 Exceptions, 6 Limitations and Assumptions, 7 Audit Trail, 8 Appendix (fields, SHA-256 hashes, technical parameters), 9 PDF Highlighted Regions. Every highlight has a region ID (R1, R2, …) that Section 5 links to. Generation is synchronous (`app/services/case_report_*.py`). Reports are stored in Blob Storage under `companies/{company}/cases/{case}/reports/` with a row in `case_reports` ; every export adds a new report, none are overwritten. The findings drawn onto the report's page renders are separate from the live Case Detail overlays, which never save anything, and the original files are only ever read. Run `alembic upgrade head` to create the `case_reports` table.

Highlight colours (live overlays and the report use the same convention): solid red = ELA, solid orange = copy-move, dashed blue = model judgment (visual review, signature — "approximate"), solid purple = rule-based field exception (a failed field-validation sub-check, or a cross-document mismatch, shown on **both** documents).

Field locations

Field exceptions can only be highlighted if each extracted field's position is known. At extraction time the OCR word boxes are now kept long enough to match each extracted value back onto the page, and the result is stored on the field as a normalized `bounding_box` (`app/services/field_locator_service.py`). This is text matching, so positions are approximate; a field that can't be matched simply has no highlight. Documents extracted before this existed can be backfilled (re-runs only the Azure Layout call — no LLM, no value changes):

```
cd backend
python backfill_field_locations.py CASE-XXXXXXXX      # or no args for every document
```

Tests

```
cd backend
pytest
```

Tests run against an in-memory SQLite DB, independent of the Postgres container, so `pytest` works without `docker compose up`. The Row-Level Security, PgBouncer and rate-limiter suites need the real services — see `docs/deployment/local-setup.md`.

Project layout

```
backend/    FastAPI app, Celery tasks, Alembic migrations, tests
frontend/   React + TypeScript SPA (Vite)
docs/       Architecture, pipeline, database, API and deployment docs
sample-documents/   Sample PDFs for testing
docker-compose.yml  Local PostgreSQL + Redis + PgBouncer (+ optional Linux workers)
pgbouncer/          PgBouncer config for local development
```

See `docs/README.md` for the full documentation index.

FDDT documentation

FDDT — Fraud Document Detection Tool authenticates third-party documents (vendor invoices, school documents, quotations, travel claims and the payment evidence behind them) *before* an approval or payment is made. A user submits a **case** of PDF documents; the system extracts their data, runs a battery of automated checks, turns the results into an **explainable** risk score, and hands the case to a human reviewer (Reviewer L1, with escalation routing a case to Reviewer L2). Every step is written to an append-only audit log. Several client companies share one deployment, each isolated from the others by the application and by PostgreSQL Row-Level Security.

These docs describe the system **as implemented**, checked against the code. The capabilities included in the current build, and those outside its scope, are collected in the table in architecture/overview.md.

Start here

If you want to…	Read
Learn to use the app as a submitter, Reviewer L1 / L2 or platform admin	user-manual.md
Understand the whole system in ten minutes	architecture/overview.md
Understand companies, roles and tenant isolation (RLS)	multi-tenancy.md
Tune processing queues, worker pools and Azure rate limits	processing-queues.md
Follow one case from upload to report	architecture/data-flow.md
Run it on your machine	deployment/local-setup.md
Know what to configure and where to get keys	deployment/environment-variables.md
Understand what each check detects and its limits	the pipeline docs below
Call or change the API	api/README.md → Swagger at `http://localhost:8000/docs`
Change the database	database/schema.md, database/erd.md
Work on the UI	frontend/screens.md, frontend/components.md
Deploy to production on Azure (architecture, service tiers, costs, step-by-step runbook)	deployment/production-deployment.md
See the remaining pre-production gaps	deployment/azure-migration-notes.md
See how the Azure demo is deployed	deployment/azure-demo-deployment.md
Read what changed in the multi-tenancy / queues work, and why	MULTI_TENANCY_CHANGE_REPORT.md

Contents

User manual

- user-manual.md — role-by-role guide (submitter / Reviewer L1 / Reviewer L2 / platform admin) to using the application, in plain language: submitting a case, the review queue and case detail page, what each automated check means and how far to trust it, the risk score, approve/reject/escalate, the forensic report, company settings, platform administration, and the tool's known limitations. Also published as `docs-pdf/26-User-Manual.pdf`.

Architecture

- overview.md — system diagram, tech stack, the key design decisions and their rationale, the scope-of-the-current-build table, security posture.
- data-flow.md — sequence diagram and step-by-step walkthrough, Celery task names and queues.
- multi-tenancy.md — companies, roles, two-layer tenant isolation (application + RLS), platform-admin support access, blob layout, billing & usage.
- processing-queues.md — the three queues, fair-share scheduling, idempotent writes, calibrated Azure rate limits, database connections and PgBouncer, load tests.

Pipeline — what each stage does

Each page covers what it detects, the algorithm and libraries, the real input/output shape, known limitations and the risk rules it feeds.

1. Upload & intake
2. Extraction & normalization
3. Field validation
4. Cross-document consistency
5. Issuer verification
6. Metadata forensics
7. ELA & copy-move
8. Visual review & AI-generation
9. Signature verification
10. Duplicate detection
11. Risk scoring engine

Reference

- API + `openapi.json` (generated from the code)
- Database schema and ERD (generated from the live database by `tools/gen_db_docs.py`)
- Frontend screens and components
- Per-case forensic report

Deployment

- Local setup · Environment variables
- Production deployment — Azure Container Apps architecture, S/M/L service tiers with monthly costs (UAE North list prices), deployment runbook, operations and go-live checklist
- Azure migration notes — what is Azure-ready and the remaining gaps
- Azure demo deployment — the low-cost demo topology and operating notes

Change records

- MULTI_TENANCY_CHANGE_REPORT.md — every change, decision and problem in the multi-tenancy, processing-queue and follow-up work

Ground rules the design follows

These principles explain many choices you will meet in the code:

- **No model is trained or hosted.** Intelligence is a managed cloud API or a classical library.

- **A transparent rules engine, not a model, produces the risk score**, with versioned, admin-tunable rules.
- **One generic, template-free extraction pipeline** for every document type.
- **Original files are immutable** and hash-verified; everything else derives from them.
- **The audit log is append-only.**
- **Weak signals are labelled weak** — signature comparison, AI-generation detection and ELA are advisory, never verdicts, and never described as "verified".
- **The UI is English-only**; only extracted Arabic values need right-to-left handling.

Keeping these docs true

- The **OpenAPI snapshot** and the **database docs** are generated — regenerate them rather than editing: `api/openapi.json` (command in api/README.md); `database/*.md` with `cd backend && .venv/Scripts/python ../docs/tools/gen_db_docs.py` against the migrated local Postgres — re-run after every migration.
- The **pipeline docs** quote real constants, thresholds and result shapes. When you change an algorithm or a default rule, update its page and the rule tables (the default rules live in `backend/app/services/risk_rule_seed.py` and the platform's `risk_rule_templates`).
- When you add or change a capability listed in the overview's scope table, update that table.
- The **PDFs in `docs-pdf/`** are built from these Markdown files (and `api/openapi.json`) — never edit them by hand. After changing any doc, rebuild: `cd docs/tools && npm install && npm run build` (see tools/README.md).

FDDT User Manual

Fraud Document Detection Tool — a guide for submitters, reviewers and platform administrators.

This manual describes the application **as implemented** (see `docs/README.md` for the technical documentation this is derived from). It covers what each of the four roles can do, how a case moves from upload to decision, how to read the automated checks and the risk score, and — deliberately — what the tool does **not** do, so nobody relies on it for more than it promises.

> **One deployment, several companies.** The application serves several client companies at once. Every user belongs to exactly one company and only ever sees that company's cases, documents, settings and audit log. The platform operator's own staff (platform admins) sit outside every company.
>
> *Screenshots were captured before multi-company support was added; some labels (for example "Admin") are older than the text, but the screens work as described here.*

Who this is for

Role	Can do
User (submitter)	Submit a case with supporting documents, track its status, see the reviewer's decision. Never sees the risk score, tier or triggered reasons — only a neutral "In review" flag — so the checks that catch fraud are never exposed to the person submitting the documents.
Reviewer L1	Everything a user can do, plus: the full review queue and dashboard, every case's automated findings and risk score, approve / reject / escalate, audit history, and exporting the per-case forensic report. Once a case is escalated to L2, a Reviewer L1 can still open and read it but can no longer approve, reject or re-escalate it.
Reviewer L2	Everything a Reviewer L1 can do, plus: approve or reject cases that have been escalated to L2, and manage **your company's** Issuer Registry and Risk Rules (Settings).
Platform Admin	Belongs to no company. Creates companies and every user account, edits the default rule set new companies start from, sees billing & usage per company and the processing-queue monitor. Can **read** any company's cases, audit log and settings for support — every such view is recorded in a platform-only log — but can never approve, reject, escalate, upload, create cases or export reports.

Every rule above is enforced by the backend on every request, not just hidden in the interface — reaching a screen you're not permitted to use still returns "access required" or a 404, never data. A case or document of another company simply does not exist for you (404).

Signing in

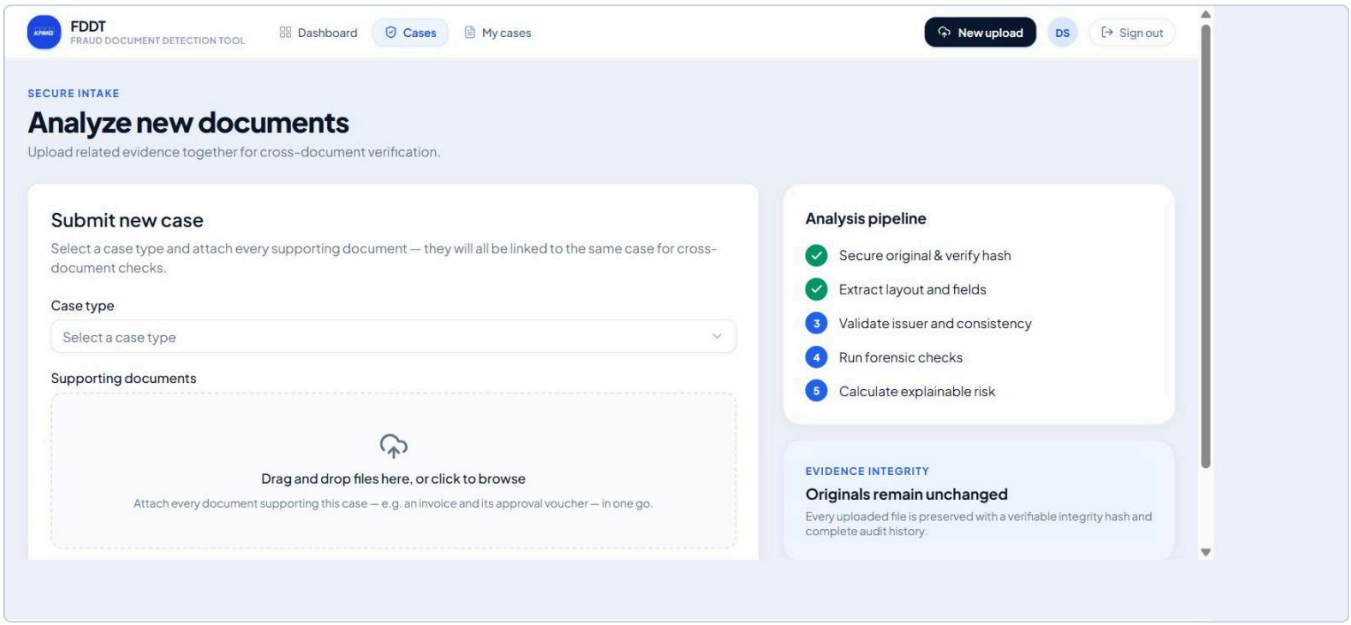
Go to the application URL and sign in with the email and password a platform admin created for you (there is no self-service sign-up, and no password-reset link — if you're locked out, ask the platform operator). The top bar shows your company and role. Sessions are a plain bearer token (8 hours) with no server-side logout. If your company is suspended, sign-in stops working for everyone in it.

> The sign-in screen also shows a "Trust this device" checkbox and a "Sign in with SAML / Okta SSO" button. Neither does anything: the checkbox isn't wired to the login form (authentication doesn't have a device-trust concept to back it), and the SSO button is disabled. Treat the screen as plain email/password sign-in only — the badges and extra controls are decorative UI that hasn't been wired up or removed yet.

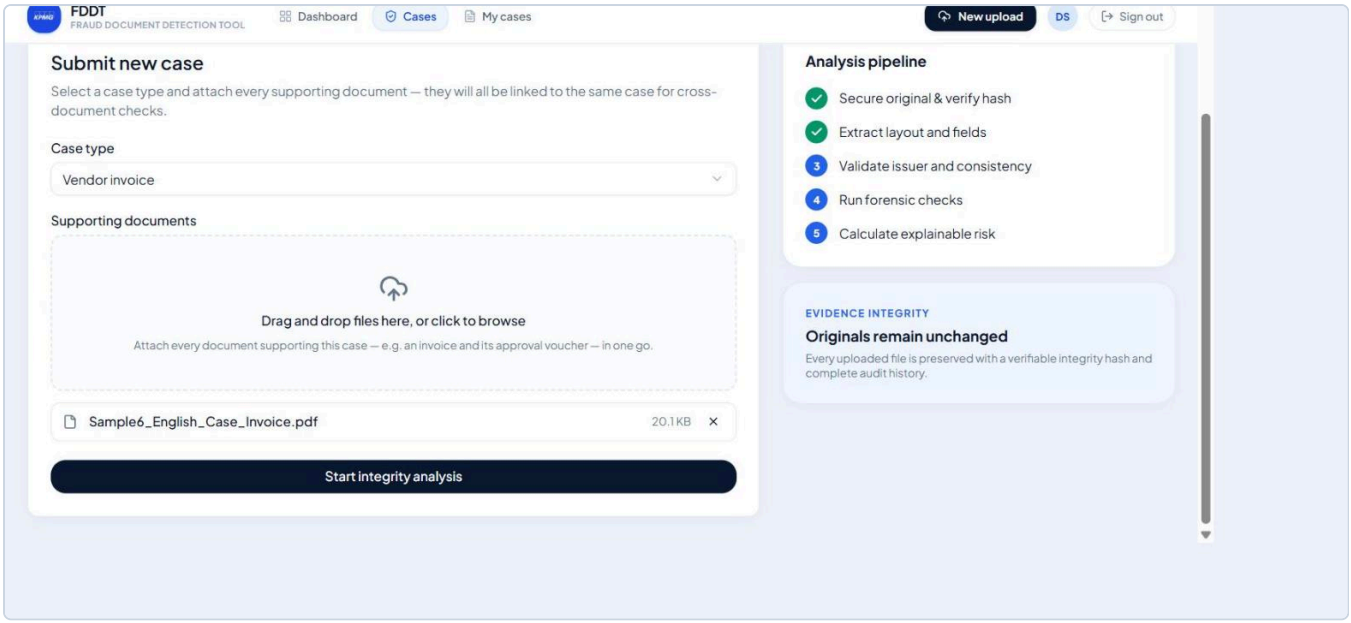
Submitting a case (User)

1. Open **New Case**.
2. Choose the case type: *School / educational document*, *Vendor invoice*, *Commercial invoice*, *Procurement documentation*, *Quotation*, *Travel / accommodation reimbursement*, or *Other*.
3. Attach the supporting document(s) — the original invoice, receipt, quotation, evidence of payment, etc. **Only PDF files are accepted, up to 10 MB each.** If you have a photo or scan as an image, save or scan it to PDF first. A file is refused immediately, with the reason shown next to it, if it is:
   - not a PDF (for example a JPEG, PNG or Word file, even if renamed to `.pdf`);
   - empty, or larger than 10 MB (the message shows the file's size and the limit);
   - **password-protected** — remove the password and upload it again (the system never asks for document passwords);
   - corrupted or cut off (for example a partly downloaded file).
4. Upload. Each file uploads independently with its own progress bar; if one fails you can fix it and retry without losing the others, remove the rejected file by clicking its `[X]` icon to proceed with the remaining documents, or click **Reset form** to clear the intake form and start fresh.
5. **Optional — set a reference signature:** for an uploaded document, you can draw a box around a signature or stamp so the system compares it against the one on other documents. Skip this and no signature comparison runs for that document.
6. The case now moves into the automated pipeline, and you're taken to its detail page.

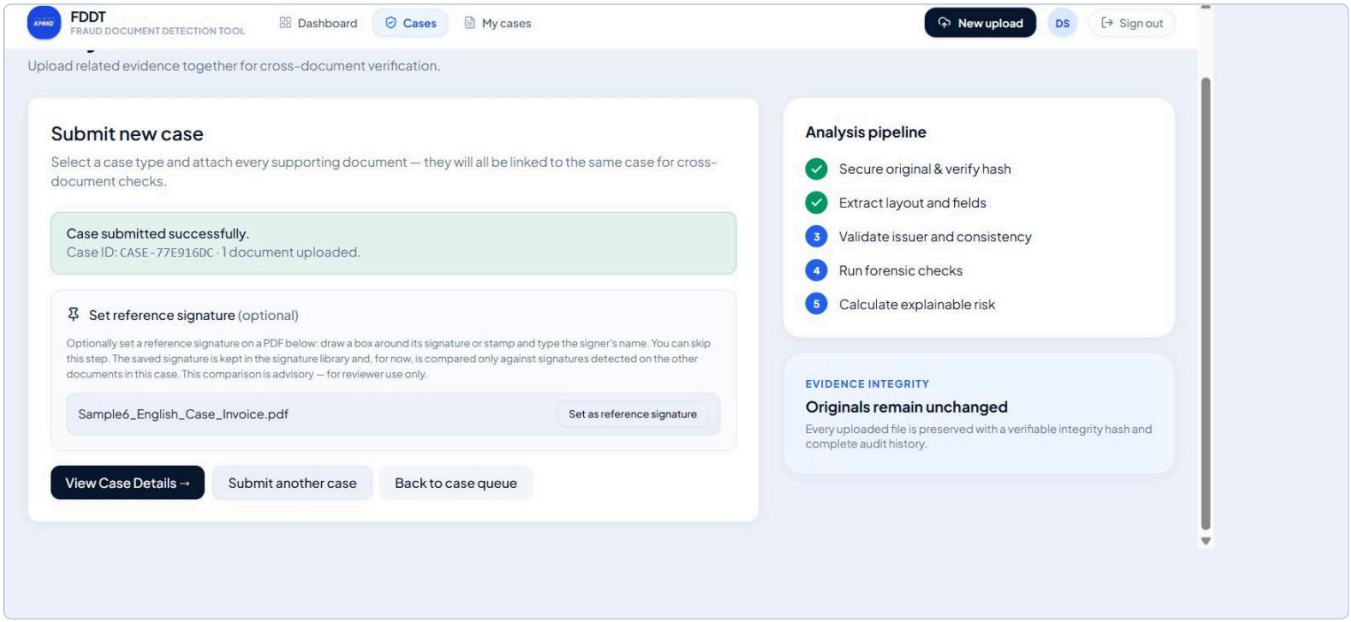
Attach every document of a case in this one submission — there is no "add a document" button on an existing case.



The New Case form



A case type chosen and a document attached, ready to submit



Confirmation after submission, with the optional reference-signature step

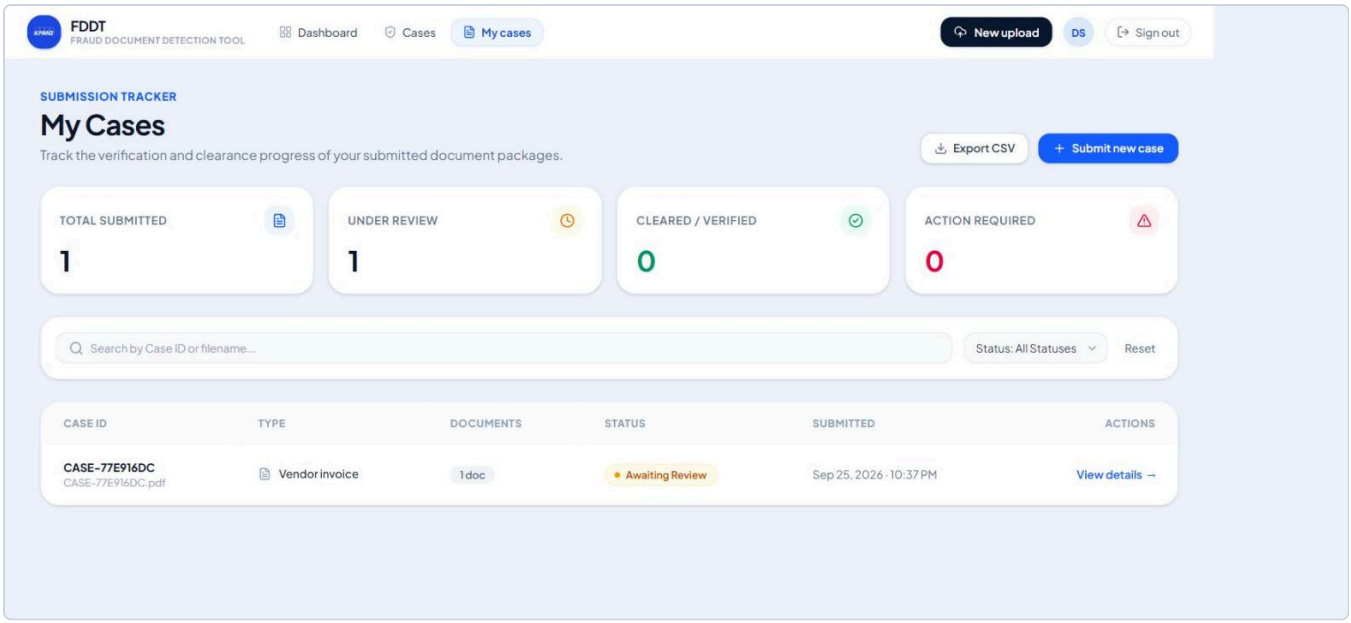
Tracking your cases (User)

My Cases lists everything you've submitted, filterable by *Under review*, *Cleared* or *Action required*, with search. You'll see the outcome (approved / rejected, with the reviewer's reason when rejected) but not the risk detail behind it — that's intentional (see below).

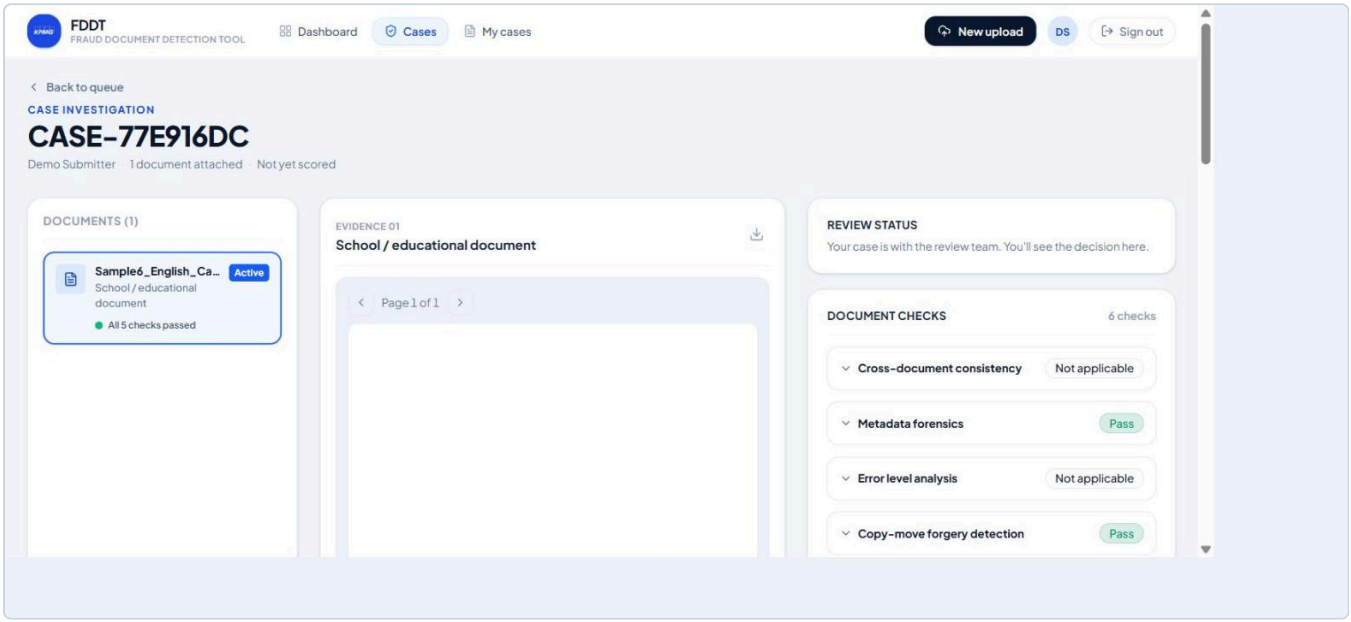
Opening one of your cases shows its documents, their extracted fields, and an activity timeline, but:

- No risk score, tier or list of triggered findings.
- No reviewer decision controls beyond a read-only history of what happened.

This is deliberate: if a submitter could see exactly which signals fired, a bad actor could learn what to avoid next time.



My Cases: a submitter's own cases and their status



A submitter's view of their own case — no risk score or tier, by design

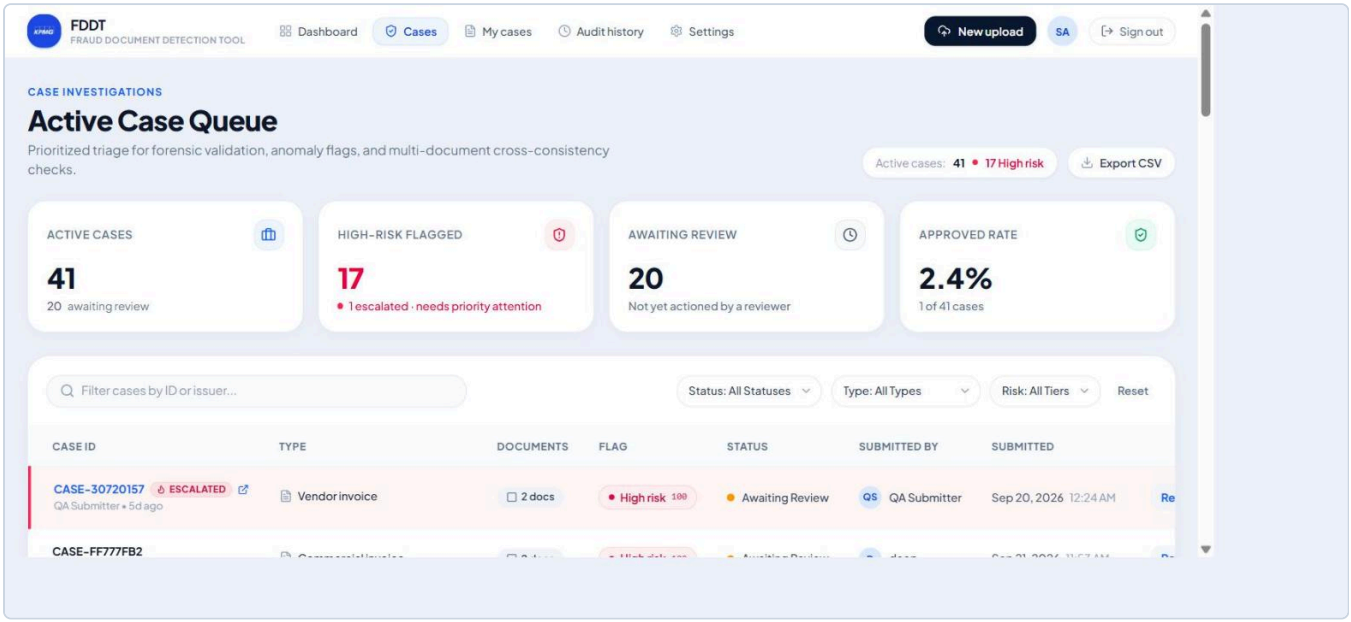
The review queue & dashboard (Reviewers)

Review Queue (also the home page, `/cases`) is the master case list: counts of open / awaiting review / cleared / escalated cases, and a table of cases with their type, status, risk flag, escalation badge, document count and submitter. Search and the status / type / flag filters run instantly in your browser.

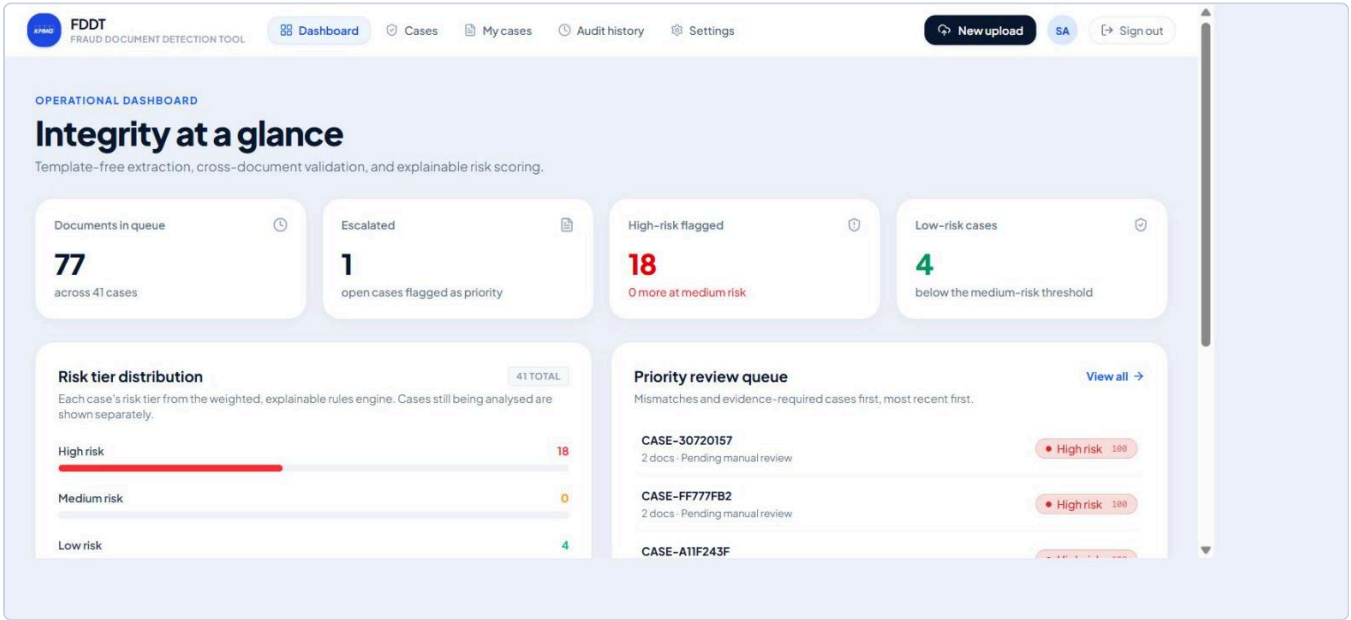
- **Reviewer L1** sees two tabs: **My queue** (the cases you can act on — the default) and **Escalated to L2 · view only** (cases handed to L2; you can open them to follow what happened, but not act on them).
- **Reviewer L2** sees **All cases** (the default — open L2 cases always sort to the top) and **Escalated queue (L2)** (only escalated cases).
- **Platform admins** pick one company at a time (company picker at the top) and see that company's list read-only.

> **"Escalated" is a reviewer tier, not a status.** An escalated case can be `submitted`, `pending_manual_review`, or already decided — escalation means the case now belongs to the Reviewer L2 tier; it doesn't change what state the case is in.

Dashboard is an at-a-glance operational summary computed entirely from the same case list: counts by risk flag, a risk-distribution panel, your top five open cases that you can act on (escalated first, then highest score, then newest), and a summary strip (total cases, documents analyzed, escalated, cleared). It is **not** an organisation-wide compliance/MIS report — there's no fraud-rate trend, average-resolution-time metric, or export here; that class of reporting is out of scope for this build (see *Known limitations*).



The Review Queue: counts, filters, and the case table (admin view)



The Dashboard: KPI cards, risk-tier distribution, and the review queue

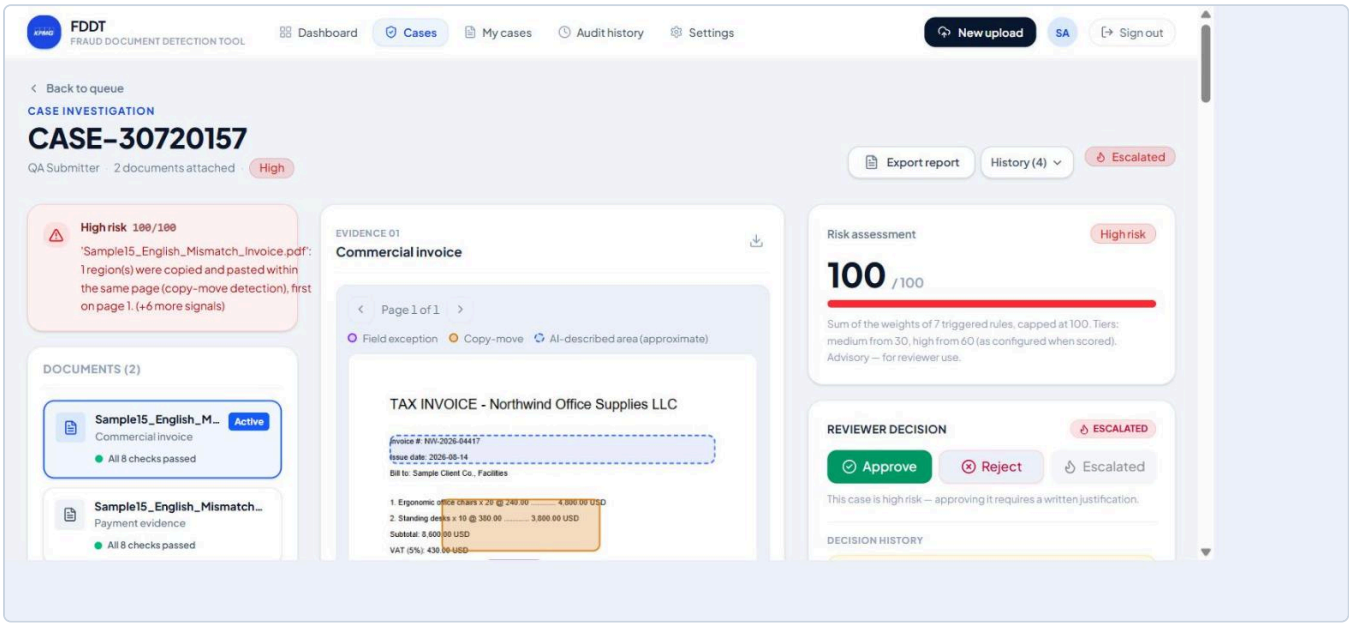
Working a case: the case detail page

Opening a case (`/cases/:caseId`) shows everything about it in three panels plus a timeline:

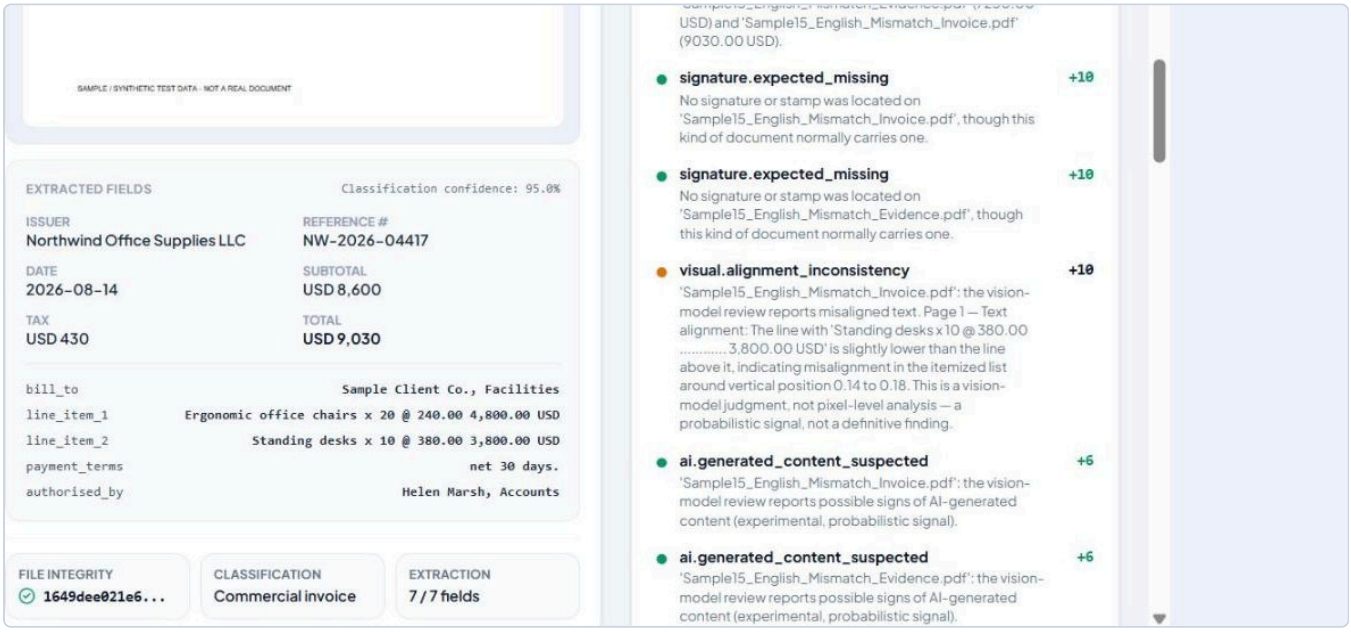
- **Left — Documents:** every file in the case, its type and processing status. Click one to inspect it.

- **Middle — Document viewer & checks:** the selected PDF with live overlays drawn on top of it showing exactly where a check found something; the document's extracted core fields (issuer, reference number, date, amounts — issuer is the only one marked **"(uncertain)"** when the model wasn't confident); and one expandable card per automated check, plus a cross-document card and the signature comparison card if applicable.
- **Right — Risk & decision:** the score, tier, and every triggered reason in plain language; the approve/reject/escalate controls; and (reviewers only) the button to export a forensic report. A platform admin sees an amber **"Support view — read-only"** banner instead of any controls.
- **Bottom — Activity timeline:** the case's full audit trail, newest first — every automated check result and every human action, in order.

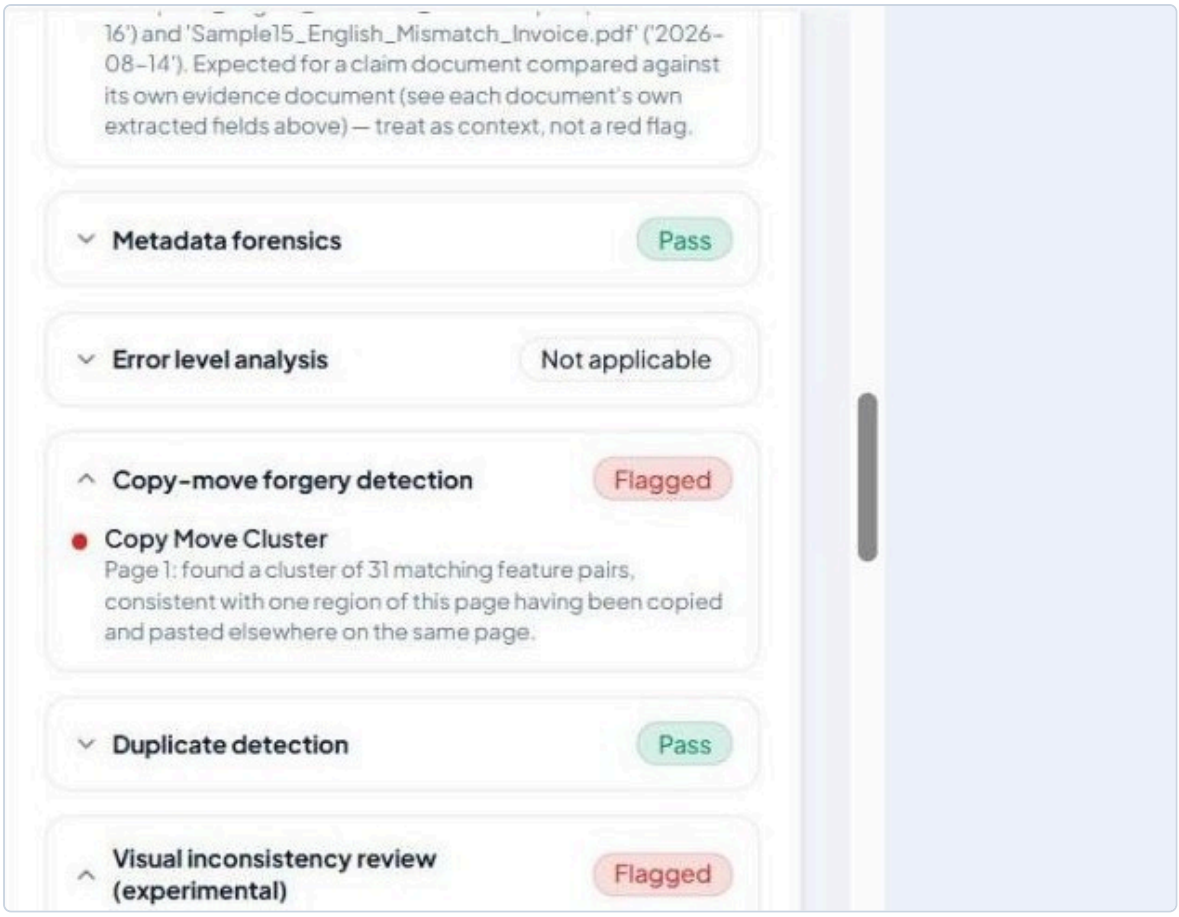
The page **polls automatically every 5 seconds** while the pipeline is still running, so you don't need to refresh — findings and the risk score simply appear as each check finishes.



A high-risk case: documents, the PDF viewer with live overlays, the score, and the decision panel



Extracted fields and the full list of triggered findings in plain language



The per-check cards — each shows its result and, when flagged, the specific finding

Understanding the automated checks

Every uploaded document goes through the same generic pipeline regardless of its type — there's no separate logic per document type, so a school certificate and a vendor invoice get exactly the same scrutiny. In plain language, here's what each check looks for and how far to trust it:

Check	What it looks for	How much to trust it
Extraction & classification	Reads the document (OCR, including printed Arabic), classifies its type, and pulls out issuer, date, amounts, reference number and any other identifiable fields — without a per-type template.	Low-confidence fields are marked; the case still goes to manual review either way.
Field validation	Rule-based checks on the extracted data: is the date in the future, does the total match the sum of line items, are required fields present.	Deterministic and reliable — it's arithmetic and date logic, not a model judgment.
Issuer verification	Fuzzy-matches the extracted issuer name against the known-issuer registry (vendors, schools, tax IDs the organisation has already vetted).	Matches on **name only** — there's no letterhead/logo comparison, and a legitimate new issuer will simply not be in the registry yet.
Cross-document consistency	Once a case has 2+ documents, compares shared fields (amount, date, vendor) pairwise across them and flags mismatches.	Deterministic comparison of extracted values; only as good as the extraction behind it.
Metadata forensics *(PDF only)*	Inspects the PDF's own internal structure: a modify date long after the create date, editing-software fingerprints (e.g., Photoshop) in the tool history, stripped metadata, multiple incremental-save markers.	A strong, mostly-deterministic signal, but presence of an edit marker doesn't by itself prove fraud — legitimate re-saves happen.

Check	What it looks for	How much to trust it
Error Level Analysis (ELA)	Looks for recompression differences that can indicate a pasted-in or edited region of an image.	**A signal, not a verdict.** Weak against a printed-and-rescanned tamper, and can false-positive on a legitimately recompressed scan — weighted lightly in scoring for this reason.
Copy-move detection	Looks for a region duplicated within the same image (e.g., a copied stamp or number).	Classical computer-vision matching — solid at finding literal copy-paste, not a semantic read of what changed.
Duplicate / near-duplicate detection	Perceptual-hashes every uploaded page and compares it against everything your company ever uploaded, including other cases.	Reliable for catching the same file (or a lightly re-saved copy of it) resubmitted — including across unrelated cases. It never compares against another company's documents.
Visual review & AI-generated-content assessment	An AI vision pass describing visual inconsistencies as a secondary check, plus one experimental question about whether the image looks AI-generated.	**Explicitly experimental** — real-world accuracy on scanned business documents is unproven. Treated as one weak signal among many, never a standalone verdict.
Signature / stamp detection & comparison	Locates a signature/stamp region on the page, and — only if a reference signature was set — gives an advisory visual comparison ("visually consistent with reference on file") against it.	**Presence/placement confirmation for a human, not an identity match.** This is a known weak point; the wording deliberately never says "verified".

You will not see a single automated "fraud / not fraud" verdict anywhere in the product — every finding above feeds the risk score as one input among several, and a human always makes the final call.

Understanding the risk score

Every case is scored by a **transparent, weighted rules engine** — not a black-box model. Every point in the score traces back to a named rule and a plain-English reason a reviewer can read.

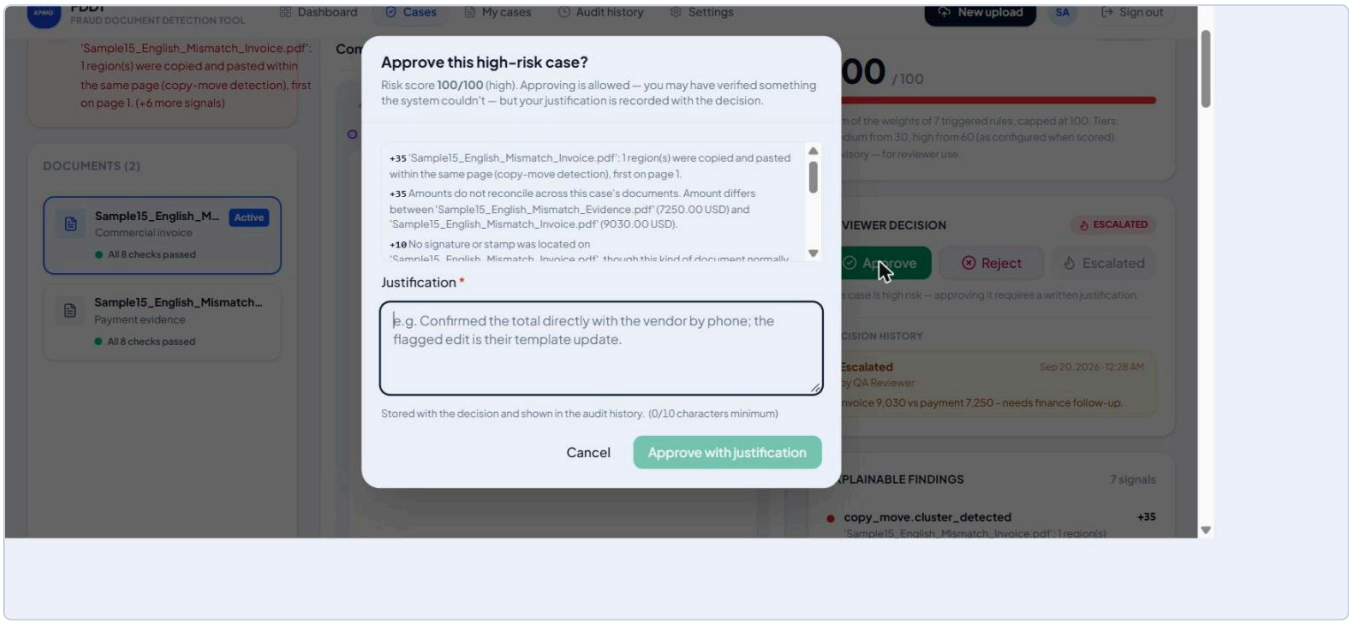
- Each rule that fires adds its configured weight; the total is capped at 100.
- The score maps to a tier: **Low** (0–29 by default), **Medium** (30–59), **High** (60–100) — your company's Reviewer L2s can retune these thresholds under Settings.
- Next to the score you'll see every **triggered reason** in plain language (e.g. "*Issuer name 'X' did not match any known vendor in the registry*"), not just a number.
- **Every scored case goes to the reviewer queue.** Nothing auto-approves, regardless of score — a low score means "no rules fired," not "cleared."
- A case can re-score itself while you're looking at it (for example, when a signature comparison finishes after you opened the case) — a small delay before the numbers settle is expected.
- **When a rule's weight is later retuned, past cases keep the score they were originally given** — the rule versions and thresholds active at scoring time are frozen with that case forever, so tuning the engine never silently rewrites history.

Making a decision: approve / reject / escalate

From the case detail page, a reviewer can do the following. On a case escalated to L2, only a Reviewer L2 can; a Reviewer L1 sees a read-only panel explaining that the case is with L2, and the backend refuses the actions too. Platform admins can never act on a case.

- **Approve.** Disabled until every automated check has finished and the case has a risk assessment (a list of what's still pending is shown). Approving a **low-risk** case is a single click. Approving anything **medium or high** opens a confirmation showing the score and every triggered reason, and requires a **written justification of at least 10 characters** — you may have verified something the system couldn't, but that reasoning is recorded with the decision and shown in the audit history.
- **Reject.** Always requires a reason. It's stored, shown in the audit trail, and visible to the submitter.
- **Escalate.** Requires a reason. This hands the case to the **Reviewer L2** tier — it does **not** change the case's status. Escalation is one-way: there is no "send back to L1". A Reviewer L2 resolves the case with the ordinary Approve / Reject. From then on, Reviewer L1s see the case read-only.

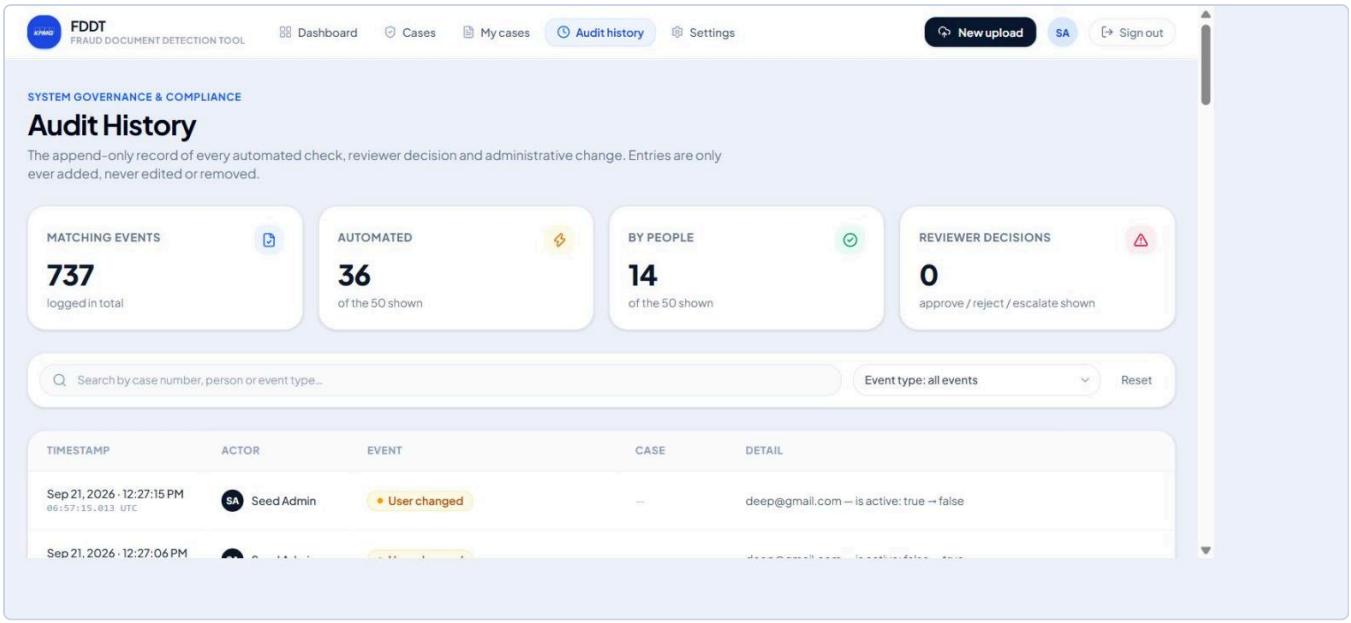
Once a case is approved or rejected, the decision panel becomes read-only.



Approving a high-risk case requires a written justification — the triggered reasons are shown alongside

Audit history

Audit History (`/audit-history` , reviewers only) is a searchable, filterable read of your company's **append-only** audit log — every automated check result and every human action, across every case of the company, written in plain sentences. Nothing in this log is ever edited or deleted; a case's own timeline (on its detail page) is the same log filtered to that case. Platform admins can read a company's log (and a separate platform-level log); their own support views are recorded only in the platform log.



The system-wide, append-only audit log with filters and search

Exporting the forensic report

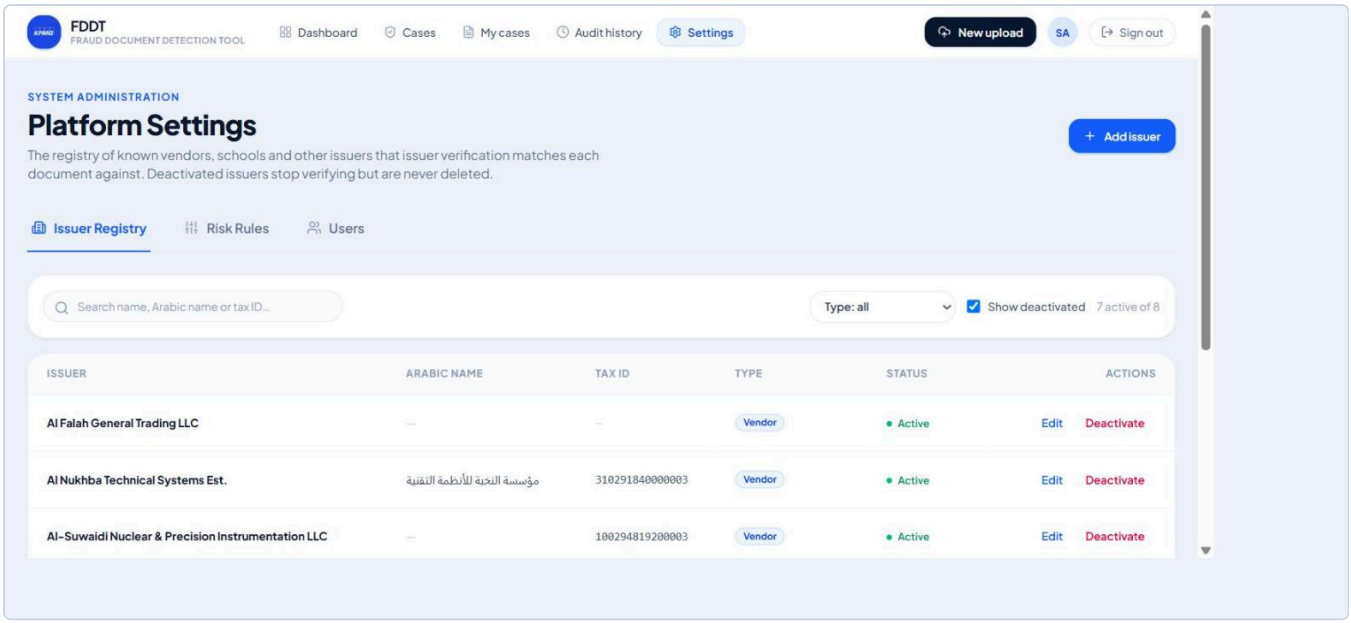
From a case's detail page, a reviewer can generate a standalone PDF report — **Export report** — built to be understood by someone who has never used the application. It contains:

1. Case details (number, type, status, submitter, documents, decision).
2. Executive summary (score, tier, decision, pointers to the sections below).
3. Every triggered risk reason, with severity and the exact region on the page it refers to.
4. The complete checklist — every check on every document, plus the cross-document and signature results.
5. Exceptions — everything that did **not** pass, spelled out concretely.
6. Limitations & assumptions — what the report is (a system-assisted analysis to support human review, not a certified forensic opinion), which checks are deterministic versus model judgment, and which checks actually ran.
7. The full audit trail (oldest first; later events beyond the first 150 are counted, not dropped).
8. An appendix: extracted fields with confidence, each original file's SHA-256 with live tamper-evidence verification, and technical parameters of each check.
9. One annotated page per flagged document page, with every finding highlighted and a legend.

Generating a report takes a few seconds and never re-runs a check or changes the score — it's a frozen, point-in-time snapshot. Every generation is kept; earlier reports for the same case remain downloadable, and two reports for the same case can legitimately show different numbers if the case was re-scored in between.

Administration: issuer registry (Reviewer L2)

Settings → **Issuer Registry** maintains **your company's** list of known vendors, schools and other issuers (name, Arabic name, tax ID, type, active/inactive) that the issuer-verification check matches against. A new company starts with an empty registry — until it is filled in, almost every document will be flagged "issuer not in registry". Deactivating an issuer keeps its history — rows are never deleted, only marked inactive.



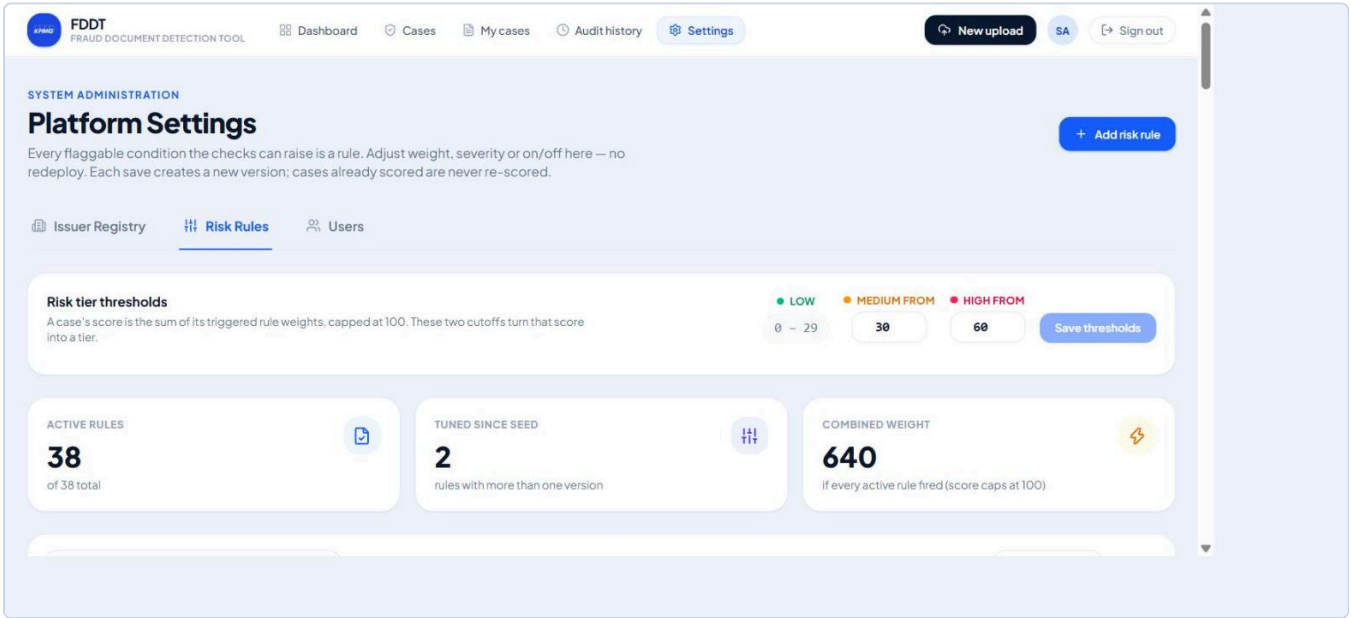
The issuer registry, including right-to-left Arabic names

Administration: risk rules & thresholds (Reviewer L2)

Settings → **Risk Rules** lets you view and tune **your company's** scoring engine without a redeploy (every company starts from the platform's default rule set and then owns its copy):

- Adjust a rule's **weight**, **severity** or **active** flag. This **inserts a new version** — the old one is kept, and every case already scored keeps the version that was active when it was scored.
- View each rule's change history.
- Adjust the **Low / Medium / High thresholds** (default 30 / 60).
- Add a new rule from the catalog of match types the engine understands — you pick a condition and its parameters from a guided list; nobody hand-writes rule logic, and a combination the engine can't evaluate is rejected outright.

Changes apply to your company's cases scored **from that point on**. They are never applied retroactively, and never affect another company.



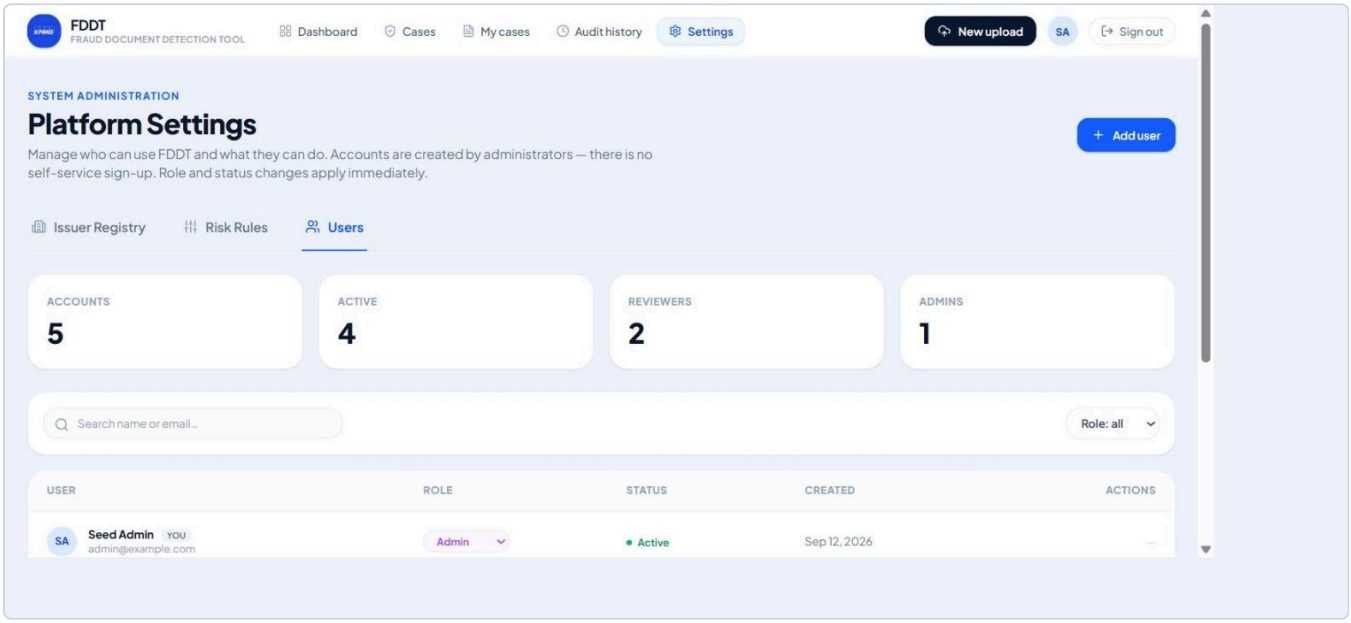
Risk-tier thresholds and the active rule count — tunable without a redeploy

Platform administration (Platform Admin)

The **Platform** area is visible only to platform admins:

- **Companies** — create a company (it receives the default rule set and thresholds; its issuer registry starts empty), rename it, or suspend / reactivate it. Suspending blocks sign-in for all its users.
- **Users** — create and manage every account; there's no self-service sign-up and companies cannot manage their own users. Each account gets a company and a role (User, Reviewer L1, Reviewer L2), or is a Platform Admin with no company. Moving a user to another company signs them out. A platform admin cannot change their own role or deactivate their own account.
- **Rule templates** — the default rules every **new** company starts from. Editing them never changes an existing company's rules.
- **Billing & usage** — per company: cases created, documents uploaded, files stored and storage used, for this month, last month, the last 30 days, this year, all time or a custom range; **Reconcile now** recounts from the source data.
- **Processing queues** — live view of the three processing queues (waiting, running, oldest waiting task, waits, rate-limiter activity, outstanding work per company).

For support, a platform admin can also open any company's case list, cases, audit log and settings (choosing the company at the top of the page). These views are **read-only** and every one is recorded in the platform-only audit log; the company's users don't see those records. A platform admin may change a company's issuer registry or risk rules on its behalf — such a change is recorded in that company's own audit log.



Account management — roles and status, no self-service sign-up

Case statuses & flags — quick reference

What you see	Meaning
Analyzing	The automated pipeline is still running (`submitted` / `under_automated_review`).
Awaiting Review	The pipeline finished; the case is in the reviewer queue (`pending_manual_review`).
Approved	A reviewer approved it.

What you see	Meaning
Rejected	A reviewer rejected it, with a recorded reason.
Escalated · L2 (badge, not a status)	A reviewer handed it to the Reviewer L2 tier; it keeps whatever status it already had.
Low / Medium / High (flag, reviewers only)	The risk tier from the scoring engine. A submitter sees only a neutral "In review" instead.

Known limitations — what this tool does not do

Being direct about this is part of using the tool responsibly:

- **No signature is ever "verified."** Signature/stamp checks confirm presence and placement, and give an advisory visual comparison at best — never a reliable identity match.
- **AI-generated-content detection is experimental.** It's one weak signal inside the visual review, not a proven detector, and its real-world accuracy on scanned business documents hasn't been validated.
- **ELA is a signal, not a verdict** — weak against print-and-rescan tampering, and can flag a legitimately recompressed scan.
- **Only PDFs (up to 10 MB, not password-protected) can be uploaded**, because every forensic check works on the PDF itself. There is no page-count limit and no virus scan.
- **A failed check contributes nothing to the score, silently.** If a check errors out, the case still scores — the failure is visible in the audit log and the check's own card, but no rule fires because of it.
- **The absence of a flagged finding does not guarantee a document is genuine.**
- Not built in this version: notifications/email, SLA tracking and deadlines, organisation-wide MIS/compliance reporting, letterhead/logo matching, cross-case signature-library matching, integration with an external ticketing platform (ServiceNow, Jira, etc.), password reset, and company-managed user accounts.
- **The interface is English-only.** Documents may be in English or printed/typed Arabic; **handwritten Arabic is not reliably read.** Only extracted Arabic field values render right-to-left — the application itself is not translated.

Glossary

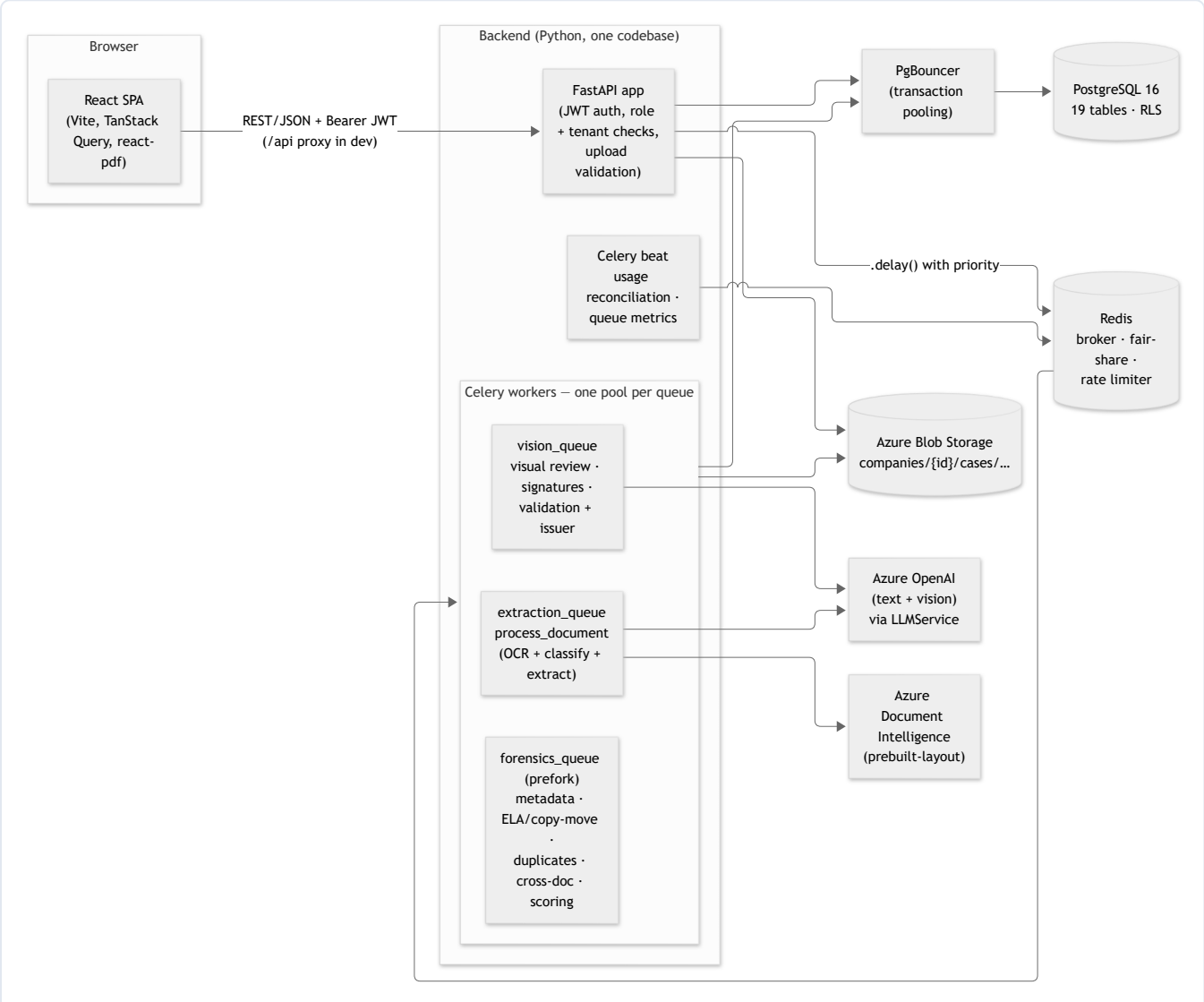
Term	Meaning
Company	One client organisation (tenant). Its users, cases, settings and audit log are invisible to every other company.
Case	One submission, grouping one or more related documents (e.g., an invoice plus its payment evidence) under one `case_number` (`CASE-XXXXXXXX`).
Document	One uploaded PDF within a case.
Tier	Low / Medium / High — the risk-score band that drives the case flag and whether approval needs a justification.
Triggered reason	One rule that fired on a case, shown as a plain-language sentence with its severity and weight.
Tier (L1 / L2) / Escalated	Which reviewer tier owns a case. Every case starts at L1; escalating moves it to L2 for good — not a status.
Pipeline	The set of automated Celery jobs (OCR, classification, extraction, validation, forensics, scoring) that run on every uploaded document.
Audit log	The append-only record of every automated result and human action, never edited or deleted.

Architecture overview

FDDT (Fraud Document Detection Tool) authenticates third-party documents — vendor invoices, school documents, quotations, travel claims and the payment evidence that supports them — *before* a payment or approval is made. A user submits a **case** containing one or more PDF documents. The system extracts their data, runs a battery of automated checks, turns the results into an explainable risk score, and hands the case to a human reviewer (L1, escalating to L2 when needed) who approves, rejects or escalates it. Every step is written to an append-only audit log.

Several client **companies** share one deployment; each company's users see only that company's data, enforced twice — in the application and by PostgreSQL Row-Level Security (multi-tenancy.md).

System diagram



Everything intelligent is a managed cloud API or a classical library. **No model is trained or hosted** (a design rule of the platform).

Tech stack

Layer	Technology	Used for
API	FastAPI 0.121, Pydantic 2, Uvicorn	REST endpoints, validation, OpenAPI
Auth	PyJWT (HS256), bcrypt	Email/password login, 8 h bearer tokens carrying `role`, `company_id`, `is_platform_admin`
Database	PostgreSQL 16, SQLAlchemy 2, Alembic, psycopg2	Persistence, migrations (14), Row-Level Security per company
Connection pooling	PgBouncer (transaction mode)	Multiplexes API + worker connections onto ≤ 60 server connections
Queue	Celery 5.4 + Redis 7	Three queues (extraction / vision / forensics), fair-share priorities, beat jobs
Rate limiting	Redis sliding windows (`services/rate_limiter.py`)	Global caps on Azure calls across every worker
File store	Azure Blob Storage (`azure-storage-blob`)	Immutable originals, signature crops, report PDFs (company-prefixed paths)
OCR	Azure Document Intelligence `prebuilt-layout`	Text, tables, key-values and word geometry
LLM	Azure OpenAI (chat completions, strict JSON-schema output)	Classification, field extraction, issuer semantic match, visual review, signature detect/compare
PDF forensics	pikepdf, PyMuPDF	Upload validation, metadata, xref/incremental-update analysis, page rendering, report PDF
Image forensics	OpenCV (headless), Pillow, NumPy	ELA, copy-move (BRISK), signature crop/pixel comparison
Duplicates	imagehash (pHash)	Near-duplicate page detection (within a company)
Fuzzy match	rapidfuzz	Issuer registry lookup, signer-name comparison
Workflow	`transitions`	Case state machine
Frontend	React 19, TypeScript 6, Vite 8, React Router 7	SPA, route-level code splitting
UI kit	Tailwind CSS 4, Radix primitives, shadcn-style `ui/` components, lucide-react	Styling and base components
Data	TanStack Query 5	Server state, polling
Forms	React Hook Form + Zod	Form state and validation
PDF view	react-pdf (pdf.js)	Live viewer with overlays, signature-region drawing

Key architectural decisions

1. Modular monolith

One FastAPI app and the Celery workers share one codebase (`backend/app/`), one database and one settings object. Modules are separated by responsibility — `api/` (HTTP), `services/` (logic, mostly pure functions), `tasks/` (Celery wrappers that own DB sessions and audit events), `models/` , `schemas/` , `db/` (sessions and tenancy) — not by deployable unit. The same image runs as API, as each worker pool and as beat, differing only in the start command.

The convention that makes this work: **services contain the algorithm and are testable without a database or network; tasks wrap them with I/O.** For example `services/forensics/ela.py` `run_ela_check(pages)` is pure; `tasks/tampering_checks_task.py` downloads the file, renders it, calls the pure function and stores the result.

2. Azure OpenAI behind an `LLMService` abstraction

Classification, field extraction, issuer semantic matching, visual review and signature analysis all use Azure OpenAI. `services/llm_service.py` defines the `LLMService` interface and one implementation, `AzureOpenAILLMService`. Callers depend only on the interface, so switching to another LLM provider means adding one implementation class, not rewriting call sites. The same pattern guards `StorageService` (Azure Blob) and `OCRService` (Document Intelligence). All LLM calls request **strict JSON-schema output** (`response_format: json_schema` , `strict: true`), so every response is validated by a Pydantic model (`DocumentAnalysis` , `PageVisualAnalysis` , `SignatureComparisonResult` , …). Vision calls require a vision-capable deployment (`gpt-4o` , `gpt-4.1` , `gpt-4.1-mini` , …) and raise `LLMConfigurationError` otherwise.

3. Three queues, fair-share, globally rate-limited

Uploads return immediately. The upload endpoint validates the file, commits the document row and enqueues **one task per document per step** — six of them. Each task goes to the queue of the service it depends on: `extraction_queue` (Document Intelligence), `vision_queue` (Azure OpenAI) or `forensics_queue` (local CPU, prefork). Each queue has its own worker pool, so they scale independently. Dispatch is **fair-share per company**: a task's priority comes from how much work its company already has outstanding, so one company's 100-document batch can't starve another company's single upload. Azure calls pass through a **Redis-backed global rate limiter** (requests and tokens per minute, calibrated against real usage) shared by every worker. Tasks commit before every slow step (no transaction is held across a download, an Azure call or CPU work), and check results are written **idempotently** (one row per document + check), so a redelivered task overwrites rather than duplicates. The browser polls `GET /cases/{id}` (every 5 s while the pipeline is incomplete). Details: processing-queues.md.

3b. Event-driven case workflow

State changes go through `services/workflow_service.py` , which drives a `transitions` state machine and **emits** `WorkflowEvent` s (`case_status_changed` , `case_approved` , `case_rejected` , `case_escalated`) to subscribers instead of calling side effects directly. The only subscriber today writes the audit log. A future external case-management integration would be one more `subscribe()` call plus the already-present nullable `cases.external_ref_id` column — no change to the state machine.

4. Immutable, validated originals

Only PDFs are accepted, and every upload is validated **before** it is stored: at most 10 MB, not empty, a PDF by its header bytes, not corrupted, not password-protected (pipeline/01-upload-intake.md). A valid file is written to Blob Storage exactly once, at `companies/{company_id}/cases/{case_id}/documents/{document_id}_{sha256}{ext}` , with `overwrite=False` ; `StorageService` exposes no update or delete. Its SHA-256 is stored on `documents.file_hash` so integrity can be re-verified later — the PDF report does exactly that and prints the result. Nothing downstream mutates the original: forensics render pages **in memory**; the signature crop and the report are *new* blobs (`…/signatures/…` , `…/reports/…`). The private container means the API never hands out the raw blob URL; it mints short-lived SAS URLs (15 minutes by default, 30 for OCR), and refuses to sign a path under another company's prefix.

5. One normalized bounding-box convention

Every located thing — ELA/copy-move regions, visual-review findings, extracted-field positions, signature/stamp regions, signature references — is stored as `{"page": 1-based int, "x", "y", "width", "height"}` where the four numbers are **fractions (0–1) of the page's width/height, origin top-left**. It is resolution-independent, so the same box works on a 200-DPI server render, a browser canvas, and the report. The frontend draws boxes live over the real

PDF (`PdfOverlayViewer`) and never saves an annotated image; only the exported report burns findings into *its own* rendered pages.

6. A transparent rules engine, not a model

Risk scoring is a weighted rules engine (`services/risk_scoring_service.py`). Each rule is a row in the company's `risk_rules` describing *what to match* in already-stored check results; there is no per-rule Python. Rules are **immutable and versioned** — editing inserts `version + 1` — and every scoring run (`case_risk_assessments`) freezes the rendered reason text, the exact rule versions and the thresholds it used, so retuning never changes a historical score. New companies start from the platform rule template (`risk_rule_templates`). See pipeline/11-risk-scoring-engine.md.

7. Template-free extraction

One generic pipeline handles every document type: Azure Layout OCR → a single Azure OpenAI call that both classifies the document and extracts core fields plus arbitrary additional fields as `{name, value}` pairs. No per-type extractors exist, so new document types need no engineering.

8. Two-layer tenant isolation

Every tenant-owned table carries `company_id` . The application binds each request's (and each task's) database session to one company and filters every ORM query automatically; independently, PostgreSQL Row-Level Security refuses any row of another company even to SQL with no filter at all. No database role has BYPASSRLS; the platform path sees all rows through an explicit `platform_all` policy. A cross-company id is a 404. See multi-tenancy.md.

9. Explainability over accuracy claims

Weak signals are labelled as such everywhere in code and UI: ELA is a "signal, not a verdict"; signature/stamp detection is presence-and-placement only; signature comparison uses advisory wording ("visually consistent with reference on file", never "verified"); the visual review's AI question is "experimental". Weights reflect that.

Scope of the current build

What each capability area does today, and what is intentionally outside the current scope.

#	Area	Implementation	Where
1	Classification & extraction	Azure OpenAI via `LLMService` , one combined call per document	`services/llm_service.py`
2	Signature/stamp detection	Azure OpenAI vision model locates signature and stamp regions (presence/placement only)	`services/signature_detection_service.py`
3	AI-generated-content assessment	One question inside the visual-review vision call; experimental, one signal among several. There is no separate content-detection service	`services/visual_inconsistency_service.py`
4	Metadata forensics	pikepdf + PyMuPDF read the PDF's own object model	`services/forensics/metadata_forensics.py`
5	Accepted file types	**PDF only**, ≤ 10 MB, validated at upload (type by header bytes, corruption, password). Every forensic check is PDF-based	`services/upload_validation.py`

#	Area	Implementation	Where
6	Error Level Analysis	Runs only on pages containing an embedded raster image; born-digital text pages return `not_applicable`	`services/forensics/ela.py`
7	Case workflow	Statuses in use: `submitted`, `pending_manual_review`, `approved`, `rejected`. Every scored case goes to the reviewer queue; nothing auto-approves. Escalation moves the case to the L2 reviewer tier (`assigned_tier`), not a status	`services/workflow_service.py`
8	Low-confidence fields	The per-field `uncertain` flag is stored and displayed to the reviewer; it does not change routing, because every case reaches manual review	`services/llm_service.py`
9	Multi-tenancy	Companies, `company_id` everywhere, application filter + PostgreSQL RLS, platform admin with audited read-only support access	`db/tenancy.py`, `api/tenant_access.py`
10	Billing & usage	Per-company counters (cases, documents, files, storage) with nightly reconciliation; platform-admin dashboard	`services/usage_service.py`
11	Processing queues	Three queues, fair-share per company, global Azure rate limiter, queue monitor, JSON queue metrics + `queue_alert` log line	`tasks/celery_app.py`, `tasks/fairshare.py`
12	Notifications	Not part of the current scope (no notifications table, no email sending)	—
13	SLA tracking	Not part of the current scope (no deadline column, no overdue checks)	—
14	Organisation-wide MIS reporting	Not part of the current scope. The *per-case* forensic report is fully built	`services/case_report_service.py`
15	Frontend libraries	Tables are hand-built per page; no charting library is used	`frontend/package.json`
16	Signature verification	Detection + reviewer-created references + in-case vision comparison + a classical pixel-identical-reuse check. Cross-case library comparison is not built (the `is_library` flag only stores data)	`services/signature_*`
17	Evidence export	Per-case PDF report with file hashes and annotated renders; no bundle of the original files	`services/case_report_service.py`
18	Arabic values	Extracted Arabic field values render with the default text direction on the case detail page; Arabic issuer names render right-to-left in the Issuer Registry settings page	`frontend/src/pages/`
19	Document labels	`school_document, vendor_invoice, commercial_invoice, procurement_documentation, quotation, travel_invoice, payment_evidence, other`. Note the **case** type is `travel_reimbursement` while the **document** label is `travel_invoice`	`services/classification_service.py`, `models/case.py`
20	Cloud deployment	Not built — local development only. The production plan and costs are in deployment/production-deployment.md	—

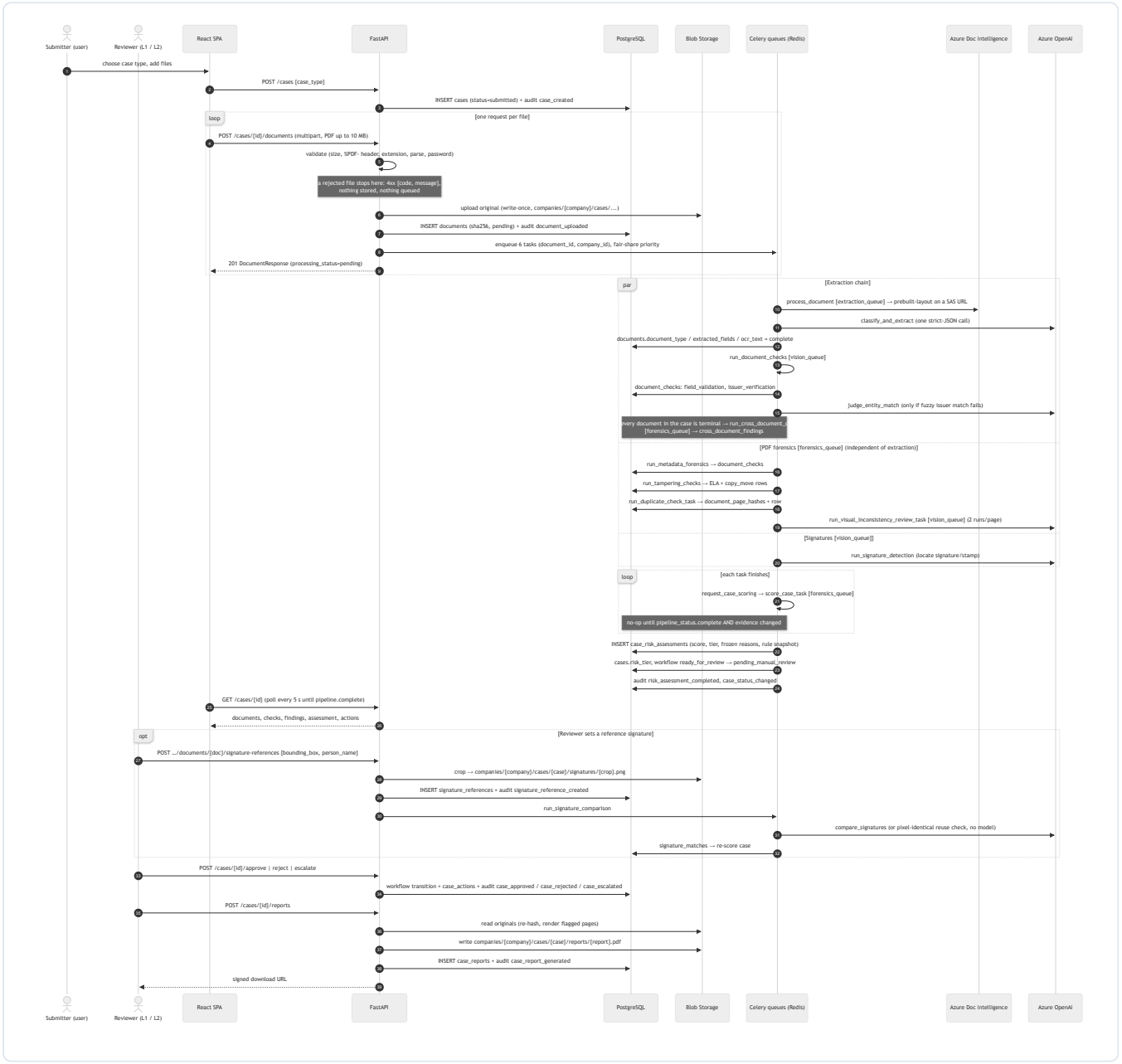
Security posture

- **Two mandatory checks on every endpoint:** a role check (`require_company_role` / `require_reader` with a minimum rank — `user` < `reviewer_l1` < `reviewer_l2` — or `require_platform_admin`) and a tenant check (the request's session is bound to the caller's company by `get_tenant_db`). The user is re-read from the database on every request; a token whose company or platform flag no longer matches, or whose company is suspended, is refused. Case actions additionally check the case's tier (`app/api/case_access.py` `can_act_on_case`).
- **Tenant isolation is enforced twice** — application-layer filtering and PostgreSQL Row-Level Security — and a cross-company id is a **404**. Platform admins belong to no company, are read-only on company data, and every one of their accesses writes a platform-only audit row that no company user can see.
- **Submitters are deliberately blinded:** `GET /cases` and `GET /cases/{id}` return a neutral "In review" flag and no risk tier/assessment to a `user` , and the case timeline (`GET /cases/{id}/audit-log`) omits the scoring event and strips result details (results, finding and match counts, score, tier, rules fired) from the automated check events — so a bad actor cannot learn the score. A case owned by someone else is a **404, not 403**. *Note:* the per-document check cards on the case page (e.g. "Issuer verification: Flagged") are still shown to the submitter — see the change report R6 for this open decision.
- **Case-content routes are guarded** (`api/case_access.py`). Document upload and `GET …/documents/{id}/file-url` require that the caller is the case's **owner or a reviewer of its company** — anyone else gets a **404** (`load_visible_case`), and a rejected upload writes nothing to storage. The three `signatures` endpoints follow the same rule: the **uploader sets the reference signature during upload**, so the case owner may use them, as may any reviewer; a stranger gets a 404. Covered by `tests/test_case_authorization.py` . A submitter can still read their own case's signature comparison results (`GET …/signature-matches`), unlike the risk assessment, which is withheld.
- **Uploads are validated before storage** (PDF only, ≤ 10 MB, not corrupted, not password-protected), so a malformed file never reaches Blob Storage, a worker or an Azure API.
- Tokens are stored in `localStorage` and there is no server-side logout, refresh or password-reset endpoint. `JWT_SECRET_KEY` defaults to `changeme-in-.env` — production must set it (see deployment/production-deployment.md).

End-to-end data flow

This walks one case from upload to exported report. Names in `monospace` are real Celery task names, functions, audit `event_type`s or table names; every one can be found by grepping the code.

Sequence



Every request runs bound to the caller's company and every task to the document's company, so each arrow that touches the database is confined to one company by the application filter **and** by PostgreSQL Row-Level Security (multi-tenancy.md). Azure calls from any worker go through the global rate limiter (processing-queues.md).

Step by step

1. Case creation

`POST /cases` creates a `cases` row with `status=submitted`, `assigned_tier=l1`, and a case number `CASE-` + the first 8 hex characters of the case UUID (unique without a counter table), stamped with the submitter's `company_id`. Audit event `case_created`; the company's `cases_created` usage counter is incremented in the same transaction.

2. Upload (one file per request)

The frontend (`NewCasePage`) calls the upload endpoint once per file with an XHR so it can show progress (and refuses non-PDF, empty and over-10 MB files before sending). The endpoint first **validates** the file (PDF only by its header bytes, at most 10 MB, not corrupted, not password-protected) and rejects a bad one with a specific `{code, message}` before anything is stored or queued. A valid file is hashed (SHA-256), stored in Blob Storage, its `documents` row inserted, `document_uploaded` written and the company's usage counters bumped; then **six tasks** are enqueued (each carrying the company, each with a fair-share priority) and the endpoint returns. See 01-upload-intake.

3. Extraction chain (`process_document` → `run_document_checks`)

`process_document` marks the document `processing`, gets a 30-minute SAS URL, runs Azure Layout OCR, then one Azure OpenAI call that classifies the document and extracts fields. It stores `document_type`, `ocr_text` and `extracted_fields` (each field with a normalized value, confidence and — best effort — a bounding box) and marks the document `complete` (or `failed` with `processing_error`). On success it enqueues `run_document_checks`, which writes two `document_checks` rows: `field_validation` and `issuer_verification`. See 02, 03, 05.

4. Cross-document check (case level)

After each `run_document_checks` — and also after a failed extraction — the task checks whether **every** document in the case has reached `complete` or `failed`; if so it enqueues `run_cross_document_checks`, which compares amount/date/issuer pairwise across completed documents and writes `cross_document_findings` (deleting and recomputing any previous rows for the case). Needs at least two completed documents. See 04.

5. Forensics (in parallel)

Independently of extraction, four tasks read only the file: `run_metadata_forensics`, `run_tampering_checks` (renders once, runs ELA and copy-move) and `run_duplicate_check_task` on the CPU-bound `forensics_queue`, and `run_visual_inconsistency_review_task` on `vision_queue`; plus `run_signature_detection` (`vision_queue`). Uploads are PDF-only, so every document gets all of them (documents stored before that rule may be images; the forensic tasks skip those silently). See 06, 07, 08, 09, 10.

6. Scoring

Every task ends by calling `request_case_scoring(case_id)`. `score_case` does nothing until `pipeline_status(case).complete` — extraction terminal for every document, `field_validation` and `issuer_verification` for every completed document, and the five forensic checks for every PDF, plus a cross-document check newer than the newest upload. It then evaluates every current rule, and — unless the evidence fingerprint equals the latest assessment's — inserts a new `case_risk_assessments` row, sets `cases.risk_tier`, and on the first assessment moves the case to `pending_manual_review` (`case_status_changed`). See 11.

7. Reviewer decision

The case detail page polls until the assessment exists, then unlocks the decision panel. `POST /cases/{id}/approve` requires a finished pipeline and — above low risk — a justification of at least 10 characters; `reject` requires a reason; `escalate` (reason required) moves `assigned_tier` to `l2`, after which only a `reviewer_l2` of the company may act on it (an L1 keeps read access). Platform admins can never act on a case. Each writes a `case_actions` row **and** an

`audit_log` row via the workflow event subscriber. Decided cases (approved/rejected/closed) reject further decisions with 409.

8. Report export

`POST /cases/{id}/reports` synchronously builds the nine-section PDF, stores it under `companies/{company}/cases/{case}/reports/`, and inserts a `case_reports` row (with the PDF's SHA-256); the company's `files_stored` / `storage_bytes` counters go up. See reports/forensic-report.md.

Task names

The names in the diagram are the Python functions. The worker registers each under a Celery name (`@celery_app.task(name=...)`); three differ from the function name, which matters when reading worker logs or calling `celery_app.send_task`. The queue comes from `TASK_QUEUES` in `tasks/celery_app.py` (anything unlisted goes to `forensics_queue`, which always has workers):

Python function	Registered Celery name	Queue	Module
`process_document`	`process_document`	`extraction_queue`	`tasks/document_processing.py`
`run_document_checks`	`run_document_checks`	`vision_queue`	`tasks/document_checks.py`
`run_cross_document_checks`	`run_cross_document_checks`	`forensics_queue`	`tasks/document_checks.py`
`run_metadata_forensics`	`run_metadata_forensics`	`forensics_queue`	`tasks/metadata_forensics_task.py`
`run_tampering_checks`	`run_tampering_checks`	`forensics_queue`	`tasks/tampering_checks_task.py`
`run_duplicate_check_task`	**`run_duplicate_check`**	`forensics_queue`	`tasks/duplicate_check_task.py`
`run_visual_inconsistency_review_task`	**`run_visual_inconsistency_review`**	`vision_queue`	`tasks/visual_inconsistency_task.py`
`run_signature_detection`	`run_signature_detection`	`vision_queue`	`tasks/signature_detection_task.py`
`run_signature_comparison`	`run_signature_comparison`	`vision_queue`	`tasks/signature_comparison_task.py`
`score_case_task`	**`score_case`**	`forensics_queue`	`tasks/risk_scoring_task.py`
`reconcile_usage_stats`	`reconcile_usage_stats` (beat, nightly)	`forensics_queue`	`tasks/usage_tasks.py`
`log_queue_metrics`	`log_queue_metrics` (beat, every minute)	`forensics_queue`	`tasks/usage_tasks.py`

Timing and idempotency notes

- **Everything is at-least-once** (`task_acks_late`: a task whose worker crashed is redelivered). Each task is safe to repeat: check results are upserted (one `document_checks` row per (document, check), one page hash per

(document, page), one signature match per (reference, document, scope), all backed by unique constraints); `run_cross_document_checks` deletes then recomputes; `score_case` skips when the evidence fingerprint is unchanged (one assessment per fingerprint is a unique constraint).

- `signature_stamp_detection` **is deliberately not in the "required checks" list**, because it postdates older cases (which would otherwise never score). Its completion still re-triggers scoring.
- **Signature comparison is not part of the pipeline.** It runs only after a reviewer creates a reference. It re-scores the case when it finishes, so a case can be re-assessed after the reviewer has already seen a score.
- **The audit trail is the source of truth for "what happened when".** Distinct event types seen in a working install: `case_created`, `document_uploaded`, `document_processing_completed/failed`, `document_checks_completed`, `cross_document_check_completed`, `metadata_forensics_completed`, `tampering_checks_completed`, `duplicate_check_completed`, `visual_inconsistency_review_completed`, `signature_stamp_detection_completed`, `signature_reference_created`, `signature_comparison_completed`, `risk_assessment_completed`, `case_status_changed`, `case_approved`, `case_rejected`, `case_escalated`, `case_report_generated`, the settings events `issuer_created/updated/deactivated/reactivated`, `risk_rule_created/updated`, `risk_thresholds_updated`, `user_created/updated` (written into the **user's company's** log, actor shown to the company as "Platform team"; a platform admin account's own creation is platform-level), and the platform-level events (`company_id` NULL, visible only to platform admins) `company_created/updated`, `risk_rule_template_created/updated`, `usage_stats_reconciled` and `platform_admin_access`. A failed forensic task writes `<check>_failed` (e.g. `tampering_checks_failed`). A rejected upload writes no event. The three decision events carry `actor_role` (the actor's role at the time, e.g. `reviewer_l1`); `case_escalated` also carries `from_tier` / `to_tier` (`l1` → `l2`) and the required `reason`.

Multi-tenancy

FDDT serves several client companies from one deployment. Each company's users see **only** that company's data; the platform operator's own team (platform admins) sits outside every company.

Roles

Role	Company	Can
`user`	exactly one	submit cases/documents, track own cases
`reviewer_l1`	exactly one	review the company's queue; approve / reject / escalate (L1 tier)
`reviewer_l2`	exactly one	everything L1 can, plus escalated (L2) cases, plus the company's own **Settings › Issuer Registry** and **Settings › Risk Rules**
`platform_admin`	**none**	create companies; create every user and assign company + role; edit the **platform risk-rule template** new companies start from; billing & usage dashboard; processing-queue monitor; **read-only, audited** support access to any company's cases, audit log and settings (and may edit a company's issuer registry/risk rules on its behalf, also audited)

- Company roles are ranked (`has_rank` / `hasRank`); `platform_admin` is deliberately *not* on that ladder, so it never inherits "reviewer" powers — it cannot approve, reject, escalate, upload, create cases or generate reports.
- There is no company-level user management: **Settings › Users** is platform-admin only.
- The former single-tenant `admin` role was migrated to `platform_admin` .

Data model

- `companies (id, name, is_active, created_at, updated_at)` . `is_active = false` suspends a company: its users can't sign in and existing tokens stop working on the next request.
- `company_id` (FK, indexed) on every tenant-owned table: `users` , `cases` , `documents` , `document_checks` , `document_page_hashes` , `cross_document_findings` , `case_actions` , `case_risk_assessments` , `risk_scores` , `risk_rules` , `risk_settings` , `issuer_registry` , `signature_references` , `signature_matches` , `case_reports` , `audit_log` .
  - NOT NULL everywhere except `users` (NULL = platform admin; a CHECK constraint enforces `role = 'platform_admin' ⇔ company_id IS NULL`) and `audit_log` (NULL = platform-level event such as *company created*, *template edited*, *usage reconciled*, *platform-admin access*). *User created / updated* for a company user is written into **that company's** log; the company sees the actor as "Platform team" (the platform admin's own user row is invisible to a company session).
  - Extracted fields are a column on `documents` and duplicate-detection hashes live in `document_page_hashes` — both covered by those tables' `company_id` .
- `risk_rules` , `risk_settings` (tier thresholds) and `issuer_registry` are **per company**. Rule uniqueness is `(company_id, rule_id, version)` . The issuer registry starts empty.

- `risk_rule_templates` (platform-level, **no** `company_id`) holds the default rule set. Platform admins edit it under **Platform › Rule templates**. When a company is created, every **active** template row is copied into that company's own `risk_rules` (version 1, `company_id` stamped) plus default thresholds (`seed_risk_rules(db, company_id)`). From then on the copy is the company's own: editing the template never changes an existing company — only companies created afterwards (the same "no retroactive change" principle as rule versioning). The migration seeded the template from the Default Company's current rules (38 rules — its 42 rows were 38 rules plus 4 older versions). Only `fddt_platform` has access to the table.
- `company_usage_stats (company_id, usage_date, cases_created, documents_uploaded, files_stored, storage_bytes)` — see Billing & usage.

Tenant isolation — two independent layers

1. Application layer (`app/db/tenancy.py`, `app/api/auth.py`, `app/api/tenant_access.py`)

- The JWT carries `role`, `company_id` (null for platform admins) and `is_platform_admin`. Every request re-reads the user from the database; if the token's company or platform flag no longer matches (user moved to another company, role changed across the platform boundary) the token is refused (401), and a suspended company's users are refused too.
- Every protected endpoint applies **two mandatory checks**: the role check (`require_company_role`, `require_reader`, `require_platform_admin`) **and** the tenant check — the request's data session is bound to the caller's company by the `get_tenant_db` dependency, so a route cannot forget it.
- Every session is in one of three states:
  - **unbound** — refuses to query or write any tenant-owned table (raises `TenantContextError`);
  - **company** — every ORM SELECT/UPDATE/DELETE touching a tenant model gets `company_id = <bound company>` added automatically (`with_loader_criteria`, including relationship loads and `session.get()`); every INSERT is stamped with that company and a row of another company is refused at flush; objects of other companies are evicted from the identity map on binding;
  - **platform** — the explicit RLS-bypassing path (below), no automatic filter.
- On top of that backstop, service functions that read tenant data take `company_id` explicitly and filter by it themselves (`load_current_rules`, `get_risk_settings`, `match_issuer`, `run_duplicate_check`, `score_case`, `pipeline_status`, `latest_assessment`, `get_case_flags`, …).
- An id that belongs to another company is a **404** — never a silent empty result and never a 403 that would confirm the id exists.
- Celery tasks are enqueued as `task.delay(document_id, company_id)` and open a session bound to that company (`app/tasks/tenant.py`), so workers are confined exactly like requests.
- Duplicate detection compares a document only against **its own company's** history (matching across companies would leak another tenant's filenames and case numbers).

2. Database layer — PostgreSQL Row-Level Security

The migration `a7c1e9d3f5b2_multi_tenancy` creates two login roles and the policies:

Role	Attributes	Used for
`fddt_app` (`DATABASE_APP_USER`)	NOSUPERUSER **NOBYPASSRLS**	every company-scoped request and pipeline task
`fddt_platform` (`DATABASE_PLATFORM_USER`)	NOSUPERUSER **NOBYPASSRLS** — full access through an explicit `platform_all` policy on every table	sign-in/token identity lookup, platform-admin screens, usage reconciliation, resolving a legacy task's company

Role	Attributes	Used for
owner (`DATABASE_URL`)	owns the tables; needs CREATEROLE, **not** superuser	Alembic migrations and `seed.py` only

No role has BYPASSRLS, and no superuser is needed anywhere. Platform access is policy-based: `CREATE POLICY platform_all ON <table> FOR ALL TO fddt_platform USING (true) WITH CHECK (true)` on each RLS table — creating a policy only needs table ownership, so this works unchanged on managed PostgreSQL (Azure Flexible Server). Verified by running the **entire** migration chain as a NOSUPERUSER/CREATEROLE owner on a fresh database and re-running the RLS suite there. (PostgreSQL 16 requires superuser even to *mention* `SUPERUSER` / `BYPASSRLS` in `ALTER ROLE`, so those attributes are set on `CREATE ROLE` only and the migration asserts afterwards that the roles really are unprivileged.)

- RLS is enabled on every tenant-owned table (plus `companies` and `company_usage_stats`) with a `tenant_isolation` policy for `fddt_app`: `USING (company_id = fddt_current_company_id()) WITH CHECK (same)`, where `fddt_current_company_id()` reads the transaction-local setting `app.current_company_id`.
- The app sets that setting at the start of **every** transaction of a company-bound session (`after_begin` hook, `set_config(..., is_local => true)`), so a pooled connection never carries one request's company into the next. No setting ⇒ no rows.
- A bug that forgets the company filter therefore still cannot read, update, delete or insert another company's rows — the database refuses. `tests/test_rls_postgres.py` proves this with raw SQL that has **no** company filter at all.
- Grants are least-privilege and also make the append-only/immutable tables read+insert-only **in the database** for both app roles: `audit_log`, `risk_rules` (versioned), `case_risk_assessments`, `case_reports`. `users` is SELECT-only for `fddt_app` (user management is platform-only).
- The platform engine is reachable only through a session explicitly bound to platform mode (`get_system_db`, `system_session()`); the default request session is unbound and then company-bound.

Adding a tenant-owned table later: give it `TenantScopedMixin`, and in its migration `ENABLE ROW LEVEL SECURITY`, create the `tenant_isolation` policy (`enable_tenant_rls()` in the multi-tenancy migration shows how) and grant the two app roles explicitly. There are deliberately no default privileges, so a new table is unreachable by the app until that is done.

Platform-admin support access (audited)

A platform admin can open any company's case, case list, audit log or settings. The route resolves which company the resource belongs to through the platform session, binds the request to **that** company (so the rest of the request is confined by both layers), and writes a `platform_admin_access` audit row, e.g.:

> Platform admin ops@example.com viewed Company B / Case CASE-1A2B3C4D

These rows are platform-only. They are written through the platform session with `company_id = NULL` (the target company is in `event_data.company_id`), so:

- RLS hides them from every company session — a company query cannot return them even without a filter;
- every company-facing view (Audit History, its event-type filter, a case's timeline, the case report) also excludes the event type explicitly;
- company users have no indication that platform access happened.

Platform admins see every row — in the platform-level log, and in a company's log view (which shows the company's own events plus the platform-access rows about it). **Every access is recorded individually**: `PLATFORM_ACCESS_AUDIT_DEDUP_SECONDS` defaults to `0` (a value > 0 would collapse identical reads).

Rows written before this change were moved to `company_id = NULL` by migration `c9e3a1b5d7f4` (no row removed, no content changed).

Platform admins have no cross-company case list: the case queue takes one company at a time.

Blob storage layout

```
companies/{company_id}/cases/{case_id}/documents/{document_id}_{sha256}{ext}
companies/{company_id}/cases/{case_id}/reports/{report_id}.pdf
companies/{company_id}/cases/{case_id}/signatures/{crop_id}.png
```

`ensure_company_blob()` refuses to sign a URL under another company's prefix. Files written before the migration keep their original paths (`{case_id}/…` , `reports/…` , `signatures/…`) — originals are immutable and are not moved; their database rows are company-scoped like everything else.

Billing & usage

Platform › Billing & usage shows one row per company (never a mixed list of cases): cases created, documents uploaded, files stored (documents + generated reports), storage added in the period, and each company's current all-time storage and file count, for this month / last month / last 30 days / this year / all time / a custom UTC date range.

- Counters live in `company_usage_stats` , one row per company per UTC day, bumped with an atomic `INSERT … ON CONFLICT DO UPDATE SET n = n + delta` **in the same transaction** that creates the case, document or report (`app/services/usage_service.py`). The dashboard sums a few small rows — never a COUNT/SUM over the case tables.
- Storage is the exact byte size recorded when each file was written (`documents.file_size_bytes` , `case_reports.file_size_bytes`). Signature crops are small derived images and are not counted.
- `reconcile_usage_stats` (Celery beat, nightly at `USAGE_RECONCILIATION_HOUR_UTC` , default 02:00 UTC) recomputes every (company, day) from the source tables and corrects any drift, writing a platform-level `usage_stats_reconciled` audit row when it changed something. **Reconcile now** on the dashboard runs the same job on demand.
- Nothing deletes cases, documents or reports today, so counters only increase. If deletion is added it must decrement them (the nightly job would otherwise correct it the next night).

Migrating an existing single-tenant database

`alembic upgrade head` (as the owner role):

1. creates `companies` / `company_usage_stats` and one company named `DEFAULT_COMPANY_NAME` ("Default Company");
2. adds `company_id` everywhere and assigns **every** existing row to that company;
3. turns existing `admin` accounts into `platform_admin` (no company); everyone else keeps their role in the default company;
4. backfills the usage counters from the existing data;
5. creates `fddt_app` / `fddt_platform` with the configured passwords, grants, and the RLS policies (tenant isolation for the app role, `platform_all` for the platform role).

Then `c9e3a1b5d7f4` (production hardening): removes BYPASSRLS from an existing `fddt_platform` (only if it has it), adds the `platform_all` policies, moves platform-access audit rows to platform-only, creates and seeds

`risk_rule_templates`, and enforces one row per check / page hash / assessment / signature match (collapsing duplicates earlier retries left: on the dev database 6 check rows, 1 page hash, 1 assessment).

The migrations need a table-owner role with **CREATEROLE**; no superuser.

`alembic downgrade` reverses it (keeping only the oldest company's rules/thresholds, since a single tenant can't hold several companies' copies).

Tests

Spec check	Where
Platform admin creates companies A and B and their users	`tests/test_multi_tenancy.py::tenants` fixture
A's reviewer_l2 sees A's cases, issuers, rules; 404 for every B resource by id (case detail/audit/file URL/signatures/reports/actions/issuer PATCH/rule history, `company_id=B` params)	`test_reviewer_l2_sees_own_company_and_gets_404_for_the_other`, `test_a_rule_history_of_another_company_is_a_404`
A's reviewer_l2 cannot reach Settings › Users	`test_reviewer_l2_cannot_reach_settings_users`
Platform admin opens B's case; a platform-only audit row is recorded (visible to platform admins, never to A or B)	`test_platform_admin_reads_any_case_and_each_access_is_audited`
Billing per company matches uploads; manual cross-check; drift corrected	`test_billing_dashboard_separates_companies_and_matches_uploads`
RLS alone blocks cross-tenant rows with no company filter in the SQL	`tests/test_rls_postgres.py` (needs `FDDT_RLS_TEST_DATABASE_URL`)
No BYPASSRLS anywhere; platform access via policies; platform-only audit rows invisible to the app role	`tests/test_rls_postgres.py` (`test_no_role_has_bypassrls`, `test_every_rls_table_has_the_platform_policy`, …)
Company users never see platform-access rows (Audit History, filter, timeline)	`tests/test_multi_tenancy.py::test_company_users_never_see_platform_admin_access`
Every idempotent write (page hashes, check rows, usage counters) works under the least-privilege app role	`tests/test_rls_postgres.py::test_every_upsert_the_pipeline_uses_works_under_the_app_role`
New company gets the template; template edits don't change existing companies	`tests/test_rule_templates.py`
Tenant context never leaks through PgBouncer transaction pooling	`tests/test_pgbouncer_rls.py` (needs PgBouncer)
Pre-migration data intact and attributed to the default company	verified against the real dev database (see the change summary)

Processing queues, fairness, rate limits and database connections

Design

- **One task per document per step.** An upload enqueues six independent tasks for that document (extraction, metadata forensics, ELA + copy-move, duplicate hashing, visual review, signature detection), each carrying the document's company. No task ever processes a whole case or batch.
- **Three shared queues, one per external dependency**, each with its own worker pool:

Queue	Tasks	Limited by	Pool
`extraction_queue`	`process_document` (Document Intelligence layout **+ the Azure OpenAI classification call**)	`AZURE_DOCUMENT_INTELLIGENCE_MAX_CALLS_PER_SECOND` (10; Azure allows 15) — and, for the classification call, the Azure OpenAI caps	threads
`vision_queue`	visual review, signature detection/comparison, field validation + issuer check	`AZURE_OPENAI_MAX_REQUESTS_PER_MINUTE`, `AZURE_OPENAI_MAX_TOKENS_PER_MINUTE`	threads
`forensics_queue` (+ legacy `celery`)	metadata forensics, ELA/copy-move, duplicate hashing, cross-document check, risk scoring, housekeeping	**nothing** — local CPU	**prefork**, one process per CPU of the container

- **Fair-share per company** (`app/tasks/fairshare.py`). Each task is published with a priority computed from how much work *its company* already has outstanding (queued + running) on that queue: `priority = min(9, outstanding // FAIR_SHARE_BUCKET_SIZE)` (bucket default 5). Celery's Redis transport keeps one list per level (`<queue>` , `<queue>:1` … `<queue>:9`) and workers always drain level 0 first, so a company with little outstanding work is served before the tail of another company's big batch. Equal share, no fixed tiers: priorities are recomputed for every task from live counters, and a company's new tasks move forward again as its backlog drains. Counters are incremented on publish and decremented in `task_postrun` ; a redelivered task (worker crash) is counted once; the per-minute metrics job resets a queue's counters whenever that queue is empty (self-healing). If Redis is unreachable a task is published without priority (FIFO) — scheduling degrades, processing doesn't stop.
- **Fair dispatch per worker**: `worker_prefetch_multiplier=1` + `task_acks_late` — a worker holds only the task it is running.
- **Idempotent writes** (`app/services/check_store.py`): exactly one `document_checks` row per (document, check_type), written by `INSERT … ON CONFLICT DO UPDATE` ; likewise one page hash per (document, page), one assessment per (case, evidence fingerprint), one signature match per (reference, document, scope) — all enforced by unique constraints. A task redelivered after a worker crash overwrites instead of duplicating; the full run history stays in the append-only audit log.
- **No transaction held during slow work.** Tasks commit (ending their read transaction) before every slow step — blob download, Azure call, CPU forensics — and task sessions keep loaded attributes after a commit (`expire_on_commit=False`) so touching them doesn't silently reopen one. Through PgBouncer in transaction mode an open transaction pins a server connection; the first Linux load test showed 32 threads holding transactions across Azure calls exhausting the pool and stalling the API. The upload endpoint likewise ends its read transaction before reading the body and writing the blob.

- **Independently scalable workers**: `start-worker.sh extraction|vision|forensics`, or the `*-worker` services in `docker-compose.yml` (`--profile workers`, `--scale forensics-worker=N`). The forensics worker runs prefork with one process per CPU of **its container's cgroup limit** (`cpus:` / `FORENSICS_WORKER_CPUS`), not the host's core count — `nproc` ignores container limits and would oversubscribe (2 replicas each starting 4 processes on a 4-core host). Pool sizes: `EXTRACTION_WORKER_CONCURRENCY`, `VISION_WORKER_CONCURRENCY`, `FORENSICS_WORKER_CONCURRENCY` (0 = per CPU of the limit).
- **Monitoring**: Platform › Processing queues (`GET /platform/queues`) — waiting (all priority levels), running, oldest waiting task, average / p95 / max wait, run time, consumers online, limiter counters (throttled calls, seconds waited, Azure 429s) and **outstanding work per company** (the fair-share counters). A beat task logs one structured line per queue every minute, `queue_metrics {"queue": …, "waiting": …, "running": …, "oldest_waiting_seconds": …, "avg_wait_seconds": …, "p95_wait_seconds": …, "started_in_window": …, "window_seconds": 300}`, and a **WARNING** `queue_alert {…same fields…, "alert": "oldest_waiting_exceeded", "threshold_seconds": …}` line when a queue's oldest waiting task is older than `QUEUE_ALERT_OLDEST_WAITING_SECONDS` (default 300; 0 = off). Shipping these logs to a monitoring platform and alerting on `queue_alert` is a deployment step (not built yet: local dev only). For example, on Azure Container Apps with a Log Analytics workspace, a log alert on `ContainerAppConsoleLogs_CL | where Log_s has "queue_alert"` (production-deployment.md).

Azure rate limits

Global limiter (`app/services/rate_limiter.py`): Redis sliding windows shared by every worker, applied per Azure call (Celery's own `rate_limit` is per worker). A 429 that still gets through pauses every worker for the Retry-After period (the SDKs retry first).

This deployment's quota (from the `x-ratelimit-*` headers of `gpt-4.1-mini`): 100 requests/min, **100,000 tokens/min**. `backend/.env` caps at **80 %** (`80` RPM, `80000` TPM). This was 85 % until 2026-10-03. It was lowered because the real-quota load test saw one 429 in 123 calls at full saturation (two independent sliding windows can race). The load-test figures below were measured at 85 %.

Token reservations are calibrated against real usage (`scripts/calibrate_llm_tokens.py`). Azure charges each request its prompt tokens + its full `max_tokens` up front. Every call now logs its real `usage` next to the reservation (`llm_usage …` log line, Redis `fddt:llm_usage:<call type>`). Measured on the 12 sample documents (English + Arabic, 96 calls):

Call type	Real prompt (avg)	Old estimate	New estimate	Real max output	`max_tokens` old → new
classification + extraction	2,105	1,270 (**0.60×**)	2,762 (1.31×)	1,207	4,000 → 4,000
visual review (2 per page)	2,904	1,564 (**0.54×**)	3,622 (1.25×)	204	1,500 → 640
signature/stamp detection	1,418	1,392	1,967 (1.39×)	144	800 → 480
signature comparison	1,055	2,649 (2.5×)	1,558 (1.48×)	47	800 → 320
issuer entity match	400	353	768 (1.92×)	155	4,000 → 512

- The old estimator (4 chars/token, flat 1,100 tokens/image) **under-reserved the two most frequent calls by 36–46 %**: at full load the limiter would have let traffic exceed the real quota.
- New model: text at **1.93 chars/token** (the most token-dense observed — Arabic; English prompts are reserved a little high, never low) + images at **1.62 tokens per 32-px patch** (how gpt-4.1-family models bill images; full pages

and small crops now cost what they really cost) + **5 % margin**. Every measured call is reserved at ≥ 1.05× its real prompt.

- Output reservations were sized from measured outputs with ample headroom for the four fixed-shape calls; classification keeps 4,000 because its output grows with the number of fields on a document.
- **Real-Azure check**: 40 vision calls from 8 threads through the limiter at the 85 % caps → **40 × HTTP 200, 0 × HTTP 429, 0 SDK retries**; the token limiter held 15–20 calls back and settled at 19 calls/min.
- Re-run the calibration when the model, the prompts or the page render size change.

Throughput ceiling at the current quota (what the limiter reserves per one-page document: classification 6.8k + visual review 2 × 4.3k + signature detection ~3 × 2.4k + issuer match 1.3k ≈ 23.9k tokens): **≈ 3.6 documents/minute (~210/hour) for the whole platform.** The earlier "5–6/minute" figure was based on the under-estimates. The `max_tokens` reductions raised the real ceiling by ~26 % (from ≈ 2.8/min). Raising the deployment's TPM is the main lever beyond this.

Database connections

Without PgBouncer (direct)

Each process keeps a SQLAlchemy pool **per role it uses** (app + platform): up to `DB_POOL_SIZE + DB_MAX_OVERFLOW` = 5 + 10 = **15 per role**, opened on demand.

Process	Concurrency that can hold a connection	Max server connections
API (uvicorn worker)	FastAPI sync endpoints run in a 40-thread pool → capped by the pool	2 roles × 15 = **30**
extraction / vision worker (threads, T threads)	T	≈ min(T, 15) app + a few platform
forensics worker process (prefork)	1 task at a time	≈ 1–2

Example: 2 API replicas × 4 uvicorn workers (8 × 30 = 240) + extraction 16 + vision 16 + 2 forensics replicas × 2 processes (≈ 8) ≈ **280** — beyond Postgres' default `max_connections = 100`. Hence:

With PgBouncer (transaction pooling) — the recommended setup

Set `DATABASE_POOLER_HOST` / `DATABASE_POOLER_PORT`: the two app roles connect to PgBouncer (`pgbouncer/pgbouncer.ini`, service `pgbouncer` in `docker-compose.yml`) and **each process keeps no pool of its own** (`NullPool`), so idle slack isn't held twice. Client connections are cheap; PgBouncer multiplexes them onto a small set of server connections:

PgBouncer setting	Value	Meaning
`pool_mode`	`transaction`	a server connection is lent per transaction
`default_pool_size`	20	server connections per (database, user): `fddt_app`, `fddt_platform`
`reserve_pool_size`	5 (after 3 s)	burst headroom per pool
`max_db_connections`	60	hard cap of server connections to the database
`max_client_conn`	1000	client connections accepted (API + workers)

Postgres `max_connections` required: PgBouncer's 60 + direct owner connections for migrations/seed (≈ 2) + admin/monitoring (≈ 5) + superuser reserve (3) ⇒ **≥ 70; the default 100 is enough.** Raise `default_pool_size` /

`max_db_connections` (and `max_connections` with them) only if PgBouncer's `SHOW POOLS` shows clients waiting (`cl_waiting`).

Tenant context is safe with transaction pooling because the company is set with `set_config('app.current_company_id', …, is_local => true)` at the start of every transaction and dies with it. `tests/test_pgbouncer_rls.py` proves it on a single-server-connection pool (forced reuse): a *session*-level setting **does** leak to the next client (negative control — the test can detect leaks), the transaction-local one never does, including 32 concurrent clients × 25 transactions alternating companies. Never set tenant state with plain `SET` / `set_config(..., false)`.

Production: `userlist.txt` holds local dev passwords only — use SCRAM secrets from a vault.

Load tests

`backend/loadtest/` drives the real API, real Celery workers, Postgres (RLS) and Redis; only Document Intelligence, Azure OpenAI and Blob Storage are stubbed (they simulate latency and Azure's own quotas, answering 429 above them). `docker-compose.loadtest.yml` runs it all in **Linux containers through PgBouncer**.

Linux, final run (forensics 2 replicas × 2 prefork processes; Azure queues 16 threads; Azure OpenAI quota simulated at 10× so the run takes minutes)

Check	Result
1 — 27 concurrent single uploads, 3 companies	interleaved (`ABCABCABCACBB…`), Kendall τ vs submission 1.0 / 0.86 / 0.95 (extraction / vision / forensics)
2 — A uploads 100 docs, B uploads 1 right after	**B waited 0.0–0.2 s on forensics (A's median 30 s) and 1.2 s on visual review (A's median 44 s)**; started after 43–45 of A's 100 instead of all 100
3 — no 429 storms	Document Intelligence peaked at exactly 10 calls/s (quota 15), **0 × 429**; Azure OpenAI **0 × 429**, token limiter held 242 calls back
4 — monitor reflects reality	peaks 20 / 146 / 299 waiting, tracked every 2 s for the whole run
Drain	everything drained in 92 s; forensics finished before vision (97 s vs 130 s)

Linux, real quota (85 RPM / 85k TPM app caps vs 100 / 100k simulated Azure), 30-document batch

Queue	Last task started	Oldest wait
forensics	**35 s**	16 s
extraction	69 s	50 s
vision	315 s	288 s

At the real quota **forensics is the fastest-draining queue by far** and the Azure OpenAI quota is the long pole — as it should be. Fair-share: B started after 10 of A's 30 on forensics and vision. One simulated 429 occurred in 123 calls while the token limiter was fully saturated (103 calls held back) — a boundary race between the app's window and the stub's independent window; the real SDK retries such a 429 after Retry-After (the stub doesn't), and the real-Azure burst at the same caps gave none. For more margin set the caps to 80 % of quota.

Comparison with the first (Windows, strict FIFO) run

	Windows threads, FIFO	Linux prefork, fair-share
B's wait behind A's 100-doc batch (forensics / vision)	144 s / 70 s, after **100/100**	0.2 s / 1.2 s, after 43–45/100
Forensics oldest waiting	264 s	47 s (10× load), 16 s (real quota)
429s	0	0

Open findings

- `process_document` **couples Document Intelligence and the Azure OpenAI classification in one task.** When the token limiter is saturated, extraction threads sit blocked on the *OpenAI* quota, so OCR for a newly arrived document waits for a free thread (B waited 14 s / 50 s on extraction in the two runs while A's OCR barely waited). Fix: move classification + extraction into its own `vision_queue` task (needs the OCR word boxes persisted between the two). Not done in this change.
- **A rate-limited task holds its worker thread while it waits for the limiter.** Fair-share decides the dequeue order, but a newly prioritised task still needs a free thread. Sizing the Azure pools to just saturate the caps (not far above) keeps that wait short.
- **On one machine, forensics competes with the API for CPU.** Prefork saturates its CPU allowance by design; run workers and the API on separate nodes (or cap `FORENSICS_WORKER_CPUS`) so uploads stay fast.

01 — Upload & intake

What it does: accepts one PDF at a time into a case, validates it, stores the original immutably, records its hash, and fans out the automated pipeline. It detects nothing itself; it is the chain-of-custody starting point.

Code: `api/documents.py` (`upload_document`), `services/upload_validation.py` , `services/storage_service.py` , `services/usage_service.py` , `models/document.py` , frontend `pages/NewCasePage.tsx` + `components/upload/FileDropzone.tsx` + `lib/uploadLimits.ts` .

Flow

1. The UI creates the case (`POST /cases`), then uploads each selected file with its own `POST /cases/{case_id}/documents` (multipart field `file`) using `XMLHttpRequest` for progress. Files upload in parallel; a failed file can be retried without recreating the case, dismissed using the remove icon (`[X]`), or discarded completely via `Reset form` which cleanly restores the intake form and unlocks case type selection.
2. The endpoint 404s on an unknown case (or one of another company, or — for a `user` — one they did not submit; platform admins get 403), then **validates the file synchronously** (next section). A rejected file is never stored, never gets a `documents` row or audit event, never counts toward usage and never queues a task, so it costs no worker slot and no Azure call. A valid file's `sha256(content)` is computed.
3. It writes the bytes to Blob Storage at **`companies/{company_id}/cases/{case_id}/documents/{document_id}_{sha256}{original_extension}`** with `overwrite=False` . The path embeds the hash and the document's own id, so collisions are impossible even if a filename repeats. A storage failure is a **502**.
4. It inserts a `documents` row (`processing_status = pending` , `content_type = application/pdf`) and flushes, writes the audit event `document_uploaded` with `{original_filename, file_hash, file_size_bytes}` , bumps the company's usage counters (`documents_uploaded` , `files_stored` , `storage_bytes`) in the same transaction, and commits.
5. It enqueues six Celery tasks — each as `(document_id, company_id)` with a fair-share priority — and returns **201**:

Task	Queue	Module	Depends on	Notes
`process_document`	`extraction_queue`	`tasks/document_processing.py`	—	OCR + classify + extract; chains into `run_document_checks`
`run_metadata_forensics`	`forensics_queue`	`tasks/metadata_forensics_task.py`	file only	
`run_tampering_checks`	`forensics_queue`	`tasks/tampering_checks_task.py`	file only	ELA + copy-move share one render
`run_duplicate_check_task`	`forensics_queue`	`tasks/duplicate_check_task.py`	file only	compares within the company only
`run_visual_inconsistency_review_task`	`vision_queue`	`tasks/visual_inconsistency_task.py`	file only	also carries the AI-generation question
`run_signature_detection`	`vision_queue`	`tasks/signature_detection_task.py`	file only	

6. The response contains a fresh signed download URL (if signing fails it falls back to the durable URL; the upload itself has already succeeded).

Afterwards the UI offers an **optional** step to draw a reference signature (see 09).

Upload validation

`services/upload_validation.py`, called before anything is stored. **Only PDF documents are accepted** (decision of 2026-10-03: every forensic check is PDF-based, so an image would only ever get OCR and no tamper analysis). The checks run in this order and the first failure wins. Each rejection returns `detail: {"code", "message", ...}`, and the UI shows `message` as-is next to the file.

#	Check	HTTP	`code`	Message / extra fields
1	0 bytes	400	`file_empty`	"The file is empty (0 bytes)."
2	Over `MAX_UPLOAD_SIZE_BYTES` (default **10 MB** = 10,485,760 bytes)	413	`file_too_large`	States both sizes, e.g. "The file is 12.0 MB; the maximum allowed size is 10.0 MB."; `size_bytes`, `max_size_bytes`
3	Content is not a PDF, judged by the **header bytes** (`%PDF-`), never the name or declared MIME type	415	`unsupported_file_type`	"This file is a JPEG image. Only PDF files are accepted — please save or scan the document as a PDF." (the type is named when recognised: JPEG, PNG, TIFF, GIF, BMP images, ZIP/Office files)
4	A PDF whose name carries another type's extension (`.png`, `.docx`, …)	415	`file_type_mismatch`	`detected_type`, `extension`
5	The PDF needs a password to open	422	`file_password_protected`	"This file is password-protected. Please remove the password and re-upload."
6	The PDF can't be parsed locally (PyMuPDF)	422	`file_corrupted`	"The file appears to be corrupted or unreadable."

Details:

- **Magic bytes.** `%PDF-` anywhere in the first 1024 bytes (the PDF spec allows junk before the header; Acrobat and MuPDF accept it). An image renamed to `.pdf` is refused as an image.
- **Extensions.** Only a *recognised* file-type extension (images, Office, text, archives, …) is compared with the content. An unrecognised suffix (the ".2026" in "Invoice 12.03.2026") and a name with no extension are accepted on the content alone; `.PDF` in upper case is fine.
- **Parse.** PyMuPDF — the same library the pipeline renders pages with — opens the file and parses every page's content stream. MuPDF silently "repairs" a truncated file, so two cases count as corrupted: a file it repairs into **0 pages**, and a file it had to repair that has **no `%%EOF` marker in its last 1024 bytes** (a cut-off file can otherwise yield a few pages and silently lose the rest). A merely sloppy PDF with wrong cross-reference offsets still ends in `%%EOF` and is accepted. All 38 sample PDFs pass, including the 12 tampered samples (incremental saves).
- **Password policy: reject-and-ask** (🔶 decision, 2026-10-03). The server never asks for or applies a password, so the system never handles other people's document passwords. A PDF with only an **owner** password (print/copy restrictions; it opens without a password) is accepted. To revisit: the alternative is a password field on upload plus server-side decryption.
- **Stored `content_type`** is always `application/pdf` (the detected type), not the browser-declared one.
- The endpoint reads at most limit + 1 bytes into memory and parses in a worker thread, so the event loop isn't blocked. Uvicorn still receives (and spools to a temp file) the *whole* request body before the endpoint runs — cap the request size at the ingress / reverse proxy in production too (production deployment).

- **Frontend:** `lib/uploadLimits.ts` refuses files that are not named/typed as PDF, empty files and files over 10 MB before sending (the server check is the one that counts). The file picker only offers PDFs and the drop zone states the rules ("PDF only · up to 10.0 MB each · not password-protected"). Keep `MAX_UPLOAD_BYTES` in step with `MAX_UPLOAD_SIZE_BYTES`.
- Tests: `tests/test_upload_validation.py`.

Output shape

`DocumentResponse` (`schemas/document.py`):

```
{
  "id": "uuid", "case_id": "uuid",
  "original_filename": "Invoice.pdf",
  "content_type": "application/pdf",       // detected from the file's header bytes
  "file_size_bytes": 184223,
  "file_hash": "e3b0c4…",                  // SHA-256 hex, 64 chars
  "file_url": "https://…?sv=…&sig=…",      // short-lived SAS URL
  "uploaded_at": "2026-09-20T17:08:40Z",
  "processing_status": "pending"           // pending | processing | complete | failed
}
```

`documents.blob_storage_path` stores the **durable** (unsigned) blob URL. The container is private, so anything that needs the file mints a SAS URL with `StorageService.get_download_url()` (15 min default; 30 min for OCR); `ensure_company_blob()` refuses to sign a path under another company's prefix.

Invariants

- **Originals are write-once.** `StorageService` has `upload()` and `download_bytes()` but no overwrite/delete path. Every later stage only *reads* the original.
- **`file_hash` is the integrity anchor.** It is recomputed and compared when a report is generated (`HashVerification` in the report appendix).
- **Byte-identical re-uploads are not blocked.** `file_hash` is indexed but there is no uniqueness constraint; exact duplicates are caught by the *perceptual* check (10), not at upload.

Known limitations

- No page-count limit and no antivirus scan (type, size, corruption and password checks exist; see above).
- Any company user can create a case, but only its owner or a reviewer of the company can add to it.
- Documents stored before the PDF-only rule may be images; they were OCR'd but no forensic check runs on them, and `pipeline_status` only requires forensic checks for PDFs, so such a case can still be scored.
- One task failing to enqueue (Redis down) after the DB commit leaves a `pending` document that never progresses; there is no sweeper.
- If every file of a new case is rejected, the case itself (created first) remains, with no documents.

Risk rules fed

None directly. It produces the documents every rule evaluates.

02 — Extraction & normalization

What it does: turns an uploaded PDF into structured, comparable data — a document type and a set of normalized fields — with no per-document-type template.

Where it runs: `process_document` is on `extraction_queue`, in a session bound to the document's company. Both Azure calls pass through the global Redis rate limiter: Document Intelligence at `AZURE_DOCUMENT_INTELLIGENCE_MAX_CALLS_PER_SECOND` (10), the Azure OpenAI call within `AZURE_OPENAI_MAX_REQUESTS_PER_MINUTE` / `AZURE_OPENAI_MAX_TOKENS_PER_MINUTE` (each call reserves its estimated prompt tokens + `max_tokens` 4,000). The task commits before each slow step, so it never holds a database transaction across an Azure call (processing-queues.md).

Code: `tasks/document_processing.py` (`process_document`), `services/ocr_service.py`, `services/classification_service.py`, `services/extraction_service.py`, `services/llm_service.py`, `services/field_locator_service.py`.

Algorithm

1. **Status:** `documents.processing_status = processing`.
2. **OCR / layout.** `OCRService.analyze_url(sas_url)` calls Azure Document Intelligence **`prebuilt-layout`** on a 30-minute SAS URL (Azure fetches the file itself). Printed Arabic is supported; **handwritten Arabic is not reliable** (an Azure limitation). Returns an `OCRResult`:

   ```
   @dataclass OCRResult:
       text: str                              # full text, reading order
       tables: list[OCRTable]                 # row_count, column_count, cells
       key_value_pairs: dict[str, str]
       pages: list[OCRPage]                   # page_number, words[], lines[]
   @dataclass OCRWord: text, x, y, width, height   # normalized 0-1 page fractions
   ```

3. **Classify + extract in one call.** `extraction_service.classify_and_extract(llm, ocr_text)` → `LLMService.classify_and_extract`, an Azure OpenAI chat completion at `temperature=0` with **strict JSON-schema output**, returning a `DocumentAnalysis` (below). The label list is passed in from `classification_service.DOCUMENT_TYPE_LABELS`, so adding a type is a one-line change.
4. **Locate fields on the page (best effort).** `field_locator_service.attach_field_locations(pages, extracted_fields)` matches each extracted value back onto the OCR words and stores a normalized `bounding_box` on the field. It is text matching, so the position is approximate; ambiguous values (a total that also appears in a summary line) are disambiguated by nearby keywords ("total", "subtotal", "VAT", "date"); a field that cannot be found simply has no box. **A failure here never fails extraction.**
5. **Persist.** `document_type`, `ocr_text`, `extracted_fields`; `processing_status = complete`; audit `document_processing_completed` `{document_type, document_type_confidence, fields_located}`; then `run_document_checks.delay(document_id, company_id)` (on `vision_queue`).
6. **On any exception:** `processing_status = failed`, `processing_error` (≤ 4000 chars), audit `document_processing_failed`, and the case-level completion check and scoring are still triggered so a failed document cannot block the case forever.

Labels

`school_document, vendor_invoice, commercial_invoice, procurement_documentation, quotation, travel_invoice, payment_evidence, other`.

`get_document_role()` maps them to a **role** used by the cross-document check: `payment_evidence` → `evidence`, `other` → `other`, anything else → `claim`, `None` → `pending`. (A claim and its own evidence are *expected* to differ on issuer and date.) The frontend mirrors this in `types/case.ts`'s `getDocumentRole`; the two are kept in sync **by hand**.

Output shape

The LLM returns a `DocumentAnalysis` (Pydantic, `services/llm_service.py`):

```
class DocumentAnalysis:
    document_type: str
    document_type_confidence: float
    issuer: FieldValue             # value: str|None, confidence: float, uncertain: bool
    reference_number: FieldValue
    date: DateFieldValue           # value = ISO 8601 (YYYY-MM-DD) or None; raw_text kept
    amount: AmountFieldValue       # value: float|None, currency: str|None, raw_text
    subtotal: AmountFieldValue
    tax_amount: AmountFieldValue
    tax_rate: NumericFieldValue
    additional_fields: list[AdditionalField]   # {field_name, value, confidence, uncertain}
```

Amounts are **normalized at extraction time** to plain floats (Western digits, no separators or symbols) with a separate ISO currency code, and dates to ISO 8601, regardless of the document's own script or numeral system — so an Arabic-Indic-numeral invoice and a Western-numeral receipt compare directly. `raw_text` keeps what was actually seen, for display only; **logic must read `.value`, never `.raw_text`**.

What is stored in `documents.extracted_fields` (JSONB):

```
{
  "document_type_confidence": 0.97,
  "core_fields": {
    "issuer":           {"value": "Northgate Engineering", "confidence": 0.95, "uncertain": false,
                         "bounding_box": {"page":1,"x":0.05,"y":0.06,"width":0.30,"height":0.02}},
    "reference_number": {"value": "INV-NGT-2026-1049", "confidence": 0.99, "uncertain": false,
"bounding_box": {…}},
    "date":             {"value": "2026-08-25", "raw_text": "٢٥ أغسطس ٢٠٢٦", "confidence": 0.9,
"uncertain": false, …},
    "amount":           {"value": 197500.0, "currency": "AED", "raw_text": "197,500.00", …},
    "subtotal": {…}, "tax_amount": {…}, "tax_rate": {…}
  },
  "additional_fields": [
    {"field_name": "Payment terms", "value": "30 days", "confidence": 0.8, "uncertain": false,
"bounding_box": {…}}
  ]
}
```

`additional_fields` is how the pipeline stays template-free under Azure's strict-schema rule (which cannot express an open-ended object): arbitrary fields are a list of `{field_name, value}` pairs.

Known limitations

- **Low-confidence routing is not implemented.** `uncertain` is shown on the case detail page but nothing acts on it.
- The extraction model sees **text only** (OCR output), never the page image, so it cannot use visual layout cues beyond what OCR text order conveys.
- Only the first attempt runs; there is no retry of a failed extraction.
- Field locations are approximate and documents extracted before the locator existed have none (`backend/backfill_field_locations.py` re-runs OCR only, no LLM).
- LLM output is validated against a JSON schema but its *content* is not verified against the source.

Risk rules fed

None directly. Extraction feeds 03, 04 and 05.

03 — Field validation

What it detects: internally inconsistent documents — the kind of arithmetic or date mistake that falsified or hand-edited documents often contain. It is **pure rule logic over one document's own extracted fields**: no I/O, no model, no database.

Code: `services/field_validation_service.py` (`validate_fields`), wrapped by `tasks/document_checks.py` (`run_document_checks`), which stores a `document_checks` row with `check_type = field_validation` . Highlight regions come from `services/field_regions.py` and `services/field_exception_service.py` .

Sub-checks

`validate_fields(extracted_fields) -> {"result": "pass"|"flag", "details": {...}}` runs four independent sub-checks; the overall result is `flag` if **any** sub-check is `flag` .

Sub-check	Flags when	Skipped when
`date_in_future`	the document's `date` core field is after today	no parseable date
`subtotal_line_item_consistency`	`subtotal` ≠ sum of additional fields that look like per-line-item totals (name contains `item` , value hint `total` / `amount`)	no subtotal, or no such fields
`total_tax_consistency`	`amount` ≠ `subtotal + tax_amount` (or ≠ `subtotal` when no tax line is shown)	no subtotal
`reference_number_format`	the reference number is not a plausible identifier (regex `^[A-Za-z0-9](?:[A-Za-z0-9\-/_. #]*[A-Za-z0-9])?$`)	—

Line-item detection is deliberately loose because extraction is template-free: line items have no fixed field name.

Output shape (real example)

```
{
  "result": "flag",
  "details": {
    "date_in_future": {"status": "pass", "reason": "Document date 2026-08-25 is not in the future."},
    "total_tax_consistency": {
      "status": "flag",
      "reason": "Total 197500.00 does not match subtotal 180000.00 + tax 9000.00 (expected 189000.00).",
      "regions": [
        {"field": "amount", "label": "Total amount", "value": "197,500.00 AED",
         "caption": "Field exception: Total amount (197,500.00 AED vs expected 189,000.00)",
         "bounding_box": {"page": 1, "x": 0.7821, "y": 0.5081, "width": 0.1321, "height": 0.0119}}
      ]
    },
    "reference_number_format": {"status": "pass", "reason": "…"},
    "subtotal_line_item_consistency": {"status": "pass", "reason": "Subtotal matches the sum of 3 line-item field(s)."}
  }
}
```

Unlike the forensic checks, `details` is a **dict keyed by sub-check**, not a list of findings. Each sub-check is `{status, reason, regions?}` where `status` is `pass` , `flag` or `skipped` (no value to check, or unparseable); `regions` (only on `flag`) are drawn as **purple field exception** boxes over the PDF. If a field's position could not be located, the exception

is still reported without a box. Rows stored before locations existed are re-derived at read time (`with_field_regions`) without modifying the stored row.

Known limitations

- Depends entirely on extraction quality; a wrong extracted value produces a wrong flag (or a miss).
- Both the subtotal-vs-line-items and the total-vs-subtotal+tax comparisons tolerate `max(0.01, 1 % of the compared amount)` ; a document whose rounding differs by more than that will flag.
- `date_in_future` compares against `date.today()` on the worker's machine, not the case's submission date, so a worker with a wrong clock can mis-flag.

Risk rules fed

`rule_id`	Sub-check	Weight	Severity
`field.date_in_future`	`date_in_future`	25	high
`field.subtotal_line_item_mismatch`	`subtotal_line_item_consistency`	15	medium
`field.total_tax_mismatch`	`total_tax_consistency`	20	high
`field.reference_number_malformed`	`reference_number_format`	6	low

(Weights are the template defaults; each company's Reviewer L2s can change them for their company — see 11.)

04 — Cross-document consistency

What it detects: documents in the same case that disagree on facts they should share — most importantly, an invoice total that does not match the payment that supposedly settled it.

Code: `services/cross_document_service.py` (`find_cross_document_mismatches`), run by `tasks/document_checks.py` (`run_cross_document_checks`). Persisted to `cross_document_findings` (a **case-level** table — `document_checks` has no case concept).

When it runs

`run_document_checks` (and the failure path of `process_document`) calls `_maybe_enqueue_cross_document_check` , which enqueues `run_cross_document_checks` once **every** document in the case has a terminal `processing_status` (`complete` or `failed`). The check itself:

1. loads the case's documents and keeps only those that are `complete` ;
2. **deletes any existing findings for the case** and recomputes (this table is *not* append-only, unlike `audit_log`), so a re-run cannot accumulate stale rows;
3. if fewer than **2** completed documents exist, writes no findings;
4. writes one `cross_document_findings` row per mismatch, audit event `cross_document_check_completed {finding_count}` (written even when zero), then requests scoring.

`pipeline_status` waits for a `cross_document_check_completed` event newer than the newest upload before a multi-document case can be scored.

Algorithm

Every unordered pair of completed documents (`itertools.combinations`) is compared on three fields, reading the **normalized** `.value` s (see 02):

Field	Mismatch when	Notes
`amount`	currencies are both present and differ; **or** `abs(a − b) > max(0.01, 1% of the larger)`	Skipped if either side has no amount
`date`	both parse to a date and the dates differ	ISO dates from extraction; fallback formats exist
`issuer`	`rapidfuzz.fuzz.WRatio` (with `default_process`) < **85**	Case/punctuation-insensitive; skipped if either name is empty

Severity depends on the pair's roles

`get_document_role()` classifies each document as `claim` (invoice/quotation/…), `evidence` (`payment_evidence`) or `other` . A claim and its own evidence are **expected** to differ on issuer (vendor vs. bank) and date (invoice date vs. payment date), so:

Field	claim ↔ evidence	any other pair
`amount`	**high** (always)	**high**
`date` , `issuer`	**low**, with an "expected — treat as context" sentence appended	**medium**

The scoring rules only fire on `medium` / `high`, so an expected claim-vs-evidence date/issuer difference stays visible to a reviewer but adds no risk points.

Output shape

A row in `cross_document_findings` (model `models/cross_document_finding.py`):

```
{
  "case_id": "uuid",
  "field_name": "amount",                          // amount | date | issuer
  "finding_type": "cross_document_consistency",
  "severity": "high",                              // low | medium | high
  "description": "Amount differs between 'Invoice.pdf' (9030.00 USD) and 'Evidence.pdf' (7250.00 USD).",
  "document_ids": ["<uuid a>", "<uuid b>"]
}
```

The API returns these as `CrossDocumentFindingSummary`, and derives **one highlight region per involved document** at read time (`field_exception_service.cross_document_regions`), each captioned with what the *other* document said, so the reviewer sees both sides without switching pages.

Known limitations

- Only three fields. Other shared facts (reference numbers, line items, tax IDs) are not compared.
- Pairwise on *all* pairs: a case with three inconsistent documents yields three findings per field.
- A document that failed extraction is excluded, so a case whose only mismatching document failed will not show the mismatch. `pipeline_status` still lets it score.
- Amount tolerance is a flat 1%; there is no per-currency rounding logic and no currency conversion (a currency mismatch is reported, not reconciled).

Risk rules fed

`rule_id`	Field	Fires on severities	Weight	Severity
`cross_doc.amount_mismatch`	`amount`	medium, high	35	high
`cross_doc.date_mismatch`	`date`	medium, high	12	medium
`cross_doc.issuer_mismatch`	`issuer`	medium, high	15	medium

A case-level rule fires **once per case**, with `count` = number of matching findings.

05 — Issuer verification

What it detects: a document issued by someone who is not a known, active vendor/school/issuer — including fabricated or look-alike issuer names.

Code: `services/issuer_service.py` (`match_issuer`), `models/issuer_registry.py`, run by `tasks/document_checks.py` (`run_document_checks`), which stores a `document_checks` row with `check_type = issuer_verification`. Registry administration: `api/settings.py` `/settings/issuers` and the *Settings › Issuer Registry* page (a company's Reviewer L2s; platform admins on a company's behalf).

The registry is per company. Each company maintains its own list of known issuers; a document is only ever matched against **its own company's** registry (`issuer_registry.company_id`, enforced by the tenant-bound session and Row-Level Security).

Algorithm

`match_issuer(db, company_id, issuer_name, threshold=None, llm_service=None) -> IssuerMatchResult`:

1. Empty/blank issuer name → no match.
2. Load every **active** `issuer_registry` row **of the document's company** (deactivated issuers no longer count as known). If none → no match.
3. Build candidate strings: each row contributes its `name` **and**, if present, its `name_arabic`, so an Arabic-script extracted name can match a row whose primary name is English.
4. **Fuzzy pass:** `rapidfuzz.process.extractOne(issuer_name, candidates, scorer=fuzz.WRatio, processor=utils.default_process)` — case-, punctuation- and whitespace-insensitive. Match if the best score ≥ threshold (`ISSUER_FUZZY_MATCH_THRESHOLD`, default **85.0**).
5. **LLM fallback** (only if the fuzzy pass fails and an `LLMService` was passed — the production task passes one): `judge_entity_match(name, candidates)` asks Azure OpenAI whether the extracted name plausibly refers to the same real-world entity as any candidate — this covers transliteration and cross-script cases rapidfuzz cannot score. The model must return one of the candidates **verbatim** or `null`. An LLM failure or missing configuration is swallowed and reported as the fuzzy-only result. The call runs on `vision_queue` through the global Azure OpenAI rate limiter (`max_tokens` 512).

```
@dataclass IssuerMatchResult:
    matched: bool
    best_match_name: str | None
    best_match_score: float | None       # rapidfuzz score, rounded to 2 dp
    threshold: float
    matched_via: Literal["fuzzy", "llm"] | None
```

Output shape (real example)

`document_checks.result` for an issuer that did not match:

```
{
  "result": "flag",                                // pass | flag
  "details": {
    "issuer_name": "بنك أبوظبي الأول | First Abu Dhabi Bank",
    "threshold": 85.0,
    "match_score": 27.94,
    "matched_via": null,                           // "fuzzy" | "llm" | null
    "matched_registry_name": "Al Nukhba Technical Systems Est."   // the closest, even when not a match
  }
}
```

`details` is a **flat dict**, not a findings list. `matched_registry_name` names the closest registry entry even when the check flags, so a reviewer can see what it was nearest to.

Registry data

`issuer_registry` columns: `company_id`, `name`, `name_arabic`, `tax_id`, `type` (enum), `is_active`. Registry actions (create / update / deactivate / reactivate) each write an audit event in the company's log; **rows are never deleted**. A **new company's registry starts empty** — its Reviewer L2s fill it in. The migrations seeded a handful of **fake** test issuers into the "Default Company" created when the single-tenant database was migrated. Names must be unique case-insensitively among a company's active issuers (409 otherwise).

Known limitations

- **A new company's registry is empty, and the Default Company's seeded one is fake.** Until a company's registry holds its real vendors/schools, almost every one of its documents will flag.
- **Names only:** `tax_id` is stored and editable but not compared to the extracted document.
- The LLM fallback is a judgment call that can be wrong in either direction, and it can only choose among registry strings — it cannot verify that the issuer exists in the real world.
- No logo/letterhead matching.
- A single extracted issuer string is checked; multi-party documents (issuer + bank) may extract the "wrong" party as the issuer.

Risk rules fed

`rule_id`	Condition	Weight	Severity
`issuer.not_in_registry`	`issuer_verification` result = `flag`	15	medium

06 — Metadata forensics

What it detects: signs a PDF was edited, re-exported or assembled after it was created: editing software fingerprints, inconsistent or missing timestamps, incremental-save revisions, XMP edit history that contradicts the file, orphaned objects, embedded JavaScript, and a stripped metadata block.

Code: `services/forensics/metadata_forensics.py` (`analyze_pdf_metadata`), run by `tasks/metadata_forensics_task.py` (`run_metadata_forensics`); stores `check_type = metadata_forensics` .

What is read

Using **pikepdf** on the PDF's own object model (never a page render):

- the **Info dictionary** (`/Title /Author /Creator /Producer /CreationDate /ModDate`);
- the separate **XMP metadata stream**, including custom namespaces and the `xmpMM:History` revision chain;
- the **document ID chain** (`/ID`);
- the file's real **incremental-update / xref-revision structure** (`_walk_xref_chain`);
- **reachability**: which objects are unreachable from `/Root` / `/Info` (orphans);
- JavaScript / `OpenAction` , optional content groups, and digital-signature fields.

Findings

Every finding is `{finding, severity, description, data?}` . `severity` is `info | low | medium | high` . `info` findings are context (they show the raw metadata) and never affect the result.

`finding`	Severity	Meaning
`pdf_unreadable`	high	The file could not be opened as a PDF
`metadata_entirely_stripped`	high	No Info and no XMP metadata at all
`editing_software_detected`	high	Producer/Creator/ `softwareAgent` matches an image/graphics editor (`METADATA_FORENSICS_EDITING_SOFTWARE_NAMES` : photoshop, gimp, illustrator, affinity, coreldraw, paint.net, pixlr, canva, inkscape)
`mod_date_after_creation_date`	high	ModDate later than CreationDate by more than `METADATA_FORENSICS_MOD_DATE_THRESHOLD_SECONDS` (default 1 s — only jitter tolerance; being modified at all is the anomaly)
`info_xmp_creation_date_mismatch`	high	Info and XMP creation dates differ by > 1 minute
`info_xmp_mod_date_mismatch`	high	Info and XMP modification dates differ by > 1 minute
`info_xmp_producer_mismatch`	medium	Info `/Producer` ≠ XMP producer
`incremental_updates_present`	medium (1 update) / high (> 1)	The file has been saved incrementally
`history_editing_tool_anomaly`	medium	An XMP history entry names an editing tool
`history_edit_after_final_date`	high	An XMP history event is later than the file's modified date (+1 min)

finding	Severity	Meaning
history_shorter_than_revisions	medium	Fewer history entries than xref revisions
history_absent_despite_revisions	medium	Revisions exist but the history is empty
orphaned_objects	medium (orphans found) / info	Objects present but unreachable — typical of deleted content
javascript_or_openaction	medium (flagged) / info	Embedded JS or auto-run action
optional_content_groups	low / info	Layers present
digital_signature	info	Signature field present / signed / unsigned / absent
info_dictionary, xmp_metadata, xmp_history, revision_count, document_id_chain	info	Raw context

The check `result` is `flag` **if any finding is** `medium` **or** `high`, else `pass`.

Output shape (real, truncated)

```
{
  "result": "flag",
  "details": [
    {"finding": "orphaned_objects", "severity": "medium",
     "description": "Found 25 object(s) present in the file but not reachable from /Root or /Info — likely left…",
     "data": {"count": 25, "object_numbers": [2, 3, "…"]}},
    {"finding": "info_dictionary", "severity": "info", "description": "Info dictionary has 6 field(s).",
     "data": {"/Creator": "Adobe Photoshop 2024 (Windows)", "/ModDate": "D:20260902164500Z",
              "/CreationDate": "D:20260801090000Z", "/Producer": "Adobe Photoshop 2024 (Windows)", "…": "…"}}
  ]
}
```

`details` is a **list** of findings (the same shape ELA, copy-move, duplicate and visual review use, so one findings-list UI renders them all). These findings carry no `page` / `bounding_box`; metadata is not positional.

Known limitations

- **PDF only.** Images and Word files are not analyzed.
- Many legitimate tools leave incremental updates or editor fingerprints (a scanned-then-annotated invoice, for example). A finding is a reason to look, not proof.
- A determined forger can rewrite or strip metadata; `metadata_entirely_stripped` is itself scored because clean-room generators rarely strip everything.
- Uploads are PDF-only, so every new document is analysed; a non-PDF stored before that rule is a quiet no-op.

- The visual-review task cross-references this result (`metadata_forensics_correlation`) to strengthen or explain its own findings; that cross-reference is deliberately **not scored** to avoid double counting.

Risk rules fed

`rule_id`	Finding	Fires on	Weight	Severity
`metadata.pdf_unreadable`	`pdf_unreadable`	any	20	high
`metadata.stripped`	`metadata_entirely_stripped`	any	12	medium
`metadata.editing_software_detected`	`editing_software_detected`	any	30	high
`metadata.creation_date_mismatch`	`info_xmp_creation_date_mismatch`	any	18	high
`metadata.mod_date_mismatch`	`info_xmp_mod_date_mismatch`	any	18	high
`metadata.producer_mismatch`	`info_xmp_producer_mismatch`	any	10	medium
`metadata.modified_after_creation`	`mod_date_after_creation_date`	any	15	high
`metadata.incremental_update_single`	`incremental_updates_present`	medium	8	medium
`metadata.incremental_update_multiple`	`incremental_updates_present`	high	20	high
`metadata.history_tool_anomaly`	`history_editing_tool_anomaly`	any	12	medium
`metadata.history_edit_after_final`	`history_edit_after_final_date`	any	22	high
`metadata.history_shorter_than_revisions`	`history_shorter_than_revisions`	any	8	medium
`metadata.history_absent_despite_revisions`	`history_absent_despite_revisions`	any	8	medium
`metadata.orphaned_objects`	`orphaned_objects`	medium, high	8	medium
`metadata.javascript_or_openaction`	`javascript_or_openaction`	medium, high	10	medium

Note the two severities: a rule's `severity` (how it is displayed) is independent of the *finding's* severity (which the rule may filter on with `severity_in`).

07 — Error Level Analysis & copy-move detection

What they detect: *pixel-level* evidence that part of a page was edited after the rest was last saved (ELA), or that a region was copied from one place on a page to another (copy-move). Together they look for pasted-over totals, cloned signatures/stamps and patched line items.

Code: `services/forensics/ela.py` (`run_ela_check`), `services/forensics/copy_move.py` (`run_copy_move_check`), `services/forensics/pdf_render.py` (`render_pdf_pages`), run by `tasks/tampering_checks_task.py` (`run_tampering_checks`). Two `document_checks` rows result: `error_level_analysis` and `copy_move_detection` .

Shared render

`render_pdf_pages(pdf_bytes) -> list[RenderedPage]` rasterizes every page **once** at **200 DPI** with PyMuPDF into BGR `numpy` arrays and hands the same list to both checks (the render is the expensive part, so `run_tampering_checks` does it once). The duplicate-detection, visual-review and signature tasks render the PDF separately with the same function.

```
@dataclass RenderedPage:
    page_number: int          # 1-based
    image: np.ndarray         # BGR uint8
    width: int; height: int
    has_image_content: bool   # page has ≥1 embedded raster image XObject
```

200 DPI is chosen to resemble a real scan: rendering far higher would exaggerate ELA's block artifacts and make its known print-and-rescan weakness worse.

ELA (Error Level Analysis)

Idea: JPEG-recompress the page at a fixed quality and diff it against the original. A region that was edited after the last save recompresses differently from the rest of the page.

Algorithm (adapted from an internal reference tool): recompress at JPEG quality **75**; take the non-linear `sqrt(diff)` error with a gain; threshold at the **99.7th percentile** with an absolute floor of 60; group into regions; keep regions between **0.2 %** and **60 %** of the page area. `_detect_anti_forensics` separately flags a page that is *suspiciously clean*: a median-blur noise variance below 5.0 ("possible anti-forensic smoothing") or a Laplacian standard deviation below 1.5 ("possible over-compression") — the signature of someone smoothing or re-compressing a page to defeat ELA.

Gate: only pages with `has_image_content` are analyzed. No such page → `not_applicable` .

Findings: `recompression_error_region` (medium) — with `page` , `bounding_box` and `data.region_area_fraction` ; `anti_forensic_signal` (high).

Copy-move detection

Idea: find a region duplicated elsewhere on the *same page* — a common way to hide or fabricate content.

Algorithm (adapted from an internal reference tool, pipeline unchanged apart from one fixed bug):

1. **BRISK** keypoints on the grayscale page (detector threshold 200), keeping those whose normalized response is above `100 – 90` .

2. **Radius matching** of each keypoint's descriptor against all others (Hamming distance ≤ 20 % of 255 ≈ 51), excluding self-matches.
3. **Displacement filter:** discard matches whose two points are closer than `0.15 × min(page height, width) / 2` — nearby points are not "moved" content. That same distance bounds the grouping in the next step.
4. **Cluster** matches that share a similar displacement and whose endpoints sit near each other; keep clusters of **≥ 30** matches.
5. **De-duplicate** overlapping clusters (IoU > 0.3) and keep regions covering ≥ **2 %** of the page.

It runs on **every** page, image-bearing or not: a duplicated paragraph or logo shows up as a keypoint cluster whether the source is a photo or vector text.

Safety valves: if a page yields more than 4000 keypoints after filtering, or more than 4000 matches after the displacement filter, the page is skipped and returns **no clusters** (the clustering loop is $O(n^2)$). A very dense page therefore *cannot* produce a copy-move finding.

Finding: `copy_move_cluster` (high) with `page`, `bounding_box` and `data.matched_pairs`. No visualization image is generated; the frontend draws boxes live.

Output shape (real)

Both checks return `{"result": "pass"|"flag", "details": [Finding, …]}`; ELA can also return `"not_applicable"`.

```
// error_level_analysis
{"result": "flag", "details": [{
  "finding": "recompression_error_region", "severity": "medium", "page": 1,
  "description": "Page 1: a localized area shows a higher JPEG recompression error than the rest of the
page",
  "bounding_box": {"page": 1, "x": 0.5526, "y": 0.4951, "width": 0.3688, "height": 0.0338},
  "data": {"region_area_fraction": 0.0094}}]}

// copy_move_detection
{"result": "flag", "details": [{
  "finding": "copy_move_cluster", "severity": "high", "page": 1,
  "description": "Page 1: found a cluster of 58 matching feature pairs, consistent with one region of this
p…",
  "bounding_box": {"page": 1, "x": 0.348, "y": 0.3164, "width": 0.1469, "height": 0.4271},
  "data": {"matched_pairs": 58}}]}
```

The `Finding` dataclass is `{finding, severity, description, page?, bounding_box?, data?}`.

In the UI

Boxes are drawn live over the PDF by `PdfOverlayViewer` — **ELA solid red, copy-move solid orange** — and in the report's Section 9. Nothing annotated is ever stored by the live UI.

Known limitations

- **ELA is a signal, not a verdict.** It is weak against print-and-rescan tampering and prone to false positives on legitimately recompressed scans. This is why the seeded weights are moderate.
- **Copy-move flags repeated legitimate structure** (table rows, repeated logos or headers, identical stamps) — it has no notion of "expected" repetition.
- Both are PDF-only and both render at a fixed DPI; very small edits can fall below the region-area floors.
- Copy-move silently produces nothing on very dense pages (the 4000-keypoint / 4000-match caps above).

- Findings on documents with no raster content will never come from ELA (`not_applicable`); a pure-text forged PDF is caught, if at all, by metadata, copy-move, visual review and field checks.

Risk rules fed

`rule_id`	Check / finding	Weight	Severity
`ela.tamper_region_detected`	ELA / `recompression_error_region`	25	medium
`ela.anti_forensic_signal`	ELA / `anti_forensic_signal`	30	high
`copy_move.cluster_detected`	copy-move / `copy_move_cluster`	35	high

08 — Visual inconsistency review & AI-generation assessment

What it detects: whether a page *looks* edited or synthetic to a vision model — inconsistent fonts, misaligned text, patchy colour/contrast, uneven sharpness or lighting — and whether it appears AI-generated. It is a **secondary, judgment-based** signal that can catch what pixel forensics miss, or corroborate them.

Code: `services/visual_inconsistency_service.py` (`run_visual_inconsistency_review`), `services/llm_service.py` (`analyze_page_visual_consistency` , `PageVisualAnalysis`), run by `tasks/visual_inconsistency_task.py` ; stores `check_type = visual_inconsistency_review` .

Algorithm

1. Render every page of the PDF (`render_pdf_pages` , 200 DPI) and encode as PNG, longest side capped at **1536 px** (`MAX_IMAGE_DIMENSION`). PNG, not JPEG, so glyph edges stay crisp and compression artifacts are not introduced.
2. Send each page to the vision model **twice, independently** (`RUNS_PER_PAGE = 2`), at `temperature = 0.4` — deliberately non-zero, unlike every other call in `LLMService` , so the runs are genuinely independent samples. Each run answers six structured questions: font consistency, text alignment, colour/contrast, resolution/sharpness, shadow/lighting — each a `VisualCategoryFinding {consistent, description, confidence, bounding_box}` — plus the `AIGenerationAssessment {likely_ai_generated, description, confidence}` .
3. **Corroboration rule** (the reliability mechanism): a category is reported as a real finding only if **both runs** flagged it, or **either** run flagged it with `high` confidence. A single low/medium one-off is kept but labelled "treated as noise" at severity `info` .
4. **Severity:** for a confirmed finding, `high` if both runs agreed *and* at least one was high-confidence, else `medium` ; everything not confirmed (both runs consistent, or an uncorroborated one-off) is `info` . The check's `result` is `flag` if any finding is `medium` or `high` , else `pass` .
5. The task appends `metadata_forensics_correlation` (info) cross-referencing the metadata result, so a reviewer sees editing-tool evidence next to visual evidence. It is **not scored** (avoids double counting), as is `experimental_signal_notice` (an info notice always added to the result).

Findings

`finding`	Meaning
`visual_font_consistency`	Fonts vary within the page
`visual_text_alignment`	Text or table alignment is inconsistent
`visual_color_contrast_consistency`	Colour/contrast differs across regions
`visual_resolution_sharpness_consistency`	Sharpness differs across regions
`visual_shadow_lighting_consistency`	Shadows/lighting inconsistent
`ai_generation_assessment`	The page may be AI-generated/synthetic
`experimental_signal_notice` , `metadata_forensics_correlation`	Info only, never scored

Output shape (real, truncated)

```
{
  "result": "flag",                          // flag if any corroborated finding, else pass
  "details": [
    {"finding": "experimental_signal_notice", "severity": "info",
     "description": "This check uses a vision-capable language model's visual judgment, run twice
independently…"},
    {"finding": "visual_font_consistency", "severity": "high", "page": 1,
     "description": "Page 1 — Font consistency: The font weight and style for the item descriptions … This
is a vision-model judgment, not pixel-level analysis — a probabilistic signal, not a definitive finding.",
     "bounding_box": {"page": 1, "x": 0.05, "y": 0.3, "width": 0.9, "height": 0.15},
     "data": {
       "category": "font_consistency",
       "run_agreement": "2-of-2 runs",       // | "1-of-2 runs (high confidence)" | "consistent in both
runs" | "1-of-2 runs, not corroborated" | "not flagged"
       "run_1": {"consistent": false, "confidence": "high", "description": "…", "bounding_box": {"x": 0.05,
"y": 0.3, "width": 0.9, "height": 0.15}},
       "run_2": {"consistent": false, "confidence": "high", "description": "…", "bounding_box": {…}}}}
  ]
}
```

`data` keeps **both raw runs**, so a reviewer or developer can see exactly what was said and why a finding did or did not survive. The `bounding_box` (from the first run that supplied one) is drawn as a **dashed blue "AI-described area (approximate)"** overlay — deliberately styled to look less precise than the red/orange pixel-forensic boxes.

Known limitations

- **It is a language model's opinion.** Runs vary; corroboration reduces but does not remove hallucination. Boxes are approximate (models are poor at localisation).
- **AI-generation detection is unproven** on scanned/business documents. It is weighted lowest of any visual signal (weight 6) and every description says "experimental, probabilistic".
- **Every page costs two vision calls.** A 10-page PDF is 20 calls; there is no page cap on this task (unlike signature detection, capped at 10 pages).
- Legitimate documents made of mixed sources (a template with a pasted logo, mixed fonts) can flag.
- PDF only.

Risk rules fed

Each rule fires only on `medium` / `high` findings (`severity_in`).

`rule_id`	Finding	Weight	Severity
`visual.font_inconsistency`	`visual_font_consistency`	12	medium
`visual.alignment_inconsistency`	`visual_text_alignment`	10	medium
`visual.color_contrast_inconsistency`	`visual_color_contrast_consistency`	12	medium
`visual.sharpness_inconsistency`	`visual_resolution_sharpness_consistency`	10	medium
`visual.lighting_inconsistency`	`visual_shadow_lighting_consistency`	8	low
`ai.generated_content_suspected`	`ai_generation_assessment`	6	low

09 — Signature & stamp verification

What it does: three separate, deliberately modest things.

1. **Detection** (automatic, every PDF): locates where a handwritten signature or stamp/seal *appears* and whether a document of this kind would normally carry one. Presence and placement only.
2. **Comparison** (on demand): after a reviewer draws a *reference* signature, compares it to the same kind of region on every other document in the case — a qualitative vision-model opinion.
3. **Reuse check** (on demand, no model): notices when the very same signature *image* appears under two documents — which a real hand never produces.

None of these verifies who signed. They give a human reviewer something to look at.

Code: `services/signature_detection_service.py`, `services/signature_comparison_service.py`, `tasks/signature_detection_task.py`, `tasks/signature_comparison_task.py`, `api/signatures.py`, `models/signature_reference.py`, `models/signature_match.py`, frontend `components/upload/SignatureReferenceCreator.tsx`.

1. Detection

`run_signature_detection` (PDF only) renders the pages and calls `detect_signatures(llm, pages, pdf_bytes)`, analysing at most the first **10** pages. For each page it sends a copy of the render with a light **labelled 0.1 grid** overlay — vision models are poor at estimating "where" as a fraction of a page (observed off by up to ~0.3), so the grid lets them read coordinates off the image — and receives a `PageSignatureDetection`:

```
class PageSignatureDetection:
    signature_expected: bool                    # would this kind of document normally carry one?
    regions: list[DetectedSignatureRegion]      # kind: "signature"|"stamp", description,
                                                # confidence: low|medium|high, bounding_box
```

Box refinement (accuracy fix)

The model's box is only a first guess, so each region is refined before it is stored:

1. **`embedded_image`** — if the PDF has an embedded raster image near the model's box (the usual case for a pasted or scanned signature/stamp), the image's exact placement rectangle replaces the guess. Boxes and images are paired nearest-first, one-to-one, so a signature and a stamp never claim the same image. Page-sized images (> 35 % of the page) and rotated pages are ignored.
2. **`second_look`** — otherwise the model is shown a zoomed window around its own box (grown by 10 % of the page width and 8 % of its height on each side) and asked again; the answer is mapped back to page coordinates. Skipped if the window is blank.
3. **`none`** — neither worked; the (padded) first-pass box is kept.

Refined boxes are trimmed to the ink inside them (+0.4 % margin). The chosen method is recorded per region in `refinement`. Without this, boxes were seen to land on the printed name under a signature instead of the signature itself.

Result (stored as a `document_checks` row, `check_type = signature_stamp_detection`)

```
{
  "result": "pass",                       // "flag" only when a signature was expected but none found
  "details": {
    "signature_expected": true,
    "detected": [
      {"kind": "signature", "confidence": "high", "refinement": "embedded_image",
       "description": "Handwritten signature in blue ink under the label …",
       "bounding_box": {"page": 1, "x": 0.130, "y": 0.493, "width": 0.177, "height": 0.053}},
      {"kind": "stamp", "confidence": "high", "refinement": "embedded_image", "description": "Round company
stamp …",
       "bounding_box": {"page": 1, "x": 0.441, "y": 0.485, "width": 0.118, "height": 0.087}}
    ],
    "bounding_box": {"page": 1, "x": 0.130, "y": 0.493, "width": 0.177, "height": 0.053}   // the "primary"
region, or null
  }
}
```

`bounding_box` is the primary region: highest confidence, signature preferred on a tie. `detected` keeps every region with its `kind`; results stored before that field existed have only `bounding_box`. Presence alone is a **pass**.

2. Reference creation

`POST /cases/{case_id}/documents/{document_id}/signature-references` with a `SignatureReferenceCreate`:

```
{"person_name": "Sultan Al-Dhaheri",       // typed by the uploader/reviewer; NEVER auto-filled
 "bounding_box": {"page": 1, "x": 0.10, "y": 0.49, "width": 0.22, "height": 0.05},
 "is_library": true}                        // default true
```

Allowed for the case's owner (who sets the reference while uploading) and any reviewer of the company; anyone else gets a 404, and platform admins are read-only. The endpoint renders the PDF page, crops the normalized box, uploads a PNG to `companies/{company_id}/cases/{case_id}/signatures/{crop_id}.png`, inserts the `signature_references` row and enqueues `run_signature_comparison` (`vision_queue`). If the crop fails (render error) the row is still created with a null image URL and comparisons record `cannot_determine`. The box is drawn in the UI on a `react-pdf` viewer (`SignatureReferenceCreator`), offered after upload. References, like everything else, belong to one company: the library flag never makes a reference visible to another company.

3. Comparison (`run_signature_comparison`)

For the new reference, the task:

1. **Infers the reference's kind** (`signature` or `stamp`) by overlapping its box with the source document's detected regions (IoU > 0.1). Unknown if nothing overlaps.
2. For each **other** document in the case, picks the detected region of the **same kind** (a signature is never compared with a stamp); documents with no region of that kind are skipped. If the kind is unknown, the document's primary region is used. Documents whose detection has not finished are skipped and picked up when it does (the detection task re-triggers this one; it is idempotent and skips already-compared pairs).
3. Crops the target region (fresh blob name per attempt).
4. **Reuse check** (signature references only): see below. A match is stored and the model is **not** called.
5. Otherwise calls `LLMService.compare_signatures(ref_image, target_image)` — a strict-JSON vision call at `temperature = 0` returning `{reasoning, result}` (`reasoning` first, so the model commits its thinking before naming a verdict). Advisory wording only.

6. Stores a `signature_matches` row, audits `signature_comparison_completed` `{comparisons_attempted, comparisons_stored, reference_kind, identical_reuse_found}`, and re-scores the case.

If the vision model is not configured, non-reuse targets get `cannot_determine`; the reuse check still runs because it needs no model.

The reuse check (classical CV, no model)

A model asked "do these look alike?" calls two identical images "consistent" — reassurance, when the real signal is the opposite: nobody signs pixel-for-pixel the same twice, so an exact match means the same image file was inserted into both documents. That can be legitimate (a stored e-signature image), so on its own it is a modest flag.

`find_signature_reuse` trims both crops to their ink, requires similar proportions (± 6 %), resamples to 256 px wide, blurs, and takes the **maximum normalized cross-correlation** allowing ±4 px of shift. Measured on real files: identical copies score 0.999, a 60 % rescale 0.97, JPEG recompression 0.999, a wobbled re-signing 0.40, a different signature 0.17. The threshold is **0.93**.

On a match it reads the **printed line beneath each signature** from the PDF's own text layer (`printed_label_below`, the first text line within 7 % of page height below the box) and compares them with `rapidfuzz.fuzz.token_set_ratio` (< 60 = different people):

Verdict	Meaning
`reused_different_signer`	identical image **and** the printed signer text differs
`identical_reuse`	identical image; printed text similar or unreadable (e.g. a scan)

It never runs for **stamp** references — a company's rubber stamp is legitimately identical every time.

Output shape

Verdicts (`signature_match_result` enum) and their UI labels:

`result`	Source	Label
`consistent`	vision model	Visually consistent with reference on file
`possibly_consistent`	vision model	Possibly consistent with reference on file
`inconsistent`	vision model	Visual appearance inconsistent with reference
`cannot_determine`	vision model / failure	Cannot determine — insufficient image quality
`identical_reuse`	pixel check	Pixel-identical to reference — same image appears to be reused
`reused_different_signer`	pixel check	Pixel-identical to reference, but printed signer differs

`SignatureMatchResponse` (`schemas/signature.py`):


```
{"id": "uuid", "document_id": "uuid", "case_id": "uuid", "signature_reference_id": "uuid",
 "reference_person_name": "Sultan Al-Dhaheri", "comparison_scope": "in_case",
 "result": "reused_different_signer", "result_label": "Pixel-identical to reference, but printed signer
differs",
 "reasoning": "The signature here is pixel-for-pixel identical to the reference … reference: 'م . سلطان
الظاهري'; this document: 'سعيد المنصوري' …",
 "compared_at": "2026-09-20T23:10:15Z"}
```


There is **no numeric score** on the model verdicts, by design.

Known limitations

- **Weak, advisory signal.** Handwriting varies; a model's visual opinion is not identity matching. Copy must never claim verification.
- **Reviewer-initiated.** With no reference, no comparison happens. A reference is per case; the library is data collection only.
- **Detection depends on a vision model** and, for non-embedded marks, on its localisation.
- The reuse check needs a **text layer** to compare signer names (scans fall back to `identical_reuse`), and only sees marks that were cropped in the first place.
- The reference's kind is inferred from overlap; a box that overlaps neither region leaves it unknown (and skips the reuse check).
- The task and endpoints are PDF-only.

Risk rules fed

`rule_id`	Condition	Weight	Severity
`signature.expected_missing`	detection `flag` (expected but none located)	10	low
`signature.inconsistent_with_reference`	verdict `inconsistent`	25	medium
`signature.possibly_consistent_only`	verdict `possibly_consistent`	5	low
`signature.identical_reuse`	verdict `identical_reuse`	15	medium
`signature.reused_different_signer`	verdict `reused_different_signer`	35	high

`cannot_determine` and `consistent` fire no default rule (a company's Reviewer L2 can add one for `cannot_determine`).

10 — Duplicate & near-duplicate detection

What it detects: a page that looks the same as a page in a document seen before — a resubmitted invoice, the same quotation re-uploaded under a slightly different vendor name, a re-scan, a re-saved copy. It catches what a byte-level SHA-256 comparison (`documents.file_hash`) cannot: re-saving, re-compressing or trivially editing a file changes its bytes but not how it *looks*.

Code: `services/forensics/duplicate_check.py` (`run_duplicate_check`), run by `tasks/duplicate_check_task.py` (`run_duplicate_check_task`); model `models/document_page_hash.py` (`document_page_hashes`); stores `check_type = duplicate_detection` .

Algorithm

1. Render each page (`render_pdf_pages` , 200 DPI) and compute a **perceptual hash**: `imagehash.phash` (64-bit, `hash_size = 8`) of the RGB page image → a 16-character hex string, one per page (`compute_page_hashes`).
2. `find_matches` loads **every stored page hash of every other document of the same company** (joined to filename, case id and case number) and, for each new page, finds the single **closest** prior page whose Hamming distance is ≤ `DUPLICATE_HASH_HAMMING_THRESHOLD` (default **5** of 64 bits — deliberately conservative: near- identical resubmissions, not merely similar layouts such as two invoices from one template). One match per new page at most.
3. `store_page_hashes` writes this document's own hashes **regardless of outcome**, so future uploads can match against it — as an upsert, one row per (document, page), so a redelivered task can't duplicate them. (The hash is a side effect of the check; there is no separate step.)
4. `result` is `flag` if any page matched, else `pass` .

Match details (filename, case number) are captured **at check time**, not resolved later, so a finding stays readable even if the matched document's case is one the current viewer has never opened.

Output shape (real)

```
{
  "result": "flag",
  "details": [{
    "finding": "near_duplicate_page",
    "severity": "high",
    "page": 1,
    "description": "Page 1 is a near-identical match (perceptual hash Hamming distance 0) to page 1 of
\"Sample15_Arabic_Case_Evidence.pdf\" in case CASE-8D8C00E1 — a possible duplicate or near-duplicate
resubmission.",
    "data": {
      "matched_document_id": "uuid", "matched_document_filename": "Sample15_Arabic_Case_Evidence.pdf",
      "matched_case_id": "uuid", "matched_case_number": "CASE-8D8C00E1",
      "matched_page": 1, "distance": 0
    }
  }]
}
```

Same `{finding, severity, description, page, data}` shape as ELA/copy-move (no `bounding_box` — the whole page is the match), so the shared findings-list UI renders it. In the UI the finding links to the matched case.

`document_page_hashes` : `id` , `company_id` , `document_id` (FK), `page_number` (1-based), `phash` (`varchar(32)` , indexed), timestamps; unique on (`document_id` , `page_number`).

Known limitations

- **It matches against everything the company ever stored**, including your own test uploads. Re-uploading the same sample file in a new case (or uploading the two halves of one case that share a page) **will flag**. This is correct behaviour, and the most common source of "unexpected" flags during testing.
- **Never across companies.** The same invoice submitted to two different client companies is not detected: matching across tenants would reveal another company's filenames and case numbers. This is a deliberate privacy choice.
- **Same-case matches count** (see the note at the top) — despite the rule's `cross_case` name.
- Full-scan comparison per page: cost grows linearly with the number of stored pages. There is no index (e.g. BK-tree/LSH); fine for thousands of pages, not millions.
- pHash is sensitive to layout, not content: two documents that differ only in a changed amount will hash almost identically (that is the intent, and why it is high severity), whereas a heavy crop or rotation can defeat it.
- No cleanup: hashes of documents whose cases are closed stay forever. Two uploads processed at exactly the same moment by parallel forensics processes may not see each other.
- PDF only (as are uploads).

Risk rules fed

`rule_id`	Finding	Weight	Severity
`duplicate.cross_case_match`	`near_duplicate_page`	40	high

This is the heaviest single seeded rule: with the default thresholds (medium ≥ 30, high ≥ 60) one duplicate page alone makes a case **medium** risk; with any other signal it approaches **high**.

11 — Risk scoring engine

What it does: turns every stored check result into a single **explainable** 0–100 score, a tier (`low` / `medium` / `high`) and a list of human-readable reasons. It is a **transparent, weighted rules engine — not a model**. Every point in a score traces to a named rule and a sentence a reviewer can read.

Code: `services/risk_scoring_service.py` (`score_case` , `evaluate_rules` , `pipeline_status`), `tasks/risk_scoring_task.py` (`score_case_task`), `services/risk_rule_seed.py` (the default rules and `seed_risk_rules`), `services/risk_rule_catalog.py` (what a rule editor may build), `api/settings.py` (rule and threshold administration), `api/platform.py` (the platform rule template), models `risk_rules` , `risk_settings` , `risk_rule_templates` , `case_risk_assessments` .

Rules and thresholds are per company. Each company has its own `risk_rules` and `risk_settings` , edited by its Reviewer L2s under *Settings › Risk Rules*; a case is always scored with its own company's rules. A new company starts from the platform's **rule template** (see Rule templates).

When a case is scored

Every pipeline task ends with `request_case_scoring(case_id)` , which enqueues `score_case_task` on `forensics_queue` (and never raises if the broker is down). `score_case(db, company_id, case_id)` then:

1. **Gate —** `pipeline_status(case)` must be `complete` . For every document: extraction terminal; for every `complete` document: `field_validation` and `issuer_verification` terminal; for every **PDF**: `metadata_forensics` , `error_level_analysis` , `copy_move_detection` , `duplicate_detection` , `visual_inconsistency_review` terminal (`completed` or `failed`); and, when the case has ≥ 2 documents, a `cross_document_check_completed` event **newer than the newest upload**. Anything missing is listed in `pipeline.pending` and shown to the reviewer. (`signature_stamp_detection` is intentionally **not** required — see data-flow.) Scoring a half-checked case would under-report and could let it be approved.
2. **Gather evidence:** the **latest completed** check per (document, check_type), the case's `cross_document_findings` , and all `signature_matches` .
3. **Evaluate every current rule of the case's company** (highest `version` per `rule_id` , inactive included but never fired).
4. **Fingerprint** = SHA-256 of the sorted document ids + the sorted `(rule_id, document_id, count)` of fired rules. If it equals the latest assessment's `evidence_fingerprint` , **stop** — no new row.
5. **Score:** `raw_score = Σ weight` of fired rules; `score = clamp(round(raw_score), 0, 100)` ; `tier` from the company's thresholds (`score ≥ high → high` , `≥ medium → medium` , else `low` ; defaults **30 / 60**).
6. **INSERT** a `case_risk_assessments` row (one per (case, fingerprint) — a unique constraint, so a redelivered task can't insert it twice), set `cases.risk_tier` , write `risk_assessment_completed` , and — if the case is still `submitted` / `under_automated_review` — transition it to `pending_manual_review` via the workflow service.

The fingerprint deliberately excludes weights, so a re-trigger can never quietly re-score an old case under newly tuned weights: a case is only re-scored when its *evidence* changes.

Rules

A rule is a row in the company's `risk_rules` — configuration, not code:

Column	Meaning
`rule_id`	Stable key, `family.name` (e.g. `metadata.editing_software_detected`)
`category`	`forensics`, `consistency`, `verification` or `duplication`
`check_type`	Which check's output it reads (used for display/grouping)
`condition`	JSON: **what to match** (below) — created by the catalog, not hand-written
`weight`	Points added when it fires (−100…100)
`severity`	`info` / `low` / `medium` / `high` — shown beside the reason
`reason_template`	Sentence with `{placeholders}` the reviewer reads
`is_active`	Off = evaluated but never fires
`version`, `effective_from`, `updated_by`, `change_note`	Versioning trail

Condition kinds (`condition.match`)

Kind	Matches	Fires
`finding`	Findings in a check's `details` **list** whose `finding` name equals `condition.finding` and (optionally) whose severity ∈ `severity_in`	**Once per document**, `count` = number of matching findings
`check_result`	The check's overall `result` equals `condition.result` (e.g. `flag`)	Once per document
`sub_check`	One sub-check in a **dict**-shaped `details` (field validation) with `status == flag`	Once per document
`cross_document`	A `cross_document_findings` row for `condition.field_name`, optionally filtered by severity	**Once per case**, `count` = matching findings
`signature_match`	A `signature_matches` row whose `result` equals `condition.result`	Once per target document

A rule fires **once per hit, regardless of how many findings a document has** — five separate `orphaned_objects` findings on one document add the weight once, not five times.

Reason templates

`render_reason` fills the template from the hit's context. Available placeholders: `{document}` (filename), `{count}`, `{page}`, `{description}`, `{reason}`, `{issuer}`, `{person}`, `{field}`, plus any key in a finding's `data`. A placeholder with no value renders as an **empty string** (never an error). The **rendered** text is what is stored and shown.

The default rules (38)

These are the rules in the platform template, i.e. what a new company starts with. Full tables of `rule_id`, condition and default weight are in each stage's doc: 03 (4), 04 (3), 05 (1), 06 (15), 07 (3), 08 (6), 09 (5), 10 (1). Heaviest single rules: `duplicate.cross_case_match` (40), `copy_move.cluster_detected` / `cross_doc.amount_mismatch` / `signature.reused_different_signer` (35). Because weights sum, a case with an amount mismatch (35) plus a duplicate page (40) reaches 75 → **high**.

Unscored findings and the coverage warning

Medium/high findings that **no rule covers** are logged as warnings ("No risk rule covers …") so a new finding type cannot silently go unscored. `metadata_forensics_correlation` and `experimental_signal_notice` are excluded on purpose (see 08).

Rule templates — how a new company gets its rules

- `risk_rule_templates` is a platform-level table (no `company_id`; only the platform database role can reach it). Platform admins edit it under **Platform › Rule templates** (`/platform/risk-rule-templates`): weight, severity, active flag, or a new rule. Each template edit is a platform audit event (`risk_rule_template_created/updated`, old → new values).
- Creating a company (`POST /platform/companies`) calls `seed_risk_rules(db, company_id)`, which copies every **active** template row into that company's `risk_rules` as version 1, plus default thresholds (30 / 60) into `risk_settings`.
- From then on the copy is the company's own. **Editing the template never changes an existing company** — only companies created afterwards. This is the same "no retroactive change" principle as rule versioning below.
- The template was seeded from the Default Company's current rules (38).

Versioning — history is never rewritten

- **Rules are immutable.** `PATCH /settings/risk-rules/{rule_id}` (weight, severity, `is_active`) INSERTs a new row with `version + 1` in the company's rules; the old row is untouched (the database grants only SELECT + INSERT on `risk_rules`). The current rule is the highest version — there is no mutable "is_current" flag.
- **Creating a rule** (`POST /settings/risk-rules`) inserts version 1; it applies to cases scored from then on. The editor (a Reviewer L2) picks a match kind and parameters from the catalog (`GET /settings/risk-rule-options`); nobody writes JSON, and a choice the engine cannot evaluate is a 422. Wording (`reason_template`) and `condition` cannot be edited after creation.
- **Every assessment freezes what it used**: the rendered reason text, and `risk_rules_version_snapshot = {"rules": [{rule_pk, rule_id, version, weight, severity}, …], "thresholds": {"medium": …, "high": …}}`. Retuning weights or thresholds later cannot change a historical case's score or reasons.

Output shape

Stored — `case_risk_assessments` (insert-only), the full record:

```
{
  "id": "uuid", "case_id": "uuid", "computed_at": "2026-09-20T17:09:12Z",
  "score": 75, "raw_score": 75.0, "tier": "high",
  "triggered_reasons": [                  // sorted by weight desc, then rule_id
    {"rule_id": "duplicate.cross_case_match", "rule_version": 1, "category": "duplication",
     "check_type": "duplicate_detection", "severity": "high", "weight": 40.0,
     "reason": "Page 1 of 'Invoice.pdf' is a near-identical match … to page 1 of 'X.pdf' in case CASE-….",
     "document_id": "uuid", "document_filename": "Invoice.pdf"},   // both null for case-level rules
    {"rule_id": "cross_doc.amount_mismatch", "rule_version": 1, "weight": 35.0, "…": "…"}
  ],
  "risk_rules_version_snapshot": {        // exactly which immutable versions produced the score
    "rules": [{"rule_pk": "uuid", "rule_id": "duplicate.cross_case_match", "version": 1, "weight": 40.0,
"severity": "high"}, "…"],
    "thresholds": {"medium": 30, "high": 60}
  },
  "evidence_fingerprint": "9f2c…"         // SHA-256 used to skip unchanged re-scores
}
```

Exposed by the API — `RiskAssessmentSchema` on `GET /cases/{id}` (`assessment`; reviewers and platform admins only, `null` for a submitter) is a **subset**: `{id, score, raw_score, tier, computed_at, triggered_reasons[], thresholds}`, where each reason is a `RiskReasonSchema` with the fields shown above and `thresholds` is copied from the snapshot. The rule-version snapshot and the evidence fingerprint stay internal (the snapshot is printed in the report's Section 6, the fingerprint is never shown).

What the tier does

Tier	Effect in the code
any	Case flag/badge colour; case moves to `pending_manual_review`; report Executive Summary
`low`	Reviewer may approve without a written justification
`medium` / `high`	Approving needs a justification of ≥ 10 characters (422 otherwise); rejecting/escalating always possible
—	**No** auto-approval, **no** automatic escalation to L2, **no** SLA (none of these exist)

Submitters see only a neutral "In review" flag, never the tier, score or reasons — so a bad actor cannot learn what to evade.

Known limitations

- **Weights are judgment, not calibration.** The seeded values were chosen to be sensible, not fitted to data; tune them against real cases (Settings › Risk Rules, per company; or the platform template for future companies). The thresholds default to 30/60.
- Rules only see what checks store. A check that fails (status `failed`) contributes nothing, and the case still scores — a `failed` forensic task under-reports risk silently (the failure is in the audit log and shown as a failed check, but no rule fires on it).
- Additive scoring: many weak signals can outweigh one strong one, and signals are not de-correlated (an edited PDF may trip metadata, ELA and visual rules for one root cause).
- The default rules encode this deployment's judgment about weak signals (ELA, AI-generation and signature checks are deliberately light).
- A case re-scores when evidence changes (for example a signature comparison finishing), so the score a reviewer first saw can change while they are looking at the case.

Database schema

PostgreSQL, 19 application tables plus Alembic's `alembic_version` (current head `c9e3a1b5d7f4`). All primary keys are UUIDs generated by the application. The entity-relationship diagram is in erd.md; tenant isolation is described in multi-tenancy.md.

Conventions

- **Multi-tenant.** Every tenant-owned table has `company_id` (FK → `companies.id`). Row-Level Security is enabled on them with a `tenant_isolation` policy for `fddt_app` (`company_id = fddt_current_company_id()`, the transaction-local `app.current_company_id`) and a `platform_all` policy for `fddt_platform`. No role has BYPASSRLS.
- **Insert-only tables** (the app roles are granted only SELECT + INSERT): `audit_log`, `case_risk_assessments`, `case_reports`, `risk_rules` (rule edits insert a new version).
- **JSON columns** (`jsonb`) hold check output and snapshots; their shapes are documented in the pipeline docs.
- **Bounding boxes** anywhere in JSON use one convention: `{page (1-based), x, y, width, height}` as 0-1 fractions of the page, origin top-left.
- **Enums** are native Postgres enums; their values are listed under each table and again in the index at the bottom.

Table	Tenancy	`fddt_app`	Purpose
`companies`	RLS (own row)	SELECT	One row per client company (tenant). `is_active = false` suspends it: its users can't sign in and existing tokens stop working.
`company_usage_stats`	tenant · RLS	SELECT, INSERT, UPDATE	Per-company, per-UTC-day usage counters (cases, documents, files, storage bytes) for the platform billing dashboard; bumped in the same transaction as each upload/case/report and reconciled nightly.
`users`	tenant · RLS	SELECT	Accounts (email + bcrypt-hashed password). Company users have a `company_id` and a ranked role (`user` / `reviewer_l1` / `reviewer_l2`); platform admins have `role = platform_admin` and no company (CHECK constraint).
`cases`	tenant · RLS	SELECT, INSERT, UPDATE	The unit of work: one submission of one or more documents. Carries workflow `status`, the owning reviewer tier `assigned_tier` and the latest `risk_tier`.
`documents`	tenant · RLS	SELECT, INSERT, UPDATE	One row per uploaded PDF. The original bytes live in Blob Storage (immutable); this row holds the SHA-256, the extraction results (`extracted_fields`, `ocr_text`) and `processing_status`.
`document_checks`	tenant · RLS	SELECT, INSERT, UPDATE	One row per (document, check type) — the latest run of each automated check (extraction stages, every forensic check, signature detection…), upserted so a redelivered task never duplicates it. `result` is the check's raw JSON output; the run history is in `audit_log`.
`document_page_hashes`	tenant · RLS	SELECT, INSERT, UPDATE	Perceptual hash (pHash) of every rendered PDF page, one per (document, page), used to find duplicate/near-duplicate pages within the same company.
`cross_document_findings`	tenant · RLS	SELECT, INSERT, UPDATE, DELETE	Case-level reconciliation results: a shared field (amount, date, issuer) that disagrees between documents in the same case.

Table	Tenancy	`fddt_app`	Purpose
`issuer_registry`	tenant · RLS	SELECT, INSERT, UPDATE	A company's known vendors/schools/etc. that extracted issuer names are fuzzy-matched against. Rows are deactivated, never deleted. A new company starts empty.
`risk_rules`	tenant · RLS	SELECT, INSERT	A company's configurable scoring rules. Rows are immutable and versioned: an edit inserts `version + 1`; the current rule is the highest version. Seeded from `risk_rule_templates` when the company is created.
`risk_rule_templates`	platform-only	—	PLATFORM-LEVEL (no `company_id`): the default rule set copied into every NEW company's `risk_rules`. Edited by platform admins; edits never change existing companies. Only `fddt_platform` can access it.
`risk_settings`	tenant · RLS	SELECT, INSERT, UPDATE	One row per company holding its `medium` / `high` score thresholds.
`case_risk_assessments`	tenant · RLS	SELECT, INSERT	One row per scoring run of a case (insert-only history; unique per (case, evidence fingerprint)): score, tier, frozen reason text and a snapshot of the exact rule versions and thresholds used.
`risk_scores`	tenant · RLS	SELECT	LEGACY. Created by the initial migration before the versioned engine existed; nothing in the application reads or writes it. `case_risk_assessments` replaced it.
`case_actions`	tenant · RLS	SELECT, INSERT	Reviewer-facing history of human actions on a case (approve / reject / escalate …) with the actor's role at the time. Complements, and is also mirrored into, `audit_log`.
`case_reports`	tenant · RLS	SELECT, INSERT	History of generated per-case forensic PDF reports (insert-only). The PDF lives in Blob Storage under `companies/{company}/cases/{case}/reports/`; this row stores its SHA-256, size and the assessment it reflects.
`audit_log`	tenant · RLS	SELECT, INSERT	Append-only system of record: every automated check result, human action and settings change is inserted here and never updated or deleted. `company_id` NULL = a platform-level event (company created, template edits, usage reconciliation, platform-admin access), visible only to platform admins; a company user's *user created/updated* goes into that company's log.
`signature_references`	tenant · RLS	SELECT, INSERT, UPDATE	A signature/stamp region drawn on a document, cropped and stored as a PNG, used as the reference for comparisons.
`signature_matches`	tenant · RLS	SELECT, INSERT, UPDATE	Result of comparing one reference against the signature region of another document in the same case (vision-model verdict or pixel-identical reuse verdict); one row per (reference, document, scope).

companies

One row per client company (tenant). `is_active = false` suspends it: its users can't sign in and existing tokens stop working.

Column	Type	Null	Default	Notes
`id`	uuid	no		PK

Column	Type	Null	Default	Notes
`name`	varchar(255)	no		unique
`is_active`	boolean	no	`true`	
`created_at`	timestamptz	no	`now()`	
`updated_at`	timestamptz	no	`now()`	

- **Index** `companies_name_key` on (name) — unique
- **Row-Level Security** on: `platform_all` (ALL to fddt_platform); `tenant_isolation` (ALL to fddt_app)
- **Grants** `fddt_app` : SELECT · `fddt_platform` : SELECT, INSERT, UPDATE, DELETE

company_usage_stats

Per-company, per-UTC-day usage counters (cases, documents, files, storage bytes) for the platform billing dashboard; bumped in the same transaction as each upload/case/report and reconciled nightly.

Column	Type	Null	Default	Notes
`company_id`	uuid	no		PK, FK → `companies.id`
`usage_date`	date	no		PK
`cases_created`	integer	no	`0`	
`documents_uploaded`	integer	no	`0`	
`files_stored`	integer	no	`0`	
`storage_bytes`	bigint	no	`'0'::bigint`	
`updated_at`	timestamptz	no	`now()`	

- **Row-Level Security** on: `platform_all` (ALL to fddt_platform); `tenant_isolation` (ALL to fddt_app)
- **Grants** `fddt_app` : SELECT, INSERT, UPDATE · `fddt_platform` : SELECT, INSERT, UPDATE, DELETE

users

Accounts (email + bcrypt-hashed password). Company users have a `company_id` and a ranked role (`user` / `reviewer_l1` / `reviewer_l2`); platform admins have `role = platform_admin` and no company (CHECK constraint).

Column	Type	Null	Default	Notes
`id`	uuid	no		PK
`created_at`	timestamptz	no		
`updated_at`	timestamptz	no		
`email`	varchar(320)	no		unique
`hashed_password`	varchar(255)	no		
`full_name`	varchar(255)	yes		
`role`	enum `user_role`	no		

Column	Type	Null	Default	Notes
`is_active`	boolean	no		
`company_id`	uuid	yes		FK → `companies.id`

- **Index** `ix_users_company_id` on (company_id)
- **Index** `ix_users_email` on (email)
- **Index** `users_email_key` on (email) — unique
- **Check** `ck_users_platform_admin_has_no_company` : `CHECK (((role = 'platform_admin'::user_role) = (company_id IS NULL)))`
- **Enum** `user_role` : `user` , `reviewer_l1` , `reviewer_l2` , `admin` , `platform_admin`
- **Row-Level Security** on: `platform_all` (ALL to fddt_platform); `tenant_isolation` (ALL to fddt_app)
- **Grants** `fddt_app` : SELECT · `fddt_platform` : SELECT, INSERT, UPDATE, DELETE

cases

The unit of work: one submission of one or more documents. Carries workflow `status` , the owning reviewer tier `assigned_tier` and the latest `risk_tier` .

Column	Type	Null	Default	Notes
`id`	uuid	no		PK
`created_at`	timestamptz	no		
`updated_at`	timestamptz	no		
`case_number`	varchar(64)	no		unique
`submitted_by_user_id`	uuid	no		FK → `users.id`
`status`	enum `case_status`	no		
`risk_tier`	enum `risk_tier`	yes		
`external_ref_id`	varchar(255)	yes		
`case_type`	enum `case_type`	no		
`assigned_tier`	enum `case_tier`	no	`'l1'::case_tier`	
`company_id`	uuid	no		FK → `companies.id`

- **Index** `cases_case_number_key` on (case_number) — unique
- **Index** `ix_cases_case_number` on (case_number)
- **Index** `ix_cases_company_id` on (company_id)
- **Enum** `case_status` : `submitted` , `under_automated_review` , `pending_manual_review` , `auto_approved` , `escalated` , `under_investigation` , `approved` , `rejected` , `closed`
- **Enum** `risk_tier` : `low` , `medium` , `high`
- **Enum** `case_type` : `school_document` , `vendor_invoice` , `commercial_invoice` , `procurement_documentation` , `quotation` , `travel_reimbursement` , `other`
- **Enum** `case_tier` : `l1` , `l2`
- **Row-Level Security** on: `platform_all` (ALL to fddt_platform); `tenant_isolation` (ALL to fddt_app)

- **Grants** `fddt_app`: SELECT, INSERT, UPDATE · `fddt_platform`: SELECT, INSERT, UPDATE, DELETE

documents

One row per uploaded PDF. The original bytes live in Blob Storage (immutable); this row holds the SHA-256, the extraction results (`extracted_fields` , `ocr_text`) and `processing_status` .

Column	Type	Null	Default	Notes
`id`	uuid	no		PK
`created_at`	timestamptz	no		
`updated_at`	timestamptz	no		
`case_id`	uuid	no		FK → `cases.id`
`uploaded_by_user_id`	uuid	no		FK → `users.id`
`original_filename`	varchar(512)	no		
`blob_storage_path`	varchar(1024)	no		
`file_hash`	varchar(64)	no		
`content_type`	varchar(255)	yes		
`file_size_bytes`	integer	yes		
`document_type`	varchar(128)	yes		
`processing_status`	enum `document_processing_status`	no		
`extracted_fields`	jsonb	yes		
`ocr_text`	text	yes		
`processing_error`	text	yes		
`company_id`	uuid	no		FK → `companies.id`

- **Index** `ix_documents_case_id` on (case_id)
- **Index** `ix_documents_company_id` on (company_id)
- **Index** `ix_documents_file_hash` on (file_hash)
- **Enum** `document_processing_status`: `pending` , `processing` , `complete` , `failed`
- **Row-Level Security** on: `platform_all` (ALL to fddt_platform); `tenant_isolation` (ALL to fddt_app)
- **Grants** `fddt_app`: SELECT, INSERT, UPDATE · `fddt_platform`: SELECT, INSERT, UPDATE, DELETE

document_checks

One row per (document, check type) — the latest run of each automated check (extraction stages, every forensic check, signature detection…), upserted so a redelivered task never duplicates it. `result` is the check's raw JSON output; the run history is in `audit_log` .

Column	Type	Null	Default	Notes
`id`	uuid	no		PK

Column	Type	Null	Default	Notes
`created_at`	timestamptz	no		
`updated_at`	timestamptz	no		
`document_id`	uuid	no		FK → `documents.id`
`check_type`	enum `document_check_type`	no		
`status`	enum `document_check_status`	no		
`result`	jsonb	yes		
`confidence`	double precision	yes		
`error_message`	varchar(2048)	yes		
`company_id`	uuid	no		FK → `companies.id`

- **Index** `ix_document_checks_company_id` on (company_id)
- **Index** `ix_document_checks_document_id` on (document_id)
- **Index** `uq_document_checks_document_check_type` on (document_id, check_type) — unique
- **Enum** `document_check_type`: `ocr_layout`, `classification`, `field_extraction`, `issuer_verification`, `signature_stamp_detection`, `metadata_forensics`, `error_level_analysis`, `copy_move_detection`, `duplicate_detection`, `ai_content_detection`, `field_validation`, `visual_inconsistency_review`
- **Enum** `document_check_status`: `pending`, `running`, `completed`, `failed`
- **Row-Level Security** on: `platform_all` (ALL to fddt_platform); `tenant_isolation` (ALL to fddt_app)
- **Grants** `fddt_app`: SELECT, INSERT, UPDATE · `fddt_platform`: SELECT, INSERT, UPDATE, DELETE

document_page_hashes

Perceptual hash (pHash) of every rendered PDF page, one per (document, page), used to find duplicate/near-duplicate pages within the same company.

Column	Type	Null	Default	Notes
`id`	uuid	no		PK
`created_at`	timestamptz	no		
`updated_at`	timestamptz	no		
`document_id`	uuid	no		FK → `documents.id`
`page_number`	integer	no		
`phash`	varchar(32)	no		
`company_id`	uuid	no		FK → `companies.id`

- **Index** `ix_document_page_hashes_company_id` on (company_id)
- **Index** `ix_document_page_hashes_document_id` on (document_id)
- **Index** `ix_document_page_hashes_phash` on (phash)
- **Index** `uq_document_page_hashes_document_page` on (document_id, page_number) — unique
- **Row-Level Security** on: `platform_all` (ALL to fddt_platform); `tenant_isolation` (ALL to fddt_app)

- **Grants** `fddt_app`: SELECT, INSERT, UPDATE · `fddt_platform`: SELECT, INSERT, UPDATE, DELETE

cross_document_findings

Case-level reconciliation results: a shared field (amount, date, issuer) that disagrees between documents in the same case.

Column	Type	Null	Default	Notes
`id`	uuid	no		PK
`created_at`	timestamptz	no		
`updated_at`	timestamptz	no		
`case_id`	uuid	no		FK → `cases.id`
`field_name`	varchar(128)	no		
`finding_type`	varchar(128)	no		
`severity`	enum `finding_severity`	no		
`description`	varchar(2048)	no		
`document_ids`	jsonb	yes		
`company_id`	uuid	no		FK → `companies.id`

- **Index** `ix_cross_document_findings_case_id` on (case_id)
- **Index** `ix_cross_document_findings_company_id` on (company_id)
- **Enum** `finding_severity`: `info`, `low`, `medium`, `high`
- **Row-Level Security** on: `platform_all` (ALL to fddt_platform); `tenant_isolation` (ALL to fddt_app)
- **Grants** `fddt_app`: SELECT, INSERT, UPDATE, DELETE · `fddt_platform`: SELECT, INSERT, UPDATE, DELETE

issuer_registry

A company's known vendors/schools/etc. that extracted issuer names are fuzzy-matched against. Rows are deactivated, never deleted. A new company starts empty.

Column	Type	Null	Default	Notes
`id`	uuid	no		PK
`created_at`	timestamptz	no		
`updated_at`	timestamptz	no		
`name`	varchar(255)	no		
`tax_id`	varchar(64)	yes		
`type`	enum `issuer_registry_type`	no		
`name_arabic`	varchar(255)	yes		
`is_active`	boolean	no	`true`	
`company_id`	uuid	no		FK → `companies.id`

- **Index** `ix_issuer_registry_company_id` on (company_id)
- **Index** `ix_issuer_registry_name` on (name)
- **Index** `ix_issuer_registry_name_arabic` on (name_arabic)
- **Enum** `issuer_registry_type` : `vendor` , `school` , `government` , `other`
- **Row-Level Security** on: `platform_all` (ALL to fddt_platform); `tenant_isolation` (ALL to fddt_app)
- **Grants** `fddt_app` : SELECT, INSERT, UPDATE · `fddt_platform` : SELECT, INSERT, UPDATE, DELETE

risk_rules

A company's configurable scoring rules. Rows are immutable and versioned: an edit inserts `version + 1` ; the current rule is the highest version. Seeded from `risk_rule_templates` when the company is created.

Column	Type	Null	Default	Notes
`id`	uuid	no		PK
`created_at`	timestamptz	no		
`updated_at`	timestamptz	no		
`rule_id`	varchar(64)	no		
`category`	varchar(128)	no		
`condition`	jsonb	no		
`weight`	double precision	no		
`severity`	varchar(32)	no		
`reason_template`	varchar(1024)	no		
`is_active`	boolean	no		
`check_type`	varchar(64)	no		
`version`	integer	no		
`effective_from`	timestamptz	no		
`updated_by`	uuid	yes		FK → `users.id`
`change_note`	varchar(1024)	yes		
`company_id`	uuid	no		FK → `companies.id`

- **Index** `ix_risk_rules_company_id` on (company_id)
- **Index** `ix_risk_rules_rule_id` on (rule_id)
- **Index** `uq_risk_rules_company_rule_id_version` on (company_id, rule_id, version) — unique
- **Row-Level Security** on: `platform_all` (ALL to fddt_platform); `tenant_isolation` (ALL to fddt_app)
- **Grants** `fddt_app` : SELECT, INSERT · `fddt_platform` : SELECT, INSERT

risk_rule_templates

PLATFORM-LEVEL (no `company_id`): the default rule set copied into every NEW company's `risk_rules` . Edited by platform admins; edits never change existing companies. Only `fddt_platform` can access it.

Column	Type	Null	Default	Notes
`id`	uuid	no		PK
`rule_id`	varchar(64)	no		unique
`category`	varchar(128)	no		
`check_type`	varchar(64)	no		
`condition`	jsonb	no		
`weight`	double precision	no		
`severity`	varchar(32)	no		
`reason_template`	varchar(1024)	no		
`is_active`	boolean	no	`true`	
`updated_by`	uuid	yes		FK → `users.id`
`created_at`	timestamptz	no	`now()`	
`updated_at`	timestamptz	no	`now()`	

- **Index** `risk_rule_templates_rule_id_key` on (rule_id) — unique
- **Grants** `fddt_app` : — · `fddt_platform` : SELECT, INSERT, UPDATE

risk_settings

One row per company holding its `medium` / `high` score thresholds.

Column	Type	Null	Default	Notes
`id`	uuid	no		PK
`created_at`	timestamptz	no		
`updated_at`	timestamptz	no		
`medium_threshold`	integer	no		
`high_threshold`	integer	no		
`updated_by`	uuid	yes		FK → `users.id`
`company_id`	uuid	no		FK → `companies.id`, unique

- **Index** `ix_risk_settings_company_id` on (company_id)
- **Index** `uq_risk_settings_company_id` on (company_id) — unique
- **Row-Level Security** on: `platform_all` (ALL to fddt_platform); `tenant_isolation` (ALL to fddt_app)
- **Grants** `fddt_app` : SELECT, INSERT, UPDATE · `fddt_platform` : SELECT, INSERT, UPDATE, DELETE

case_risk_assessments

One row per scoring run of a case (insert-only history; unique per (case, evidence fingerprint)): score, tier, frozen reason text and a snapshot of the exact rule versions and thresholds used.

Column	Type	Null	Default	Notes
`id`	uuid	no		PK
`case_id`	uuid	no		FK → `cases.id`
`raw_score`	double precision	no		
`score`	integer	no		
`tier`	enum `risk_tier`	no		
`triggered_reasons`	jsonb	no		
`risk_rules_version_snapshot`	jsonb	no		
`evidence_fingerprint`	varchar(64)	no		
`computed_at`	timestamptz	no		
`company_id`	uuid	no		FK → `companies.id`

- **Index** `ix_case_risk_assessments_case_id` on (case_id)
- **Index** `ix_case_risk_assessments_company_id` on (company_id)
- **Index** `ix_case_risk_assessments_computed_at` on (computed_at)
- **Index** `ix_case_risk_assessments_evidence_fingerprint` on (evidence_fingerprint)
- **Index** `uq_case_risk_assessments_case_evidence` on (case_id, evidence_fingerprint) — unique
- **Enum** `risk_tier`: `low`, `medium`, `high`
- **Row-Level Security** on: `platform_all` (ALL to fddt_platform); `tenant_isolation` (ALL to fddt_app)
- **Grants** `fddt_app`: SELECT, INSERT · `fddt_platform`: SELECT, INSERT

risk_scores

LEGACY. Created by the initial migration before the versioned engine existed; nothing in the application reads or writes it. `case_risk_assessments` replaced it.

Column	Type	Null	Default	Notes
`id`	uuid	no		PK
`created_at`	timestamptz	no		
`updated_at`	timestamptz	no		
`case_id`	uuid	no		FK → `cases.id`
`score`	integer	no		
`tier`	enum `risk_score_tier`	no		
`triggered_reasons`	jsonb	no		
`rules_snapshot`	jsonb	no		
`company_id`	uuid	no		FK → `companies.id`

- **Index** `ix_risk_scores_case_id` on (case_id)
- **Index** `ix_risk_scores_company_id` on (company_id)

- **Enum** `risk_score_tier`: `low`, `medium`, `high`
- **Row-Level Security** on: `platform_all` (ALL to fddt_platform); `tenant_isolation` (ALL to fddt_app)
- **Grants** `fddt_app`: SELECT · `fddt_platform`: SELECT, INSERT, UPDATE, DELETE

case_actions

Reviewer-facing history of human actions on a case (approve / reject / escalate …) with the actor's role at the time. Complements, and is also mirrored into, `audit_log`.

Column	Type	Null	Default	Notes
`id`	uuid	no		PK
`created_at`	timestamptz	no		
`updated_at`	timestamptz	no		
`case_id`	uuid	no		FK → `cases.id`
`actor_user_id`	uuid	yes		FK → `users.id`
`action_type`	enum `case_action_type`	no		
`notes`	varchar(4096)	yes		
`actor_role`	varchar(32)	yes		
`company_id`	uuid	no		FK → `companies.id`

- **Index** `ix_case_actions_case_id` on (case_id)
- **Index** `ix_case_actions_company_id` on (company_id)
- **Enum** `case_action_type`: `assign`, `comment`, `approve`, `reject`, `escalate`, `reopen`
- **Row-Level Security** on: `platform_all` (ALL to fddt_platform); `tenant_isolation` (ALL to fddt_app)
- **Grants** `fddt_app`: SELECT, INSERT · `fddt_platform`: SELECT, INSERT, UPDATE, DELETE

case_reports

History of generated per-case forensic PDF reports (insert-only). The PDF lives in Blob Storage under `companies/{company}/cases/{case}/reports/`; this row stores its SHA-256, size and the assessment it reflects.

Column	Type	Null	Default	Notes
`id`	uuid	no		PK
`case_id`	uuid	no		FK → `cases.id`
`generated_by_user_id`	uuid	no		FK → `users.id`
`generated_at`	timestamptz	no		
`blob_url`	text	no		
`file_size_bytes`	integer	no		
`page_count`	integer	no		
`report_sha256`	varchar(64)	no		

Column	Type	Null	Default	Notes
`risk_assessment_id`	uuid	yes		FK → `case_risk_assessments.id`
`company_id`	uuid	no		FK → `companies.id`

- **Index** `ix_case_reports_case_id` on (case_id)
- **Index** `ix_case_reports_company_id` on (company_id)
- **Index** `ix_case_reports_generated_at` on (generated_at)
- **Row-Level Security** on: `platform_all` (ALL to fddt_platform); `tenant_isolation` (ALL to fddt_app)
- **Grants** `fddt_app`: SELECT, INSERT · `fddt_platform`: SELECT, INSERT

audit_log

Append-only system of record: every automated check result, human action and settings change is inserted here and never updated or deleted. `company_id` NULL = a platform-level event (company created, template edits, usage reconciliation, platform-admin access), visible only to platform admins; a company user's *user created/updated* goes into that company's log.

Column	Type	Null	Default	Notes
`id`	uuid	no		PK
`created_at`	timestamptz	no		
`case_id`	uuid	yes		FK → `cases.id`
`document_id`	uuid	yes		FK → `documents.id`
`actor_user_id`	uuid	yes		FK → `users.id`
`event_type`	varchar(128)	no		
`event_data`	jsonb	yes		
`company_id`	uuid	yes		FK → `companies.id`

- **Index** `ix_audit_log_case_id` on (case_id)
- **Index** `ix_audit_log_company_id` on (company_id)
- **Index** `ix_audit_log_created_at` on (created_at)
- **Index** `ix_audit_log_document_id` on (document_id)
- **Index** `ix_audit_log_event_type` on (event_type)
- **Row-Level Security** on: `platform_all` (ALL to fddt_platform); `tenant_isolation` (ALL to fddt_app)
- **Grants** `fddt_app`: SELECT, INSERT · `fddt_platform`: SELECT, INSERT

signature_references

A signature/stamp region drawn on a document, cropped and stored as a PNG, used as the reference for comparisons.

Column	Type	Null	Default	Notes
`person_name`	text	no		
`signature_image_url`	text	yes		

Column	Type	Null	Default	Notes
`bounding_box`	jsonb	yes		
`source_document_id`	uuid	no		FK → `documents.id`
`source_case_id`	uuid	no		FK → `cases.id`
`created_by`	uuid	no		FK → `users.id`
`is_library`	boolean	no	`false`	
`id`	uuid	no		PK
`created_at`	timestamptz	no		
`updated_at`	timestamptz	no		
`company_id`	uuid	no		FK → `companies.id`

- **Index** `ix_signature_references_company_id` on (company_id)
- **Index** `ix_signature_references_source_case_id` on (source_case_id)
- **Index** `ix_signature_references_source_document_id` on (source_document_id)
- **Row-Level Security** on: `platform_all` (ALL to fddt_platform); `tenant_isolation` (ALL to fddt_app)
- **Grants** `fddt_app`: SELECT, INSERT, UPDATE · `fddt_platform`: SELECT, INSERT, UPDATE, DELETE

signature_matches

Result of comparing one reference against the signature region of another document in the same case (vision-model verdict or pixel-identical reuse verdict); one row per (reference, document, scope).

Column	Type	Null	Default	Notes
`id`	uuid	no		PK
`document_id`	uuid	no		FK → `documents.id`
`case_id`	uuid	no		FK → `cases.id`
`signature_reference_id`	uuid	no		FK → `signature_references.id`
`comparison_scope`	enum `comparison_scope`	no		
`result`	enum `signature_match_result`	no		
`reasoning`	text	yes		
`compared_at`	timestamptz	no		
`company_id`	uuid	no		FK → `companies.id`

- **Index** `ix_signature_matches_case_id` on (case_id)
- **Index** `ix_signature_matches_company_id` on (company_id)
- **Index** `ix_signature_matches_document_id` on (document_id)
- **Index** `ix_signature_matches_signature_reference_id` on (signature_reference_id)
- **Index** `uq_signature_matches_ref_doc_scope` on (signature_reference_id, document_id, comparison_scope) — unique

- **Enum** `comparison_scope`: `in_case`, `library`
- **Enum** `signature_match_result`: `consistent`, `possibly_consistent`, `inconsistent`, `cannot_determine`, `identical_reuse`, `reused_different_signer`
- **Row-Level Security** on: `platform_all` (ALL to fddt_platform); `tenant_isolation` (ALL to fddt_app)
- **Grants** `fddt_app`: SELECT, INSERT, UPDATE · `fddt_platform`: SELECT, INSERT, UPDATE, DELETE

Enum index

Enum	Values
`case_action_type`	`assign`, `comment`, `approve`, `reject`, `escalate`, `reopen`
`case_status`	`submitted`, `under_automated_review`, `pending_manual_review`, `auto_approved`, `escalated`, `under_investigation`, `approved`, `rejected`, `closed`
`case_tier`	`l1`, `l2`
`case_type`	`school_document`, `vendor_invoice`, `commercial_invoice`, `procurement_documentation`, `quotation`, `travel_reimbursement`, `other`
`comparison_scope`	`in_case`, `library`
`document_check_status`	`pending`, `running`, `completed`, `failed`
`document_check_type`	`ocr_layout`, `classification`, `field_extraction`, `issuer_verification`, `signature_stamp_detection`, `metadata_forensics`, `error_level_analysis`, `copy_move_detection`, `duplicate_detection`, `ai_content_detection`, `field_validation`, `visual_inconsistency_review`
`document_processing_status`	`pending`, `processing`, `complete`, `failed`
`finding_severity`	`info`, `low`, `medium`, `high`
`issuer_registry_type`	`vendor`, `school`, `government`, `other`
`risk_score_tier`	`low`, `medium`, `high`
`risk_tier`	`low`, `medium`, `high`
`signature_match_result`	`consistent`, `possibly_consistent`, `inconsistent`, `cannot_determine`, `identical_reuse`, `reused_different_signer`
`user_role`	`user`, `reviewer_l1`, `reviewer_l2`, `admin`, `platform_admin`

Schema notes

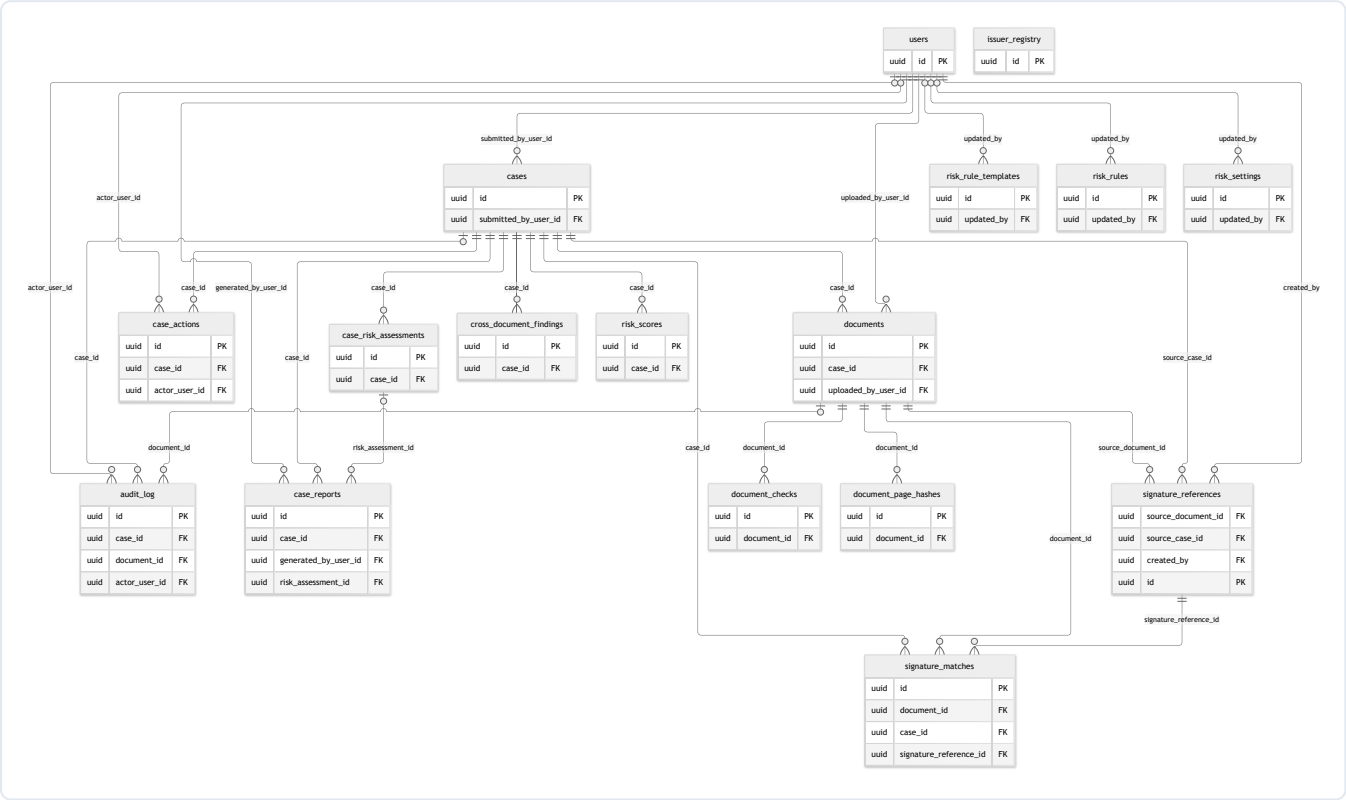
- `user_role` **still contains** `admin`**.** The single-tenant `admin` role was migrated to `platform_admin`; Postgres cannot drop an enum value, so `admin` remains in the type but no row uses it and the application never assigns it.

- `cases.case_number` **and** `users.email` carry a `UNIQUE` constraint *plus* a separate non-unique index from the original migrations; uniqueness is enforced either way.
- `risk_scores` exists but nothing in the application uses it. It is safe to leave; dropping it needs a migration.
- **No** `notifications` **table.** Notifications are not part of the current build.
- **Enum values added by later migrations** use `ALTER TYPE … ADD VALUE`; a downgrade leaves them in place.
- **New tenant-owned tables** need `TenantScopedMixin`, RLS with a `tenant_isolation` and a `platform_all` policy, and explicit grants in their migration — there are no default privileges, so a table without them is unreachable by the app.

Entity-relationship diagram

Every tenant-owned table also references `companies` through `company_id`. Those edges are left out of the first diagram so it stays readable; the second diagram shows them.

Domain relationships



Tenancy — what belongs to a company

Every table below references `companies.id` through the listed column (a table, not a diagram: drawn, the fan-out from `companies` is too wide to read).

Table	Column	Nullable	Row-Level Security
`company_usage_stats`	`company_id`	no	on
`users`	`company_id`	yes	on
`cases`	`company_id`	no	on
`documents`	`company_id`	no	on
`document_checks`	`company_id`	no	on
`document_page_hashes`	`company_id`	no	on
`cross_document_findings`	`company_id`	no	on
`issuer_registry`	`company_id`	no	on
`risk_rules`	`company_id`	no	on

Table	Column	Nullable	Row-Level Security
`risk_settings`	`company_id`	no	on
`case_risk_assessments`	`company_id`	no	on
`risk_scores`	`company_id`	no	on
`case_actions`	`company_id`	no	on
`case_reports`	`company_id`	no	on
`audit_log`	`company_id`	yes	on
`signature_references`	`company_id`	no	on
`signature_matches`	`company_id`	no	on

`users.company_id` and `audit_log.company_id` are nullable: NULL marks a platform admin and a platform-level event respectively. `risk_rule_templates` has no `company_id` — it is platform-level and only reachable by the platform database role.

API reference

The request/response detail is **not hand-written here** — it is generated from the route definitions, so it cannot drift from the code:

What	Where
Interactive docs (Swagger UI)	`http://localhost:8000/docs` while the backend runs
ReDoc	`http://localhost:8000/redoc`
Machine-readable schema	`http://localhost:8000/openapi.json`
Snapshot committed to the repo	`openapi.json` (OpenAPI 3.1, 33 paths / 41 operations / 60 schemas)

Every operation carries a `summary`, a `description` (rules, side effects, role requirements) and its error responses; every request field has a description. Import `openapi.json` into Postman, Insomnia or a client generator if you prefer.

Regenerating the snapshot

The snapshot is produced from the app object itself, no server needed:

```
cd backend
python -c "import json; from app.main import app; json.dump(app.openapi(),
open('../docs/api/openapi.json','w',encoding='utf-8'), indent=2, ensure_ascii=False)"
```

Regenerate it whenever a route or schema changes. A quick consistency check is to compare it with the live `/openapi.json`.

Trying it in Swagger UI

1. `POST /auth/login` with `{"email": "admin@example.com", "password": "ChangeMe123!"}` (the seeded **platform admin**, see local setup) and copy `access_token`. To try company endpoints, log in as a company user created by the platform admin instead.
2. Click **Authorize** (top right), paste the token (Swagger adds `Bearer` for you).
3. Every other call now sends it.

Conventions

- **Auth:** `Authorization: Bearer <JWT>`. Tokens live 8 h by default (`JWT_ACCESS_TOKEN_EXPIRE_MINUTES`) and carry `role`, `company_id` and `is_platform_admin`. The user is re-read from the database on every request; a token whose company or platform flag no longer matches, or whose company is suspended, is refused. A missing/invalid/expired token or an inactive user is **401**; a valid user with the wrong role is **403**.
- **Two mandatory checks on every endpoint:** the role check and the tenant check. A company user's request is confined to their own company — by the application and by PostgreSQL Row-Level Security.
- **Company roles (ranked):** `user` (submitter) < `reviewer_l1` < `reviewer_l2`. Each role can do everything the roles below it can; routes name the lowest role they admit. On a case escalated to L2 (`assigned_tier` = `l2`) a `reviewer_l1` can still `GET` it but gets **403** from approve/reject/escalate.
- `platform_admin` belongs to no company and is not on that ladder. It manages companies, users, the rule template, billing and the queue monitor, and has **read-only, audited** access to any company's data by passing

`company_id` (query parameter) on the company endpoints that accept it — `GET /cases`, `/audit-log`, `/settings/*` — or simply opening a case by id. It can never approve, reject, escalate, upload, create cases or generate reports (403), but may edit a company's issuer registry, risk rules and thresholds on its behalf (audited in that company's log).

- **A resource of another company is a 404**, and so is a case that is not yours for a `user` — ids cannot be probed.
- **Ids** are UUIDs. **Timestamps** are ISO-8601 with timezone.
- **Bounding boxes** are always `{page (1-based), x, y, width, height}` as 0–1 fractions of the page, origin top-left.
- **Errors** are `{"detail": "<human-readable message>"}` (or FastAPI's standard list for **422** validation errors). **Upload rejections** are structured: `{"detail": {"code": …, "message": …, …}}` (see 01-upload-intake). Status codes used: 400, 401, 403, 404, 409 (state conflict or duplicate), 413 (upload too large), 415 (not a PDF / wrong extension), 422 (validation, business rule, corrupted or password-protected PDF), 502 (Azure Blob Storage failure).
- **There is no `/api/v1` prefix in the backend.** `API_V1_PREFIX` is declared in settings but never used. In development the Vite dev server proxies `/api/*` → `http://127.0.0.1:8000/*`, so the browser calls `/api/cases` while the backend sees `/cases`. Production serves the SPA and the API behind one ingress the same way (production deployment).
- **Pagination:** only `GET /audit-log` is paged (`limit` 1–500, default 100, and `offset`; response is `{total, items}`). Other lists return everything.
- **No dedicated "checks" endpoints.** Per-document check results, extracted fields, cross-document findings, forensic findings, the risk assessment and reviewer actions are all embedded in `GET /cases/{case_id}`. Signature comparison results have their own endpoint.
- **Asynchronous work:** uploads and signature-reference creation return before processing finishes. Poll `GET /cases/{case_id}` (`pipeline.complete`, `documents[].processing_status`) and `GET /cases/{case_id}/signature-matches`.

Endpoint index

`Access` shows who may call each endpoint. "own case" means a `user` may act only on a case they submitted (anyone else gets a 404), while reviewers (L1 or L2) may act on any case of their company. "reviewer" means `reviewer_l1` or above. "PA read" means a platform admin may read it (audited). See the security notes in architecture/overview.md.

Auth — tag `auth`

Method & path	Summary	Access
`POST /auth/login`	Log in and obtain a JWT	public
`GET /auth/me`	Get the current user (with company)	any

Cases — tag `cases`

Method & path	Summary	Access
`POST /cases`	Create a case	company users
`GET /cases`	List cases (the case queue); filters `status`, `case_type`, `assigned_tier`, `actionable`; each item has `can_act`	company users (`user` sees only own); PA read with `company_id`
`GET /cases/{case_id}`	Get case detail (documents, checks, findings, assessment, actions)	company users (`user` sees a reduced view); PA read

Method & path	Summary	Access
`GET /cases/{case_id}/audit-log`	Get a case's activity timeline	own case for `user`, reviewers; PA read

Reviewer decisions — tag `case-actions`

Method & path	Summary	Access
`POST /cases/{case_id}/approve`	Approve a case (justification required above low risk)	reviewer; L2 case: `reviewer_l2`
`POST /cases/{case_id}/reject`	Reject a case (reason required)	reviewer; L2 case: `reviewer_l2`
`POST /cases/{case_id}/escalate`	Escalate a case to the L2 tier (reason required; one-way; 409 if already L2)	reviewer on an L1 case

Documents — tag `documents`

Method & path	Summary	Access
`POST /cases/{case_id}/documents`	Upload a document (multipart field `file`, one per request; **PDF only**, ≤ 10 MB, not password-protected; rejections return `detail: {code, message}`, see 01-upload-intake)	own case, or reviewer
`GET /cases/{case_id}/documents/{document_id}/file-url`	Get a fresh signed URL for the original file	own case, or reviewer; PA read

Signatures — tag `signatures`

Method & path	Summary	Access
`POST /cases/{case_id}/documents/{document_id}/signature-references`	Create a signature reference from a drawn box	own case, or reviewer
`GET /cases/{case_id}/signature-references`	List a case's references	own case, or reviewer; PA read
`GET /cases/{case_id}/signature-matches`	List a case's comparison results	own case, or reviewer; PA read

Reports — tag `case-reports`

Method & path	Summary	Access
`POST /cases/{case_id}/reports`	Generate a case report (PDF)	reviewer
`GET /cases/{case_id}/reports`	List a case's generated reports	reviewer; PA read

> Organisation-wide MIS/compliance reporting (`app/api/reports.py`) is a stub with no routes.

Audit — tag `audit`

Method & path	Summary	Access
`GET /audit-log`	Search the company's audit log (`event_type` , `case_id` , `q` , `limit` , `offset`); never shows platform-access rows to a company user	reviewer; PA with `company_id` (that company's log incl. platform-access rows about it) or without (the platform-level log)
`GET /audit-log/event-types`	List distinct audit event types	reviewer; PA

Settings — tag `settings`

Issuer registry, risk rules and thresholds are **per company**: a company's `reviewer_l2` , or a platform admin passing `company_id` (audited). Users are platform-admin only.

Method & path	Summary	Access
`GET /settings/issuers`	List issuers (`include_inactive`)	`reviewer_l2` ; PA
`POST /settings/issuers`	Create an issuer	`reviewer_l2` ; PA
`PATCH /settings/issuers/{issuer_id}`	Update or deactivate an issuer	`reviewer_l2` ; PA
`GET /settings/risk-rules`	List current risk rules	`reviewer_l2` ; PA
`POST /settings/risk-rules`	Create a risk rule	`reviewer_l2` ; PA
`GET /settings/risk-rule-options`	Options for the Add-rule form	`reviewer_l2` ; PA
`GET /settings/risk-rules/{rule_id}/history`	A rule's version history	`reviewer_l2` ; PA
`PATCH /settings/risk-rules/{rule_id}`	Change weight / severity / active flag (inserts a new version)	`reviewer_l2` ; PA
`GET /settings/risk-thresholds`	Get the tier thresholds	`reviewer_l2` ; PA
`PUT /settings/risk-thresholds`	Set the tier thresholds	`reviewer_l2` ; PA
`GET /settings/users`	List users (filter `company_id`)	PA only
`POST /settings/users`	Create a user (company + role, or a platform admin)	PA only
`PATCH /settings/users/{user_id}`	Change role / company / active flag (moving company invalidates the user's token)	PA only

Platform — tag `platform` (platform admins only)

Method & path	Summary
`GET /platform/companies`	List companies
`POST /platform/companies`	Create a company (seeds its risk rules from the template and default thresholds)
`PATCH /platform/companies/{company_id}`	Rename / suspend / reactivate a company
`GET /platform/usage`	Billing & usage per company (`period` or a custom date range)
`POST /platform/usage/reconcile`	Recompute the usage counters now
`GET /platform/queues`	Processing-queue monitor (depth, waits, workers, limiter counters, outstanding per company)
`GET /platform/risk-rule-templates`	List the platform rule template
`POST /platform/risk-rule-templates`	Add a template rule (affects companies created afterwards)
`PATCH /platform/risk-rule-templates/{rule_id}`	Change a template rule (never changes existing companies)

Health — tag `health`

Method & path	Summary	Access
`GET /health`	Liveness check (does not touch DB/Redis/Azure)	public

Not in the API (by design or not yet built)

No password reset or change-password, no self-service sign-up, no logout/refresh, no company-level user management (users are created by platform admins), no document delete (originals are immutable), no case delete, no notification endpoints, no organisation-wide reporting, no external-integration hooks.

API endpoint reference

FDDT — Fraud Document Detection Tool API · version 0.1.0 · OpenAPI 3.1.0

41 operations across 33 paths and 60 data models. This reference is generated from the API definition itself (docs/api/openapi.json).

Conventions

- **Authentication:** endpoints marked Bearer JWT need the header `Authorization: Bearer <token>`. Obtain a token from `POST /auth/login` (email and password).
- **Roles** (ranked, each includes everything below it): `user` (submitter) < `reviewer_l1` < `reviewer_l2` < `admin`. "Reviewer/admin" means `reviewer_l1` or above. On a case escalated to L2 (`assigned_tier` = `l2`) only `reviewer_l2` and `admin` may approve, reject or escalate; a `reviewer_l1` gets 403 there but can still read the case. Role rules for each endpoint are stated in its description.
- **Errors:** every operation lists its documented error responses. 422 is a request-validation error.
- **Formats:** requests and responses are JSON (file uploads are `multipart/form-data`); identifiers are UUIDs; timestamps are ISO 8601.

Endpoint index

Method	Path	Summary
Health		
GET	`/health`	Liveness check
Authentication		
POST	`/auth/login`	Log in and obtain a JWT
GET	`/auth/me`	Get the current user
Cases		
POST	`/cases`	Create a case
GET	`/cases`	List cases (the case queue)
GET	`/cases/{case_id}`	Get case detail
GET	`/cases/{case_id}/audit-log`	Get a case's activity timeline
Documents		
POST	`/cases/{case_id}/documents`	Upload a document to a case
GET	`/cases/{case_id}/documents/{document_id}/file-url`	Get a fresh signed URL for a document
Case actions (review decisions)		
POST	`/cases/{case_id}/approve`	Approve a case
POST	`/cases/{case_id}/reject`	Reject a case
POST	`/cases/{case_id}/escalate`	Escalate a case

Method	Path	Summary
Signature references & matches		
POST	`/cases/{case_id}/documents/{document_id}/signature-references`	Create a signature reference
GET	`/cases/{case_id}/signature-references`	List a case's signature references
GET	`/cases/{case_id}/signature-matches`	List a case's signature comparison results
Audit log		
GET	`/audit-log`	Search the audit log
GET	`/audit-log/event-types`	List audit event types
Case reports		
POST	`/cases/{case_id}/reports`	Generate a case report (PDF)
GET	`/cases/{case_id}/reports`	List a case's generated reports
Settings (admin)		
GET	`/settings/issuers`	List issuers
POST	`/settings/issuers`	Create an issuer
PATCH	`/settings/issuers/{issuer_id}`	Update or deactivate an issuer
GET	`/settings/risk-rules`	List current risk rules
POST	`/settings/risk-rules`	Create a risk rule
GET	`/settings/risk-rule-options`	Get the options for the Add-rule form
GET	`/settings/risk-rules/{rule_id}/history`	Get a risk rule's version history
PATCH	`/settings/risk-rules/{rule_id}`	Change a risk rule's weight, severity or active flag
GET	`/settings/risk-thresholds`	Get the risk tier thresholds
PUT	`/settings/risk-thresholds`	Set the risk tier thresholds
GET	`/settings/users`	List users
POST	`/settings/users`	Create a user
PATCH	`/settings/users/{user_id}`	Change a user's role, company or active flag
Other		
GET	`/platform/companies`	List companies
POST	`/platform/companies`	Create a company
PATCH	`/platform/companies/{company_id}`	Rename or suspend/reactivate a company
GET	`/platform/risk-rule-templates`	List the platform risk-rule template
POST	`/platform/risk-rule-templates`	Add a rule to the platform template

Method	Path	Summary
PATCH	`/platform/risk-rule-templates/{rule_id}`	Edit a platform template rule
GET	`/platform/usage`	Billing & usage by company
POST	`/platform/usage/reconcile`	Reconcile usage counters now
GET	`/platform/queues`	Processing queue monitor

Health

GET `/health`

Liveness check Public

Unauthenticated. Returns `{status: ok, environment}`; does not touch the database, Redis or Azure.

Responses

Status	Description	Body
`200`	Successful Response	map of string

Authentication

POST `/auth/login`

Log in and obtain a JWT Public

Exchanges email + password for a bearer token (HS256; lifetime from JWT_ACCESS_TOKEN_EXPIRE_MINUTES). The token carries `role`, `company_id` (null for a platform admin) and `is_platform_admin`. An unknown email, a wrong password, a deactivated account and a suspended company all return the same 401 so accounts cannot be enumerated.

Request body

LoginRequest

Responses

Status	Description	Body
`200`	Successful Response	TokenResponse
`401`	Incorrect email or password (or the account/company is inactive).	—
`422`	Validation Error	HTTPValidationError

GET `/auth/me`

Get the current user Bearer JWT

Returns the authenticated user's profile, role and company. The role and company are re-read from the database on every request rather than trusted from the token, so a demotion, deactivation or company suspension takes effect immediately.

Responses

Status	Description	Body
`200`	Successful Response	CurrentUserResponse
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—

Cases

POST /cases

Create a case Bearer JWT

Opens an empty case (status `submitted`) owned by the caller, in the caller's company; any company role may create one (platform admins may not — 403). The case number is `CASE-` plus the first 8 hex characters of the case UUID. Attach documents afterwards with POST /cases/{case_id}/documents. Writes a `case_created` audit event.

Request body

CaseCreateRequest

Responses

Status	Description	Body
`201`	Successful Response	CaseResponse
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Platform admins cannot create cases.	—
`422`	Validation Error	HTTPValidationError

GET /cases

List cases (the case queue) Bearer JWT

Only the caller's company. Reviewers see every case of their company; a `user` sees only the cases they submitted. A platform admin must pass `company_id` (one company at a time, audited) — there is no cross-company case list. For `reviewer_l2` and platform admins, open cases escalated to L2 sort first; otherwise newest first. Optional filters: `status`, `case_type`, `assigned_tier`, and `actionable=true` (only cases the caller may act on — for a `reviewer_l1` that excludes L2 cases). Each item carries `can_act`. For submitters `risk_tier` is null and `flag` is a neutral 'In review' marker, so scoring signals are never disclosed to them.

Parameters

Name	In	Type	Required	Description
`status`	query	CaseStatus (nullable)	no	
`case_type`	query	CaseType (nullable)	no	
`assigned_tier`	query	CaseTier (nullable)	no	
`actionable`	query	boolean	no	Only cases the caller may approve/reject/escalate.

Name	In	Type	Required	Description
`company_id`	query	string (uuid) (nullable)	no	Platform admins only (required for them): the company to act on. Company users are always confined to their own company.

Responses

Status	Description	Body
`200`	Successful Response	array of CaseListItem
`400`	A platform admin did not pass `company_id`.	—
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`404`	`company_id` names an unknown company (or, for a company user, not theirs).	—
`422`	Validation Error	HTTPValidationError

GET `/cases/{case_id}`

Get case detail

Bearer JWT

Everything the case detail page needs in one call: documents (with signed download URLs, extracted fields and per-document check results), cross-document findings, forensic findings, the latest risk assessment, pipeline progress (`pipeline.complete` / `pipeline.pending`) and reviewer actions. A submitter does not receive the risk assessment and sees only approve/reject actions. A case of another company, or owned by someone else, is a 404, not a 403, so ids cannot be probed. A platform admin may read any company's case; each such access is written to that company's audit log.

Parameters

Name	In	Type	Required	Description
`case_id`	path	string (uuid)	yes	

Responses

Status	Description	Body
`200`	Successful Response	CaseDetail
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`404`	No such case (or, for a `user`, not one they submitted).	—
`422`	Validation Error	HTTPValidationError

GET `/cases/{case_id}/audit-log`

Get a case's activity timeline

Bearer JWT

Chronological feed for the case detail page's activity timeline — CLAUDE.md section 3.5. Reads the same append-only `audit_log` rows every check task and case/document endpoint already writes; this adds no new events, just a way to read the ones that exist.

Parameters

Name	In	Type	Required	Description
`case_id`	path	string (uuid)	yes	

Responses

Status	Description	Body
`200`	Successful Response	array of AuditLogEntrySchema
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`404`	No such case (or, for a `user`, not one they submitted).	—
`422`	Validation Error	HTTPValidationError

Documents

POST `/cases/{case_id}/documents`

Upload a document to a case

Bearer JWT

One file per request (multipart field `file`). The original bytes are stored immutably in Blob Storage at `companies/{company_id}/cases/{case_id}/documents/{document_id}_{sha256}{ext}`, a `documents` row is written with the file's SHA-256, then the Celery pipeline is enqueued (extraction, metadata forensics, ELA/copy-move, duplicate detection, visual review, signature detection). The response returns immediately with `processing_status: pending`; poll GET /cases/{case_id} for progress. Before anything is stored or queued the file is validated: at most `MAX_UPLOAD_SIZE_BYTES` (10 MB), not empty, a PDF by its header bytes (not its name), a matching extension, and locally parseable (not corrupted, not password-protected). A rejection returns `detail: {code, message, ...}` with one of the codes `file_empty`, `file_too_large`, `unsupported_file_type`, `file_type_mismatch`, `file_corrupted`, `file_password_protected`. Each pipeline step is its own task on a shared, FIFO queue (see app/tasks/celery_app.py). A `user` may upload only to a case they submitted; reviewers to any case of their company; platform admins never (403).

Parameters

Name	In	Type	Required	Description
`case_id`	path	string (uuid)	yes	

Request body

Body_upload_document_cases__case_id__documents_post (multipart/form-data)

Responses

Status	Description	Body
`201`	Successful Response	DocumentResponse
`400`	`file_empty`: the file is 0 bytes.	—
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—

Status	Description	Body
403	Platform admins have read-only access.	—
404	No such case (or, for a `user`, not one they submitted).	—
413	`file_too_large`: over the size limit (`size_bytes`, `max_size_bytes`).	—
415	`unsupported_file_type` (not a PDF by content; images are refused) or `file_type_mismatch` (extension names a different type than the content).	—
422	`file_corrupted` (cannot be parsed) or `file_password_protected`.	—
502	Blob Storage rejected the upload.	—

GET `/cases/{case_id}/documents/{document_id}/file-url`

Get a fresh signed URL for a document Bearer JWT

Mint a fresh signed URL for a document's original file — the URL handed back at upload time expires after a few minutes, which a reviewer working through the post-upload screen can easily outlast.

Parameters

Name	In	Type	Required	Description
`case_id`	path	string (uuid)	yes	
`document_id`	path	string (uuid)	yes	

Responses

Status	Description	Body
200	Successful Response	DocumentFileUrlResponse
401	Missing, invalid or expired bearer token, or the user is inactive.	—
404	No such document in this case (or, for a `user`, not their case).	—
422	Validation Error	HTTPValidationError
502	Blob Storage failed to sign the URL.	—

Case actions (review decisions)

POST `/cases/{case_id}/approve`

Approve a case Bearer JWT

Company reviewers only (platform admins: 403). Requires the automated pipeline to have finished and a risk assessment to exist (409 otherwise). Above `low` risk a written justification (`note`, at least 10 characters) is required (422). Approving an already-decided case is a 409. Writes a `case_actions` row and an audit event.

Parameters

Name	In	Type	Required	Description
`case_id`	path	string (uuid)	yes	

Request body

ApproveRequest

Responses

Status	Description	Body
`200`	Successful Response	CaseDecisionResponse
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires a company reviewer role; a `reviewer_l1` also gets 403 on a case escalated to L2.	—
`404`	No such case in your company.	—
`409`	Case already decided, checks still running, or an invalid state transition.	—
`422`	Case is above low risk and no justification of at least 10 characters was given.	—

POST `/cases/{case_id}/reject`

Reject a case

Bearer JWT

Company reviewers only (platform admins: 403). A `reason` is required. Rejecting an already-decided case is a 409. Writes a `case_actions` row and an audit event.

Parameters

Name	In	Type	Required	Description
`case_id`	path	string (uuid)	yes	

Request body

RejectRequest

Responses

Status	Description	Body
`200`	Successful Response	CaseDecisionResponse
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires a company reviewer role; a `reviewer_l1` also gets 403 on a case escalated to L2.	—
`404`	No such case in your company.	—
`409`	Invalid state transition (e.g. already decided).	—
`422`	Validation Error	HTTPValidationError

POST `/cases/{case_id}/escalate`

Escalate a case Bearer JWT

Company reviewers only (platform admins: 403). Moves the case to the L2 tier (`assigned_tier` = `l2`); `status` is unchanged. A `reviewer_l1` then keeps read access but can no longer act on it. There is no de-escalation — an L2 reviewer approves or rejects it. A `reason` is required. Escalating an already-escalated or decided case is a 409.

Parameters

Name	In	Type	Required	Description
`case_id`	path	string (uuid)	yes	

Request body

EscalateRequest

Responses

Status	Description	Body
`200`	Successful Response	CaseDecisionResponse
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires a company reviewer role; a `reviewer_l1` also gets 403 on a case escalated to L2.	—
`404`	No such case in your company.	—
`409`	Invalid state transition (e.g. already decided).	—
`422`	Validation Error	HTTPValidationError

Signature references & matches

POST `/cases/{case_id}/documents/{document_id}/signature-references`

Create a signature reference Bearer JWT

Create a signature reference from a reviewer-drawn bounding box.

Crops the signature region from the document's PDF, uploads it to Blob Storage, inserts the signature_references row, then immediately enqueues the in-case comparison Celery task for this reference.

`person_name` is required and must be typed by the reviewer — the system does not attempt to auto-extract it (CLAUDE.md §2.3).

Parameters

Name	In	Type	Required	Description
`case_id`	path	string (uuid)	yes	
`document_id`	path	string (uuid)	yes	

Request body

SignatureReferenceCreate

Responses

Status	Description	Body
`201`	Successful Response	SignatureReferenceResponse
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Platform admins have read-only access.	—
`404`	No such case (or, for a `user`, not one they submitted), or the document is not in this case.	—
`422`	Validation Error	HTTPValidationError

GET `/cases/{case_id}/signature-references`

List a case's signature references Bearer JWT

All signature references created for this case. Used by the upload screen to list what references were set during this session.

Parameters

Name	In	Type	Required	Description
`case_id`	path	string (uuid)	yes	

Responses

Status	Description	Body
`200`	Successful Response	array of SignatureReferenceResponse
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`404`	No such case (or, for a `user`, not one they submitted).	—
`422`	Validation Error	HTTPValidationError

GET `/cases/{case_id}/signature-matches`

List a case's signature comparison results Bearer JWT

All in-case signature match results for this case, enriched with the reference person_name. Filtered to comparison_scope='in_case' only — library-scope matches are not generated yet (CLAUDE.md §2.3).

Parameters

Name	In	Type	Required	Description
`case_id`	path	string (uuid)	yes	

Responses

Status	Description	Body
`200`	Successful Response	array of SignatureMatchResponse
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—

Status	Description	Body
`404`	No such case (or, for a `user`, not one they submitted).	—
`422`	Validation Error	HTTPValidationError

Audit log

GET `/audit-log`

Search the audit log Bearer JWT

Newest-first, paged read of the append-only `audit_log`, confined to the caller's company. Filters: `event_type`, `case_id` and free-text `q` (event type, case number, actor name/email). Company reviewers, and platform admins (`company_id` = one company's log, audited; omitted = the platform-level log).

Parameters

Name	In	Type	Required	Description
`event_type`	query	string (nullable)	no	
`case_id`	query	string (uuid) (nullable)	no	
`q`	query	string (nullable)	no	Matches event type, case number, actor name/email
`company_id`	query	string (uuid) (nullable)	no	Platform admins only.
`limit`	query	integer	no	
`offset`	query	integer	no	

Responses

Status	Description	Body
`200`	Successful Response	AuditPage
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires a reviewer (L1 or L2) or platform admin role.	—
`422`	Validation Error	HTTPValidationError

GET `/audit-log/event-types`

List audit event types Bearer JWT

Distinct `event_type` values present in the caller's company's audit log (platform admins: the chosen company's, or the platform log), for the filter dropdown.

Parameters

Name	In	Type	Required	Description
`company_id`	query	string (uuid) (nullable)	no	Platform admins only.

Responses

Status	Description	Body
`200`	Successful Response	array of string
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires a reviewer (L1 or L2) or platform admin role.	—
`422`	Validation Error	HTTPValidationError

Case reports

POST `/cases/{case_id}/reports`

Generate a case report (PDF) Bearer JWT

Company reviewers only. Renders the per-case forensic report PDF, stores it in Blob Storage under the company's `companies/{company_id}/cases/{case_id}/reports/` prefix, records a `case_reports` row (with the file's SHA-256) and returns it with a signed download URL. Every call creates a new report; earlier ones are kept.

Parameters

Name	In	Type	Required	Description
`case_id`	path	string (uuid)	yes	

Responses

Status	Description	Body
`201`	Successful Response	CaseReportResponse
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires a reviewer (L1 or L2) or `admin` role.	—
`404`	No such case.	—
`422`	Validation Error	HTTPValidationError
`502`	Blob Storage failed while saving or signing the report.	—

GET `/cases/{case_id}/reports`

List a case's generated reports Bearer JWT

Report history for the case, newest first.

Parameters

Name	In	Type	Required	Description
`case_id`	path	string (uuid)	yes	

Responses

Status	Description	Body
`200`	Successful Response	array of CaseReportResponse

Status	Description	Body
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires a reviewer (L1 or L2) or `admin` role.	—
`404`	No such case.	—
`422`	Validation Error	HTTPValidationError
`502`	Blob Storage failed while signing a download URL.	—

Settings (admin)

GET `/settings/issuers`

List issuers Bearer JWT

The company's issuer registry, ordered by name. Pass `include_inactive=false` to hide deactivated entries. Reviewer L2 (own company) or platform admin (`company_id`).

Parameters

Name	In	Type	Required	Description
`include_inactive`	query	boolean	no	
`company_id`	query	string (uuid) (nullable)	no	Platform admins only (required for them): the company to act on. Company users are always confined to their own company.

Responses

Status	Description	Body
`200`	Successful Response	array of IssuerResponse
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires the Reviewer L2 role (or platform admin).	—
`422`	Validation Error	HTTPValidationError

POST `/settings/issuers`

Create an issuer Bearer JWT

Adds a known vendor/school/etc. to the company's registry used by issuer verification. Names are unique (case-insensitively) among the company's active issuers.

Parameters

Name	In	Type	Required	Description
`company_id`	query	string (uuid) (nullable)	no	Platform admins only (required for them): the company to act on. Company users are always confined to their own company.

Request body

IssuerCreate

Responses

Status	Description	Body
`201`	Successful Response	IssuerResponse
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires the Reviewer L2 role (or platform admin).	—
`409`	An active issuer with this name already exists.	—
`422`	Validation Error	HTTPValidationError

PATCH `/settings/issuers/{issuer_id}`

Update or deactivate an issuer

Bearer JWT

Partial update: only the fields sent are changed. Issuers are never deleted; set `is_active=false` to retire one. A request that changes nothing is a no-op. An issuer of another company is a 404.

Parameters

Name	In	Type	Required	Description
`issuer_id`	path	string (uuid)	yes	
`company_id`	query	string (uuid) (nullable)	no	Platform admins only (required for them): the company to act on. Company users are always confined to their own company.

Request body

IssuerUpdate

Responses

Status	Description	Body
`200`	Successful Response	IssuerResponse
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires the Reviewer L2 role (or platform admin).	—
`404`	No such issuer.	—
`422`	Validation Error	HTTPValidationError

GET `/settings/risk-rules`

List current risk rules

Bearer JWT

The CURRENT version of every rule (active or not) of the company, by category.

Parameters

Name	In	Type	Required	Description
`company_id`	query	string (uuid) (nullable)	no	Platform admins only (required for them): the company to act on. Company users are always confined to their own company.

Responses

Status	Description	Body
`200`	Successful Response	array of RiskRuleResponse
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires the Reviewer L2 role (or platform admin).	—
`422`	Validation Error	HTTPValidationError

POST `/settings/risk-rules`

Create a risk rule Bearer JWT

A new rule at version 1, for this company only. It applies to the company's cases scored from now on; nothing already scored changes (assessments freeze the versions that produced them).

Parameters

Name	In	Type	Required	Description
`company_id`	query	string (uuid) (nullable)	no	Platform admins only (required for them): the company to act on. Company users are always confined to their own company.

Request body

RiskRuleCreate

Responses

Status	Description	Body
`201`	Successful Response	RiskRuleResponse
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires the Reviewer L2 role (or platform admin).	—
`409`	A rule with this `rule_id` already exists in the company.	—
`422`	The chosen match/parameters cannot be evaluated by the engine.	—

GET `/settings/risk-rule-options`

Get the options for the Add-rule form Bearer JWT

Everything the Add-rule form offers, so it never has to hardcode (or guess) what the checks can produce.

Responses

Status	Description	Body
`200`	Successful Response	object
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires the Reviewer L2 role (or platform admin).	—

GET `/settings/risk-rules/{rule_id}/history`

Get a risk rule's version history

Bearer JWT

Every version of one of the company's rules, newest first — the per-rule audit trail (who changed what, when). The same edits are also in `audit_log`.

Parameters

Name	In	Type	Required	Description
`rule_id`	path	string	yes	
`company_id`	query	string (uuid) (nullable)	no	Platform admins only (required for them): the company to act on. Company users are always confined to their own company.

Responses

Status	Description	Body
`200`	Successful Response	array of RiskRuleResponse
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires the Reviewer L2 role (or platform admin).	—
`404`	No such rule.	—
`422`	Validation Error	HTTPValidationError

PATCH `/settings/risk-rules/{rule_id}`

Change a risk rule's weight, severity or active flag

Bearer JWT

Rules are versioned: an edit inserts a new immutable row (`version` + 1) and leaves the old one untouched, so every historical assessment stays truthful. Only `weight`, `severity` and `is_active` can be changed; at least one must differ. Only the company's own rule is affected.

Parameters

Name	In	Type	Required	Description
`rule_id`	path	string	yes	
`company_id`	query	string (uuid) (nullable)	no	Platform admins only (required for them): the company to act on. Company users are always confined to their own company.

Request body

RiskRuleUpdate

Responses

Status	Description	Body
`200`	Successful Response	RiskRuleResponse
`400`	Nothing to change.	—

Status	Description	Body
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires the Reviewer L2 role (or platform admin).	—
`404`	No such rule.	—
`409`	The rule was changed concurrently; reload and retry.	—
`422`	Validation Error	HTTPValidationError

GET `/settings/risk-thresholds`

Get the risk tier thresholds

Bearer JWT

The company's scores at which a case becomes `medium` and `high` risk (defaults 30 and 60). The row is created on first read if it does not exist.

Parameters

Name	In	Type	Required	Description
`company_id`	query	string (uuid) (nullable)	no	Platform admins only (required for them): the company to act on. Company users are always confined to their own company.

Responses

Status	Description	Body
`200`	Successful Response	RiskThresholdsResponse
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires the Reviewer L2 role (or platform admin).	—
`422`	Validation Error	HTTPValidationError

PUT `/settings/risk-thresholds`

Set the risk tier thresholds

Bearer JWT

Safe for history: each assessment embeds the thresholds it used.

Parameters

Name	In	Type	Required	Description
`company_id`	query	string (uuid) (nullable)	no	Platform admins only (required for them): the company to act on. Company users are always confined to their own company.

Request body

RiskThresholds

Responses

Status	Description	Body
`200`	Successful Response	RiskThresholdsResponse

Status	Description	Body
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires the Reviewer L2 role (or platform admin).	—
`422`	Validation Error	HTTPValidationError

GET `/settings/users`

List users Bearer JWT

Every account on the platform, ordered by company then email; filter with `company_id`. Platform admin only.

Parameters

Name	In	Type	Required	Description
`company_id`	query	string (uuid) (nullable)	no	

Responses

Status	Description	Body
`200`	Successful Response	array of UserAdminResponse
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires the platform admin role.	—
`422`	Validation Error	HTTPValidationError

POST `/settings/users`

Create a user Bearer JWT

Creates an account and assigns it to a company + role. Company roles (`user` , `reviewer_l1` , `reviewer_l2`) need a `company_id` ; a `platform_admin` must have none. Platform admin only.

Request body

UserCreate

Responses

Status	Description	Body
`201`	Successful Response	UserAdminResponse
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires the platform admin role.	—
`404`	No such company.	—
`409`	An account with that email already exists.	—
`422`	Company/role combination is invalid.	—

PATCH `/settings/users/{user_id}`

Change a user's role, company or active flag Bearer JWT

Partial update of `role`, `company_id` and/or `is_active`. A platform admin cannot change their own role or deactivate themselves (this also guarantees a platform admin always remains). Changing a user's company or role invalidates their existing tokens (they must sign in again). There is no self-service signup and no password-reset endpoint. Platform admin only.

Parameters

Name	In	Type	Required	Description
`user_id`	path	string (uuid)	yes	

Request body

UserAdminUpdate

Responses

Status	Description	Body
`200`	Successful Response	UserAdminResponse
`400`	Attempted to change your own role or deactivate yourself.	—
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires the platform admin role.	—
`404`	No such user (or company).	—
`422`	Company/role combination is invalid.	—

Other

GET `/platform/companies`

List companies Bearer JWT

Responses

Status	Description	Body
`200`	Successful Response	array of CompanyResponse
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires the platform admin role.	—

POST `/platform/companies`

Create a company Bearer JWT

Creates a client company and copies every active platform rule template into its own risk rules, plus default tier thresholds (its own independent copy, which its Reviewer L2s can then tune; later template edits don't affect it). The issuer registry starts empty. Users are added separately (Settings > Users).

Request body

CompanyCreate

Responses

Status	Description	Body
`201`	Successful Response	CompanyResponse
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires the platform admin role.	—
`409`	A company with this name already exists.	—
`422`	Validation Error	HTTPValidationError

PATCH `/platform/companies/{company_id}`

Rename or suspend/reactivate a company Bearer JWT

`is_active=false` suspends the company: its users can no longer sign in, and existing tokens stop working on the next request. Data is kept.

Parameters

Name	In	Type	Required	Description
`company_id`	path	string (uuid)	yes	

Request body

CompanyUpdate

Responses

Status	Description	Body
`200`	Successful Response	CompanyResponse
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires the platform admin role.	—
`404`	No such company.	—
`409`	Name taken.	—
`422`	Validation Error	HTTPValidationError

GET `/platform/risk-rule-templates`

List the platform risk-rule template Bearer JWT

The default rule set copied into every NEW company's own risk rules at creation time. Editing it never changes companies that already exist.

Responses

Status	Description	Body
`200`	Successful Response	array of RuleTemplateResponse
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires the platform admin role.	—

POST `/platform/risk-rule-templates`

Add a rule to the platform template Bearer JWT

Same rule builder as a company's Add-rule form. Applies to companies created from now on.

Request body

RiskRuleCreate

Responses

Status	Description	Body
`201`	Successful Response	RuleTemplateResponse
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires the platform admin role.	—
`409`	A template rule with this rule_id already exists.	—
`422`	Validation Error	HTTPValidationError

PATCH `/platform/risk-rule-templates/{rule_id}`

Edit a platform template rule Bearer JWT

Changes weight, severity, wording or active flag. Existing companies keep their own copies unchanged; only companies created afterwards start from the new values.

Parameters

Name	In	Type	Required	Description
`rule_id`	path	string	yes	

Request body

RuleTemplateUpdate

Responses

Status	Description	Body
`200`	Successful Response	RuleTemplateResponse
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires the platform admin role.	—
`404`	No such template rule.	—
`422`	Validation Error	HTTPValidationError

GET `/platform/usage`

Billing & usage by company Bearer JWT

One row per company for the period: cases created, documents uploaded, files stored (documents + generated reports) and bytes stored during the period, plus each company's current all-time storage. Read from incrementally-maintained daily counters (company_usage_stats), not live scans of the case tables; a nightly job reconciles them against the source data. Periods use UTC days.

Parameters

Name	In	Type	Required	Description
`period`	query	string: `"this_month"`, `"last_month"`, `"last_30_days"`, `"this_year"`, `"all_time"`, `"custom"`	no	
`start`	query	string (date) (nullable)	no	custom period only
`end`	query	string (date) (nullable)	no	custom period only

Responses

Status	Description	Body
`200`	Successful Response	UsageResponse
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires the platform admin role.	—
`422`	Validation Error	HTTPValidationError

POST `/platform/usage/reconcile`

Reconcile usage counters now

Bearer JWT

Recomputes every company's daily counters from the cases/documents/reports tables and corrects any drift (the same job runs nightly). Returns what was corrected; a correction is also written to the platform audit log.

Responses

Status	Description	Body
`200`	Successful Response	ReconcileResponse
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires the platform admin role.	—

GET `/platform/queues`

Processing queue monitor

Bearer JWT

Per queue (extraction / vision / forensics): tasks waiting and running, age of the oldest waiting task, average / p95 / max wait over the window, average run time, the rate limit in force and the limiter's counters (calls throttled, Azure 429s), and the worker processes consuming it. Wait times are measured from enqueue to start.

Parameters

Name	In	Type	Required	Description
`window_minutes`	query	integer	no	

Responses

Status	Description	Body
`200`	Successful Response	object

Status	Description	Body
`401`	Missing, invalid or expired bearer token, or the user is inactive.	—
`403`	Requires the platform admin role.	—
`422`	Validation Error	HTTPValidationError

Data models

ApproveRequest

Field	Type	Required	Description
`note`	string (nullable)	no	Justification. Optional for low-risk cases; at least 10 characters (after trimming) is required to approve a medium- or high-risk case.

AuditEventResponse

Field	Type	Required	Description
`id`	string (uuid)	yes	
`created_at`	string (date-time)	yes	
`event_type`	string	yes	
`actor_name`	string (nullable)	yes	
`actor_email`	string (nullable)	yes	
`case_id`	string (uuid) (nullable)	yes	
`case_number`	string (nullable)	yes	
`document_id`	string (uuid) (nullable)	yes	
`document_filename`	string (nullable)	yes	
`event_data`	object (nullable)	yes	

AuditLogEntrySchema

One `audit_log` row scoped to a case — CLAUDE.md section 3.5's "review history/timeline ... built from the same audit log." Every event_type here is a real one already written by app/services/ audit_service.py's call sites (case creation, uploads, and each Celery check task) — nothing here is synthesized for display.

Field	Type	Required	Description
`id`	string (uuid)	yes	
`event_type`	string	yes	
`actor_name`	string (nullable)	yes	
`document_id`	string (uuid) (nullable)	yes	
`event_data`	object (nullable)	yes	
`created_at`	string (date-time)	yes	

AuditPage

Field	Type	Required	Description
`items`	array of AuditEventResponse	yes	
`total`	integer	yes	

Body_upload_document_cases__case_id__documents_post

Field	Type	Required	Description
`file`	string (binary)	yes	

BoundingBoxSchema

Normalized bounding box — page-fraction (0-1) coordinates, same convention as every other bounding box in this codebase (ELA, copy-move, visual inconsistency). `page` is 1-based.

Field	Type	Required	Description
`page`	integer	yes	
`x`	number	yes	
`y`	number	yes	
`width`	number	yes	
`height`	number	yes	

CaseActionSchema

A reviewer decision/note on the case (approve/reject/escalate).

Field	Type	Required	Description
`id`	string (uuid)	yes	
`action_type`	string	yes	
`actor_name`	string (nullable)	yes	
`actor_role`	string (nullable)	no	Display label of the actor's role when they acted, e.g. 'Reviewer L1'.
`notes`	string (nullable)	yes	
`created_at`	string (date-time)	yes	

CaseCreateRequest

Field	Type	Required	Description
`case_type`	CaseType	yes	Case-level category chosen at submission. Independent of each document's own `document_type`, which the classification step fills in later.

CaseDecisionResponse

Field	Type	Required	Description
`case_id`	string (uuid)	yes	
`status`	CaseStatus	yes	
`assigned_tier`	CaseTier	yes	
`action`	CaseActionSchema	yes	

CaseDetail

Field	Type	Required	Description
`id`	string (uuid)	yes	
`case_number`	string	yes	
`case_type`	CaseType	yes	
`status`	CaseStatus	yes	
`assigned_tier`	CaseTier	yes	
`risk_tier`	RiskTier (nullable)	yes	
`submitted_by`	UserSummary (nullable)	no	
`created_at`	string (date-time)	yes	
`document_count`	integer	yes	
`flag`	CaseFlagSchema	yes	
`can_act`	boolean	yes	Whether the caller may approve/reject/escalate this case given its tier. False for submitters, and for a `reviewer_l1` on an L2-assigned case (read-only). Does not reflect status — a decided case is still `can_act` but every action is a 409.
`documents`	array of CaseDocumentSummary	yes	
`cross_document_findings`	array of CrossDocumentFindingSummary	yes	
`forensic_findings`	array of ForensicFindingSummary	yes	
`assessment`	RiskAssessmentSchema (nullable)	yes	
`pipeline`	PipelineStatusSchema	yes	
`actions`	array of CaseActionSchema	yes	

CaseDocumentSummary

A document as shown on the case-detail page. `file_url` here is a short-lived signed URL (see StorageService.get_download_url) — the storage container is private, so the durable URL alone isn't browser-fetchable.

`document_type` and `extracted_fields` are filled in by the OCR + classification + extraction Celery pipeline (app/tasks/document_ processing.py) — both stay null until `processing_status` reaches "complete". `extracted_fields` shape: {"document_type_confidence": float, "core_fields": {name: {value, confidence, uncertain}}, "additional_fields": [{field_name, value, confidence, uncertain}]}.

`checks` is filled in afterward by app/tasks/document_checks.py (field_validation, issuer_verification, and later the forensic checks) — empty until those complete.

Field	Type	Required	Description
`id`	string (uuid)	yes	
`original_filename`	string	yes	
`document_type`	string (nullable)	yes	
`processing_status`	string	yes	
`extracted_fields`	object (nullable)	yes	
`processing_error`	string (nullable)	yes	
`content_type`	string (nullable)	yes	
`file_size_bytes`	integer (nullable)	yes	
`file_hash`	string	yes	
`file_url`	string	yes	
`uploaded_at`	string (date-time)	yes	
`checks`	array of DocumentCheckSummary	yes	

CaseFlagSchema

Wire shape of app/services/case_flag_service.py's CaseFlag: the case's real risk tier (low/medium/high), or "pending" until scored.

Field	Type	Required	Description
`flag`	string: `"low"`, `"medium"`, `"high"`, `"pending"`	yes	
`label`	string	yes	
`description`	string	yes	
`score`	integer (nullable)	no	

CaseListItem

Field	Type	Required	Description
`id`	string (uuid)	yes	
`case_number`	string	yes	
`case_type`	CaseType	yes	
`status`	CaseStatus	yes	
`assigned_tier`	CaseTier	yes	
`risk_tier`	RiskTier (nullable)	yes	
`submitted_by`	UserSummary (nullable)	no	
`created_at`	string (date-time)	yes	
`document_count`	integer	yes	
`flag`	CaseFlagSchema	yes	
`can_act`	boolean	yes	Whether the caller may approve/reject/escalate this case given its tier. False for submitters, and for a `reviewer_l1` on an L2-assigned case (read-only). Does not reflect status — a decided case is still `can_act` but every action is a 409.

CaseReportResponse

One generated report. `download_url` is a short-lived signed URL (the container is private — see app/services/storage_service.py), so it is minted fresh on every response rather than stored.

Field	Type	Required	Description
`id`	string (uuid)	yes	
`case_id`	string (uuid)	yes	
`generated_by_name`	string (nullable)	yes	
`generated_at`	string (date-time)	yes	
`file_size_bytes`	integer	yes	
`page_count`	integer	yes	
`report_sha256`	string	yes	
`risk_tier`	string (nullable)	yes	
`risk_score`	integer (nullable)	yes	
`download_url`	string	yes	

CaseResponse

Field	Type	Required	Description
`id`	string (uuid)	yes	
`case_number`	string	yes	
`case_type`	CaseType	yes	
`status`	CaseStatus	yes	
`assigned_tier`	CaseTier	yes	
`risk_tier`	RiskTier (nullable)	yes	
`submitted_by_user_id`	string (uuid)	yes	
`created_at`	string (date-time)	yes	

CaseStatus

Allowed values: `"submitted"`, `"under_automated_review"`, `"pending_manual_review"`, `"auto_approved"`, `"escalated"`, `"under_investigation"`, `"approved"`, `"rejected"`, `"closed"`

CaseTier

Which reviewer tier owns the case — orthogonal to `status`. Every case starts at `l1`; escalating it moves it to `l2` (one-directional, there is no de-escalation). On an `l2` case a `reviewer_l1` keeps read access but every action endpoint returns 403; `reviewer_l2` and `admin` can act on cases of either tier.

Allowed values: `"l1"`, `"l2"`

CaseType

Case-level category, selected at submission time (CLAUDE.md section 3.6's document-type coverage list). This is separate from a document's own `document_type`, which is filled in later by the classification step — a case can hold documents of more than one type (e.g. an invoice plus its approval voucher).

Allowed values: `"school_document"`, `"vendor_invoice"`, `"commercial_invoice"`, `"procurement_documentation"`, `"quotation"`, `"travel_reimbursement"`, `"other"`

CompanyCreate

Field	Type	Required	Description
`name`	string	yes	

CompanyResponse

Field	Type	Required	Description
`id`	string (uuid)	yes	
`name`	string	yes	
`is_active`	boolean	yes	
`created_at`	string (date-time)	yes	
`user_count`	integer	yes	

CompanyUpdate

Field	Type	Required	Description
`name`	string (nullable)	no	
`is_active`	boolean (nullable)	no	

CompanyUsageRow

Field	Type	Required	Description
`company_id`	string (uuid)	yes	
`company_name`	string	yes	
`is_active`	boolean	yes	
`cases_created`	integer	yes	
`documents_uploaded`	integer	yes	
`files_stored`	integer	yes	
`storage_bytes`	integer	yes	
`total_storage_bytes`	integer	yes	
`total_files_stored`	integer	yes	

CrossDocumentFindingSummary

One `cross_document_findings` row (CLAUDE.md section 3.1, "Cross-document validation & reconciliation") — a shared field (amount/date/issuer) that disagreed between two of this case's documents. Case-level, not per-document — see app/services/ cross_document_service.py and app/tasks/document_checks.py.

Field	Type	Required	Description
`id`	string (uuid)	yes	
`field_name`	string	yes	
`finding_type`	string	yes	
`severity`	string	yes	
`description`	string	yes	
`document_ids`	array of string (nullable)	yes	
`created_at`	string (date-time)	yes	
`regions`	array of object	no	

CurrentUserResponse

Field	Type	Required	Description
`id`	string (uuid)	yes	
`email`	string (email)	yes	
`full_name`	string (nullable)	yes	Display name, if one was set.
`role`	UserRole	yes	`user` (submitter), `reviewer_l1`, `reviewer_l2` (company roles) or `platform_admin`.
`role_label`	string	yes	Display label, e.g. "Reviewer L2".
`is_active`	boolean	yes	Deactivated accounts are rejected on every request.
`is_platform_admin`	boolean	yes	True for platform admins, who belong to no company.
`company_id`	string (uuid) (nullable)	yes	The user's company; null for a platform admin.
`company_name`	string (nullable)	yes	Display name of the user's company.

DocumentCheckSummary

One `document_checks` row (CLAUDE.md sections 2.2/3.1/3.2 — field_validation, issuer_verification, and later the forensic checks). `result` holds whatever shape that check_type produces — see the individual check services (e.g. app/services/field_validation_service.py) for what's inside — the frontend's Checks panel renders it generically rather than assuming a fixed schema per check_type.

Field	Type	Required	Description
`id`	string (uuid)	yes	
`check_type`	string	yes	
`status`	string	yes	
`result`	object (nullable)	yes	
`error_message`	string (nullable)	yes	
`created_at`	string (date-time)	yes	

DocumentFileUrlResponse

A fresh short-lived signed URL for one document's original file.

Field	Type	Required	Description
`file_url`	string	yes	

DocumentResponse

Field	Type	Required	Description
`id`	string (uuid)	yes	
`case_id`	string (uuid)	yes	
`original_filename`	string	yes	
`content_type`	string (nullable)	yes	
`file_size_bytes`	integer (nullable)	yes	
`file_hash`	string	yes	
`file_url`	string	yes	
`uploaded_at`	string (date-time)	yes	
`processing_status`	string	yes	

EscalateRequest

Field	Type	Required	Description
`reason`	string	yes	Why the case needs an L2 reviewer. Required and must not be blank.

ForensicFindingSummary

One medium/high-severity finding from a flagged metadata_forensics/ error_level_analysis/copy_move_detection check — the per-document counterpart to CrossDocumentFindingSummary, both shown together in the case detail "Explainable findings" panel. See app/services/ case_flag_service.py's get_forensic_findings for what feeds this and why it's a temporary stand-in for the real risk-scoring engine.

Field	Type	Required	Description
`document_id`	string (uuid)	yes	
`document_filename`	string	yes	
`check_type`	string	yes	
`check_type_label`	string	yes	
`finding`	string	yes	
`severity`	string	yes	
`description`	string	yes	

HTTPValidationError

Field	Type	Required	Description
`detail`	array of ValidationError	no	

IssuerCreate

Field	Type	Required	Description
`name`	string	yes	English/Latin issuer name.
`name_arabic`	string (nullable)	no	Arabic-script name, matched separately during issuer verification.
`tax_id`	string (nullable)	no	Tax/VAT registration number, if known.
`type`	IssuerType	no	Kind of issuer (vendor, school, ...).

IssuerResponse

Field	Type	Required	Description
`id`	string (uuid)	yes	
`name`	string	yes	
`name_arabic`	string (nullable)	yes	
`tax_id`	string (nullable)	yes	
`type`	IssuerType	yes	
`is_active`	boolean	yes	
`created_at`	string (date-time)	yes	
`updated_at`	string (date-time)	yes	

IssuerType

Allowed values: `"vendor"`, `"school"`, `"government"`, `"other"`

IssuerUpdate

Field	Type	Required	Description
`name`	string (nullable)	no	
`name_arabic`	string (nullable)	no	
`tax_id`	string (nullable)	no	
`type`	IssuerType (nullable)	no	
`is_active`	boolean (nullable)	no	

LoginRequest

Field	Type	Required	Description
`email`	string (email)	yes	Account email (matched exactly as stored).
`password`	string	yes	Account password.

PipelineStatusSchema

Whether every automated check has finished. `pending` names what is still outstanding — shown to the reviewer when Approve is disabled.

Field	Type	Required	Description
`complete`	boolean	yes	
`pending`	array of string	yes	

ReconcileResponse

Field	Type	Required	Description
`rows_checked`	integer	yes	
`rows_corrected`	integer	yes	
`drift`	array of object	yes	

RejectRequest

Field	Type	Required	Description
`reason`	string	yes	Why the case is rejected. Required and must not be blank.

RiskAssessmentSchema

Field	Type	Required	Description
`id`	string (uuid)	yes	
`score`	integer	yes	
`raw_score`	number	yes	
`tier`	RiskTier	yes	
`computed_at`	string (date-time)	yes	
`triggered_reasons`	array of RiskReasonSchema	yes	
`thresholds`	map of integer	yes	

RiskReasonSchema

One fired rule, as frozen at scoring time (rendered text, weight and rule version) — see app/models/case_risk_assessment.py.

Field	Type	Required	Description
`rule_id`	string	yes	
`rule_version`	integer	yes	
`category`	string	yes	
`check_type`	string	yes	
`severity`	string	yes	
`weight`	number	yes	
`reason`	string	yes	
`document_id`	string (uuid) (nullable)	yes	
`document_filename`	string (nullable)	yes	

RiskRuleCreate

A brand-new rule (version 1). The admin picks a rule TYPE and its parameters from the catalog (GET /settings/risk-rule-options); the API builds and validates the engine `condition` from them — nobody writes JSON, and a choice the engine can't evaluate is rejected.

Field	Type	Required	Description
`rule_id`	string	yes	Stable key, lower-case with a dot: e.g. custom.editing_tool_seen
`category`	string: `"forensics"`, `"consistency"`, `"verification"`, `"duplication"`	yes	
`match`	string: `"finding"`, `"check_result"`, `"sub_check"`, `"cross_document"`, `"signature_match"`	yes	
`check_type`	string (nullable)	no	
`finding`	string (nullable)	no	
`severity_in`	array of string: `"low"`, `"medium"`, `"high"` (nullable)	no	
`sub_check`	string (nullable)	no	
`field_name`	string (nullable)	no	
`signature_result`	string (nullable)	no	
`weight`	number	yes	
`severity`	string: `"low"`, `"medium"`, `"high"`	yes	
`reason_template`	string	yes	
`is_active`	boolean	no	
`change_note`	string (nullable)	no	

RiskRuleResponse

One (current or historical) version of a rule.

Field	Type	Required	Description
`id`	string (uuid)	yes	
`rule_id`	string	yes	
`category`	string	yes	
`check_type`	string	yes	
`weight`	number	yes	
`severity`	string	yes	
`reason_template`	string	yes	
`is_active`	boolean	yes	
`version`	integer	yes	
`effective_from`	string (date-time)	yes	
`updated_by_name`	string (nullable)	yes	
`change_note`	string (nullable)	yes	

RiskRuleUpdate

An edit. Never mutates the existing row: the API inserts version+1.

Field	Type	Required	Description
`weight`	number (nullable)	no	Points added to the case score when the rule fires.
`severity`	string: `"low"`, `"medium"`, `"high"` (nullable)	no	Severity shown beside the rule's reason.
`is_active`	boolean (nullable)	no	Inactive rules are evaluated but never fire.
`change_note`	string (nullable)	no	Why the change was made.

RiskThresholds

Field	Type	Required	Description
`medium_threshold`	integer	yes	Score at or above which a case is `medium` risk.
`high_threshold`	integer	yes	Score at or above which a case is `high` risk. Must exceed the medium threshold.

RiskThresholdsResponse

Field	Type	Required	Description
`medium_threshold`	integer	yes	Score at or above which a case is `medium` risk.
`high_threshold`	integer	yes	Score at or above which a case is `high` risk. Must exceed the medium threshold.
`updated_at`	string (date-time)	yes	
`updated_by_name`	string (nullable)	yes	

RiskTier

Allowed values: `"low"`, `"medium"`, `"high"`

RuleTemplateResponse

Field	Type	Required	Description
`id`	string (uuid)	yes	
`rule_id`	string	yes	
`category`	string	yes	
`check_type`	string	yes	
`condition`	object	yes	
`weight`	number	yes	
`severity`	string	yes	
`reason_template`	string	yes	
`is_active`	boolean	yes	
`updated_at`	string (date-time)	yes	

RuleTemplateUpdate

Field	Type	Required	Description
`weight`	number (nullable)	no	
`severity`	string: `"low"`, `"medium"`, `"high"` (nullable)	no	
`reason_template`	string (nullable)	no	
`is_active`	boolean (nullable)	no	

SignatureMatchResponse

One signature_matches row returned by the API, enriched with the reference's person_name so the frontend can show "compared against [name]" without a separate reference lookup.

Field	Type	Required	Description
`id`	string (uuid)	yes	
`document_id`	string (uuid)	yes	
`case_id`	string (uuid)	yes	
`signature_reference_id`	string (uuid)	yes	
`reference_person_name`	string	yes	
`comparison_scope`	string	yes	
`result`	string	yes	
`result_label`	string	yes	
`reasoning`	string (nullable)	yes	
`compared_at`	string (date-time)	yes	

SignatureReferenceCreate

Request body for POST /cases/{case_id}/documents/{document_id}/ signature-references.

`person_name` is required and must be provided by the reviewer — the system never attempts to auto-extract or pre-fill it (CLAUDE.md §2.3).

Field	Type	Required	Description
`person_name`	string	yes	Name of the person whose signature this is, typed by the reviewer. Required — the system does not attempt to auto-extract this field.
`bounding_box`	BoundingBoxSchema	yes	
`is_library`	boolean	no	Defaults to True: every reference is saved to the permanent signature library. False is still accepted (case-only reference). Either way, at this phase a reference is only compared against other documents in its own case — no cross-case/library-wide comparison logic runs yet.

SignatureReferenceResponse

One signature_references row returned by the API.

Field	Type	Required	Description
`id`	string (uuid)	yes	
`person_name`	string	yes	
`signature_image_url`	string (nullable)	yes	
`bounding_box`	object (nullable)	yes	
`source_document_id`	string (uuid)	yes	
`source_case_id`	string (uuid)	yes	
`created_by`	string (uuid)	yes	
`is_library`	boolean	yes	
`created_at`	string (date-time)	yes	

TokenResponse

Field	Type	Required	Description
`access_token`	string	yes	JWT. Send as `Authorization: Bearer <token>`.
`token_type`	string	no	Always `bearer`.

UsageResponse

Field	Type	Required	Description
`period`	string: `"this_month"`, `"last_month"`, `"last_30_days"`, `"this_year"`, `"all_time"`, `"custom"`	yes	
`start`	string (date) (nullable)	yes	First UTC day included (null = from the beginning).
`end`	string (date) (nullable)	yes	Last UTC day included (null = up to today).
`companies`	array of CompanyUsageRow	yes	
`totals`	UsageTotals	yes	

UsageTotals

Field	Type	Required	Description
`cases_created`	integer	yes	
`documents_uploaded`	integer	yes	
`files_stored`	integer	yes	
`storage_bytes`	integer	yes	
`total_storage_bytes`	integer	yes	
`total_files_stored`	integer	yes	

UserAdminResponse

Field	Type	Required	Description
`id`	string (uuid)	yes	
`email`	string (email)	yes	
`full_name`	string (nullable)	yes	
`role`	UserRole	yes	
`is_active`	boolean	yes	
`created_at`	string (date-time)	yes	
`company_id`	string (uuid) (nullable)	yes	Null for platform admins.
`company_name`	string (nullable)	yes	

UserAdminUpdate

Field	Type	Required	Description
`role`	UserRole (nullable)	no	New role. Cannot be changed on your own account.
`company_id`	string (uuid) (nullable)	no	Move the user to another company (company roles only).
`is_active`	boolean (nullable)	no	False deactivates the account. Not allowed on yourself.

UserCreate

An account created by an administrator (there is no self-service sign-up). The password is set here and only ever stored hashed.

Field	Type	Required	Description
`email`	string (email)	yes	Login email. Unique regardless of letter case.
`full_name`	string (nullable)	no	Display name (optional).
`role`	UserRole	no	`user`, `reviewer_l1`, `reviewer_l2` or `platform_admin`.
`company_id`	string (uuid) (nullable)	no	Required for company roles; must be empty for `platform_admin`.
`password`	string	yes	Initial password (8-128 chars); stored hashed.

UserRole

Allowed values: `"user"`, `"reviewer_l1"`, `"reviewer_l2"`, `"platform_admin"`

UserSummary

Field	Type	Required	Description
`id`	string (uuid)	yes	
`email`	string	yes	
`full_name`	string (nullable)	yes	

ValidationError

Field	Type	Required	Description
`loc`	array of string | integer	yes	
`msg`	string	yes	
`type`	string	yes	

Frontend screens

React 19 + TypeScript SPA (`frontend/src`), routed with React Router 7 (`App.tsx`). Server state is TanStack Query (`retry: false` , `refetchOnWindowFocus: false`); the auth token lives in `localStorage` (`hooks/useAuth.tsx`). In development the Vite server proxies `/api/*` to `127.0.0.1:8000` and strips the prefix (`vite.config.ts`), so `API_BASE_URL` defaults to `/api` (override with `VITE_API_BASE_URL`). Every page except Login is loaded on demand (`React.lazy` in `App.tsx`), so the first download is the ~360 kB main bundle plus the page being opened.

Platform admins pick which company they are looking at with the **company picker** (`CompanyPicker` , `hooks/useActingCompany.tsx` , remembered per browser) on the case queue, dashboard, audit history and the company settings pages; an amber note reminds them the access is read-only and recorded in the platform audit log.

`ProtectedRoute` redirects to `/login` if there is no token or the token does not resolve to a user. `RoleRoute` (in `components/AdminRoute.tsx`) shows an "access required" panel to the wrong role. **Both are conveniences — the backend independently 401/403s every call**, so a user who reaches a URL still cannot read or change anything they are not entitled to.

Route map

Route	Screen	Access
`/login`	LoginPage	public
`/` , `/cases` , `/review-queue`	CaseQueuePage	any logged-in user (platform admin: per company, read-only)
`/cases/new`	NewCasePage	company users (hidden for platform admins)
`/cases/:caseId`	CaseDetailPage	any (reduced view for `user` ; read-only decision panel for a Reviewer L1 on an L2 case; "Support view" for a platform admin)
`/dashboard`	DashboardPage	company users
`/my-cases`	MyCasesPage	company users
`/audit-history`	AuditHistoryPage	Reviewer L1 and above; platform admins
`/settings` → `/settings/issuer-registry`	SettingsIssuerRegistryPage	Reviewer L2; platform admins (per company)
`/settings/risk-rules`	SettingsRiskRulesPage	Reviewer L2; platform admins (per company)
`/settings/users`	SettingsUsersPage	platform admins only
`/platform` → `/platform/companies`	PlatformCompaniesPage	platform admins only
`/platform/rule-templates`	PlatformRuleTemplatesPage	platform admins only
`/platform/usage`	PlatformUsagePage	platform admins only
`/platform/queues`	PlatformQueuesPage	platform admins only

Route	Screen	Access
`*`	redirect to `/`	—

The top `Nav` (`design-system/Nav.tsx`) shows each item only to who may use it (`NAV_VISIBLE`): *Dashboard* and *My cases* to company users, *Audit history* to Reviewer L1+ and platform admins, *Settings* to Reviewer L2 and platform admins, *Platform* to platform admins. It shows the company name (or "Platform") next to the user, and hides *New upload* from platform admins. Route guards (`RoleRoute minRole=… platformAdmin="allow" | "only"`) mirror this.

LoginPage — `/login`

Purpose: email/password sign-in. **Endpoints:** `POST /auth/login` , then `GET /auth/me` (via `useAuth`). **Notes:** stores the JWT in `localStorage` . Deliberately has no "trust this device", SSO button or compliance badges.

CaseQueuePage — `/` , `/cases` , `/review-queue`

Purpose: the case list / reviewer queue. Shows counts (open, awaiting review, cleared, escalated), a table with case number, type, status, flag (risk tier), escalation badge, document count and submitter. Reviewer L1 gets tabs *My queue* (`can_act`) / *Escalated to L2 · view only*; Reviewer L2 gets *All cases* / *Escalated queue (L2)*; rows the user can't act on show a "View only" pill. **Endpoints:** `GET /cases` (once; refetched by other screens via the `["cases", token]` query key). **Behaviour:** search box + status / case-type / flag filters applied **in the browser**. The status column is a UI mapping of `CaseStatus` → `Analyzing` (`submitted` , `under_automated_review`), `Awaiting Review` (`pending_manual_review` and any other open status), `Approved` (`approved` , `auto_approved` , `closed`), `Rejected` . **"Escalated" is a tier filter (L2), not a status.** For Reviewer L2 the API sorts open L2 cases first, then newest first. **Roles:** every company user, but a `user` only receives their own cases and no risk tier (the flag reads "In review"). A platform admin sees one company at a time, chosen with the company picker — never a mixed list.

NewCasePage — `/cases/new`

Purpose: submit a case: choose a case type, attach PDFs, upload. **Endpoints:** `POST /cases` , `POST /cases/{id}/documents` (once per file, XHR with progress), and, for the optional signature step, `GET …/documents/{id}/file-url` and `POST …/documents/{id}/signature-references` (through `SignatureReferenceCreator`). **Flow:** React Hook Form + Zod validate the case type. The drop zone accepts **PDF only, up to 10 MB each, not password-protected**; a file that is not a PDF, is empty or is over 10 MB is refused in the browser before anything is sent (`lib/uploadLimits.ts`). The case is created once (so a retry after a failed upload reuses it); files upload in parallel; each shows its own progress, or the server's specific rejection message (e.g. "This file is password-protected. Please remove the password and re-upload."). After upload the user can open **Set reference signature** for any uploaded document (drawing a box over a `react-pdf` page) — optional; without it no comparison runs. Then the case detail page is opened. **Roles:** any company user may create a case, upload to their own case and set reference signatures on it; reviewers may do so on any case of their company. A stranger's request is a 404; platform admins can't upload.

CaseDetailPage — `/cases/:caseId`

Purpose: everything about one case in three columns (`lg:grid-cols-12`): **documents** (3), the **selected document + its checks** (5), and **risk + decision** (4), plus the activity timeline below. **Endpoints:** `GET /cases/{id}` (polls every **5 s** while `pipeline.complete` is false or there is no assessment), `GET /cases/{id}/audit-log` , `GET /cases/{id}/signature-matches` (polled every 10 s), and, through child components, the decision endpoints and report endpoints. **Contents:**

- *Left:* the case flag card and the document list (type and processing status).
- *Middle:* `PdfOverlayViewer` showing the selected PDF with live overlays from that document's checks and cross-document findings; the document's **core** extracted fields (issuer — the only one with an "(uncertain)" marker — reference number, date, amounts; `additional_fields` are not listed here); `DocumentChecksPanel` (one expandable card per check, plus a cross-document card and the signature comparison card).
- *Right:* score/tier and the explainable findings (`SeverityFinding` per triggered reason, from `assessment.triggered_reasons`), `CaseDecisionPanel` , and `CaseReportExport` (reviewers only).
- *Bottom:* `ActivityTimeline` (the case's audit trail, newest first). **Roles:** everyone can open a case they may see (a foreign case is a 404 → "Case not found"). A `user` receives no assessment, no tier and only approve/reject actions; decision and report controls are reviewer only. A platform admin gets an amber **"Support view — read-only"** banner and no decision or report controls. The header shows an "Escalated · L2" badge on an L2 case; when the API returns `can_act: false` (a Reviewer L1 on an L2 case) the decision panel is a read-only card saying the case is with Reviewer L2, while the report export stays available.

DashboardPage — `/dashboard`

Purpose: at-a-glance operational summary. **Endpoints:** `GET /cases` only. **Everything is computed in the browser** from that list: KPI cards (counts by flag: high/medium/low/pending, escalated open cases, cleared), a risk-distribution panel, the top-5 open cases the user can act on (`can_act` ; escalated first, then highest risk score, then newest), and a "Case Summary" strip (total cases, documents analyzed, escalated, cleared). **Notes:** this is *not* an MIS/compliance dashboard (no fraud rate, resolution time, volumes over time, export). It contains no charts library.

MyCasesPage — `/my-cases`

Purpose: the signed-in user's own cases, with status filters (Under review / Cleared / Action required) and search. **Endpoints:** `GET /cases` ; client filters to cases the user submitted. **Notes:** a submitter sees decision outcomes here but not risk detail.

AuditHistoryPage — `/audit-history`

Purpose: read the company's append-only `audit_log` (platform admins: a chosen company's log, or — with the *Platform-level log* checkbox — the platform events: companies, users, template edits, usage reconciliations, platform access). **Endpoints:** `GET /audit-log` (paged; `event_type` , `q` with a debounce, `limit` / `offset`) and `GET /audit-log/event-types` (for the filter). Event descriptions are rendered by `lib/audit.ts` (`describeAuditEvent`), which turns each `event_type` + `event_data` into a sentence. **Roles:** Reviewer L1 and above, and platform admins (`RoleRoute minRole="reviewer_l1" platformAdmin="allow"` ; the API also 403s others). Company users never see platform-access rows.

SettingsIssuerRegistryPage — `/settings/issuer-registry`

Purpose: manage the company's issuer registry (name, Arabic name, tax id, type, active). **Endpoints:** `GET/POST /settings/issuers` , `PATCH /settings/issuers/{id}` (deactivate/reactivate — rows are never deleted). **Roles:** Reviewer L2 (own company); platform admins pick a company. Uses `SettingsShell` (tab bar, company picker) and `Modal` .

SettingsRiskRulesPage — `/settings/risk-rules`

Purpose: view and tune the company's scoring rules and thresholds. **Endpoints:** `GET /settings/risk-rules` , `PATCH /settings/risk-rules/{rule_id}` (weight / severity / active — inserts a new version), `GET …/{rule_id}/history` (per-

rule change trail), `GET/PUT /settings/risk-thresholds`, and via `AddRuleModal` `GET /settings/risk-rule-options` + `POST /settings/risk-rules`. **Notes:** edits apply to the company's cases scored from then on; historical assessments keep the versions they used. **Roles:** Reviewer L2 (own company); platform admins pick a company.

SettingsUsersPage — `/settings/users`

Purpose: manage every account on the platform (there is no self-service sign-up and no company-level user management). Lives under the *Platform* tabs (`PlatformShell`). **Endpoints:** `GET/POST /settings/users` (via `AddUserModal`), `PATCH /settings/users/{id}` (role, company, active). The table has a Company column and a company filter; *Add user* asks for the company and role. The role picker lists User / Reviewer L1 / Reviewer L2 / Platform Admin (`ROLE_OPTIONS` in `types/auth.ts`; API values `user` / `reviewer_l1` / `reviewer_l2` / `platform_admin`). Moving a user to another company invalidates their current token. A platform admin cannot change their own role or deactivate themselves. **Roles:** platform admins only.

PlatformCompaniesPage — `/platform/companies`

Purpose: list, create, rename, suspend and reactivate companies. **Endpoints:** `GET/POST /platform/companies`, `PATCH /platform/companies/{id}`. **Notes:** creating a company seeds its risk rules from the platform template and default thresholds; its issuer registry starts empty. Suspending blocks sign-in and existing tokens for all its users.

PlatformRuleTemplatesPage — `/platform/rule-templates`

Purpose: edit the default rule set that every **new** company starts from. **Endpoints:** `GET/POST /platform/risk-rule-templates`, `PATCH /platform/risk-rule-templates/{rule_id}` (re-uses `AddRuleModal`). **Notes:** template edits never change an existing company's rules.

PlatformUsagePage — `/platform/usage`

Purpose: billing & usage, one row per company: cases created, documents uploaded, files stored, storage added in the period, total storage/files. **Endpoints:** `GET /platform/usage` (this month / last month / last 30 days / this year / all time / custom UTC range), `POST /platform/usage/reconcile` (*Reconcile now*).

PlatformQueuesPage — `/platform/queues`

Purpose: live processing-queue monitor (refreshes every 5 s): waiting / running per queue, oldest waiting task, average / p95 / max wait, run time, consumers online, rate-limiter counters (throttled calls, seconds waited, Azure 429s) and outstanding work per company (the fair-share counters). **Endpoints:** `GET /platform/queues`. See processing-queues.md.

Frontend components

Shared building blocks, grouped by folder. All are function components styled with Tailwind; props are listed as they exist in the code. The `token` prop appears on several components because they call the API themselves (through `@/api/*`) rather than receiving data from a page.

The PDF overlay engine

`PdfOverlayViewer` — `components/case/PdfOverlayViewer.tsx`

Renders **one page at a time** of a case document with bounding boxes drawn **live on top**. Nothing annotated is ever generated or stored; the boxes vanish the moment `overlays` is empty. Wrapped in an error boundary (react-pdf can throw synchronously on an expired or blocked URL, which would otherwise unmount the whole case page). Page width is measured from the card and capped at 560 px so it never spills into the adjacent column. Uses `react-pdf` (pdf.js) with a Vite-bundled worker.

Prop	Type	Meaning
`fileUrl`	`string`	A signed SAS URL for the PDF (from `CaseDocumentSummary.file_url`)
`overlays`	`OverlayBox[]`	Boxes to draw

```
interface OverlayBox { box: BoundingBox; color: OverlayColor; label: string }
type OverlayColor = "destructive" | "warning" | "ai" | "field"
// BoundingBox = { page (1-based), x, y, width, height }  — 0-1 page fractions, origin top-left
```

The four colours are **fixed per source**, so a reviewer learns the legend once:

Colour	Style	Source
`destructive`	solid red	ELA recompression region
`warning`	solid orange	Copy-move cluster
`ai`	**dashed** blue, "approximate"	Vision-model judgment (visual review, signature comparison) — styled to look less precise
`field`	solid purple	Rule-based field exception (failed field validation, cross-document mismatch)

Because boxes are page fractions, they are resolution-independent and land correctly regardless of the rendered size.
Used in: `CaseDetailPage` (middle column).

`getCheckOverlays(checks, crossDocumentFindings?, documentId?) → OverlayBox[]` — `DocumentChecksPanel.tsx`

The **single function that turns stored check rows into boxes**, shared by the viewer and the findings list so the highlighted regions and the listed findings can never disagree (both read the same `document_checks` rows). It reads ELA/copy-move findings' `bounding_box`, flagged field-validation sub-checks' `regions`, visual-review boxes, and cross-document regions (only those belonging to `documentId`, so each document shows its own side of a mismatch).

Case components — `components/case/`

Component	Props	Purpose / where used
`DocumentChecksPanel`	`checks: DocumentCheck[]`, `crossDocumentFindings: CrossDocumentFinding[]`, `hasEnoughDocumentsForCrossCheck: boolean`, `signatureMatches?: SignatureMatch[]`	One expandable card per check with its findings, a synthetic cross-document card, and the signature-comparison card (badge colours: consistent green, possibly-consistent/identical-reuse amber, inconsistent/different-signer red, cannot-determine grey). Handles the differing `details` shapes (list of findings vs. dict of sub-checks vs. flat issuer dict). Duplicate findings link to the matched case. *CaseDetailPage.*
`CaseDecisionPanel`	`caseDetail: CaseDetail`, `role: UserRole | undefined`, `token: string`	Approve / reject / escalate with confirmation dialogs. Approve is disabled until `pipeline.complete` and an assessment exists; above low risk it requires a ≥ 10-character justification. **These are conveniences — every rule is enforced by the API.** Invalidates the `case`, `caseAuditLog` and `cases` queries on success. Escalate requires a reason and moves the case to L2. Non-reviewers get a read-only "Review status" card with the decision history instead; a Reviewer L1 on an L2 case (`caseDetail.can_act` false) gets a read-only "Escalated · L2" card. The decision history shows each actor's role label (`actor_role`). *CaseDetailPage.*
`CaseReportExport`	`caseId: string`, `token: string`	"Export report" button (`POST /cases/{id}/reports`; opens the new PDF in a tab, with an "Open PDF" link as a popup-blocker fallback) and a collapsible history of previous reports. Reviewers only (hidden in a platform admin's support view). *CaseDetailPage.*
`ActivityTimeline`	`entries: AuditLogEntry[]`, `isLoading: boolean`	The case's audit trail, **newest first** (the API returns oldest first). Sentences come from `lib/audit.ts` `describeAuditEvent`. *CaseDetailPage.*
`CaseBadges`	exports `CaseFlagBadge({flag})`, `TierBadge({tier})`, `RiskBadge`, `riskStripeClass(tier)`	The coloured risk-tier pill and "Escalated · L2" marker. *Queue, Dashboard, MyCases, Detail.*

Upload components — `components/upload/`

Component	Props	Purpose
`FileDropzone`	`files: FileWithProgress[]`, `onFilesAdded(files: File[])`, `onFileRemoved(index: number)`, `disabled?`	Drag-and-drop / click-to-browse list with per-file progress and error. `FileWithProgress = { file: File; progress?: number (0-100); error?: string }`. The picker only offers PDFs (`accept` from `lib/uploadLimits.ts`) and the zone states the rules ("PDF only · up to 10.0 MB each · not password-protected"); the page refuses non-PDF, empty and over-limit files before upload, and shows the server's specific rejection message per file. *NewCasePage.*

Component	Props	Purpose
`SignatureReferenceCreator`	`caseId`, `documentId`, `documentFilename`, `token`, `onCreated(ref: SignatureReference)`, `onClose()`	Modal that fetches a signed URL, renders the PDF with `react-pdf`, lets the reviewer **drag a box over a signature** (pointer coordinates and the box are both relative to the rendered page wrapper, so they always agree), takes a required `person_name`, then `POST`s the reference. Error-boundary wrapped. Available to whoever uploaded (the case owner) and to the company's reviewers. *NewCasePage*.

Settings components — `components/settings/`

Component	Props	Purpose
`SettingsShell`	`active: SettingsTab` (`"issuers" | "risk-rules" | "users"`), `title?`, `description`, `actions?`, `children`; also exports `StatusLine({error, ok})`	Shared page frame and tab bar for the company settings screens (Issuer Registry, Risk Rules); shows the company picker to platform admins ("Changes apply to this company only and are recorded in its audit log.")
`Modal`	`title`, `description?`, `children`, `onClose`, `busy?`, `wide?`	Generic dialog used by the settings pages
`AddRuleModal`	`onClose()`, `onCreated(rule: RiskRule)` (token comes from `useAuth`)	Builds a risk rule from `GET /settings/risk-rule-options` (no hand-written JSON), then `POST /settings/risk-rules` — or, on the Rule templates page, `POST /platform/risk-rule-templates`
`AddUserModal`	`onClose()`, `onCreated(user: AdminUser)`	`POST /settings/users` with company + role (or a platform admin); can generate a random 14-character initial password

Platform and tenancy components

Component	Props	Purpose
`PlatformShell` (`components/platform/`)	`active` tab, `title`, `description`, `actions?`, `children`	Page frame and tab bar of the platform-admin area: Companies, Users, Rule templates, Billing & usage, Processing queues
`CompanyPicker` (`components/`)	`note?`	Company selector for platform admins on company-scoped pages, with an amber read-only/audited note. Default note: "Read-only support access. Every view is recorded in the platform audit log (the company does not see it)."
`useActingCompany()` (`hooks/useActingCompany.tsx`)	—	`{isPlatformAdmin, companyId, platformCompanyParam, …}`: which company a platform admin is acting on (remembered per browser); company users always get their own. Pages pass `platformCompanyParam` as `company_id` to the API

Design system — `design-system/`

Small presentational pieces reused across pages: `Nav` (top navigation; props `active: NavItemId`, `onNewUploadClick?`; items filtered by `NAV_VISIBLE` — rank via `hasRank` for company users, explicit `is_platform_admin` checks for the Platform area; shows the company name next to the user), `PageHeader`, `StatCard` (KPI tile), `InfoChip`, `SeverityFinding` (`title`, `description`, `pointDelta`, `tone` — one triggered risk reason with its

weight), `AuditRow`. The static HTML mockups these were built against are in `stitch_docauth_document_review_platform/`.

UI primitives — `components/ui/`

`badge`, `button`, `card`, `input`, `label`, `progress`, `select` — shadcn/ui-style wrappers over Radix + Tailwind (class-variance-authority, `cn()` from `lib/utils`).

Routing and auth

- `ProtectedRoute` — token + resolved user required, else redirect to `/login`.
- `RoleRoute({minRole, platformAdmin, label})` (file `AdminRoute.tsx`) — renders `<Outlet/>` when the user's company role ranks at least `minRole` (`hasRank`), or when the user is a platform admin and `platformAdmin` is `"allow"` / `"only"` (`"only"` = platform admins exclusively); else an "access required" panel.
- `types/auth.ts` — `UserRole` (`user`, `reviewer_l1`, `reviewer_l2`, `platform_admin`), the rank table mirrored from the backend, `hasRank(role, minimum)` (always false for `platform_admin`, which is not on the ladder), `ROLE_LABELS` ("Reviewer L1" … "Platform Admin") and `ROLE_OPTIONS` (role pickers). Gate UI on `hasRank` / `is_platform_admin`, never on lists of role names.
- `hooks/useAuth.tsx` — `AuthProvider` / `useAuth()` exposing `token`, `user`, `isLoadingUser`, `login`, `logout`; token in `localStorage`; a token the backend rejects is cleared.

Data layer

- `api/client.ts` — `apiFetch<T>(path, {token, …})` (JSON, Bearer header, `ApiError(status, message, code?)`), `parseErrorDetail(body)` (reads `detail` whether it is a string or a `{code, message}` object) and `API_BASE_URL` (`VITE_API_BASE_URL` or `/api`). Uploads use `XMLHttpRequest` instead, purely for progress events.
- `api/auth.ts`, `api/cases.ts`, `api/settings.ts`, `api/platform.ts` — one function per endpoint.
- `lib/uploadLimits.ts` — `MAX_UPLOAD_BYTES` (10 MB, keep in step with the backend's `MAX_UPLOAD_SIZE_BYTES`), `ACCEPTED_UPLOAD_TYPES` (PDF), `clientUploadProblem(file)`.
- Every page except Login is a lazy route chunk (`App.tsx`).
- `types/` — hand-written TypeScript mirrors of the backend schemas (`types/case.ts` includes `SignatureMatchResult`, `getDocumentRole` — **kept in sync with the backend by hand**, so a new backend enum value needs a matching edit here, as the signature verdicts did).
- Forms use React Hook Form + Zod; the Zod schemas are hand-mirrored from the Pydantic models.

Per-case forensic report (PDF)

A reviewer (L1 or L2) of the case's company can export a standalone, printable PDF describing everything the system found about one case — what was checked, what was flagged, why the score is what it is, who did what, and how far to trust it. It is designed to be read by someone who has **not** used the application.

Trigger: `POST /cases/{case_id}/reports` (button *Export report* on the case detail page, `CaseReportExport`). **History:** `GET /cases/{case_id}/reports`. Reviewers only; a platform admin can list a company's reports (read-only, audited) but cannot generate one. **Code:** `services/case_report_service.py` (orchestration + storage), `services/case_report_data.py` (gathering — `collect_report_data → ReportData`), `services/case_report_pdf.py` (rendering — `build_report_pdf(ReportData) → bytes`, pure, no DB or network).

How it is generated

Generation is **synchronous** (a few seconds for a typical case; there is no Celery job because the codebase has no async-export pattern; move `generate_case_report` into a task if reports get heavy).

1. `collect_report_data` **reads** what the pipeline already stored: `document_checks`, `case_risk_assessments`, `cross_document_findings`, `case_actions`, `signature_matches` and the append-only `audit_log`. It **never re-runs a check, re-scores a case, regenerates a reason (the reason text is the one frozen at scoring time) or writes its own audit trail.**
2. It downloads each **original** from Blob Storage purely to (a) **re-hash** it, proving the chain of custody still matches the recorded SHA-256, and (b) render the pages that carry findings. The original is only ever read.
3. `build_report_pdf` builds the PDF in memory with PyMuPDF (HTML "story" pages + annotated page renders).
4. The PDF is uploaded to `companies/{company_id}/cases/{case_id}/reports/{report_id}.pdf`, and a `case_reports` row is inserted with `report_sha256`, `page_count`, `file_size_bytes` and the `risk_assessment_id` it reflects; the company's `files_stored` / `storage_bytes` usage counters go up in the same transaction.
5. Audit event `case_report_generated {report_id, page_count, risk_assessment_id, report_sha256}`.

Every generation adds a **new** row; earlier reports stay downloadable. A report is a point-in-time record: it reflects the evidence, assessment and rule versions as they were then, so two reports for the same case can legitimately differ. Storage failure is a 502 and nothing is recorded.

Live overlays vs. the report's annotated pages

These are different mechanisms, on purpose. The live UI draws boxes **on top of** a rendered page in the browser and stores nothing. The report **burns** findings into *its own* rendered page images. That does not violate the immutable-original rule: the original blob is only read, and the annotated renders exist only inside the new report. Both use the same colour convention (red = ELA, orange = copy-move, dashed blue = model judgment, purple = rule-based field exception).

The nine sections

#	Section	What it contains	Data source (`ReportData` field → tables)
1	**Case Details**	Case number, type, status, submitter, submission time, document list and types, who generated the report and when (with their role label), the recorded decision / escalation, each "by Name (Reviewer L1/L2)" from `case_actions.actor_role`	`case_*` scalars, `documents`, `decision: DecisionInfo`, `escalation: EscalationInfo` ← `cases`, `case_actions`, `users`

#	Section	What it contains	Data source (`ReportData` field → tables)
2	**Executive Summary**	Short: tier and score, the decision, how many documents/exceptions, pointers to Sections 3, 4, 5 and 9. Notes when the case had not finished analysis (`pipeline_pending`)	`risk: RiskSummary` , `pipeline_pending` , `exceptions` ← `case_risk_assessments` , `pipeline_status`
3	**Explainable Findings**	Each triggered risk reason: severity, weight, check, document, page, the frozen reason text, and region IDs (R1, R2 …)	`findings: list[FindingRow]` ← `case_risk_assessments.triggered_reasons`
4	**All Checks**	The complete checklist: every check on every document with its result and a one-line summary, the cross-document table, and the signature comparison	`documents[].check_rows: CheckRow` , `cross_document: list[CrossRow]` (None if < 2 documents) ← `document_checks` , `cross_document_findings` , `signature_matches`
5	**Exceptions**	Everything that did **not** pass, written as concrete items ("Total amount 197,500 does not match subtotal + tax"), each linked to the highlighted region that shows it	`exceptions: list[ExceptionItem]` derived from Section 4's data (failed sub-checks, flagged findings, cross-document mismatches, missing required fields, signature results other than *consistent*)
6	**Limitations and Assumptions**	What the report is and is not (system-assisted analysis to support human review, not a certified forensic opinion); which checks are deterministic vs. model judgment; **which checks actually ran on this case (coverage)**; the rule versions and thresholds behind the score; assumptions specific to this case	`coverage: list[CoverageRow]` , `assumptions` , `risk.rule_versions` , `risk.thresholds`
7	**Audit Trail**	Chronological events for the case, **oldest first, capped at the first 150**; any later events are counted ("N later event(s) omitted"). Human actors are shown as "Name (Reviewer L1)" etc., using the role recorded in the event at the time (falling back to the user's current role for older events). Platform-admin access rows are never included (they are platform-only)	`audit: list[AuditRow]` , `audit_truncated` ← `audit_log`
8	**Appendix**	**8.1** extracted fields per document with confidence and whether the field was located; **8.2** SHA-256 of each original with a live **verification** status (`match` / `mismatch` / `unverified` — the original could not be fetched); **8.3** technical parameters of each check	`documents[].fields` , `documents[].hash: HashVerification` , `technical: list[TechnicalBlock]`
9	**PDF Highlighted Regions**	One consolidated page per flagged document page: the original page render with every highlight burned in, a legend, and the list of findings on it	`documents[].annotations: list[PageAnnotation]` + a re-render of the original page

Region IDs

Every highlight gets a report-wide ID **R1, R2, …** (assigned in document order, then page order by `assign_region_ids`). Sections 3 and 5 cite them, and each citation is an internal **PDF link** to the Section 9 page that shows it. Annotation kinds map to the colours above (`AnnotationKind`).

Wording rules

The report follows the same restraint as the rest of the product: model-judgment items are labelled "approximate", the AI-generation and signature results are described as advisory/experimental, and the limitations section states plainly that **the absence of a flagged finding does not guarantee a document is genuine**. It never uses "verified" for a signature.

Output record

`case_reports` row / `CaseReportResponse` (`schemas/case_report.py`):

```
{"id": "uuid", "case_id": "uuid", "generated_by_name": "Seed Admin",
 "generated_at": "2026-09-20T18:02:11Z", "file_size_bytes": 412331, "page_count": 14,
 "report_sha256": "…64 hex…", "risk_tier": "high", "risk_score": 75,
 "download_url": "https://…?sv=…&sig=…"}      // short-lived SAS URL, minted on each list/create call
```

Known limitations

- Only the **first 150 audit events** (oldest first) appear in Section 7; on a long-lived case the *later* events — often the reviewer's decision and the report itself — are omitted from the excerpt and only counted. The full history is in the audit log (`GET /cases/{id}/audit-log`).
- A report of a case whose original is missing or unreadable degrades gracefully: that document's Section 9 page shows a banner explaining the problem, and 8.2 reports `unverified` instead of failing the export.
- Annotated pages exist only for pages that carry findings and whose original could be rendered — PDFs only.
- Generation cost grows with the number of documents and flagged pages, since it is synchronous.
- The report reflects the **latest assessment at generation time**; a case that is re-scored afterwards (for example a signature comparison completing) is not reflected until a new report is generated.

Local setup

Locally FDDT runs as: **Postgres**, **Redis** and **PgBouncer** (in Docker), the **FastAPI** app, the **Celery workers** (one per queue, or one that serves all three), **Celery beat**, and the **Vite dev server**. Examples are Windows/PowerShell first (the project is developed on Windows); bash equivalents are in comments. For production, see production-deployment.md.

Prerequisites

Tool	Version	Notes
Docker Desktop	any recent	Postgres 16, Redis 7, PgBouncer 1.23
Python	3.11+ (developed and tested on **3.14**; the Docker image uses 3.12)	
Node.js	20+ (developed on **24**)	
Azure resources	—	Storage account, Document Intelligence, Azure OpenAI (vision-capable deployment). Needed for the *pipeline*, not to start the app — see environment-variables.md

1. Start Postgres, Redis and PgBouncer

From the repo root:

```
docker compose up -d        # postgres, redis, pgbouncer
docker compose ps           # postgres and redis "healthy", pgbouncer "running"
```

This creates `docauth-postgres` (port 5432), `docauth-redis` (6379) and `docauth-pgbouncer` (6432) with named volumes (`postgres_data` , `redis_data`). The owner user/password/database are all `docauth` . Data persists across restarts; `docker compose down -v` deletes it. PgBouncer is optional locally (see step 2).

2. Backend

```
cd backend
python -m venv .venv
.\.venv\Scripts\Activate.ps1          # bash: source .venv/bin/activate
pip install -r requirements.txt       # several minutes (OpenCV, PyMuPDF, pikepdf, Azure SDKs)
copy .env.example .env                # bash: cp .env.example .env
```

Edit `backend/.env` : set `JWT_SECRET_KEY` and the Azure values (storage, Document Intelligence, OpenAI, and the Azure OpenAI rate caps at ~80 % of your deployment's quota). The Postgres/Redis defaults already match `docker-compose.yml` . To route the app through PgBouncer as production does, also set `DATABASE_POOLER_HOST=localhost` (port 6432); leave it unset to connect directly. **Always run backend commands from `backend/`** — the `.env` path is relative to the working directory.

Create the schema, the database roles and the first account:

```
alembic upgrade head        # 14 migrations → 19 tables, RLS policies, the app roles fddt_app /
                            # fddt_platform (passwords from DATABASE_APP_PASSWORD /
DATABASE_PLATFORM_PASSWORD),
                            # the "Default Company" with 38 rules, thresholds 30/60 and a few FAKE test
                            # issuers, and the platform rule template
python seed.py              # creates the PLATFORM ADMIN admin@example.com / ChangeMe123! (idempotent;
                            # re-running resets that password — set SEED_ADMIN_EMAIL / SEED_ADMIN_PASSWORD)
```

The migrations run as the **owner** (`DATABASE_URL`), which needs CREATEROLE but not superuser. The app itself connects as `fddt_app` / `fddt_platform` , never as the owner.

The seeded account is a **platform admin**: it belongs to no company and cannot submit or review cases. After logging in, under **Platform**: create a company (or use "Default Company"), then under **Platform › Users** create that company's `user` , `reviewer_l1` and `reviewer_l2` accounts. Sign in as those to exercise the case workflow.

3. Run the API

```
uvicorn app.main:app --reload        # http://localhost:8000
```

Check: `http://localhost:8000/health` → `{"status":"ok","environment":"local"}` ; interactive docs at `http://localhost:8000/docs` . Smoke-test login:

```
curl.exe -X POST http://localhost:8000/auth/login -H "Content-Type: application/json" `
     -d '{\"email\":\"admin@example.com\",\"password\":\"ChangeMe123!\"}'
# (use curl.exe, not `curl`: in Windows PowerShell 5.1 `curl` is an alias for Invoke-WebRequest)
# → {"access_token":"eyJ…","token_type":"bearer"}
```

4. Run the Celery workers and beat

The pipeline uses three queues (processing-queues.md). Locally the simplest setup is **one worker that serves all three** (second terminal):

```
cd backend
.\.venv\Scripts\Activate.ps1
celery -A app.tasks.celery_app worker --loglevel=info --pool=threads --concurrency=8 `
       -Q extraction_queue,vision_queue,forensics_queue,celery
```

Windows cannot use the prefork pool — use `--pool=threads` (or `--pool=solo` , one task at a time). On Linux/macOS run one worker per queue exactly as production does:

```
sh start-worker.sh extraction     # threads, EXTRACTION_WORKER_CONCURRENCY (4)
sh start-worker.sh vision         # threads, VISION_WORKER_CONCURRENCY (4)
sh start-worker.sh forensics      # prefork, one process per CPU (FORENSICS_WORKER_CONCURRENCY=0)
```

or all of them in Linux containers: `docker compose --profile workers up -d` (add `--scale forensics-worker=N`). The worker lists its tasks (`process_document` , `run_document_checks` , `run_cross_document_checks` , `run_metadata_forensics` , `run_tampering_checks` , `run_duplicate_check` , `run_visual_inconsistency_review` , `run_signature_detection` , `run_signature_comparison` , `score_case` , `reconcile_usage_stats` , `log_queue_metrics`) and then `ready` . **Without a worker, uploads succeed but every document stays `pending` forever** — the API only enqueues.

Optionally run **beat** (third terminal) for the nightly usage reconciliation and the per-minute queue metrics line — run exactly one:

```
celery -A app.tasks.celery_app beat --loglevel=info      # bash: sh start-beat.sh
```

> **Restart the workers after every backend code change.** Unlike `uvicorn --reload`, Celery does not auto-reload; it keeps running the old code.

5. Run the frontend

```
cd frontend
npm install                # or `npm ci` for an exact lockfile install
npm run dev                # http://localhost:5173
```

No env file is needed: the dev server proxies `/api/*` → `http://127.0.0.1:8000/*` (prefix stripped), so no CORS setup is required. Sign in with `admin@example.com` / `ChangeMe123!` (platform admin) or a company account you created.

Other scripts: `npm run build` (type-check + production build into `dist/`, route-level code splitting), `npm run lint` (oxlint), `npm run preview`.

6. Try the pipeline

1. Sign in as a company `user` or reviewer → **New case** → pick a case type → drop PDFs from `sample-documents/` (e.g. the `Sample9`, `Sample11` Arabic pairs, or the PDFs inside `tampered_test_samples/tampered_test_samples.zip`; the `_Invoice` / `_Evidence` pairs share a case). Only PDFs up to 10 MB are accepted; images, password-protected and corrupted PDFs are refused with a specific message.
2. Open the case; the page polls every 5 s until *Analyzing* becomes a risk tier and *Approve* unlocks.
3. Optionally, on the upload screen, **Set reference signature** to trigger signature comparison.
4. As the platform admin, watch **Platform › Processing queues** and **Billing & usage**.

Re-uploading a sample you already uploaded *in the same company* **will** flag duplicates (10). The `stitch_docauth_document_review_platform/` folder holds static HTML design mockups, not part of the app.

Tests

```
cd backend
pytest                     # ~480 tests, ~4 minutes; external services are mocked (SQLite)
```

The database-level tests need a real, migrated Postgres and are skipped otherwise:

```
$env:FDDT_RLS_TEST_DATABASE_URL = "postgresql+psycopg2://docauth:docauth@localhost:5432/docauth"
pytest tests/test_rls_postgres.py       # Row-Level Security, role privileges, upserts as the app role
$env:FDDT_PGBOUNCER_HOST = "localhost"
pytest tests/test_pgbouncer_rls.py      # tenant context never leaks through transaction pooling
```

The rate-limiter tests need Redis on `localhost:6379`. `backend/loadtest/` holds the load-test harness (`docker-compose.loadtest.yml`).

Housekeeping and troubleshooting

Symptom	Cause / fix
Documents stuck at `pending`	No worker is consuming that queue, or Redis is unreachable. Check **Platform › Processing queues**
Upload rejected with a message	Working as designed: PDF only, ≤ 10 MB, not empty, not corrupted, not password-protected (01)
Upload returns 503	`AZURE_STORAGE_CONNECTION_STRING` missing/invalid
Document shows `failed` with an Azure error	Check the document's `processing_error`, and the Document Intelligence / OpenAI keys, endpoint and deployment name
Visual review / signature checks fail, extraction works	The OpenAI deployment is not vision-capable (must contain `gpt-4o`, `gpt-4.1`, `gpt-4-turbo` or `gpt-4-vision`)
Azure 429s / slow vision queue	Raise the deployment's TPM, or lower `AZURE_OPENAI_MAX_*_PER_MINUTE` to ~80 % of the real quota
Case stuck at *Analyzing*	A pipeline task has not finished; `GET /cases/{id}` → `pipeline.pending` lists what it is waiting for
"permission denied for table …" in a worker log	A write needs a privilege the app role lacks — every tenant table's grants are explicit (see schema)
Login works but a company user sees nothing	They belong to another company than the data, or the company is suspended
Port 5432/6379/6432 in use	Another Postgres/Redis/PgBouncer; stop it or set `POSTGRES_PORT` / `REDIS_PORT` / `PGBOUNCER_PORT` (and the URLs)
`alembic` cannot connect	Run it from `backend/`, with the containers healthy and `DATABASE_URL` matching
Backend code change has no effect on processing	Restart the Celery workers
Wrong/empty `.env` values	Confirm you started the process from `backend/`

Useful commands:

```
docker exec docauth-postgres psql -U docauth -d docauth -c "select name, is_active from companies"
python backfill_field_locations.py [CASE-XXXXXXXX]   # re-locate extracted fields (re-runs OCR only, no LLM)
python -m scripts.calibrate_llm_tokens               # re-measure token usage after a model/prompt change
cd ..\docs\tools; ..\..\backend\.venv\Scripts\python gen_db_docs.py   # regenerate the schema docs
```

Environment variables

The backend reads settings through `backend/app/core/config.py` (Pydantic `BaseSettings`). Values come from the process environment, then from `backend/.env`. The env file path is **relative to the working directory**, so start `uvicorn`, `celery`, `alembic` and `pytest` from `backend/`. Names are case-insensitive. Copy `backend/.env.example` to `backend/.env`; never commit `.env`.

Required below means "the pipeline does not work without it", not "the process refuses to start" — most services are built lazily, so a missing value surfaces when first used.

Backend — required for a working local pipeline

Variable	Default	Purpose	Where to get it	If missing
`DATABASE_URL`	`postgresql+psycopg2://docauth:docauth@localhost:5432/docauth`	The **owner** connection — used only by Alembic and `seed.py` (needs CREATEROLE, not superuser). Its host/port/database are also used for the two app roles below	Matches the Postgres in `docker-compose.yml` (default works as-is)	Migrations fail; API/worker can't build their URLs
`DATABASE_APP_USER` / `DATABASE_APP_PASSWORD`	`fddt_app` / `fddt_app_local`	The role every company request and pipeline task connects as (NOBYPASSRLS — Row-Level Security applies). The migration creates it with this password	Choose a long random password **before** running `alembic upgrade head`	Every request fails to connect
`DATABASE_PLATFORM_USER` / `DATABASE_PLATFORM_PASSWORD`	`fddt_platform` / `fddt_platform_local`	The role for sign-in, platform-admin screens and usage reconciliation (all rows through the `platform_all` policy; no BYPASSRLS)	Same	Login and platform screens fail
`CELERY_BROKER_URL`	`redis://localhost:6379/0`	Celery broker	The Redis in `docker-compose.yml`	Uploads succeed but nothing is processed; documents stay `pending`
`CELERY_RESULT_BACKEND`	`redis://localhost:6379/1`	Celery results	Same Redis, DB 1	Worker errors on result storage
`JWT_SECRET_KEY`	`changeme-in-.env`	Signs login tokens (HS256)	Generate a long random string, e.g. `python -c "import secrets; print(secrets.token_urlsafe(48))"`	**Works with an insecure default — always change it**; changing it logs everyone out
`AZURE_STORAGE_CONNECTION_STRING`	*(none)*	Blob Storage account for originals, signature crops and reports	Azure portal → Storage account → *Access keys* → connection string	Upload and file-url endpoints return **503** with an explanatory message

Variable	Default	Purpose	Where to get it	If missing
`AZURE_STORAGE_CONTAINER_NAME`	`documents`	Blob container (must be **private**)	Create it in the storage account	Uploads fail
`AZURE_DOCUMENT_INTELLIGENCE_ENDPOINT`	*(none)*	OCR/layout (`prebuilt-layout`)	Azure portal → Document Intelligence resource → *Keys and Endpoint*	Every document ends `failed` at extraction
`AZURE_DOCUMENT_INTELLIGENCE_KEY`	*(none)*	Same	Same	Same
`AZURE_OPENAI_KEY`	*(none)*	Classification, extraction, issuer semantic match, visual review, signature detect/compare	Azure portal (or AI Foundry) → your OpenAI resource → *Keys*	Extraction fails (document `failed`); visual review/signature checks fail
`AZURE_OPENAI_ENDPOINT`	*(none)*	Resource endpoint. Accepts either `https://<res>.openai.azure.com/` or an AI Foundry project URL `https://<res>.services.ai.azure.com/api/projects/<name>` (normalized to the origin)	Same	Same
`AZURE_OPENAI_DEPLOYMENT_NAME`	*(none)*	The model **deployment** to call. **Must be vision-capable**: its name must contain `gpt-4o` , `gpt-4.1` , `gpt-4-turbo` or `gpt-4-vision` , or the visual review and signature features raise `LLMConfigurationError`	Name you gave the deployment in Azure	Same
`AZURE_OPENAI_MAX_REQUESTS_PER_MINUTE`	`60`	Global cap on Azure OpenAI requests across every worker (Redis limiter). Set to **~80 %** of the deployment's RPM	Azure portal → deployment → *Rate limit*, or the `x-ratelimit-limit-requests` response header	Default 60 may under- or over-shoot your quota (429s or idle capacity)
`AZURE_OPENAI_MAX_TOKENS_PER_MINUTE`	`0` (off)	Global token cap (prompt + `max_tokens` reservation per call). Usually the binding limit. Set to **~80 %** of the deployment's TPM	`x-ratelimit-limit-tokens` header / portal	Without it, bursts exceed TPM → 429s

There is **no local/offline substitute** for storage, OCR or the LLM (no Azurite/mock backend is wired in); a real Azure account is needed to exercise the pipeline. The unit tests mock all three, so they do not call Azure.

Backend — optional / tuning

Variable	Default	Purpose
`ENVIRONMENT`	`local`	Label shown in `/health` and the OpenAPI description
`JWT_ALGORITHM`	`HS256`	JWT algorithm

Variable	Default	Purpose
`JWT_ACCESS_TOKEN_EXPIRE_MINUTES`	`480` (8 h)	Login token lifetime
`AZURE_OPENAI_API_VERSION`	`2024-10-21`	Azure OpenAI REST API version
`AZURE_OPENAI_REQUEST_TIMEOUT_SECONDS`	`60.0`	Per-call LLM timeout. Kept short on purpose: a hung call holds a worker thread (and its rate-limiter reservation)
`AZURE_DOCUMENT_INTELLIGENCE_MAX_CALLS_PER_SECOND`	`10.0`	Global cap on Document Intelligence calls (Azure's default S0 limit is 15 TPS)
`DATABASE_POOLER_HOST` / `DATABASE_POOLER_PORT`	*(unset)* / `6432`	When set, the two app roles connect through **PgBouncer** (transaction pooling) and each process keeps no pool of its own. Recommended in production (processing-queues.md)
`DB_POOL_SIZE` / `DB_MAX_OVERFLOW`	`5` / `10`	SQLAlchemy pool per role per process when connecting **directly** (no PgBouncer)
`MAX_UPLOAD_SIZE_BYTES`	`10485760` (10 MB)	Largest accepted PDF. The frontend enforces the same 10 MB in `frontend/src/lib/uploadLimits.ts` — keep them in step
`DEFAULT_COMPANY_NAME`	`Default Company`	Name of the company the multi-tenancy migration assigns pre-existing data to (used once, by the migration)
`PLATFORM_ACCESS_AUDIT_DEDUP_SECONDS`	`0`	0 = every platform-admin read of company data writes its own (platform-only) audit row; > 0 collapses identical reads within that window
`EXTRACTION_WORKER_CONCURRENCY` / `VISION_WORKER_CONCURRENCY`	`4` / `4`	Threads per extraction / vision worker (`start-worker.sh`). Size to just saturate the Azure caps
`FORENSICS_WORKER_CONCURRENCY`	`0`	Prefork processes per forensics worker; 0 = one per CPU of the container's limit
`FAIR_SHARE_BUCKET_SIZE`	`5`	A company's tasks drop one priority level per this many outstanding tasks on a queue
`QUEUE_ALERT_OLDEST_WAITING_SECONDS`	`300`	The per-minute `queue_metrics` job logs a WARNING `queue_alert {json}` line when a queue's oldest waiting task is older than this; 0 = off
`USAGE_RECONCILIATION_HOUR_UTC`	`2`	Hour (UTC) of the nightly usage-counter reconciliation (Celery beat)
`SEED_ADMIN_EMAIL` / `SEED_ADMIN_PASSWORD`	`admin@example.com` / `ChangeMe123!`	Read by `seed.py` (not by the app): the platform admin it creates — **and resets on every run** (`start-api.sh` runs it). Set both in production
`CELERY_POOL`	per queue	Read by `start-worker.sh` : overrides the pool type (`threads` for extraction/vision, `prefork` for forensics)
`CELERY_BEAT_SCHEDULE_FILE`	`/tmp/celerybeat-schedule`	Read by `start-beat.sh`

Variable	Default	Purpose
`ISSUER_FUZZY_MATCH_THRESHOLD`	`85.0`	rapidfuzz WRatio score (0–100) an extracted issuer must reach to match the registry (05)
`METADATA_FORENSICS_MOD_DATE_THRESHOLD_SECONDS`	`1.0`	Jitter tolerance before "modified after created" flags (06)
`METADATA_FORENSICS_EDITING_SOFTWARE_NAMES`	`photoshop,gimp,illustrator,affinity,coreldraw,paint.net,pixlr,canva,inkscape`	Comma-separated, case-insensitive substrings that mark a Producer/Creator as an editing tool
`DUPLICATE_HASH_HAMMING_THRESHOLD`	`5`	Max pHash Hamming distance (0–64) for a near-duplicate (10)

Risk **weights, severities and tier thresholds are not environment variables** — they live in the database, per company, and are edited in Settings › Risk Rules (and the platform template under Platform › Rule templates; see 11).

Backend — declared but UNUSED (do not need a value)

Variable	Would be for	Status
`AZURE_AI_VISION_ENDPOINT`, `AZURE_AI_VISION_KEY`	Signature/stamp detection, logo matching	Never read — detection uses the Azure OpenAI vision model
`AI_CONTENT_DETECTION_ENDPOINT`, `AI_CONTENT_DETECTION_KEY`	Content Safety / Hive	Never read — see 08
`EMAIL_SERVICE_CONNECTION_STRING`, `EMAIL_FROM_ADDRESS`	Transactional email	Never read — notifications are not built
`API_V1_PREFIX` (`/api/v1`)	URL prefix	Never read — routes have no prefix
`REDIS_URL`	Redis	Never read — Celery, the rate limiter, the fair-share counters and the queue monitor all use `CELERY_BROKER_URL`
`ANTHROPIC_API_KEY`, `ANTHROPIC_MODEL`	Claude API	Never read — the `LLMService` implementation in use is Azure OpenAI

docker-compose (optional overrides)

Read by `docker-compose.yml` from the shell or a root `.env`; defaults shown.

Variable	Default	Purpose
`POSTGRES_USER` / `POSTGRES_PASSWORD` / `POSTGRES_DB`	`docauth` / `docauth` / `docauth`	Postgres container credentials. If you change them, update `DATABASE_URL` to match
`POSTGRES_PORT`	`5432`	Host port
`REDIS_PORT`	`6379`	Host port
`PGBOUNCER_PORT`	`6432`	Host port of PgBouncer (`pgbouncer/pgbouncer.ini`, `pgbouncer/userlist.txt` — dev passwords only)
`FORENSICS_WORKER_CPUS`	`2`	CPUs per `forensics-worker` replica (`--profile workers`)

Frontend

Variable	Default	Purpose
`VITE_API_BASE_URL`	`/api`	Base URL the SPA calls (build time). The default relies on a proxy that maps `/api/*` → the API with the prefix stripped — the Vite dev server locally, the ingress in production (production-deployment.md). Setting it to a different origin instead requires CORS on the backend, which currently has **no** CORS middleware

No other frontend variables exist. Login state is a JWT in `localStorage`.

Secrets checklist

`AZURE_STORAGE_CONNECTION_STRING`, `AZURE_DOCUMENT_INTELLIGENCE_KEY`, `AZURE_OPENAI_KEY`, `JWT_SECRET_KEY`, the owner database password (in `DATABASE_URL`), `DATABASE_APP_PASSWORD`, `DATABASE_PLATFORM_PASSWORD` and `SEED_ADMIN_PASSWORD` are the secrets (plus PgBouncer's `userlist.txt`). Keep them out of the repo (`.env` is git-ignored) and, on Azure, in Key Vault referenced from the container apps (see production-deployment.md).

Azure migration notes

> **Status:** background and open items. The concrete production plan — architecture, service tiers, costs and the step-by-step runbook — is production-deployment.md, which chooses **Azure Container Apps**. The repo ships container images (`backend/Dockerfile` , `frontend/Dockerfile` + `nginx.conf.template`) but no infrastructure-as-code or CI/CD yet.

What is already Azure-shaped

Area	State
File storage	Already Azure Blob Storage (`AzureBlobStorageService`), private container, SAS URLs, write-once uploads. Behind a `StorageService` interface. Nothing to migrate
OCR	Already Azure Document Intelligence (`prebuilt-layout`)
LLM	Already Azure OpenAI (text + vision) behind `LLMService` ; accepts a classic resource endpoint or an AI Foundry project URL
Configuration	All secrets, endpoints and thresholds come from environment variables (`core/config.py`); no hardcoded local paths. The same image can run anywhere given the right env
Statelessness	The API and the workers hold no local state. Files go to Blob, everything else to Postgres/Redis. Rendering is done in memory, so no local scratch disk is required
Schema management	Alembic migrations apply cleanly to an empty database (verified), so a new environment is bootstrapped by `alembic upgrade head`
Managed PostgreSQL	No role needs SUPERUSER or BYPASSRLS; the migration owner needs only CREATEROLE — which the Flexible Server admin has. The whole chain was verified as a non-superuser owner
Connection pooling	Tenant context is transaction-local, so the built-in PgBouncer (transaction mode) of Flexible Server works as-is
Images	`backend/Dockerfile` (API, every worker, beat, migrations — the command differs) and `frontend/Dockerfile` (nginx serving the SPA and proxying `/api`)

What is still local-only (and what to do about it)

Component today	Local form	Azure target (proposed)	Notes
PostgreSQL	Docker `postgres:16-alpine` + PgBouncer container	**Azure Database for PostgreSQL – Flexible Server** (v16) with the **built-in PgBouncer** (General Purpose tier)	`DATABASE_URL` with `?sslmode=require` ; private access; PITR backups. Run migrations as a one-off job per release. Set `SEED_ADMIN_PASSWORD` (the seed resets it)
Redis (broker, fair-share counters, rate limiter)	Docker `redis:7-alpine` , no auth	**Azure Managed Redis**, *Non-Clustered* policy	Azure Cache for Redis no longer accepts new customers and retires 2028-09-30. Use `rediss://…:10000/0` for **both** Celery URLs (database 0 only)

Component today	Local form	Azure target (proposed)	Notes
FastAPI app	`uvicorn --reload` on the host	Container App with internal ingress (`backend/Dockerfile`)	`uvicorn … --workers 2`; not `start-api.sh` (it migrates and re-seeds on every start)
Celery workers	one local worker	Three Container Apps (`start-worker.sh extraction	vision
Frontend	Vite dev server + `/api` proxy	Container App `fddt-web` (`frontend/Dockerfile`: nginx serves the build and proxies `/api/*` to the API's internal FQDN)	One origin, so no CORS. nginx caps request bodies at 11 MB
Secrets	`backend/.env` file	**Key Vault** referenced by the Container Apps	`JWT_SECRET_KEY`, owner / `fddt_app` / `fddt_platform` passwords, storage connection string, Document Intelligence and OpenAI keys, Redis key, seed password
Azure credentials	Long-lived keys / connection string	**Managed identity** + RBAC (Storage Blob Data Contributor, Cognitive Services User)	The code authenticates with keys/connection strings only; switching to `DefaultAzureCredential` is a code change in `storage_service.py`, `ocr_service.py`, `llm_service.py` (each already sits behind an interface). Note the SAS generation currently needs an account key — with managed identity use *user delegation* SAS
Health/readiness	`GET /health` (does not touch dependencies)	Probe target	Consider a readiness probe that checks Postgres and Redis
Auth	Email/password JWT, in `localStorage`	Same for v1; **Entra ID SSO** is a possible later upgrade	Not started
Email/notifications	Not built	Azure Communication Services (planned)	`EMAIL_*` settings are declared but unused
CI/CD	None	GitHub Actions or Azure DevOps	Suggested stages: lint (`oxlint`), `tsc -b`, `pytest`, build images, migrate, deploy

Things that must change before any production exposure

These are pre-deployment blockers found while documenting, not deployment tasks per se:

1. ~~**Authorization gaps** on upload, file-url and the signature endpoints~~ — **fixed** (details). Still worth a security review of the remaining routes before exposure.
2. ~~**Remove** `/__debug_new_case`~~ — **done** (the public route outside `ProtectedRoute` no longer exists).
3. **TLS + single origin.** Terminate TLS at the Container Apps ingress (or Front Door) and keep the SPA and API on one origin (the nginx proxy) — the backend has no CORS middleware. Blob Storage needs a CORS rule for the app origin (the PDF viewer reads SAS URLs directly).
4. **JWT hardening.** Change `JWT_SECRET_KEY`; consider shorter lifetimes, refresh tokens, and server-side revocation (there is no logout or password reset endpoint). Tokens in `localStorage` are exposed to XSS.

5. **Login abuse.** No rate limiting or lockout on `POST /auth/login`; no password policy beyond "8+ characters" on admin-created accounts.
6. **Issuer registry.** Each company's registry starts empty (the migrated "Default Company" has fake test issuers); load the client's real data, or most documents will fail issuer verification.
7. **Upload limits.** PDF only (by header bytes), 10 MB, corruption and password-protection checks exist (`services/upload_validation.py`), and the nginx proxy refuses bodies over 11 MB. Still missing: a page-count limit, a malware scan (e.g. Defender for Storage malware scanning) and a Blob lifecycle/quarantine policy.
8. **Stuck-work recovery.** If Redis is down at upload time the document stays `pending` forever; there is no sweeper or retry. Add a periodic reconciliation task.
9. **Observability.** Python `logging`, the audit log, the queue monitor and the per-minute `queue_metrics` / `queue_alert` log lines exist. Ship container logs to Log Analytics and create the alerts in production-deployment.md; Application Insights/OpenTelemetry tracing is not wired in.

Scaling and cost considerations

- **Azure OpenAI is the throughput limit; Document Intelligence is the larger cost.** Per document: one extraction call, an optional issuer-match call, **two vision calls per PDF page** (visual review), plus signature detection and, on demand, comparison calls — about 1 cent of gpt-4.1-mini per 2-page document, against 2 cents of Document Intelligence. Full model: production-deployment.md › Costs. Set the TPM quota for the volume and the app's caps to 80 % of it.
- **Duplicate detection is a linear scan** of every stored page hash of the company per new page. Fine for thousands of pages; move to an index (BK-tree/LSH) or a pre-filter before millions.
- **Worker memory:** a multi-page PDF is rendered to full-resolution arrays (200 DPI); large documents benefit from a memory-limited, low-concurrency worker rather than many parallel tasks.
- **Report generation is synchronous** in the API request; if reports get heavy, move `generate_case_report` into a Celery task and poll (the function is already the unit to move).
- **Pipeline tasks are not retried automatically.** A transient Azure failure marks the document/check `failed`; there is no retry policy or dead-letter handling. Consider Celery `autoretry_for` with backoff on the LLM/OCR calls.
- **Data retention and privacy.** Documents and extracted text (including personal data) are stored indefinitely. Define retention, encryption-at-rest (Azure default), region/residency, and who may read the audit log. Azure OpenAI abuse-monitoring/data-handling terms should be reviewed for the client's data classification.

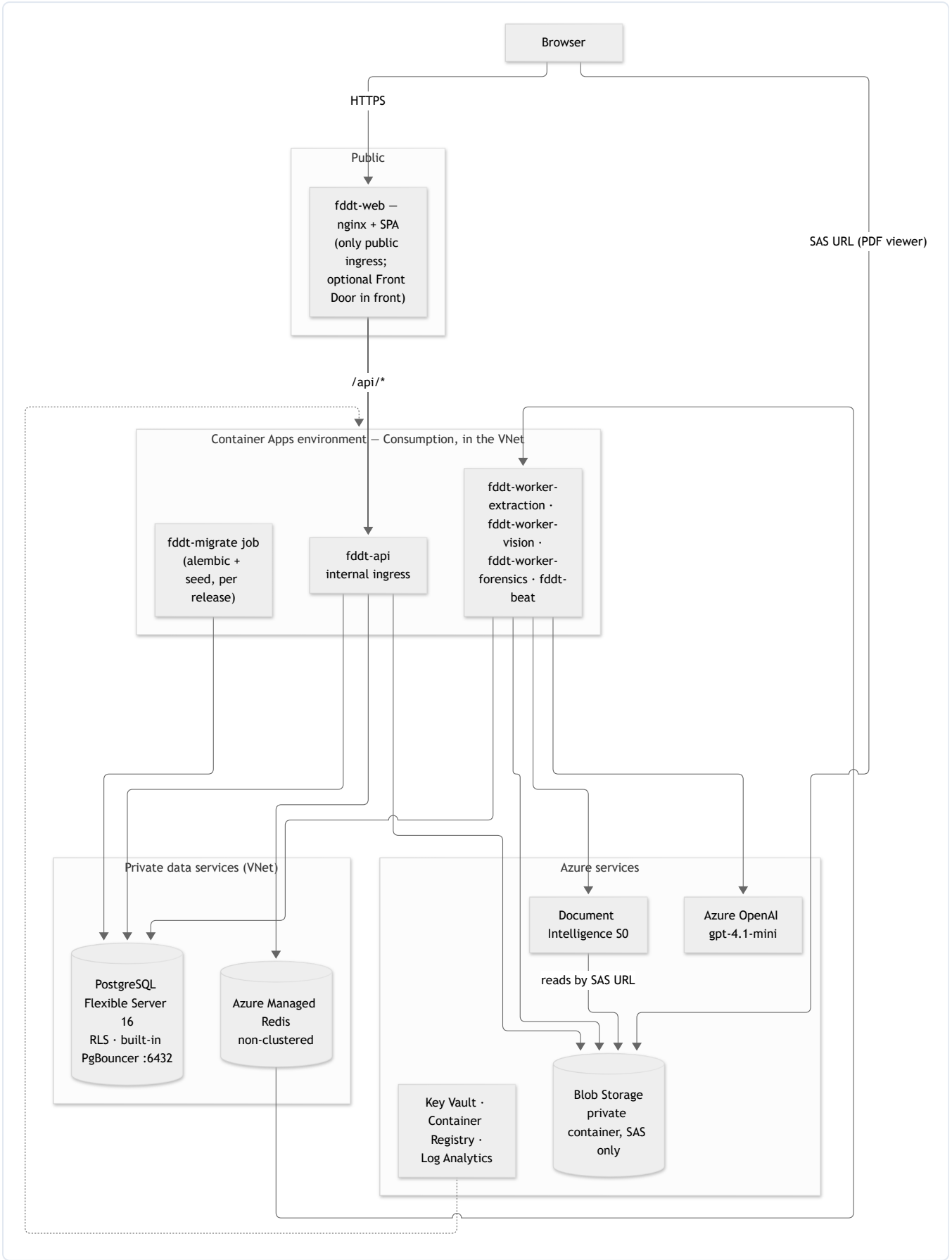
Suggested order

1. Follow production-deployment.md section 4 (network, registry, PostgreSQL, Redis, Storage, AI resources, Key Vault, environment, migration job, apps, domain).
2. Work through the blockers above that apply (login abuse, JWT hardening, stuck-work recovery).
3. Add CI/CD (suggested stages: `oxlint`, `tsc -b`, `pytest` incl. the Postgres RLS suite, build images, run the migration job, roll the apps) and infrastructure-as-code (Bicep/Terraform) for the resources.
4. Load real issuer data, tune risk weights on real cases, and validate the weak signals (ELA, signature and AI-generation checks) against known-genuine and known-forged documents before relying on them.

Production deployment (Azure)

> **Status:** this is the deployment plan and runbook. Per the project rules no infrastructure-as-code or CI/CD is committed yet — the repository ships the two container images (`backend/Dockerfile`, `frontend/Dockerfile` + `nginx.conf.template`) and everything below is done with the Azure CLI or portal. Prices are **list prices in USD for UAE North, retrieved 2026-10-03** from the Azure Retail Prices API; they exclude VAT and any EA/CSP discount. Re-check them in the pricing calculator before quoting.

1. Target architecture



Component	Azure service	Why
SPA + reverse proxy	Container App `fddt-web` (frontend image), **public** ingress	One origin for the SPA and `/api`, so no CORS is needed; nginx caps request bodies at 11 MB
API	Container App `fddt-api` (backend image), **internal** ingress only	Never exposed directly; reached only through `fddt-web`
Workers	Container Apps `fddt-worker-extraction`, `-vision`, `-forensics` (backend image), no ingress	One per queue so each scales on its own (processing-queues.md)
Scheduler	Container App `fddt-beat`, no ingress, **min = max = 1**	Nightly usage reconciliation, per-minute queue metrics. Two beats would run every job twice
Migrations	Container Apps **job** `fddt-migrate` (manual trigger)	Runs `alembic upgrade head` + `seed.py` once per release, as the database owner — not on every API start
Database	Azure Database for PostgreSQL **Flexible Server 16**, private access (VNet)	Row-Level Security, roles without superuser/BYPASSRLS (verified to work without superuser); built-in PgBouncer on General Purpose tiers
Broker / limiter / fair-share	**Azure Managed Redis**, *Non-Clustered* policy	Azure Cache for Redis no longer accepts new customers (since 2026-04-01) and retires 2028-09-30
Files	Storage account (Standard LRS/ZRS, hot), one **private** container	Originals (write-once), signature crops, report PDFs; accessed only through short-lived SAS URLs
OCR	Document Intelligence **S0**	`prebuilt-layout`
LLM	Azure OpenAI / Foundry, **gpt-4.1-mini** deployment	Classification, extraction, visual review, signatures, issuer match
Secrets	Key Vault, referenced as Container Apps secrets	No secret in images or the repo
Images	Container Registry (Basic)	Backend and frontend images
Observability	Log Analytics workspace (Container Apps logs) + Azure Monitor alerts	`queue_alert`, failed documents, database metrics
Edge (optional)	Front Door Standard	Global edge, caching of static assets, managed certificate. Container Apps ingress already provides TLS and a free managed certificate, so this is optional; a WAF needs **Front Door Premium** ($330/month base)

Region and data residency

- **UAE North** is used for everything that *stores* data (Storage, PostgreSQL, Redis, Document Intelligence processing). The documents are mostly UAE/Saudi.
- **Azure OpenAI gpt-4.1-mini is only offered as a *Global* deployment in UAE North** (no regional or Data Zone SKU there). With a Global deployment, prompts — the OCR text and page images — may be **processed in any Azure region**; data at rest stays in the resource's region. If the client's data classification forbids that, use a region/model combination that offers a regional deployment, or have the client accept Global processing in writing. (In West Europe, a Data Zone EU deployment costs 10 % more: $0.44 / $1.76 per 1M tokens.)
- Review the Azure OpenAI data-handling and abuse-monitoring terms against the client's policy; request modified abuse monitoring if required.

2. Service tiers

Three sizes. All three use the same images and the same configuration keys — only replica counts, CPU and SKUs differ, so moving between them is a scaling change, not a redeployment.

	S — Pilot	**M — Production**	**L — Scale**
Volume (2-page documents/month)	≈ 3,000	≈ 20,000	≈ 100,000
Companies	1–3	up to ~20	20+
`fddt-web`	1 × 0.25 vCPU / 0.5 GiB	2 × 0.25 / 0.5	2 × 0.25 / 0.5
`fddt-api`	1 × 0.5 / 1	2 × 1 / 2	3 × 1 / 2
`fddt-worker-extraction`	1 × 0.5 / 1	1 × 1 / 2	2 × 1 / 2
`fddt-worker-vision`	1 × 0.5 / 1	1 × 1 / 2	4 × 1 / 2
`fddt-worker-forensics`	1 × 1 / 2	2 × 2 / 4	4 × 2 / 4
`fddt-beat`	1 × 0.25 / 0.5	1 × 0.25 / 0.5	1 × 0.25 / 0.5
PostgreSQL	Burstable **B2s** (2 vCore, 4 GiB), 64 GB, no HA, direct connections	General Purpose **D2ds_v5** (2 vCore, 8 GiB), 128 GB, **zone-redundant HA**, built-in PgBouncer	General Purpose **D4ds_v5** (4 vCore, 16 GiB), 512 GB, zone-redundant HA, built-in PgBouncer
Redis	Managed Redis **B0**, no HA	Managed Redis **B0**, HA	Managed Redis **B3**, HA
Azure OpenAI quota to request (TPM)	100k	150k	500k
Availability	single zone, no failover	zone-redundant DB + Redis, 2+ replicas for web/API	same as M

Notes on sizing:

- **Forensics memory:** pages are rendered at 200 DPI in memory; give each forensics *process* ~2 GiB (`start-worker.sh` runs one prefork process per vCPU of the container's limit).
- **Vision/extraction** pools are I/O-bound threads; their real ceiling is the Azure OpenAI quota. Throughput ≈ (TPM cap) ÷ ~32k tokens *reserved* per 2-page document — e.g. 80k TPM ≈ 2.5 documents/min, 400k TPM ≈ 12/min. Quota itself costs nothing; request it before go-live (processing-queues.md).
- Workers keep `minReplicas ≥ 1` (they poll Redis). Autoscaling with KEDA on queue length is possible but must sum all ten priority lists per queue (`<queue>` , `<queue>\x06\x163` , …); fixed replica counts are the simpler, predictable choice.

3. Costs

Unit prices used (UAE North, USD, list)

Service	Meter	Price
Container Apps (Consumption)	vCPU active / idle	$0.000024 / $0.000003 per vCPU-second (≈ $63.07 / $7.88 per vCPU-month)
	Memory	$0.000003 per GiB-second (≈ $7.88 per GiB-month)
	Free grant per subscription per month	180,000 vCPU-s + 360,000 GiB-s + 2M requests (≈ $5.40)
	Environment management (only if environment private endpoint or planned maintenance is enabled)	$0.143/hour (≈ $104/month) — **not used** in this design
PostgreSQL Flexible Server	Burstable B2s	$0.079/hour
	General Purpose Ddsv5	$0.1085 per vCore-hour (D2ds_v5 $0.217, D4ds_v5 $0.434); 1-year reserved $571 per vCore-year (≈ 40 % less), 3-year $1,141 per vCore (≈ 60 % less)
	Storage / backup storage beyond the free 100 %	$0.138 / $0.105 per GB-month (HA doubles compute **and** storage)
Azure Managed Redis (Balanced)	B0 / B3 per instance	$0.0726 / $0.09295 per hour (HA = 2 instances). *The UAE North AMR meters look inconsistent (B1 is listed above B3; West Europe lists B0 at $0.019) — confirm in the calculator; the figures here are the higher, conservative ones*
Blob Storage	Hot LRS data / writes / reads	$0.0203 per GB-month / $0.06 per 10k / $0.0048 per 10k
Document Intelligence S0	Layout (pre-built) pages	**$10 per 1,000 pages**; commitment tiers from 20k pages for $190/month (overage $9.50/1k) and 100k pages for $900/month ($9/1k)
Azure OpenAI gpt-4.1-mini (Global)	Input / cached input / output	**$0.40 / $0.10 / $1.60 per 1M tokens**; Batch API half price (not usable for this interactive pipeline)
Container Registry	Basic	$0.1666/day (≈ $5/month)
Key Vault	Standard operations	$0.0396 per 10,000
Log Analytics	Analytics Logs ingestion (first 5 GB/month free)	$3.29/GB; Basic Logs $0.66/GB
Azure Monitor alerts	Log alert at 5-minute frequency	$1.50/month each
Front Door Standard (optional)	Base fee + requests/data	$35/month + usage

Service	Meter	Price
Bandwidth	Internet egress	First 100 GB/month free, then $0.075–0.12/GB

AI cost per document (measured)

Token use is taken from the app's own measurements of real calls (`fddt:llm_usage:*` , 12 English + Arabic sample documents):

Call	Per	Real input tokens	Real output tokens
Classification + extraction	document	2,105	889
Visual review (2 runs)	page	2 × 2,904	2 × 177
Signature/stamp detection	document (~3 calls)	3 × 1,418	3 × 84
Issuer semantic match	document, only if fuzzy match fails (assumed always)	400	126
Signature comparison	on demand only	1,055	47

Document size	Azure OpenAI	Document Intelligence	**AI cost per document**
1 page	$0.0076	$0.0100	**$0.018**
2 pages	$0.0105	$0.0200	**$0.031**
3 pages	$0.0134	$0.0300	**$0.043**
5 pages	$0.0192	$0.0500	**$0.069**

Document Intelligence (per page) is the larger part; Azure OpenAI is ~1 cent per document.

Monthly estimate per tier (2-page documents)

	S — Pilot	M — Production	L — Scale
Container Apps (all apps, always-active rate)	$231	$684	$1,394
PostgreSQL	$67	$352	$775
Managed Redis	$53	$106	$136
Blob Storage	$1	$4	$19
Container Registry, Key Vault	$6	$9	$21
Log Analytics + alerts	$12	$87	$252
Front Door Standard (optional)	—	$40	$40
Infrastructure	**≈ $369**	**≈ $1,283**	**≈ $2,636**
Document Intelligence	$60	$400	$2,000
Azure OpenAI	$32	$210	$1,051
AI (usage-based)	**≈ $92**	**≈ $610**	**≈ $3,051**
Total per month	**≈ $460**	**≈ $1,893**	**≈ $5,687**
Per document	≈ $0.15	≈ $0.095	≈ $0.057

How the figures were computed (`costs.py`-style model, 730 h/month): every container is billed at the **active** rate (an upper bound — a replica sitting at its minimum count with < 0.01 vCPU is billed the idle vCPU rate, which can cut a quiet pilot's Container Apps bill by half or more); Blob assumes ~0.5 MB per document retained plus ~4 writes and ~40 reads per document; log volume S 8 GB, M 30 GB, L 80 GB per month. Not included: VAT, support plans, Defender for Cloud, Front Door Premium/WAF, egress beyond 100 GB, signature comparisons, development/staging environments (a staging copy of tier S costs about the same as tier S).

Ways to reduce cost

- **PostgreSQL reserved capacity:** 1-year reservation ≈ 40 % off compute (M: −$127/month, L: −$254).
- **Document Intelligence commitment tier** once volume is steady: at 200k pages/month (tier L), the 100k-page commitment ($900) + 100k overage at $9/1k = $1,800 instead of $2,000.
- **Logs:** send container console logs to **Basic Logs** ($0.66/GB) and keep only the alerting tables (`queue_alert`, errors) in Analytics; set retention to 30 days.
- **Pilot:** scale the web/API/beat apps to their minimum, let workers idle at the idle rate, and use Managed Redis without HA.
- Keep pages per document honest: the cost scales with pages, and every page is OCR'd and reviewed twice.

4. Deployment steps

All commands are Azure CLI (bash). Replace `<...>` placeholders. Run them in this order the first time; later releases only repeat steps 10 and 12.

4.1 Prerequisites and naming

```
az login
az extension add --name containerapp --upgrade
az provider register --namespace Microsoft.App
az provider register --namespace Microsoft.OperationalInsights

RG=rg-fddt-prod
LOC=uaenorth
PREFIX=fddtprod                    # lowercase, globally unique where needed
az group create -n $RG -l $LOC
```

4.2 Logging

```
az monitor log-analytics workspace create -g $RG -n $PREFIX-logs -l $LOC --retention-time 30
LA_ID=$(az monitor log-analytics workspace show -g $RG -n $PREFIX-logs --query customerId -o tsv)
LA_KEY=$(az monitor log-analytics workspace get-shared-keys -g $RG -n $PREFIX-logs --query primarySharedKey
-o tsv)
```

4.3 Network

```
az network vnet create -g $RG -n $PREFIX-vnet -l $LOC --address-prefixes 10.20.0.0/16
# Container Apps (workload-profiles environment needs a /27 or larger, delegated)
az network vnet subnet create -g $RG --vnet-name $PREFIX-vnet -n aca --address-prefixes 10.20.0.0/23 \
  --delegations Microsoft.App/environments
# PostgreSQL private access (delegated)
az network vnet subnet create -g $RG --vnet-name $PREFIX-vnet -n pg --address-prefixes 10.20.2.0/24 \
  --delegations Microsoft.DBforPostgreSQL/flexibleServers
# Private endpoints (Redis)
az network vnet subnet create -g $RG --vnet-name $PREFIX-vnet -n pe --address-prefixes 10.20.3.0/24
```

4.4 Container registry and images

```
az acr create -g $RG -n ${PREFIX}acr --sku Basic
az acr build -r ${PREFIX}acr -t fddt-backend:<version> ./backend
az acr build -r ${PREFIX}acr -t fddt-web:<version> ./frontend
```

The frontend image is built with the default `VITE_API_BASE_URL=/api`; nginx proxies `/api/*` to the API.

4.5 PostgreSQL

```
az postgres flexible-server create -g $RG -n $PREFIX-pg -l $LOC --version 16 \
  --tier GeneralPurpose --sku-name Standard_D2ds_v5 --storage-size 128 \
  --high-availability ZoneRedundant \
  --vnet $PREFIX-vnet --subnet pg --private-dns-zone $PREFIX-pg.private.postgres.database.azure.com \
  --admin-user fddtowner --admin-password '<OWNER_PASSWORD>' --backup-retention 35
# Tier S instead: --tier Burstable --sku-name Standard_B2s --storage-size 64 (no --high-availability)

az postgres flexible-server db create -g $RG -s $PREFIX-pg -d fddt
# Built-in PgBouncer (General Purpose / Memory Optimized only — NOT available on Burstable)
az postgres flexible-server parameter set -g $RG -s $PREFIX-pg -n pgbouncer.enabled --value true
az postgres flexible-server parameter set -g $RG -s $PREFIX-pg -n metrics.pgbouncer_diagnostics --value on
```

- The admin login (`fddtowner`) is the **owner** role the migrations run as. It has CREATEROLE through `azure_pg_admin` but is not a superuser — exactly what the migrations need (they create `fddt_app` and `fddt_platform` themselves, without BYPASSRLS).
- TLS is enforced by default (`require_secure_transport = on`); always add `?sslmode=require` to `DATABASE_URL`. The app keeps that query string when it derives the two app-role URLs and the pooler URL from it.
- Built-in PgBouncer: port **6432**, transaction pooling (the default), `default_pool_size` 50 per user/database. The app sets the tenant only transaction-locally, so it is safe behind transaction pooling (`tests/test_pgbouncer_rls.py`). It authenticates against the database, so no `userlist.txt` is used in production.

4.6 Redis

```
az redisenterprise create -g $RG -n $PREFIX-redis -l $LOC --sku Balanced_B0 \
  --clustering-policy NoCluster --high-availability Enabled \
  --public-network-access Disabled --access-keys-authentication Enabled
# Tier S: --high-availability Disabled. Tier L: --sku Balanced_B3
az network private-endpoint create -g $RG -n $PREFIX-redis-pe --vnet-name $PREFIX-vnet --subnet pe \
  --private-connection-resource-id $(az redisenterprise show -g $RG -n $PREFIX-redis --query id -o tsv) \
  --group-id redisEnterprise --connection-name redis
REDIS_KEY=$(az redisenterprise database list-keys -g $RG --cluster-name $PREFIX-redis --query primaryKey -o
tsv)
```

(The `az redisenterprise` command group manages Azure Managed Redis; flags vary between CLI versions. `NoCluster` exists in the current API, but some CLI versions only accept `OSSCluster` / `EnterpriseCluster` — in that case create the cache in the **portal** and pick *Non-Clustered*. **The clustering policy cannot be changed after the database is created** — choose it correctly the first time.) Requirements for Celery:

- **Clustering policy *Non-Clustered***. Celery's Redis transport and the fair-share priority lists use multi-key commands across `<queue>`, `<queue>:1…9`; the default *OSS* policy needs a cluster-aware client, and *Enterprise* can reject cross-slot multi-key commands.
- Only database **0** exists — point **both** Celery URLs at `/0` (`rediss://:<KEY>@<name>.<region>.redis.azure.net:10000/0`; port 10000, TLS). Result keys are prefixed, so sharing the database is safe.
- Add a private DNS zone link for the private endpoint (`privatelink.redis.azure.net`) so the apps resolve it inside the VNet.

4.7 Storage

```
az storage account create -g $RG -n ${PREFIX}files -l $LOC --sku Standard_ZRS --kind StorageV2 \
  --min-tls-version TLS1_2 --allow-blob-public-access false --https-only true
az storage container create --account-name ${PREFIX}files -n documents --auth-mode login   # private
az storage cors add --account-name ${PREFIX}files --services b --methods GET HEAD OPTIONS \
  --origins https://<your-app-domain> --allowed-headers '*' --exposed-headers '*' --max-age 3600
```

- The account **must keep its public endpoint reachable**: the browser fetches each PDF directly by its short-lived SAS URL (the case viewer), and Document Intelligence fetches the file by SAS URL too. Access is protected by the SAS signature, and anonymous blob access is disabled. Locking the account to private endpoints would require code changes (stream files through the API; send bytes to Document Intelligence).
- The CORS rule lets the viewer read PDFs from the web app's origin; add the Front Door hostname too if used.
- Consider a lifecycle rule moving blobs older than e.g. 180 days to *Cool* (cheaper storage; the files are immutable).

4.8 Document Intelligence and Azure OpenAI

```
az cognitiveservices account create -g $RG -n $PREFIX-docintel -l $LOC --kind FormRecognizer --sku S0 \
  --custom-domain $PREFIX-docintel
az cognitiveservices account create -g $RG -n $PREFIX-openai -l $LOC --kind AIServices --sku S0 \
  --custom-domain $PREFIX-openai
az cognitiveservices account deployment create -g $RG -n $PREFIX-openai --deployment-name gpt-4.1-mini \
  --model-name gpt-4.1-mini --model-version 2025-04-14 --model-format OpenAI \
  --sku-name GlobalStandard --sku-capacity 150      # capacity in thousands of TPM (tier M)
```

- The deployment name must contain `gpt-4.1` (or `gpt-4o`, …) — the app checks it is vision-capable.
- Set the app's caps to **80 %** of the granted quota: `AZURE_OPENAI_MAX_TOKENS_PER_MINUTE` = 0.8 × TPM, `AZURE_OPENAI_MAX_REQUESTS_PER_MINUTE` = 0.8 × RPM (read both from the portal or the `x-ratelimit-limit-*`

response headers). Document Intelligence S0 allows 15 requests/s; keep `AZURE_DOCUMENT_INTELLIGENCE_MAX_CALLS_PER_SECOND=10`.

- After the first real traffic, run `python -m scripts.calibrate_llm_tokens` once against the production deployment (costs a few cents) to confirm the token reservations.

4.9 Key Vault and secrets

```
az keyvault create -g $RG -n $PREFIX-kv -l $LOC --enable-rbac-authorization true
for s in jwt-secret owner-db-url app-db-password platform-db-password redis-url storage-conn \
         docintel-key openai-key seed-admin-password; do
  az keyvault secret set --vault-name $PREFIX-kv -n $s --value '<value>'
done
```

Secret	Value
`jwt-secret`	`python -c "import secrets; print(secrets.token_urlsafe(48))"`
`owner-db-url`	`postgresql+psycopg2://fddtowner:<OWNER_PASSWORD>@<pg-host>:5432/fddt?sslmode=require`
`app-db-password` / `platform-db-password`	Two long random passwords (they become the passwords of `fddt_app` / `fddt_platform`)
`redis-url`	`rediss://:<REDIS_KEY>@<redis-host>:10000/0`
`storage-conn`	The storage account connection string (the code signs SAS URLs with the account key)
`docintel-key`, `openai-key`	Resource keys
`seed-admin-password`	The first platform admin's password

URL-encode special characters inside passwords that are embedded in URLs.

4.10 Container Apps environment, identity and shared configuration

```
az containerapp env create -g $RG -n $PREFIX-env -l $LOC \
  --infrastructure-subnet-resource-id $(az network vnet subnet show -g $RG --vnet-name $PREFIX-vnet -n aca
--query id -o tsv) \
  --logs-workspace-id $LA_ID --logs-workspace-key $LA_KEY --enable-workload-profiles
```

Every backend app uses one **user-assigned managed identity** with *AcrPull* on the registry and *Key Vault Secrets User* on the vault. Secrets are referenced as `--secrets jwt=keyvaultref:https://$PREFIX-kv.vault.azure.net/secrets/jwt-secret,identityref:<identity-id> …` and mapped into environment variables with `secretref:`.

Backend environment (identical for API, workers, beat and the migration job):

Variable	Value
`ENVIRONMENT`	`production`
`DATABASE_URL`	`secretref:owner-db-url`
`DATABASE_APP_PASSWORD` / `DATABASE_PLATFORM_PASSWORD`	`secretref:app-db-password` / `secretref:platform-db-password`

Variable	Value
`DATABASE_POOLER_HOST` / `DATABASE_POOLER_PORT`	`<pg-host>` / `6432` (tiers M/L). Tier S (Burstable): leave unset and set `DB_POOL_SIZE=3`, `DB_MAX_OVERFLOW=5`
`CELERY_BROKER_URL`, `CELERY_RESULT_BACKEND`	both `secretref:redis-url`
`JWT_SECRET_KEY`	`secretref:jwt-secret`
`AZURE_STORAGE_CONNECTION_STRING`, `AZURE_STORAGE_CONTAINER_NAME`	`secretref:storage-conn`, `documents`
`AZURE_DOCUMENT_INTELLIGENCE_ENDPOINT` / `_KEY`	`https://$PREFIX-docintel.cognitiveservices.azure.com/` / `secretref:docintel-key`
`AZURE_OPENAI_ENDPOINT` / `_KEY` / `_DEPLOYMENT_NAME`	`https://$PREFIX-openai.openai.azure.com/` / `secretref:openai-key` / `gpt-4.1-mini`
`AZURE_OPENAI_MAX_REQUESTS_PER_MINUTE` / `_MAX_TOKENS_PER_MINUTE`	80 % of the deployment's RPM / TPM
`AZURE_DOCUMENT_INTELLIGENCE_MAX_CALLS_PER_SECOND`	`10`
`MAX_UPLOAD_SIZE_BYTES`	`10485760`
`QUEUE_ALERT_OLDEST_WAITING_SECONDS`	`300`
`EXTRACTION_WORKER_CONCURRENCY` / `VISION_WORKER_CONCURRENCY`	`4` / `4` (per replica)
`FORENSICS_WORKER_CONCURRENCY`	`0` (one process per vCPU of the container)
`SEED_ADMIN_EMAIL` / `SEED_ADMIN_PASSWORD`	the first platform admin / `secretref:seed-admin-password` — **migration job only**

Full reference: environment-variables.md.

4.11 Database migration job (every release, before the apps)

```
az containerapp job create -g $RG -n $PREFIX-migrate --environment $PREFIX-env \
  --trigger-type Manual --replica-timeout 1800 --replica-retry-limit 0 \
  --image ${PREFIX}acr.azurecr.io/fddt-backend:<version> --cpu 0.5 --memory 1Gi \
  --command sh --args "-c" "alembic upgrade head && python seed.py" \
  --env-vars <the backend variables above> ...
az containerapp job start -g $RG -n $PREFIX-migrate
az containerapp job execution list -g $RG -n $PREFIX-migrate -o table     # wait for Succeeded
```

- **Take a database backup/point-in-time marker before migrating** (Flexible Server keeps automatic backups; note the time so a PITR restore target is known).
- `seed.py` creates the platform admin and **resets that account's password to `SEED_ADMIN_PASSWORD` on every run**. Remove `&& python seed.py` from the job after the first release (or keep the variable at the real password).
- Do **not** use `start-api.sh` as the API command in production: it migrates and re-seeds on every start, and several API replicas would race.

4.12 Applications

```
IMG=${PREFIX}acr.azurecr.io/fddt-backend:<version>
# API — internal ingress only
az containerapp create -g $RG -n fddt-api --environment $PREFIX-env --image $IMG \
  --ingress internal --target-port 8000 --cpu 1 --memory 2Gi --min-replicas 2 --max-replicas 4 \
  --command uvicorn --args app.main:app --host 0.0.0.0 --port 8000 --workers 2 --proxy-headers --forwarded-
allow-ips '*' \
  --env-vars <backend variables>
# Workers — no ingress
az containerapp create -g $RG -n fddt-worker-extraction --environment $PREFIX-env --image $IMG \
  --cpu 1 --memory 2Gi --min-replicas 1 --max-replicas 1 --command sh --args start-worker.sh extraction --
env-vars <…>
az containerapp create -g $RG -n fddt-worker-vision --environment $PREFIX-env --image $IMG \
  --cpu 1 --memory 2Gi --min-replicas 1 --max-replicas 1 --command sh --args start-worker.sh vision --env-
vars <…>
az containerapp create -g $RG -n fddt-worker-forensics --environment $PREFIX-env --image $IMG \
  --cpu 2 --memory 4Gi --min-replicas 2 --max-replicas 2 --command sh --args start-worker.sh forensics --
env-vars <…>
# Beat — exactly one
az containerapp create -g $RG -n fddt-beat --environment $PREFIX-env --image $IMG \
  --cpu 0.25 --memory 0.5Gi --min-replicas 1 --max-replicas 1 --command sh --args start-beat.sh --env-vars
<…>
# Web — the only public app
az containerapp create -g $RG -n fddt-web --environment $PREFIX-env \
  --image ${PREFIX}acr.azurecr.io/fddt-web:<version> --ingress external --target-port 80 \
  --cpu 0.25 --memory 0.5Gi --min-replicas 2 --max-replicas 3 \
  --env-vars API_UPSTREAM=https://fddt-api.internal.<environment-default-domain>
```

(Add `--registry-server`, `--registry-identity` and `--user-assigned` for the managed identity on each command; the replica counts above are tier M — take them from section 2.)

- `API_UPSTREAM` is the API's **internal FQDN** (`az containerapp show -n fddt-api --query properties.configuration.ingress.fqdn`). nginx strips `/api`, forwards to it, and refuses bodies over **11 MB** (`client_max_body_size`) before they reach Uvicorn.
- Workers need no probes beyond the default; give the API a liveness probe on `GET /health`.
- Rolling update order for a release: build images → run `fddt-migrate` → update `fddt-api` → update the workers and beat → update `fddt-web`. Migrations are additive within a release, so old workers keep working during the roll.
- **Restart the workers whenever their code changes** (a new revision does this).

4.13 Domain and TLS

```
az containerapp hostname add -g $RG -n fddt-web --hostname app.<client-domain>
az containerapp hostname bind -g $RG -n fddt-web --hostname app.<client-domain> \
  --environment $PREFIX-env --validation-method CNAME     # free managed certificate
```

Or put Front Door Standard in front (managed certificate on Front Door, origin = `fddt-web`) and restrict `fddt-web` ingress to the Front Door service tag. Add the final origin to the Blob CORS rule (4.7).

4.14 First login and smoke test

1. Sign in as the seeded platform admin; change nothing else until the checks below pass.
2. **Platform › Companies** → create the client company (it receives the 38 template rules and 30/60 thresholds; its issuer registry is empty). **Platform › Users** → create its reviewers and users.

3. As a company user, upload one sample PDF; confirm it reaches a risk tier, the PDF renders in the case viewer (Blob CORS works) and **Platform › Processing queues** shows all three queues with consumers.
4. Upload an image and a password-protected PDF — both must be refused with their specific messages.
5. Generate a report as a reviewer and open it.
6. As the company's Reviewer L2, load the company's real **issuer registry** (until then almost every document flags `issuer.not_in_registry`).

5. Operations

Monitoring and alerts

Alert	Query / metric	Suggested threshold
Queue backlog	`ContainerAppConsoleLogs_CL | where Log_s has "queue_alert"`	any row in 5 min (the app logs it when the oldest waiting task > 300 s)
Worker errors	`ContainerAppConsoleLogs_CL | where ContainerAppName_s startswith "fddt-worker" and Log_s has_any ("ERROR", "Traceback")`	> 5 in 15 min
Database permission errors	`… Log_s has "permission denied"`	any (a write needs a grant the app role lacks)
Azure throttling	`… Log_s has "429"`	> 10 in 15 min — raise TPM or lower the caps
Failed documents	`audit_log` events `document_processing_failed` / `*_failed` (query through the platform audit log)	review daily
PostgreSQL	CPU > 80 %, storage > 80 %, `client_connections_waiting` (PgBouncer) > 0 for 5 min	
Redis	Used memory > 70 %, server load > 80 %	
API	`fddt-api` 5xx rate, replica restarts	

`queue_metrics {json}` lines (one per queue per minute) can be charted directly in Log Analytics (`parse_json(extract("queue_metrics (.*)", 1, Log_s))`).

Backups and disaster recovery

Data	Protection	Restore
PostgreSQL	Automatic backups, 35-day point-in-time restore; zone-redundant HA (M/L) fails over automatically	PITR to a new server, repoint `DATABASE_URL` / pooler host
Blob Storage	ZRS; enable **soft delete** (14 days) and **versioning** for defence in depth (the app never deletes or overwrites)	Undelete / previous version
Redis	Holds only queued tasks, fair-share counters and rate-limiter windows	A lost broker loses queued tasks: affected documents stay `pending` — re-upload them (there is no sweeper yet)
Configuration	Key Vault (soft delete + purge protection on) and the image tags	Redeploy

Recovery objectives with tier M as designed: RPO ≈ 5 minutes (PostgreSQL log backups), RTO ≈ 1 hour (restore + repoint), excluding in-flight pipeline tasks.

Routine tasks

- **Secret rotation:** rotate storage/Document Intelligence/OpenAI keys with the two-key pattern; rotating `JWT_SECRET_KEY` signs everyone out. To rotate `fddt_app` / `fddt_platform`, `ALTER ROLE … PASSWORD` as the owner, then update the Key Vault secrets and restart the apps.
- **Usage reconciliation** runs nightly (02:00 UTC); **Billing & usage** feeds client invoicing.
- **Re-calibrate token reservations** (`scripts.calibrate_llm_tokens`) after any model, prompt or render change; re-check the 80 % caps after any quota change.
- **Schema docs:** regenerate `docs/database/*.md` with `docs/tools/gen_db_docs.py` after each migration.

6. Go-live checklist

Must do

- ☐ `JWT_SECRET_KEY`, both app-role passwords, the owner password and `SEED_ADMIN_PASSWORD` are long random values in Key Vault; nothing secret is in an image or the repo.
- ☐ The migration job ran (`alembic upgrade head` → head `c9e3a1b5d7f4` or later) and the seed step was removed or pinned to the real password.
- ☐ `DATABASE_URL` ends in `?sslmode=require`; tiers M/L connect through the built-in PgBouncer (6432).
- ☐ Redis is Non-Clustered and both Celery URLs use `/0` over `rediss://`.
- ☐ Azure OpenAI quota granted for the tier; the app caps at 80 % of it.
- ☐ Exactly one `fddt-beat` replica.
- ☐ Blob container private, anonymous access off, CORS limited to the app origin, soft delete on.
- ☐ Only `fddt-web` has public ingress.
- ☐ Alerts from section 5 created and routed to an on-call address.
- ☐ Data-residency decision for the Global Azure OpenAI deployment recorded with the client.
- ☐ The client's issuer registry loaded; risk weights reviewed with the client.
- ☐ A full browser walkthrough of every role (submitter, Reviewer L1, Reviewer L2, platform admin).

Known gaps to accept or schedule (from azure-migration-notes.md)

- No login rate limiting or lockout, no password reset, tokens in `localStorage`.
- No malware scan and no page-count limit on uploads (type, size, corruption and password are checked).
- No automatic retry of a failed Azure call and no sweeper for documents left `pending`.
- Keys and connection strings, not managed identity, for Storage / Document Intelligence / OpenAI.

Sources

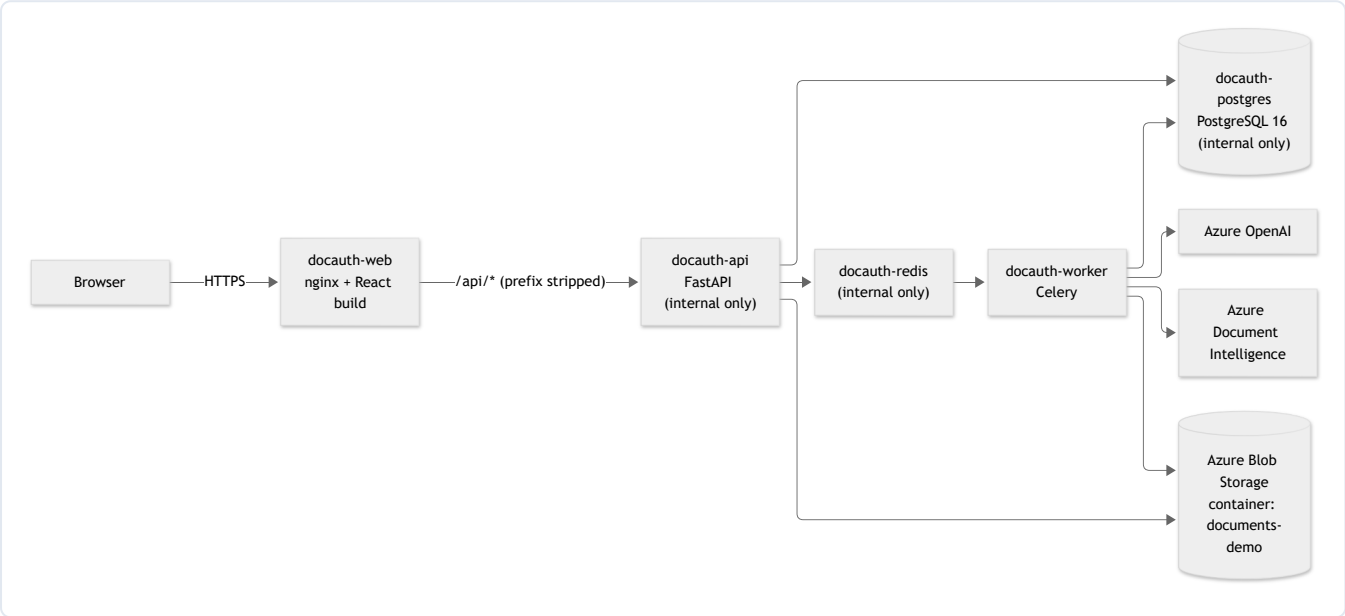
- Azure Retail Prices API — `https://prices.azure.com/api/retail/prices` (UAE North and West Europe, retrieved 2026-10-03)
- Billing in Azure Container Apps
- PgBouncer in Azure Database for PostgreSQL Flexible Server
- Azure Cache for Redis retirement FAQ
- Azure Managed Redis architecture (clustering policies)
- Azure Managed Redis pricing

- Azure Front Door pricing

Azure demo deployment

A small, low-cost Azure deployment used to demonstrate the platform. It is **not** the production design (see production-deployment.md for that). It predates multi-tenancy and the three processing queues — redeploying it with the current images needs the changes listed under *Updating the demo to the current build* below.

Topology



Only `docauth-web` has a public address. The API, Redis and PostgreSQL are reachable only from inside the Container Apps environment, so the database is never exposed to the internet. The browser talks to a single origin, so no CORS configuration is needed.

Resources

Resource	Purpose
Azure Container Apps: `docauth-web`, `docauth-api`, `docauth-worker`, `docauth-redis`, `docauth-postgres`	The five runtime components (1 replica each)
Azure Container Registry	Holds the backend and frontend images
Azure Blob Storage container `documents-demo`	Original documents, signature crops and report PDFs (kept separate from development data)
Azure Document Intelligence, Azure OpenAI	Existing resources, used as in development

Images

Both images are built from the repository:

Image	Source	Notes
Backend	`backend/Dockerfile`	One image for both the API (`start-api.sh`) and the worker (`start-worker.sh`)

Image	Source	Notes
Frontend	`frontend/Dockerfile`, `frontend/nginx.conf.template`	Static build served by nginx; `API_UPSTREAM` selects the backend address

`start-api.sh` applies the Alembic migrations, seeds the administrator account, then starts the API, so a new environment bootstraps itself.

Configuration

All settings are environment variables (see environment-variables.md). Secrets (database URL, JWT secret, storage connection string, Azure keys, seeded administrator password) are stored as Container Apps secrets and referenced by the apps; none are in the images or the repository. The seeded administrator email and password come from `SEED_ADMIN_EMAIL` / `SEED_ADMIN_PASSWORD`.

Operating notes

- **PDF viewer requirements.** The browser loads the pdf.js worker from the web app and fetches each PDF directly from Blob Storage. This needs (1) the web server to serve `.mjs` files as `application/javascript` (configured in `frontend/nginx.conf.template`) and (2) a Blob Storage CORS rule allowing `GET`, `HEAD` and `OPTIONS` from the web app's address. Add the rule again if the web app's address changes.
- The Celery worker ran the default prefork pool with two processes (Linux) when the demo was set up, before the queue split (see *Updating the demo to the current build*).
- Redis and PostgreSQL run as containers on the container's temporary disk (no persistent volume). If either restarts, in-flight jobs (Redis) or all case data (PostgreSQL) are lost. Restart the API app to re-apply the migrations and re-seed the administrator, then re-upload the sample documents.
- This deployment is intended for demonstrations. For production use production-deployment.md (managed PostgreSQL and Redis with private networking, persistent storage, Key Vault, monitoring).

Updating the demo to the current build

- **Workers:** the pipeline now uses three queues. A single `docauth-worker` must consume all of them: `celery -A app.tasks.celery_app worker -Q extraction_queue,vision_queue,forensics_queue,celery --pool=threads --concurrency=8` (or run `start-worker.sh extraction|vision|forensics` as three apps). `start-worker.sh` with no argument serves **only** `forensics_queue`, so documents would never be OCR'd.
- **Beat:** add one `start-beat.sh` app for the nightly usage reconciliation and queue metrics (optional for a demo).
- **Database roles:** the migrations now create `fddt_app` / `fddt_platform`; set `DATABASE_APP_PASSWORD` / `DATABASE_PLATFORM_PASSWORD` as secrets on every backend app. The Postgres container's `docauth` user is a superuser, which is fine for a demo.
- **Seeded account:** `admin@example.com` is now a **platform admin** — create a company and its users under *Platform* before submitting cases.
- **Uploads:** PDF only, 10 MB; the new nginx template caps request bodies at 11 MB.

Tear-down

Delete the five `docauth-*` container apps, the container registry and the `documents-demo` container. The demo creates no other resources.

Multi-Tenancy & Processing Queues — Full Change Report

Date: 2026-10-02 / 03 · **Scope:** convert FDDT from single-tenant to multi-tenant, and split the document pipeline into three fair, rate-limited queues. **Round 2 (2026-10-03, production hardening)** is at the top; the original round follows from §0.

This file records **everything that was done, every decision taken (and why), every bottleneck or problem found, and what to change before production**. Items marked

> ⚠️ **PRODUCTION ACTION** — must be reviewed/changed before going live 🔶 **DECISION FOR YOU** — I chose a default; you may want it different 🐞 **PROBLEM FOUND** — a bug or limit hit during the work, and how it was handled

Detailed design docs: `multi-tenancy.md`, `processing-queues.md`.

R6. Browser test of the reviewer workflow (2026-10-03)

Chrome, company *QA Alpha Trading*, three cases processed end-to-end on real Azure (one English invoice, one school invoice, one Arabic hotel invoice). GIF of the escalate → L2 approve flow: `reviewer_escalate_and_l2_approve.gif` (browser downloads).

Step	Result
Submitter tries approve / reject / escalate (API)	✅ 403 each; risk tier hidden (`null`)
L1 approves a **high**-risk case	✅ dialog lists score + every triggered reason; button disabled until ≥ 10 characters (2/10 → disabled); approved; history "Approved · by Alpha Reviewer L1 · Reviewer L1" with the justification
L1 rejects	✅ "Reject case" disabled without a reason; API with empty reason → 422 "A reason is required..."; rejected with reason
L1 escalates	✅ dialog explains the one-way hand-off; status stays `pending_manual_review`, tier `l1 → l2`; "Escalated · L2" badge; L1 queue moves it to "Escalated to L2 · view only"
L1 on the escalated case (API)	✅ approve / reject / escalate → 403
Acting on decided cases	✅ 409 ("Cannot reject a case that is 'approved'", "This case is already rejected")
L2	✅ "Escalated queue (L2)" tab, escalated case sorted first; Escalate disabled; medium risk also needs a justification; approved; history shows both the L1 escalation and the L2 approval with roles
Audit log	✅ `case_approved` / `case_rejected` / `case_escalated` with `actor_role`, escalation `l1 → l2`
Submitter afterwards	✅ My cases: 2 Cleared, 1 Action Required; rejection reason visible
🐞 **Submitter's case timeline leaked the risk score**	✅ Fixed — `GET /cases/{id}/audit-log` returned `risk_assessment_completed` (tier, score, rules fired) and check results to the submitter. Now omitted/stripped for non-reviewers; test `test_submitter_timeline_hides_risk_score_and_check_results`

Step	Result
🐞 Escalated panel said "only an L2 reviewer **or admin**"	✅ Fixed — "only a Reviewer L2" (platform admins can't act)
🔶 Submitter still sees per-document check cards ("Issuer verification: Flagged" with match details)	**Open decision** — the case-detail API returns each document's checks to the submitter. Hiding them would fully blind submitters to which signals fired, but also hides useful feedback (e.g. a missing signature). Not changed
⚠️ Dev server only: case page flickers "Loading…" and remounts every ~1 s for a while after load	Not in the production build (verified with `vite preview`: stable, one fetch). Tied to Vite's dev server + the pdf.js module worker (respawned each cycle); my code-splitting change is **not** the cause (reproduced with eager imports). Use the production build when demoing
Minor	"My cases" shows a made-up subtitle `CASE-XXXX.pdf` under each case id instead of a real filename

R5. Browser test of PDF uploads and multi-tenancy (2026-10-03)

Run in Chrome against the local stack (real Azure for OCR/LLM/Blob), with two fresh companies (*QA Alpha Trading*, *QA Beta Schools*) and four users — accounts in `sample-documents/upload-tests/TEST_ACCOUNTS.md`, test files from `sample-documents/upload-tests/make_files.py`.

Area	Result
Upload — refused in the browser before sending	✅ `photo.jpg` ("only PDF files are accepted"), `empty.pdf` ("the file is empty")
Upload — refused by the server, specific message per file	✅ JPEG renamed `.pdf` ("This file is a JPEG image…"), text renamed `.pdf`, password-protected, truncated; 12 MB → 413 with both sizes; empty → 400
Upload — valid PDFs	✅ stored and processed end-to-end (score 87/high); the case holds exactly 2 documents, billing counts exactly their bytes
Isolation	✅ Beta L2 → Alpha case/timeline/file URL/signatures/reports/approve/reject/escalate/upload: all **404**; `company_id=<Alpha>` → 404; Beta sees 0 cases and no Alpha audit rows; UI shows "Case not found"
New company defaults	✅ empty issuer registry, 38 template rules
Roles	✅ L1: Settings hidden, page shows "Reviewer L2 access required", API 403; L2: own settings, user management 403; submitter: no Audit/Settings/Platform
Platform admin	✅ amber read-only Support view, no actions; every view logged (12 rows for one case visit); company users see none of them (UI and API)
Duplicate detection fix (R4)	✅ "Duplicate check complete" for both documents
🐞 Validation errors showed "Unprocessable Content"	✅ Fixed — `parseErrorDetail` now renders FastAPI 422 lists (e.g. "email: value is not a valid email address…")
🐞 "user created" rows in a company's log showed actor "System pipeline"	✅ Fixed — an actor the company session can't see (a platform admin) is now "Platform team"; test `test_company_log_names_platform_actions_platform_team`. Docs corrected: user_created/updated live in the user's company log
PDF viewer from `http://127.0.0.1:5173`	⚠️ "Couldn't load this file" — environment, not code: the storage account's CORS rule allows `http://localhost:5173` (works there). Add `127.0.0.1` to the rule if you use that origin

R4. Uploads, a permission bug, documentation and the production plan (2026-10-03)

Item	Status	What changed	Proof
Upload validation	✅ Done	Synchronous checks in the upload endpoint before anything is stored or queued: empty, > 10 MB, not a PDF by header bytes, extension mismatch, password-protected, corrupted (PyMuPDF; a repaired file without `%%EOF` counts as truncated). Distinct `{code, message}` per reason; frontend shows the message and refuses non-PDF / empty / > 10 MB before sending	`tests/test_upload_validation.py` ; all 38 sample PDFs (incl. 12 tampered) still pass
PDF only	✅ Done	Images (JPEG/PNG/TIFF) are no longer accepted — every forensic check is PDF-based. The message names the detected type ("This file is a JPEG image…")	same tests
🔶 Password policy	Decision	**Reject and ask** the uploader to remove the password; the server never handles document passwords. Owner-password-only PDFs (open without a password) are accepted	`test_password_protected_pdf_…` , `test_owner_password_only_pdf_is_accepted`
nginx body cap	✅ Done	`frontend/nginx.conf.template` `client_max_body_size` 50m → **11m**, so an oversized upload is refused at the proxy	—
🐞 **Duplicate detection failed in Postgres since round 2**	✅ Fixed	`store_page_hashes` used `ON CONFLICT DO UPDATE` , which needs UPDATE privilege; `fddt_app` has only SELECT + INSERT on `document_page_hashes` , so **every** duplicate check failed with "permission denied" (dev DB: 299 failed, last success 2026-10-02 18:35). The SQLite unit tests can't see grants. Now `ON CONFLICT DO NOTHING` (a page hash is a pure function of the immutable file) — no grant change, table stays insert-only	New `test_rls_postgres.py::test_every_upsert_the_pipeline_uses_works_under_the_app_role` runs every upsert twice as `fddt_app` ; it fails with the production error against the old code (negative control)
🐞 UI said platform access is "recorded in that company's audit log"	✅ Fixed	Three strings (company picker, Audit History, case Support view) now say the platform audit log, which the company does not see	build passes
Documentation	✅ Done	Every doc re-checked against the code: architecture, data flow, pipeline 01–11, API (33 paths / 41 operations), frontend, report, local setup, environment variables, Azure notes, user manual, READMEs. Database docs regenerated from the live DB by the new `docs/tools/gen_db_docs.py` (19 tables, RLS, grants). OpenAPI snapshot regenerated. PDFs rebuilt, including the multi-tenancy, queues, production-deployment and this report	—
Production plan	✅ Done	`deployment/production-deployment.md` : Container Apps architecture, S/M/L tiers, measured per-document AI cost, monthly costs from Azure list prices (UAE North), CLI runbook, monitoring, DR, go-live checklist	—

⚠️ **Re-run the failed duplicate checks** on the dev database if those cases matter (re-upload, or re-queue `run_duplicate_check` for the affected documents) — they will not re-run on their own.

R3. Follow-up fixes (2026-10-03)

Three of the four "should fix soon" items from R2.3 are done. The fourth (splitting classification out of `process_document`) is **deliberately not done** in this pass and is still open (R2.3).

Item	Status	What changed	Proof
Rate-limiter caps 85 % → 80 %	✅ Done	`backend/.env`: `AZURE_OPENAI_MAX_REQUESTS_PER_MINUTE=80`, `AZURE_OPENAI_MAX_TOKENS_PER_MINUTE=80000` (quota 100 / 100,000). `.env.example` and `config.py` comments now say ~80 %. Code defaults are unchanged (60 RPM / TPM off) because they don't assume a quota.	`tests/test_rate_limiter.py` passes. ⚠️ **Restart the vision and extraction workers** to pick it up. Expected cost: ~6 % less peak Azure OpenAI throughput (≈ 3.4 instead of ≈ 3.6 one-page docs/min).
Frontend code-splitting	✅ Done	`src/App.tsx`: every page except Login is `React.lazy` with one `Suspense` ("Loading…") around the routes.	`npm run build` (tsc + vite) and `oxlint` are clean. Main bundle **1.39 MB → 359 kB** (gzip 111 kB). Pages are now 4–106 kB chunks; shared Nav chunk 110 kB. The one chunk still > 500 kB is pdf.js's prebuilt `pdf.worker.min` (631 kB). It is a separate file that is only fetched when a PDF is shown, and it can't be split further. ⚠️ Not click-tested in a browser yet.
Queue metrics → monitoring + alerting	✅ App side done	`log_queue_metrics` (every minute) now logs `queue_metrics {json}` per queue, plus a **WARNING** `queue_alert {json}` line when the oldest waiting task > `QUEUE_ALERT_OLDEST_WAITING_SECONDS` (new setting, default 300, 0 = off). Format in `processing-queues.md` › Monitoring.	New `tests/test_queue_metrics_log.py` (2 tests): JSON per queue, alert only past the threshold, 0 disables. ⚠️ **Still a deployment step:** ship worker logs to Azure Monitor / Log Analytics and create an alert rule on `queue_alert`. That is not built, because CLAUDE.md says no cloud config yet.

Files: `backend/.env`, `backend/.env.example`, `backend/app/core/config.py`, `backend/app/tasks/usage_tasks.py`, `backend/tests/test_queue_metrics_log.py`, `frontend/src/App.tsx`, `docs/processing-queues.md`, this report.

R2. Round 2 — production hardening (2026-10-03)

Safety first: code backup `scratchpad/pre-round2-backup.tgz`, DB dump `scratchpad/docauth-pre-round2.dump`; the migration was run on a restored copy before the real dev DB. New migration: `c9e3a1b5d7f4`.

R2.0 Status of the eight requested fixes

#	Fix	Status	Proof
1	No BYPASSRLS — policy-based platform access	✅ Done	No role has BYPASSRLS; 18 `platform_all` policies; **whole migration chain run as a NOSUPERUSER/CREATEROLE owner** on a fresh DB; RLS suite 15/15 on both that DB and the real one
2	Forensics on Linux, prefork, horizontally scaled	✅ Done	Containers run `--pool=prefork`; `--scale` verified (2 replicas); at the real quota forensics drains first (35 s vs vision 315 s)
3	Platform-admin audit rows platform-only, no dedup	✅ Done	Rows stored with `company_id NULL` (RLS hides them from every company); company views (Audit History, case timeline, report PDF) exclude them; dedup default 0; tests (re-verified 2026-10-03, see §5)

#	Fix	Status	Proof
4	Fair-share per company	✅ Done	Q-2 now passes: B waited 0.0–0.2 s (forensics) / 1.2 s (vision) vs A's median 30 s / 44 s
5	Global risk-rule templates seeding new companies	✅ Done	`risk_rule_templates` = Default Company's 38 rules; new company populated immediately; template edits don't touch existing companies; Platform › Rule templates screen
6	Idempotent check writes	✅ Done	Upserts + unique constraints; simulated worker crash and redelivery leave exactly one row
7	Connection pooling + PgBouncer (transaction mode)	✅ Done	PgBouncer service; NullPool behind it; leak test incl. negative control passes; arithmetic documented
8	Calibrate image-token estimate on real usage	✅ Done	96 real calls measured; estimator replaced; real-Azure burst 40 × 200, **0 × 429**

Tests: **452 passed** (twice, full suite including Postgres RLS, PgBouncer and live-Redis tests; nothing skipped).

R2.1 Decisions I made (🔶 change if you disagree)

- 🔶 **Platform-access rows are stored with** `company_id = NULL` (target company in `event_data`), not just filtered out of company views. That makes them invisible to company users at the **database** level (RLS), not only in the UI. Existing rows (0 on the dev DB) were moved by the migration — no row deleted, no content changed.
- 🔶 **Platform admins see platform-access rows in both** the platform-level log and a company's log view.
- 🔶 **The template = the current version of each of Default Company's rules: 38 rules.** (The "42" in the brief were 38 rules + 4 older versions of some of them.) Inactive template rules are not copied to new companies. Templates are plain editable rows (no version history of their own); every edit is in the platform audit log with old → new values.
- 🔶 **Fair-share bucket = 5** (`FAIR_SHARE_BUCKET_SIZE`): a company's tasks drop one priority level per 5 outstanding tasks on a queue, 10 levels. Priority is computed when a task is published (from live counters), not by re-sorting already-queued tasks — simpler and robust; the load tests show it meets Q-2.
- 🔶 **Duplicates collapsed by the migration** (from earlier retries): 6 check rows, 1 page hash, 1 assessment on the dev DB — newest check/hash kept; for assessments the one a report references (else the earliest) kept.
- 🔶 `max_tokens` **lowered** for the four fixed-shape LLM calls (entity match 4000→512, visual review 1500→640, signature detection 800→480, comparison 800→320) based on measured outputs (max 155 / 204 / 144 / 47). Classification stays 4,000 (its output grows with the document). +26 % throughput.
- 🔶 **Behind PgBouncer, processes keep no pool of their own** (NullPool). Without PgBouncer the default per-role pool stays 5 + 10, because a 16-thread worker would starve on a smaller one.

R2.2 Problems found during this round (🐞) and how they were fixed

🐞 Problem	Fix
The old token estimate UNDER-reserved the two most common calls (visual review 0.54×, classification 0.64× of real) — the limiter would have allowed more than the real quota at full load	Patch-based image estimate + measured chars/token (1.93, Arabic-dense) + 5 % margin; every call now ≥ 1.05× real
Flat per-image cost is wrong: full pages vs. signature crops differ ~5×	Image tokens from pixel size (32-px patches × 1.62)

🐞 Problem	Fix
Tasks held a DB transaction across Azure calls/downloads → in PgBouncer transaction mode each pinned a server connection; 32 threads exhausted the pool and **stalled the API** (uploads 93 s for 100, monitor timeouts)	Tasks commit before every slow step; task sessions keep attributes after commit; upload endpoint releases its transaction before the blob upload. After the fix: uploads 50 s, drain 92 s (was 120 s), no timeouts
PostgreSQL 16 rejects `ALTER ROLE … NOSUPERUSER/NOBYPASSRLS` from a non-superuser (found by the non-superuser migration test)	Attributes set on `CREATE ROLE` only; `ALTER` sets the password; migration asserts the roles are unprivileged
`nproc` **ignores container CPU limits** → 2 forensics replicas each started 4 processes on a 4-core host	`start-worker.sh` reads the cgroup quota (`cpu.max`); compose sets `cpus:` per replica
`platform_admin_access` rows were visible to company reviewers (by design in round 1)	Now platform-only (above)
Celery's Redis priorities split each queue into 10 lists — the queue monitor would have under-counted	Monitor sums all priority lists; oldest-waiting checks each

R2.3 ⚠️ What to know / do before production (new in this round)

- ⚠️ **Real throughput ceiling at the current quota is ≈ 3.6 one-page documents/minute (~210/hour)** — *lower* than round 1's "5–6/min", which rested on the under-estimate. Raise the Azure OpenAI TPM (e.g. to 500k–1M) for more. The calibrated limiter will keep using exactly what you configure.
- ⚠️ `process_document` **couples Document Intelligence with the Azure OpenAI classification call.** When the token limiter saturates, extraction threads wait on the OpenAI quota and a new document's OCR waits for a free thread (B waited 14–50 s on extraction while A barely did). Recommended next change: split classification into its own `vision_queue` task (persist OCR word boxes between them). **Still open** (not done in R3).
- ⚠️ **One simulated 429 in 123 calls** at full token-limiter saturation in the real-quota load test (a race between two independent sliding windows). The real SDK retries it; the real-Azure burst had none. For extra margin set the caps to **80 %** of quota instead of 85 %. ✅ **Done in R3.**
- ⚠️ **Run workers and the API on separate nodes.** On one 4-core box prefork forensics takes the CPU the API needs (by design it uses all of its allowance) — cap `FORENSICS_WORKER_CPUS` if they must share.
- ⚠️ **PgBouncer** `userlist.txt` **holds dev passwords** — use SCRAM secrets from a vault in production; keep Postgres `max_connections` ≥ 70 (default 100 is fine) with the shipped PgBouncer sizing.
- ⚠️ **Re-run** `python -m scripts.calibrate_llm_tokens` whenever the model, prompts or page render size change (costs a few cents).
- ⚠️ The **migration owner needs CREATEROLE** (no superuser). On Azure Flexible Server the admin user has it.

R2.4 New / changed files

Backend: `app/tasks/fairshare.py` , `app/services/check_store.py` , `app/models/risk_rule_template.py` , migration `c9e3a1b5d7f4_production_hardening.py` , `scripts/calibrate_llm_tokens.py` , `scripts/azure_limiter_burst.py` , tests `test_pgbouncer_rls.py` , `test_idempotent_tasks.py` , `test_rule_templates.py` , `test_fairshare.py` ; changed: migration `a7c1e9d3f5b2` (no BYPASSRLS, CREATE ROLE attributes), `tenant_access.py` , `audit.py` , `cases.py` , `platform.py` , `db/session.py` , `core/config.py` , `celery_app.py` , `usage_tasks.py` , `queue_monitor.py` , `llm_service.py` , `issuer_service.py` , `risk_rule_seed.py` , `risk_scoring_service.py` , `case_report_data.py` , `duplicate_check.py` , every task module (`open_task_session` , commit before slow steps, upserts), models (unique constraints), `start-worker.sh` , `loadtest/*` , `.env.example` . Frontend: `pages/PlatformRuwhen

Round-1 #	Status now
1 Azure OpenAI quota	Measured & calibrated; real ceiling ≈ 3.6 docs/min (see R2.3); still needs a TPM increase

Round-1 #	Status now
2 FIFO fairness	✅ Fair-share built and verified
3 Forensics bottleneck	✅ Linux prefork + scaling; fastest queue at the real quota
4 BYPASSRLS on Azure	✅ Removed everywhere
5 Dev DB passwords	⚠️ Still to set in production (now also PgBouncer `userlist.txt`)
6 Audit dedup	✅ 0 (every access logged), rows platform-only
7 Limiter fails open	🔶 Unchanged — no fail-open event occurred in any test
8 User manual / browser walkthrough	⚠️ Still open

0. Summary of the most important findings (round 1)

#	Finding	Severity	What to do
1	**Azure OpenAI quota caps the whole platform at ~5–6 documents/minute** (gpt-4.1-mini: 100 RPM, **100,000 TPM**). Tokens, not requests, are the limit.	⚠️ High	Raise TPM on the Azure deployment and/or lower `max_tokens` on LLM calls (see §6.1)
2	**Strict FIFO cannot let a small upload jump an existing big backlog** (Load test 2: Company B waited behind all 100 of A's docs).	🔶 Decision	Approve "fair-share per company" scheduling if wanted (see §6.2)
3	**Forensics queue is the throughput bottleneck** on the local 4-core Windows box.	⚠️ Medium	Run forensics on Linux with prefork, scale containers (see §6.3)
4	**Managed Azure PostgreSQL may not allow `BYPASSRLS`**, which the platform DB role uses.	⚠️ High (for Azure deploy)	Use the policy-based alternative in §6.5
5	Default DB role passwords (`fddt_app_local`, `fddt_platform_local`) are dev-only.	⚠️ High	Set long random `DATABASE_APP_PASSWORD` / `DATABASE_PLATFORM_PASSWORD`
6	Platform-admin audit rows are **de-duplicated for 5 minutes** for identical reads.	🔶 Decision	Set `PLATFORM_ACCESS_AUDIT_DEDUP_SECONDS=0` if every request must be logged
7	The rate limiter **fails open** if Redis is down (calls go through un-throttled).	🔶 Decision	Keep (pipeline keeps running) or change to fail-closed (§6.7)
8	`docs/user-manual.md` and some other docs still describe the old `admin` role. Frontend not click-tested in a browser.	⚠️ Medium	Update manual; do a UI walkthrough

1. Safety steps taken before changing anything

- The project is **not a git repository**, so a full code backup was made first: `scratchpad/pre-multitenant-backup.tgz` (backend, frontend, docs, compose, CLAUDE.md, README).
- The real dev database was dumped before migration: `scratchpad/docauth-pre-multitenant.dump` (`pg_restore` to roll back).

- The migration was first run on a **restored copy** (`docauth_mtest`), including a downgrade → upgrade round trip, and the RLS tests were run there before touching the real DB. The copy was dropped afterwards.

> ⚠️ **PRODUCTION ACTION:** put the project under git before further work, and take a DB backup before running `alembic upgrade head` on any shared/production database.

---

2. Role model (replaces the old one)

Role	Company	Permissions
`user`	exactly one	submit cases/documents, track own cases
`reviewer_l1`	exactly one	review company queue, approve/reject/escalate (L1)
`reviewer_l2`	exactly one	L1 + escalated cases + **own company's** Issuer Registry & Risk Rules
`platform_admin`	**none**	create companies, create **all** users (company + role), billing dashboard, queue monitor, read-only audited support access to any company

Decisions:

- 🔶 `platform_admin` **is NOT on the rank ladder.** It never inherits reviewer powers: it **cannot approve, reject, escalate, upload, create cases or generate reports**. Spec said "full read access for support" — I interpreted that as read-only. (If support needs to act on cases, add it explicitly.)
- 🔶 **Platform admins CAN edit a company's Issuer Registry and Risk Rules** (with `company_id`), and every such change is audited in that company's log. Reason: support/onboarding needs it. Remove if not wanted.
- **No company-level user management** at all — `Settings › Users` is platform-admin only (as specified).
- The old `admin` role was migrated to `platform_admin`.
- Moving a user to another company, or changing them across the platform/company boundary, **invalidates their existing JWT** (they must log in again). Plain role changes inside a company take effect on the next request (role is always re-read from the DB).
- Suspending a company (`is_active=false`) blocks login and existing tokens for all its users.

---

3. Database changes

Migration: `backend/app/db/migrations/versions/a7c1e9d3f5b2_multi_tenancy.py`

- New table `companies` (id, name unique, is_active, created_at, updated_at).
- New table `company_usage_stats` (company_id, usage_date, cases_created, documents_uploaded, files_stored, storage_bytes) — one row per company per UTC day.
- `company_id` **added to every tenant table:** users, cases, documents, document_checks, document_page_hashes, cross_document_findings, case_actions, case_risk_assessments, risk_scores, risk_rules, risk_settings, issuer_registry, signature_references, signature_matches, case_reports, audit_log.
  - NOT NULL everywhere except `users` (NULL = platform admin, enforced by CHECK constraint `(role='platform_admin') = (company_id IS NULL)`) and `audit_log` (NULL = platform-level event).

- 🔶 The spec listed `extracted_fields` and `duplicate_matches` tables — **they don't exist**. Extracted fields are a column on `documents`; duplicate hashes live in `document_page_hashes`. Both are covered by their table's `company_id`. I also added `company_id` to tables the spec didn't list but which hold company data (case_actions, risk_settings, cross_document_findings, risk_scores, document_page_hashes).
- Risk rules unique key changed to `(company_id, rule_id, version)`; `risk_settings` is one row per company.
- `user_role` enum gained `platform_admin` (the old `admin` value stays in the Postgres enum because Postgres can't drop enum values; nothing uses it).

Data migration (done on your real dev DB):

- One company **"Default Company"** created (`DEFAULT_COMPANY_NAME`); all 42 cases, 78 documents, 514 checks, 750 audit rows, 42 rules, 8 issuers, 14 reports etc. assigned to it.
- `admin@example.com` → `platform_admin`, no company. 4 reviewers and 2 users kept their roles.
- Verified **byte-for-byte**: every case's id, number, status, risk tier, document count, size and file hashes are identical before/after.
- Usage counters back-filled from existing data.

🐞 **`seed.py` resets the `admin@example.com` password to `ChangeMe123!`** every time it runs (it always did; `start-api.sh` runs it on every start).

> ⚠️ **PRODUCTION ACTION:** set `SEED_ADMIN_EMAIL` / `SEED_ADMIN_PASSWORD`, or remove `seed.py` from `start-api.sh` in production.

---

4. Tenant isolation — two layers

4.1 Application layer

Files: `app/db/tenancy.py`, `app/db/session.py`, `app/api/auth.py`, `app/api/tenant_access.py`, `app/api/case_access.py`

- JWT now carries `role`, `company_id`, `is_platform_admin`; the token is re-validated against the DB on every request.
- Every endpoint has **two mandatory checks**: role check + tenant check (request session bound to the caller's company by the `get_tenant_db` dependency — a route can't forget it).
- Every DB session is **unbound / company / platform**:
  - *unbound* → refuses any query or write on tenant tables (raises an error);
  - *company* → every ORM query on a tenant table gets `company_id = X` added automatically, inserts are stamped with X, a row of another company is refused at flush;
  - *platform* → explicit bypass path.
- Service functions also take `company_id` explicitly (`load_current_rules`, `get_risk_settings`, `match_issuer`, `run_duplicate_check`, `score_case`, `pipeline_status`, `latest_assessment`, `get_case_flags` …).
- 🔶 **Cross-company id → 404** (not 403), so ids can't be probed. Matches the existing submitter rule.
- Celery tasks now receive `(document_id, company_id)` and open a company-bound session. Old queued messages without company are resolved once via the platform session.

🐞 **PROBLEM FOUND — identity-map leak:** `session.get()` returns cached objects without running a query, so the automatic filter couldn't see it. If one session had loaded company B rows and was then re-bound to A, `get()` could return a B row. **Fixed:** binding to a company now evicts other companies' objects from the session cache (test:

`test_company_session_never_returns_another_companys_rows_even_without_a_filter`). Production sessions are bound once per request anyway, and RLS would still block it.

4.2 Database layer — PostgreSQL Row-Level Security

- Two new login roles created by the migration:
  - `fddt_app` — NOSUPERUSER, **NOBYPASSRLS** — used by every company request and every pipeline task.
  - `fddt_platform` — **BYPASSRLS** — used only for login/identity lookup, platform-admin screens, usage reconciliation.
  - The owner role (`DATABASE_URL`, `docauth`) is now used **only** by Alembic and `seed.py`.
- RLS enabled on all tenant tables + `companies` + `company_usage_stats`, policy `company_id = fddt_current_company_id()` (reads the transaction-local `app.current_company_id`). No company set ⇒ **zero rows**.
- The app sets the company at the start of **every transaction** (`set_config(..., is_local=true)`), so a pooled connection never leaks one request's company into the next.
- Least-privilege grants, which also make these tables **append-only in the database itself**: `audit_log`, `risk_rules`, `case_risk_assessments`, `case_reports` (SELECT + INSERT only). `users` is SELECT-only for `fddt_app`.
- 🔶 No default privileges: any **future tenant table is unreachable** by the app until its migration adds RLS + grants. (Forces developers to think about isolation; see `enable_tenant_rls()` in the migration.)

> ⚠️ **PRODUCTION ACTION:** set strong `DATABASE_APP_PASSWORD` and `DATABASE_PLATFORM_PASSWORD` (defaults `fddt_app_local` / `fddt_platform_local` are for local dev only).

> ⚠️ **PRODUCTION ACTION:** the app now opens **two connection pools** (app + platform). Size Postgres `max_connections` accordingly (each engine uses SQLAlchemy defaults: 5 + 10 overflow per process, times API workers + Celery worker processes).

---

5. Platform admin support access (audited, platform-only)

> **Updated in round 2 (R2.0 #3).** In round 1 these rows went into the company's own audit log and the company's reviewers could see them. That is **no longer the case**. Below is the current behaviour, verified on 2026-10-03.

- Opening any company's case / case list / audit log / settings writes a `platform_admin_access` row: "*Platform admin ops@x.com viewed Company B / Case CASE-1A2B3C4D*", with method and path.
- **Only platform admins can see these rows. No company user (User, Reviewer L1, Reviewer L2) can see them, in any view.** Company users have no way to tell that platform access happened.
  - **Database layer:** the row is written through the platform session with `company_id = NULL`. The target company goes in `event_data.company_id`. Row-Level Security hides NULL-company rows from the `fddt_app` role that every company request uses. So even a raw SQL query with no filter cannot return them (`app/api/tenant_access.py`).
  - **Application layer (defence in depth):** every company-facing read also excludes the event type explicitly. That covers Audit History (`app/api/audit.py` `_scope_filter`), its event-type filter list, the case timeline (`app/api/cases.py`, GET `/cases/{id}/audit-log`) and the per-case forensic report PDF (`app/services/case_report_data.py` `audit_rows`).
  - **Platform admins** see the rows in the platform-level log, and in a company's log view (filtered by `event_data.company_id`). Opening a company's log is itself audited.

- Platform admins never get a mixed cross-company case list. The queue takes one company at a time (company picker in the UI).
- **No de-duplication:** every access is logged (`PLATFORM_ACCESS_AUDIT_DEDUP_SECONDS=0` , the default). A value > 0 would collapse identical reads by the same admin within that window. *(Round 1 used 300 s.)*
- The platform-level audit log (company created, user created, usage reconciled, platform access) is a separate view (checkbox in Audit History for platform admins).
- Tests that prove it:
  - `tests/test_multi_tenancy.py::test_company_users_never_see_platform_admin_access` covers every company role: Audit History, the `event_type` filter, the event-type list and the case timeline.
  - `tests/test_rls_postgres.py::test_platform_admin_access_rows_are_invisible_to_the_app_role` checks raw SQL as `fddt_app` with no event-type filter.
  - `tests/test_multi_tenancy.py::test_platform_admin_access_is_read_only` checks the platform-admin side.
  - Re-run 2026-10-03: all three pass (the RLS one against the real Postgres).

---

6. Bottlenecks found and how to solve them

6.1 ⚠️ Azure OpenAI quota (gpt-4.1-mini) — the real throughput ceiling

- I read the deployment's real limits from the `x-ratelimit-*` headers (one ~10-token test call): **`x-ratelimit-limit-requests = 100` / min, `x-ratelimit-limit-tokens = 100,000` / min.**
- Azure charges each request **prompt tokens + the full `max_tokens` reservation** against TPM up front. Our calls reserve: classification **4,000**, visual review **1,500 (run twice per page)**, signature detection/comparison **800**, plus ~1,100 tokens per page image.
- **≈ 13–15k tokens per one-page document ⇒ ~5–6 documents/minute (~350/hour) for the whole platform**, no matter how many workers you add.
- Implemented: a **token-weighted global limiter** in addition to the request limiter, set in `backend/.env` to 85%: `AZURE_OPENAI_MAX_REQUESTS_PER_MINUTE=85` , `AZURE_OPENAI_MAX_TOKENS_PER_MINUTE=85000` . Without it, an RPM-only limiter would have caused 429 storms at load.

> ⚠️ **PRODUCTION ACTION (pick one or more):**
>
> 1. **Increase the deployment's TPM quota** in Azure (e.g. 500k–1M TPM) and update the two env vars to ~80% of the new values (R3).
> 2. **Lower `max_tokens`** where outputs are small: classification 4000 → ~1500, visual review 1500 → ~800 (check real response sizes first). This alone can roughly double throughput.
> 3. Consider a **Provisioned/PTU** deployment or a second deployment/region for vision calls.
> 4. The per-image estimate (`_IMAGE_TOKEN_ESTIMATE = 1100` in `llm_service.py`) is an approximation; tune it from real `usage` data.

6.2 🔶 Fairness: strict FIFO vs. "small upload jumps big backlog"

- Every document step is its own task, all companies share FIFO queues, no priority (as specified).
- **Load test 1 passed:** 27 concurrent uploads from 3 companies were processed interleaved (Kendall τ = 0.96 vs submission order).
- **Load test 2 did NOT meet the expectation:** Company B's single document, submitted right after A's 100-document upload, started after **all 100** of A's on every queue (waited 15 s extraction, 70 s vision, 144 s forensics). A FIFO queue cannot let a later job jump work that is already queued.

- **Solution (not built — needs your approval):** *fair-share per company* — order each new task by how much work its company already has waiting (e.g. Celery Redis priority computed from the company's queued count, or one queue per company drained round-robin). No company gets permanent priority; each gets an equal share. It does reorder the queue, which you said should be your decision.

6.3 ⚠️ Forensics queue is CPU-bound

- In the load test the forensics queue reached **649 waiting / oldest 264 s** while the Azure queues idled. Cause: 4 threads on a 4-core **Windows** machine (Windows can't use Celery's prefork pool; threads share the Python GIL).
- **Solution:** run forensics workers on **Linux** with `start-worker.sh forensics` (prefork, one process per core) and scale containers: `docker compose --profile workers up -d --scale forensics-worker=N`. This queue has no rate limit — more cores = more throughput.

6.4 Celery's built-in `rate_limit` is per worker, not global

- Celery's own `rate_limit` stops being a ceiling once you run 2+ worker containers.
- **Solved:** Redis-backed sliding-window limiter (`app/services/rate_limiter.py`) applied at **each Azure API call** (not per task), shared by all processes. On a 429 from Azure, **all** workers pause that service for the Retry-After time.
- Measured: Document Intelligence peaked at exactly **10 calls/s** (cap 10, Azure quota 15), **0 × 429**.

6.5 ⚠️ BYPASSRLS on managed Azure PostgreSQL

- The migration creates roles and grants `BYPASSRLS` ; that needs superuser. Local docker Postgres allows it. **Azure Database for PostgreSQL Flexible Server may not let you grant BYPASSRLS.**
- **Solution if it's refused:** create `fddt_platform` without BYPASSRLS and add a policy that grants it all rows instead:

  ```
  CREATE POLICY platform_all ON <table> FOR ALL TO fddt_platform USING (true) WITH CHECK (true);
  ```

  (repeat for every tenant table). Same effect, no superuser needed. Create the two roles once with the Azure admin account before running the migration.

6.6 Monitoring accuracy

- Queue monitor (Platform › Processing queues, `/platform/queues`) tracked the load spike correctly; mean difference vs. direct Redis count 1.7 (samples taken moments apart while draining).
- "Consumers online" uses Celery remote control with a 1 s timeout — can show "unknown" under heavy load.
- A beat task logs one `queue_metrics` line per queue every minute.

> ⚠️ **PRODUCTION ACTION:** ship those log lines to Azure Monitor / Log Analytics and alert on `oldest_waiting_seconds` (e.g. > 300 s). *(R3: the app now logs JSON `queue_metrics` lines and a WARNING `queue_alert` line past `QUEUE_ALERT_OLDEST_WAITING_SECONDS`. The log shipping and the alert rule are still to set up at deployment.)*

6.7 🔶 Rate limiter fails open

- If Redis is unreachable, the limiter logs a warning and lets calls through (SDK retries are then the only protection), so the pipeline doesn't stop. If you'd rather stop than risk 429s, change `_client()` in `rate_limiter.py` to raise instead.

6.8 `task_acks_late` side effects

- Enabled for fair dispatch (worker holds only the task it runs). Side effect: if a worker crashes mid-task the task is **re-run**, which can write a duplicate check row (scoring always uses the latest row, so the score is correct). Redis `visibility_timeout` (default 1 h) must stay above the longest task.

7. Billing & usage dashboard

- Platform › Billing & usage: **one row per company** — cases created, documents uploaded, files stored (documents + reports), storage added in the period, total storage/files all-time; periods: this month, last month, last 30 days, this year, all time, custom range (UTC days).
- Fast counters in `company_usage_stats`, incremented with an atomic upsert **in the same transaction** as the upload/case/report (a failed upload never inflates them).
- **Nightly reconciliation** (Celery beat, 02:00 UTC, `USAGE_RECONCILIATION_HOUR_UTC`) recomputes from source tables and fixes drift; "Reconcile now" button does it on demand. Corrections are audited.
- Storage = exact bytes recorded at write time. 🔶 Signature crop images are **not counted** (tiny, derived).
- 🔶 Nothing deletes cases/documents today. **If deletion is added, it must decrement counters** (or the nightly job corrects it next day).
- Verified in load test: Loadtest A = 10 cases / 109 docs, B = 10 / 10, C = 9 / 9 — exactly what was uploaded.

> ⚠️ **PRODUCTION ACTION:** run exactly **one** `celery beat` process (or the nightly job runs twice).

8. Blob storage

- New layout: `companies/{company_id}/cases/{case_id}/documents|reports|signatures/...`
- `ensure_company_blob()` refuses to sign a URL under another company's prefix.
- 🔶 **Existing files were NOT moved** (originals are immutable). Old paths still work; their DB rows are company-scoped. If you want a uniform layout, a one-off copy job can be written.
- 🐞 Windows 260-char path limit hit by the **load-test stub** (deep nested path) — fixed in the stub (hashed file names). Not an issue for real Azure Blob (1024-char names).

9. Processing queues — configuration

Queue	Tasks	Limit	Workers env
`extraction_queue`	OCR (Doc Intelligence) + classification	10 calls/s global	`EXTRACTION_WORKER_CONCURRENCY` (4)
`vision_queue`	visual review, signature detection/comparison, issuer check	85 RPM + 85k TPM global (80 / 80k since R3)	`VISION_WORKER_CONCURRENCY` (4)
`forensics_queue` (+ legacy `celery`)	metadata, ELA, copy-move, duplicates, cross-doc, scoring	none	`FORENSICS_WORKER_CONCURRENCY` (0 = per core)

- Start: `start-worker.sh extraction|vision|forensics`, `start-beat.sh`; Docker: `--profile workers`.
- `worker_prefetch_multiplier=1`, `task_acks_late=True`.
- Forensics worker also drains the old `celery` queue so pre-upgrade messages aren't stranded.
- 🔶 No per-company/user priority (as specified).

10. Frontend changes

- Roles: `platform_admin` added, `admin` removed; `hasRank` excludes platform admin.

- **Company picker** (platform admins) on Case queue, Dashboard, Audit History, Issuer Registry, Risk Rules — choice remembered per browser; amber "support access — audited" notice.
- Case detail: read-only **"Support view"** banner for platform admins; no actions, no report export.
- Nav shows company name + role; "New upload" hidden for platform admins.
- Settings: Users tab platform-only, with Company column, company filter, move-user, company selector in Add user.
- New pages: **Platform › Companies** (create/suspend), **Billing & usage**, **Processing queues**.
- Production build and type-check pass.

> ⚠️ **PRODUCTION ACTION:** do a full click-through of every role in a browser (not done — only build/type-checked). Bundle is 1.39 MB (Vite warns > 500 kB) — add route-level code-splitting. *(✅ R3: main bundle now 359 kB.)*

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11. Problems found during testing (and fixes)

🐞 Problem	Fix
`session.get()` could bypass the tenant filter via the identity map	Evict other companies' objects on bind (§4.1)
SQLite `CAST(.. AS DATE)` returned a number → reconciliation crash in tests	Use `date()` on SQLite; `timezone('UTC', ...)::date` on Postgres
Load-test upload 500s	Stub's Windows path length; switched to hashed filenames
Rate-limit test flaked by ~10 ms thread jitter	Limiter is correct by admission time; test window allows jitter
Old tests assumed one global `admin`	~30 test files updated; new tenant fixtures

---

12. Tests

- **430 passing**: 415 app-layer (incl. new `tests/test_multi_tenancy.py`), **11 PostgreSQL RLS** (`tests/test_rls_postgres.py`, raw SQL with **no** company filter, run against the real dev DB), **4 rate-limiter** tests against live Redis.
- RLS tests need `FDDT_RLS_TEST_DATABASE_URL`; limiter tests skip if Redis isn't reachable.
- Load test harness: `backend/loadtest/` (real API/workers/Postgres/Redis; only Azure + Blob stubbed).

Spec test	Result
MT-1 create companies & users	✅
MT-2 A's L2 sees A; B by id → 404	✅
MT-3 L2 can't reach Settings › Users	✅ (403)
MT-4 platform admin opens B case, audited	✅
MT-5 billing separated & matches	✅
MT-6 RLS alone blocks cross-tenant	✅
MT-7 pre-migration data intact	✅ (verified on real DB)

Spec test	Result
Q-1 interleaved processing	✅ τ = 0.96
Q-2 small upload near front	❌ by design of strict FIFO — see §6.2
Q-3 throttled, no 429 storms	✅ 0 × 429
Q-4 monitor reflects reality	✅

13. New / changed configuration (`backend/.env`)

```
DATABASE_APP_USER=fddt_app
DATABASE_APP_PASSWORD=...            # ⚠️ change
DATABASE_PLATFORM_USER=fddt_platform
DATABASE_PLATFORM_PASSWORD=...       # ⚠️ change
DEFAULT_COMPANY_NAME=Default Company
PLATFORM_ACCESS_AUDIT_DEDUP_SECONDS=0    # round 2: every access logged (was 300 in round 1)
AZURE_DOCUMENT_INTELLIGENCE_MAX_CALLS_PER_SECOND=10
AZURE_OPENAI_MAX_REQUESTS_PER_MINUTE=80     # measured quota 100 (85 until R3)
AZURE_OPENAI_MAX_TOKENS_PER_MINUTE=80000    # measured quota 100,000 (85000 until R3)
QUEUE_ALERT_OLDEST_WAITING_SECONDS=300      # R3: WARNING queue_alert log line; 0 = off
EXTRACTION_WORKER_CONCURRENCY=4
VISION_WORKER_CONCURRENCY=4
FORENSICS_WORKER_CONCURRENCY=0
USAGE_RECONCILIATION_HOUR_UTC=2
```

(The three quota lines were appended to your local `.env`; all are documented in `.env.example`.)

14. Production-readiness checklist

Must do before go-live

- [ ] Put code in git; back up DB before migrating.
- [ ] Strong passwords for `fddt_app` / `fddt_platform`; strong `JWT_SECRET_KEY`; secrets in Key Vault.
- [ ] On Azure Postgres: pre-create roles; use the policy alternative if BYPASSRLS is refused (§6.5).
- [ ] Raise Azure OpenAI TPM and/or lower `max_tokens` (§6.1); re-set the two env limits.
- [ ] Remove `seed.py` from startup or set `SEED_ADMIN_*` (it resets the admin password).
- [ ] Run forensics workers on Linux/prefork and scale them (§6.3); exactly one beat process.
- [ ] Alerts on queue `oldest_waiting_seconds` and on Azure 429 counters.
- [ ] Browser walkthrough of every role; update `docs/user-manual.md` (still says `admin`).
- [ ] Remove the load-test companies ("Loadtest A/B/C …") from the dev DB if it will be reused.

Your decisions

- [ ] Fair-share scheduling per company? (§6.2)
- [ ] Platform admin read-only on cases — OK? Platform admin editing company settings — OK?
- [x] Audit de-dup window 300 s or 0? → **0** (round 2); platform-access rows are platform-only (§5)
- [ ] Rate limiter fail-open or fail-closed?
- [ ] Move legacy blobs to the new company-prefixed layout?

Not built (out of scope / future)

- Deletion of cases/documents (and counter decrement), per-company priority tiers, password reset/MFA, company self-service user management (explicitly excluded), cross-company duplicate detection (deliberately excluded for privacy).

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15. Files added / changed

Added (backend): `app/models/company.py`, `app/models/company_usage_stats.py`, `app/db/tenancy.py`, `app/api/tenant_access.py`, `app/api/platform.py`, `app/services/usage_service.py`, `app/services/rate_limiter.py`, `app/services/queue_monitor.py`, `app/tasks/tenant.py`, `app/tasks/usage_tasks.py`, migration `a7c1e9d3f5b2_multi_tenancy.py`, `start-beat.sh`, `loadtest/*`, `tests/test_multi_tenancy.py`, `tests/test_rls_postgres.py`, `tests/test_rate_limiter.py`.

Changed (backend): all models (company_id), `core/config.py`, `db/session.py`, `api/auth.py`, `api/cases.py`, `api/documents.py`, `api/case_actions.py`, `api/case_reports.py`, `api/signatures.py`, `api/settings.py`, `api/audit.py`, `main.py`, risk scoring / flags / issuer / duplicate / report / workflow / audit / storage / signature / OCR / LLM services, every Celery task, `celery_app.py`, `seed.py`, `start-worker.sh`, `Dockerfile`, `.env.example`, schemas, most tests.

Frontend: added `api/platform.ts`, `hooks/useActingCompany.tsx`, `components/CompanyPicker.tsx`, `components/platform/PlatformShell.tsx`, `pages/PlatformCompaniesPage.tsx`, `PlatformUsagePage.tsx`, `PlatformQueuesPage.tsx`; changed types, routes, nav, settings pages, case queue/detail, dashboard, audit page.

Repo root/docs: `docker-compose.yml` (worker services), `CLAUDE.md`, `README.md`, `docs/multi-tenancy.md`, `docs/processing-queues.md`, `docs/README.md`, this report.